# Project S'mores

Toyota Sienna Modular Camper Conversion — Full Build Plan

Vehicle measurements surveyed and verified Aug 1, 2026. Dimensions in this document drive real cuts — re-measure your own vehicle before building.

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- **Damage to the vehicle** — trim, wiring, floor pan, upholstery, seat mechanisms and finish. Modifications may void some or all of your manufacturer warranty, and may affect insurance coverage, roadworthiness, inspection status, or resale value.
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- **Cargo becoming a projectile.** Furniture, appliances, batteries and gear that are not restrained to a genuine crash-rated standard can move, break loose, or kill occupants in a collision or rollover. Nothing in this plan is crash-rated, and no statement about straps, anchors, or load paths here should be read as a safety rating.
- **Electrical hazard.** Battery, inverter, and DC wiring work risks short circuit, arc flash, burns, lithium battery fire, and electrocution. Sizing, fusing, and installation should be checked by a qualified person.
- **Cooking, heat, and carbon monoxide.** Cooking or running any fuel-burning appliance in or near an enclosed vehicle risks fire, burns, and carbon monoxide poisoning, which can be fatal while you sleep. Ventilate, and fit working CO and smoke alarms.
- **Weight, handling and load limits.** Added weight affects braking, handling, tire load, and your vehicle's GVWR and axle ratings. Verify all figures against your own vehicle's door-jamb placard.
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**Verify every dimension on your own vehicle before cutting anything.** The measurements here were taken on one specific van and may not match yours, and errors — in measurement, in transcription, or in the drawings — are possible throughout.

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# Project S'mores

Full Build Plan — Modular Lift-Out Design

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# Project S'mores — Full Build Plan

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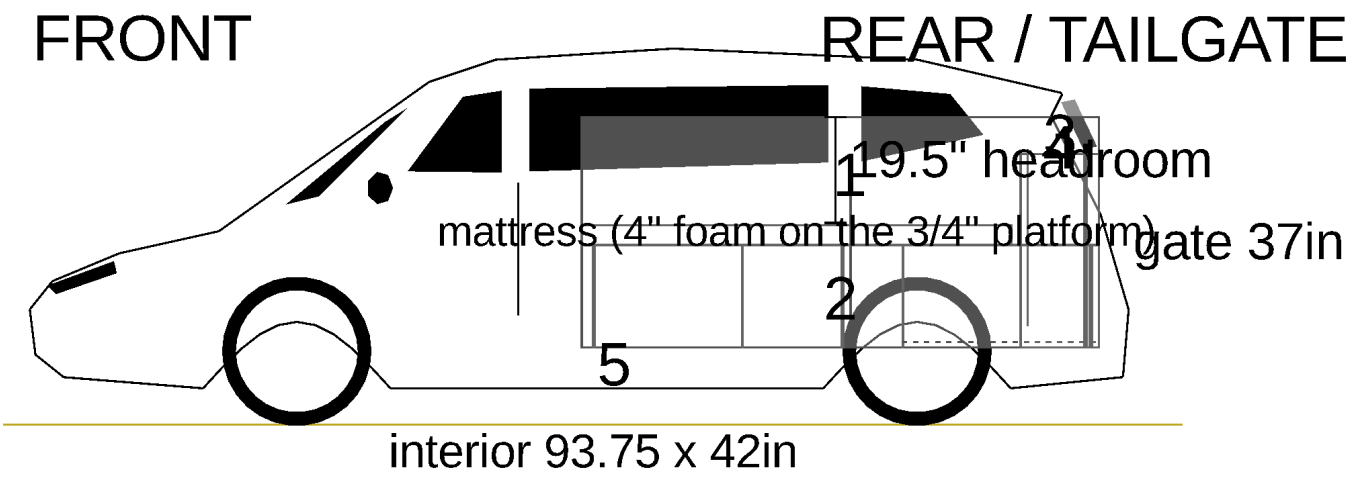
Two-person setup, 2nd row seats **removed entirely** (3rd row folded flat; seat removal/reinstall procedure incl. the SRS airbag emulators: Section 9). Layout, tailgate to front seats: **Kitchen (Panel C's fridge/kitchen void) → Rear pantry (prefab drawers on Panel C's deck) → Bed (Panel B + Panel A) → front seats.** The rear pantry is **one 27-gallon Tough Box tote** on the tailgate end of Panel C's deck, beside the microwave, the pots inside it; **one cam strap over both** to two anchors on Panel C's side rails, no cleats — nothing built, and the tote lifts straight out (Component 1; changed from an IRIS drawer cluster 30 Oct 2026). Panel A and B have no top of their own (capped by the three-piece bed frame — Panel A's section is screwed down, and Panel B's top is two centerline halves that lift out individually), while Panel C keeps its fixed top — **recessed flush into its rail plane** (the deck ply drops between the rails onto cleats; Panels A/B's rails sit at 16.25" vs Panel C's 17", so the platform on their rails lands in the same flush 18.5" plane — ¾" more headroom; the legs themselves are 1.5" taller than those figures because they are lapped past the rails, see Section 3) — forming one continuous deck with 79¼" of sleeping length for the HEST Dually Long mattress (78" x 50" x 4") (Panel A sits flush with the front seatbacks). Panel B is a bare-frame deep-storage bay — the side doors don't reach it, so it has no drawers and loads from above by lifting out either half of its **two-piece bed top**; Panel A holds a right drawer (EcoFlow DELTA 3) plus a left open-storage bay (EcoFlow WAVE 3, no drawer box); Panel C's under-deck void instead houses a real BougeRV compressor fridge and a real JAGAHAHA slide-out camp kitchen, both on heavy-duty slides pulling straight out the open tailgate — bought products, not built from plywood — plus a shallow slide-out **kitchen drawer** hung under the deck in the dead air above the kitchen unit. The fridge and its cooling fans run off the EcoFlow DELTA 3 stack (Panel A) via their own DC line, so they keep running when people are away from the van regardless of ignition state; the DELTA 3 itself gets AC-charged from the front console outlet while driving, while the induction cooktop and Power strip 1 plug into the van's REAR AC outlet (back passenger area — verified to exist), right beside them: two short cords, no seam crossings.

**Vehicle measurements** — the van was surveyed on Aug 1 2026 and every dimension in this plan comes from that survey. The full record (V1–V10 interior/opening measurements, F1–F8 rear-floor survey, and the few items still open) is **Appendix A**.

## Renders

Parametric 3D model: `platform.scad` (dimensions in `params.scad` — edit one file to regenerate every view via `./render.sh`).

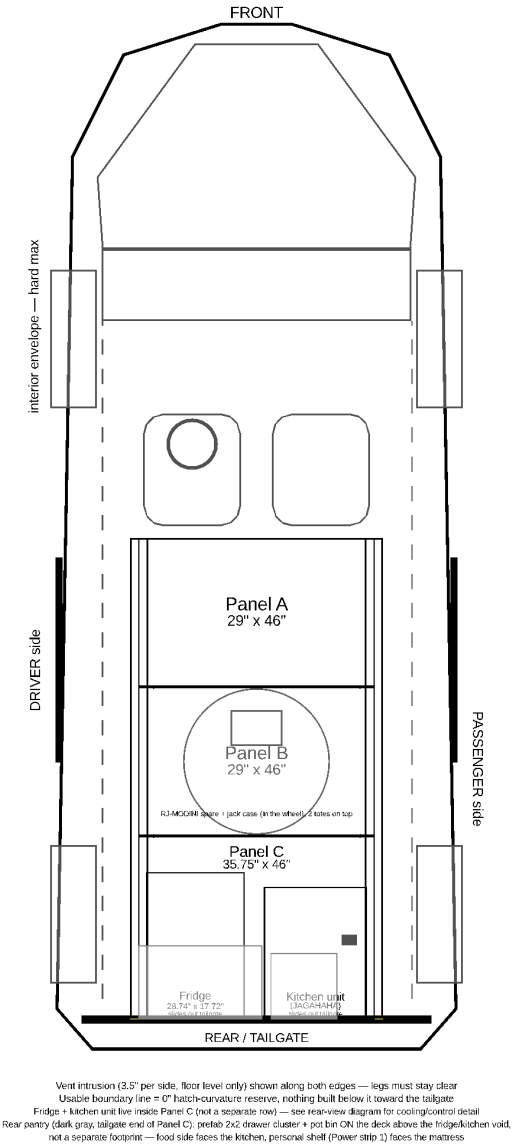
### Whole-Vehicle Overview




*Side profile of the van, front at left and tailgate at right, with the cargo envelope, liftgate opening and platform drawn to scale inside an illustrative body outline. Numbered callouts: 1 sitting headroom, mattress top to ceiling. 2 Panels A / B / C with the mattress above. 3 the rear pantry (prefab drawer cluster), 4 sitting on Panel C's deck. 5 the fridge and kitchen unit, hidden under Panel C's deck. Body outline illustrative; the interior and platform dimensions are to scale. The names were moved off the sheet — its geometry is already 258 units long, so its own labels were holding every one of them to ~5pt on the page.*

### *Overhead Floorplan*

***The anchor board** under Panel C (Section 8, and the Anchor Board shop sheet at the end of this PDF): three ¾" plywood strips tied by two steel rungs, carrying the fridge slides' risers and the kitchen unit's straps, and strapped forward and down to the 3rd-row seats' latch hoops. No holes in the van.*




*Top-down view: Panel A/B/C plus the fridge and kitchen unit living inside Panel C's footprint, drawn to scale inside the interior envelope. Front of the vehicle is at the top, tailgate at the bottom. Seam bumpers, alignment pins, and grab-handle positions are all marked.*

### *Anchor Board — Overview*

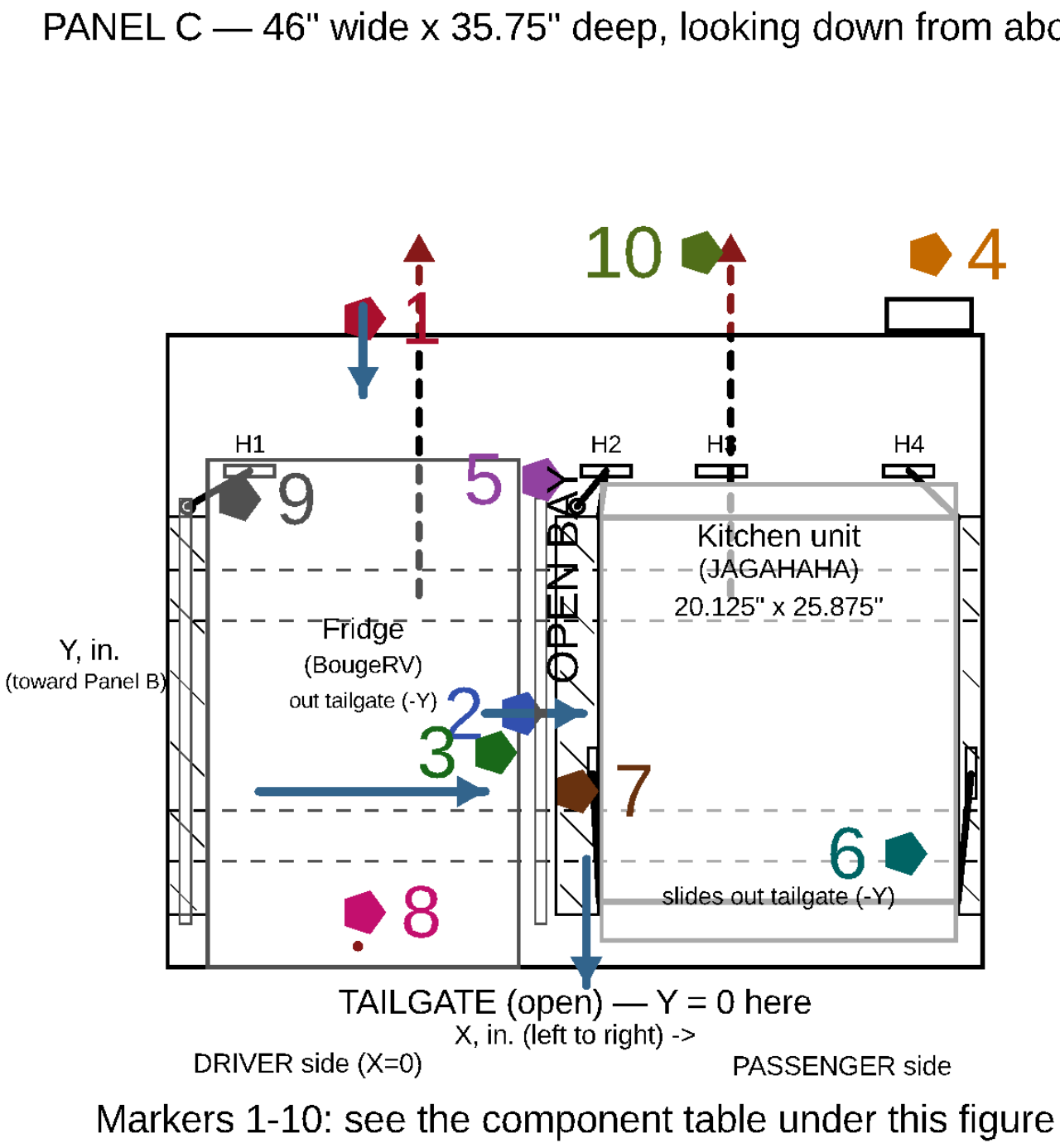
*The anchor board is the no-drill chassis under Panel C that the fridge slides and the kitchen unit's straps fasten to.* ***Its shop drawing is the Anchor Board sheet*** *(in the shop-drawing pages at the end of this PDF): plan, section, rung detail, T-nuts, cut list, hardware and build order.*

- ***Three ¾" plywood strips****, each one piece, running from* ***3" to 25½" from the hatch****: the driver strip (2½" wide, on Panel C's driver line), the middle strip (about 2½", 21⅞"–24⅜" from the driver line) and the passenger strip (1½", 44½"–46"). Each lies on a ⅛" neoprene strip.*
- ***Two steel rungs****, 3" × 3/16" flat bar 46" long, set flush into grooves in the strips' undersides at 6"–9" and 19½"–22½" from the hatch, bedded in E6000 and screwed up into each strip (holes countersunk with a ⅝" countersink bit so the heads sit flush).* ***Assembled on the bench 30 Oct 2026; van fit pending.*** *They tie the strips into one board and pass under the fridge tray (which hangs ⅜" over the seat surface) and the kitchen unit. There is* ***no crossbar****: from 26" forward the van's side trim closes to 42", and the fridge tray fills the space between the driver and middle strips for its whole length.*
- ***The fridge slides' riser angles*** *bolt down into 8 T-nuts in the driver and middle strips (Fridge Slides and Riser Angles sheets).*
- ***Anchors: the 3rd-row seats' latch hoops.*** *Four steel hoops sit in the headrest trough 28" from the hatch, their ⅝" bars 1⅞" below the folded-seat surface with the seat loaded (2⅜" bare), centered 4¾", 24¾", 31¼" and 41¾" from the driver line (H1–H4). A short chain runs forward and down from each riser's front end to* ***H1*** *(driver) and* ***H2****: a forged ¼-20 eye bolt hanging down through each riser's front end, a ¼" quick link, a short length of ¼" chain wrapped once around the hoop's ⅝" bar, closed by a second quick link. Turning the eye bolt's nut draws the chain tight, so a hard stop pulls against it and it also holds the board down. The run is only about 5", too short for any ratchet strap.*
- ***Kitchen unit:*** *a front strap in its front notches to* ***H2*** *and* ***H4****, and a rear strap in its rear notches to a short ¼-20 eye bolt in the kitchen riser's second T-nut, 11" from the hatch (the 2¼" middle strip is too narrow outboard of the riser for a footman loop) and a footman loop on the passenger strip.*
- ***Optional backup:*** *ratchet straps from the board forward to the three 3rd-row striker loops on the 2nd-row floor.*

*Nothing drills into the van, and nothing joins plywood to plywood.*

### *Fridge Installation Detail*

*The rear view above is too small-scale to label the fridge's 2 cooling fans and temperature sensor clearly. This top-down detail of Panel C zooms in on just that area, with numbered markers and a coordinate list (see Section 2 for the coordinate system and the full component coordinate table):*



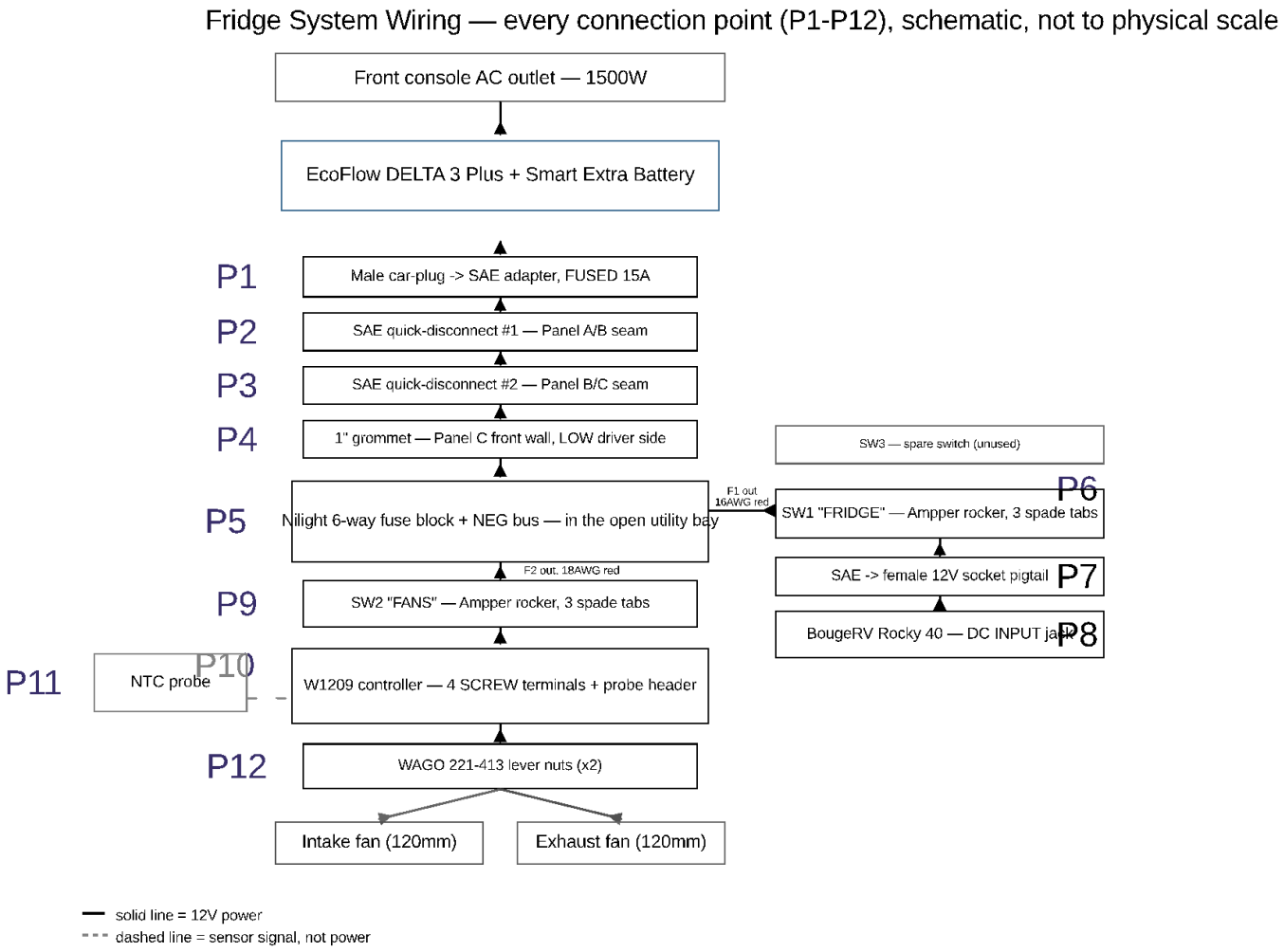

*Fridge install detail*

***Marker key.*** *Markers 1–10 on the sheet index this table.*

| # | Component | Position | Fastening |
|---|---|---|---|
| 1 | *Intake fan (120mm) — blows IN* | *X=11.11, Y=35.75 (Panel C's FRONT wall), Z=8.8* | *4× M4×20 machine screws over the wall's fan hole (see the Panel C Front Wall render)* |
| 2 | *Exhaust fan (120mm) — blows INTO the open utility bay* | *X=19.72, Y=14.4, Z=8.8* | *4× M4×20 machine screws, 105mm bolt circle, into a plywood fan ring* |
| 3 | *NTC temp sensor* | *X=18.22, Y=12.2, Z=8.8* | *Adhesive thermal pad, or 1× #4 screw through its bracket tab* |
| 4 | *Control cluster — switches + surge protector + W1209,* ***no enclosure*** | *X=40.4–45.4, Y=35.75 (Panel C's FRONT wall), Z=3.5–14.5 — on the wall's* ***Panel-B face****, standing ~2" off it* | *Screwed DIRECTLY to a backer board (3/4" offcut) on the wall; 4× #8×1" screws. Reached by lifting one of Panel B's lift-out bed-top halves* |
| 5 | *Anchor-board strips (driver and middle, under the slide rails)* | *⅛" neoprene + ¾" ply, one-piece strips tied by two steel rungs set into their undersides; the fixed rails' risers bolt to them* | *¼-20 machine screws into T-nuts from below — NO holes in the van (Anchor Board sheet)* |
| 6 | *Kitchen tie-down: 2 straps* | *Front strap down to seat-latch hoops H2 and H4; rear strap down to an eye bolt in the kitchen riser's flange and a footman loop on the passenger strip* | ***2*** *ratchet straps (400 lb WLL), each dropped into a pair of the unit's own top-edge notches (Anchor Board sheet)* |
| 7 | *WAVE 3 hose/cord hook* | *In the open bay, kitchen side — bundle the hoses so nothing swings out* | *1× heavy-duty hook, #8×1.5" screw up into the deck underside* |
| 8 | *Fridge hold-down footman loops (×2)* | *Tray side apron, near the tailgate end — hooks to the fridge's 2 end handles* | *Cam strap, snug not tight. This secures the fridge TO its tray; the anchor board secures the tray to the van. Side profile in the Fridge Slide detail* |
| 9 | *Board chains to the seat-latch hoops (×2)* | *A forged ¼-20 eye bolt in each fridge riser's front end → hoops H1 and H2 in the headrest trough, 28" from the hatch* | *¼" grade 30 chain, a quick link at the eye bolt and a second one closing the chain around the hoop's ⅝" bar; tightened with the eye bolt's nut and jam-nutted (the ~5" run is too short for any ratchet strap): forward and lift restraint. No holes in the van (Anchor Board sheet)* |
| 10 | *Striker straps (×3, optional backup)* | *From the board forward to the 3rd-row striker loops on the 2nd-row floor* | *400 lb WLL ratchet straps. Re-tension after the first drive* |

*All coordinates are measured from Panel C's tailgate-facing left corner at floor level.* ***NO-DRILL (Section 8): no new holes in the van*** *— the board straps down to the 3rd-row seats' latch hoops (Anchor Board sheet); the outline follows the October 2026 van survey, not the Appendix A F1–F8 survey. The airflow arrows on the sheet show cool air in at the front wall (fan + the low passive louver), across the fridge, and OUT through the open utility bay toward the tailgate — there is no door in the way. The CO monitor and fire extinguisher are owner-placed and deliberately not located here.*

*And the electrical side — a schematic (not to physical scale) of how the fridge, both fans, and the NTC sensor are powered and wired — down to* ***every physical wire landing****: connection points P1–P12 are numbered on the diagram, each naming the exact terminal, connector type, and crimp, with the full list repeated in Section 5's connection-point table:*




*Fridge wiring diagram*

*The schematic carries the box titles and the P-numbers; the descriptions are here.*

***Connection points P1–P12***

| | ***Connection*** |
|---|---|
| ***P1*** | *DELTA 3 12V car-power outlet → fused (15A) car-plug/SAE adapter — hand-plug, no tools* |
| ***P2*** | *SAE #1 at the A/B seam — mate the OYMSAE cord ends; zip-tie a strain loop each side* |
| ***P3*** | *SAE #2 at the B/C seam — same; both disconnects hang slack, never taut* |
| ***P4*** | *1" grommet, Panel C front wall (low, driver side) — push cord through, no termination* |
| ***P5*** | *Fuse block main studs: red → +IN, black → -IN — #10 ring terminals, M5 nuts, 20 in-lb* |
| ***P6*** | *SW1 FRIDGE: F1(10A) red → PWR spade; LOAD spade → pigtail red; GND spade → NEG bus* |
| ***P7*** | *Pigtail: SAE mates to SW1 side; female 12V socket zip-tied by the fridge's kitchen wall* |
| ***P8*** | *Fridge DC jack: fridge's own cord plug → socket; slack loop to the FIXED rail, 3 clips* |
| ***P9*** | *SW2 FANS: F2(3A) red → PWR spade; LOAD spade → W1209 +12V; GND spade → NEG bus* |
| ***P10*** | *W1209 screws: +12V, GND, K0 (jumper from +12V), K1 (fan feed) — strip 1/4", snug screws* |
| ***P11*** | *NTC probe: plug on the W1209 header; TIP zip-tied at the kitchen-facing wall by the exhaust fan* |
| ***P12*** | *WAGO 221 x2: K1 → both fan reds; fan blacks → one black → NEG bus — levers fully closed* |

***What each box on the schematic holds***

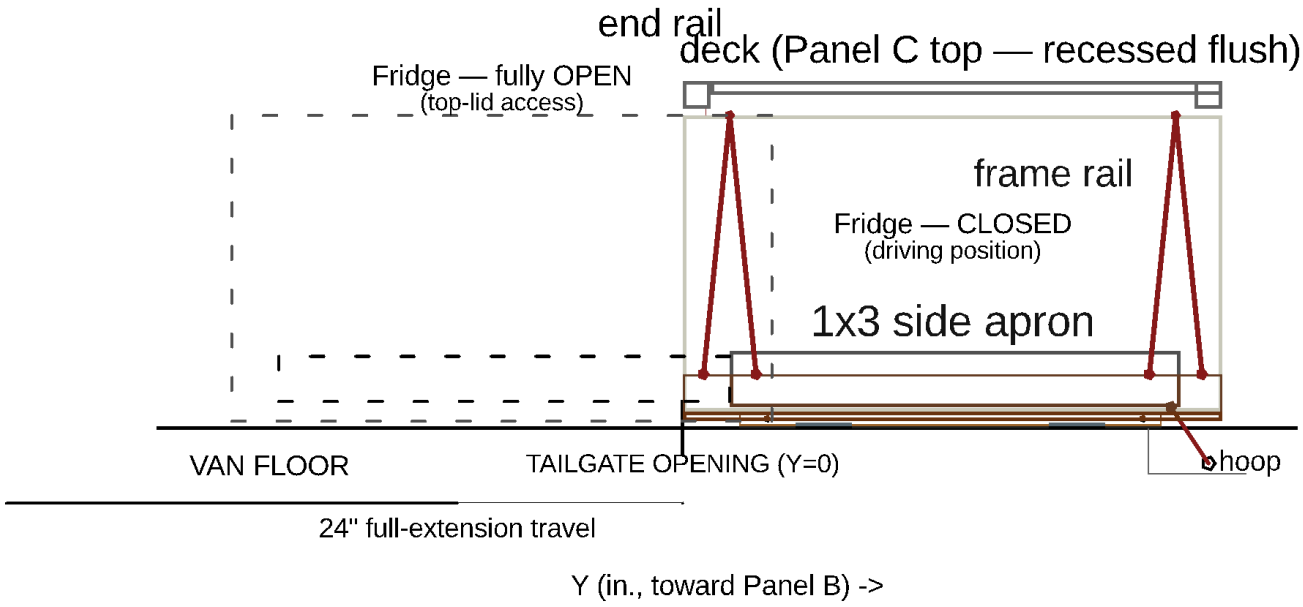
- *VERIFIED — charges DELTA 3; one 1500W inverter feeds BOTH van outlets*
- *2048Wh combined — Panel A right drawer*

- *AC in 1500W (0-100% in ~56 min) · 12V car-power DC out 12.6V/30A*
- *hand-plug into the DELTA 3's 12V car-power outlet*
- *OYMSAE cord ends mate; pull apart to lift Panel A out*
- *same cord family; pull apart to lift Panel B out*
- *cord passes into the fridge bay → open utility bay*
- *+IN / -IN main studs: strip 3/8", crimp #10 RING terminals, M5 nuts*
- *F1 = 10A ATO → fridge branch · F2 = 3A ATO → fan branch · F3-F6 spare*
- *EVERY black (-) return in the bay lands on this NEG bus (ring, screw)*
- *PWR tab ← F1 · LOAD tab → pigtail · GND tab → NEG bus*
- *(0.25" female spade crimps on all 3)*
- *zip-tied in the open bay, near the fridge's kitchen wall*
- *use the fridge's OWN 12V cord; leave a slack loop for*
- *the 24" slide, clipped to the FIXED rail (3 clips)*
- *cooktop is 120V AC — switch it at Power strip 2 instead)*
- *PWR tab ← F2 · LOAD tab → W1209 "+12V" · GND tab → NEG bus*
- *"+12V" ← SW2 LOAD · "GND" ← NEG bus (18AWG black)*
- *"K0" ← short jumper from "+12V" · "K1" → fan feed wire*
- *set: fans ON ~95F, OFF ~85F (P0=H mode, P1/P2 setpoints)*
- *plug → W1209's*
- *2-pin probe header;*
- *TIP by the exhaust fan*
- *nut A: K1 feed → both fan REDS · nut B: both fan*
- *BLACKS → one 18AWG black → NEG bus*
- *cabin air IN*
- *warm air OUT*

*Notes for this sheet:*

- *Charge + discharge simultaneously supported — running the fridge/fans*
- *while AC-charging is normal, just slows the charge rate a bit*
- *Circuit 1 (right): fridge — F1 10A, switched at SW1. Circuit 2 (left): fans — F2 3A, switched at SW2, W1209-driven off the NTC probe.*
- *CONNECTION POINTS — every wire landing, in install order*
- *Wire spec: main run 16AWG (the OYMSAE SAE cord itself); in-bay branches 18AWG red/black;*
- *crimps: heat-shrink ring #10 (block studs) + 0.25" female spades (switch tabs). Crimp, tug-test, then shrink.*

*The fridge's slide mechanism itself, closed vs. fully extended. The VADANIA rails stand vertically BESIDE the tray (side-mount): each fixed rail on a steel riser angle bolted to the no-drill anchor board's neoprene + ply rail-line strip (Section 8 — the board raises only the rails; the tray keeps its hang), each moving rail on the tray's 1x3 side apron — nothing under the hanging tray, so the slide hardware adds zero height and the mounted fridge keeps 0.33" of running clearance under Panel C's end rail:*

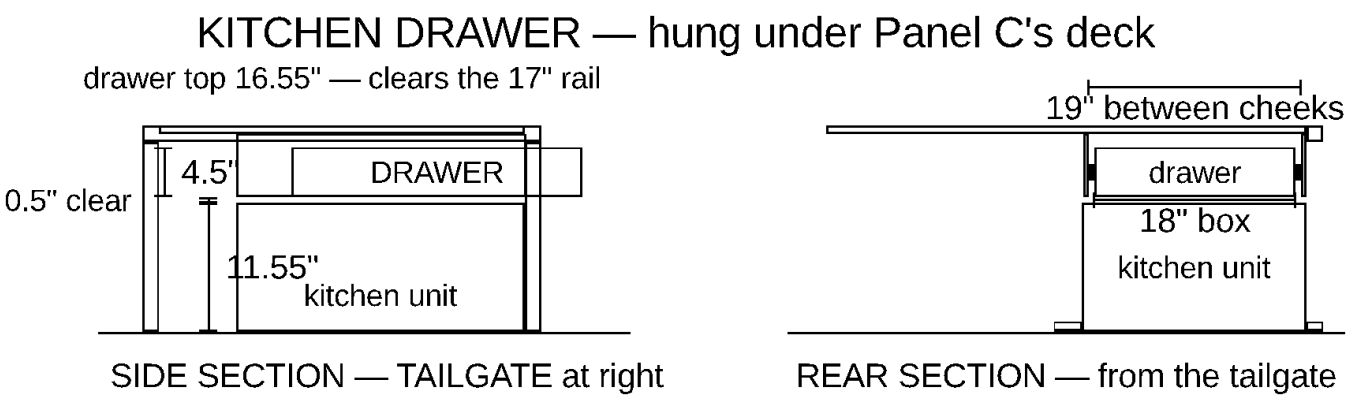



*Fridge slide detail*

*What the sheet shows, in words.*

- ***The slides** are a VADANIA VD2576 industrial pair: 24", 379 lb, locking both closed (transit) and extended (loading). A loaded fridge can hit 60–90 lb, so the rating has plenty of margin.*
- ***The tray** is 1/2" ply with two 1x3 side aprons, and hangs BETWEEN the rails on the slides' moving members — the moving member screws to the apron, not under the tray, so nothing sits beneath it. Each apron's top edge stands proud of the tray as the fridge's anti-shift lip, and the tray itself hangs 0.375" clear of the van floor.*
- ***The fixed rail stands VERTICALLY beside the tray**, screwed to a steel riser angle that bolts to the anchor-board strip. Mounted this way it adds width beside the tray and zero height under it. Never mount it undermount: laid flat under the tray it adds ~1.2" and the fridge then hits the tailgate end rail it has to slide under.*
- ***The DC line's slack clips to the FIXED rail** (3 screw-mount clips, Sections 5–6) — never to the moving tray, where it could get pinched between apron and rail as the slide comes home.*
- ***Under this rail line runs the anchor-board strip** (Section 8, no-drill): ⅛" neoprene + 3/4" ply, riser bolted through with 1/4-20 T-nuts, no holes in the van. The board is strapped forward and down to the 3rd-row seats' latch hoops (Anchor Board sheet); straps to the 3rd-row striker loops are an optional backup.*
- ***The hold-down strap** (×1, cam-buckle) runs from the fridge's end handle to 2 D-rings on the tray's side apron. It stops the fridge lifting off the TRAY — the riser and board bolting only pin the tray to the van. Snug, not tight, and it must clear the lid at full extension.*

*The kitchen drawer — a shallow slide-out drawer hung under Panel C's deck in the dead air above the kitchen unit (11.8" unit, 17" clear void), riding a 24" full-extension slide pair and pulling out the open tailgate; ~3.5" clear inside for utensils, cutting board, flat dry goods, and the cooktop's griddle plate:*




*Kitchen drawer detail*

*Notes for this sheet:*

- *anchor-board strips (mat + 3/4" ply) flank the kitchen — its tie-down straps land here (no-drill, Sec. 8)*
- *2x 1/2" ply cheeks tied by 3 ¾" spreaders into one squared unit on the bench, then screwed to the deck from above (2" screws every 6")*
- *outer cheek also screws into the side rail's inner face*
- *24" side-mount slides, 100lb pair — one per cheek*
- *The drawer box is 26" deep, riding that 24" full-extension slide pair, and pulls straight out the open tailgate.*
- *The box is 18" wide across its outside, leaving ~17" clear inside it, in the 19" between the two cheek faces.*
- *The kitchen unit below it is the bought JAGAHAHA slide-out camp kitchen, 11.8" tall, which slides out on its own rails — nothing in this drawing supports it.*
- *Box: 1/2" ply, glue + 1-1/4" screws, ~3.5" clear inside — utensils / cutting board / flat dry goods / griddle plate.*
- *Face gets a magnetic catch so nothing rattles in transit.*

### *Rear Pantry — Tote + Microwave*

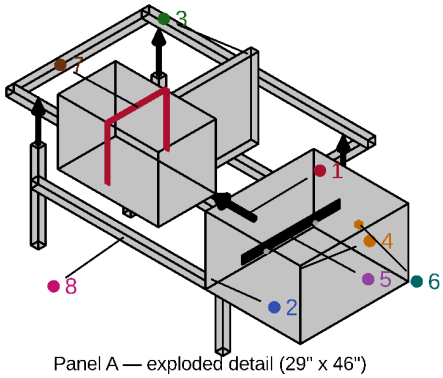
*The rear pantry is **one bought 27-gallon Tough Box tote** (Component 1) on the tailgate end of Panel C's deck, with the microwave beside it on the passenger side. One cam strap goes over both, to a footman loop on the outside face of each side top rail, 7½" from the tailgate end. **Its shop drawing is the Tote + Microwave Strap Anchors sheet** at the end of this PDF.*

*Notes for this sheet:*

- *The old rear-pantry render (IRIS drawer cluster + pot crate) is retired with the design; the model (`params.scad`, `rear_pantry_detail.scad`) has not been redrawn yet. Where they differ, the text and the strap-anchor sheet are right.*
- *Heavy things low in the tote. The strap doubles as the lid's hold-down.*

### *Floor Panel Detail*

*Each floor panel, exploded (Panel A: frame + legs + center divider + one right drawer, left bay is WAVE 3 open storage; Panel B: a bare cube frame — no divider or drawers; Panel C: no divider or drawers — the fridge/kitchen units live in its void, shown here as reserved footprint):*




Panel A — exploded detail (29" x 46")

Component

Position / fastener / material

● 1 End rails (x2) + side rails (x2)
end rails 46" (full width, outboard); side rails 26" BETWEEN them = 29" x 46" perimeter
corner brackets + 2" screws + glue

● 2 Legs (x4)
16.75" cut + leveling foot, inset 3.5" from the edge
LAPPED to the end rails, tops flush with the rail tops; 1/2" x 1.75" insert hole in each bottom (Panel B detail's inset)

● 3 Center divider
26" cut, 2x2 pine
splits the bay — drawer (right) / WAVE 3 (left)

● 4 Drawer box (right, DELTA 3 side)
19.25" x 22" x 13.75", 3/4" walls, 1/2" bottom
5 pieces, glued + 2x R3 per corner at this thickness

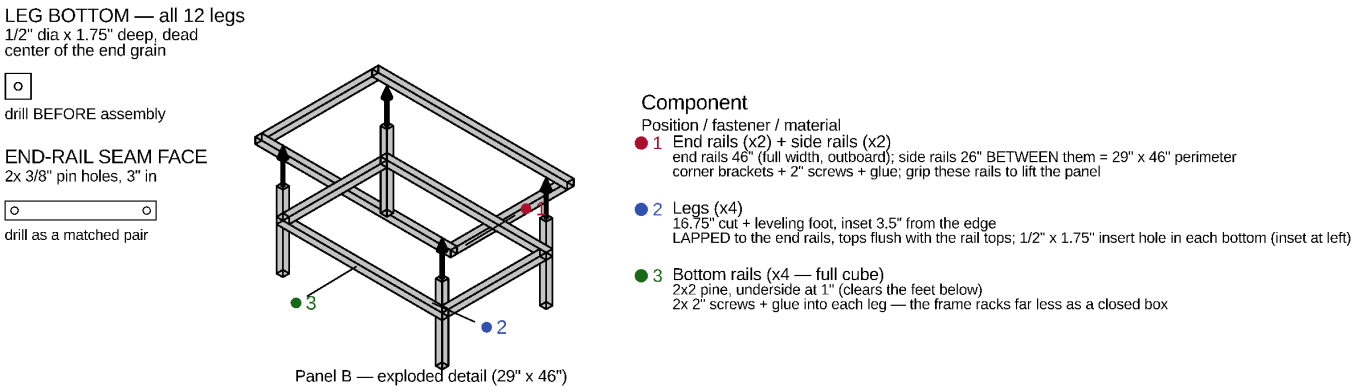
● 5 Drawer slide (right)
20" full-extension pair
box-to-rail + box-to-divider

● 6 Drawer catch (right)
friction catch or small turn latch
keeps the drawer shut in transit

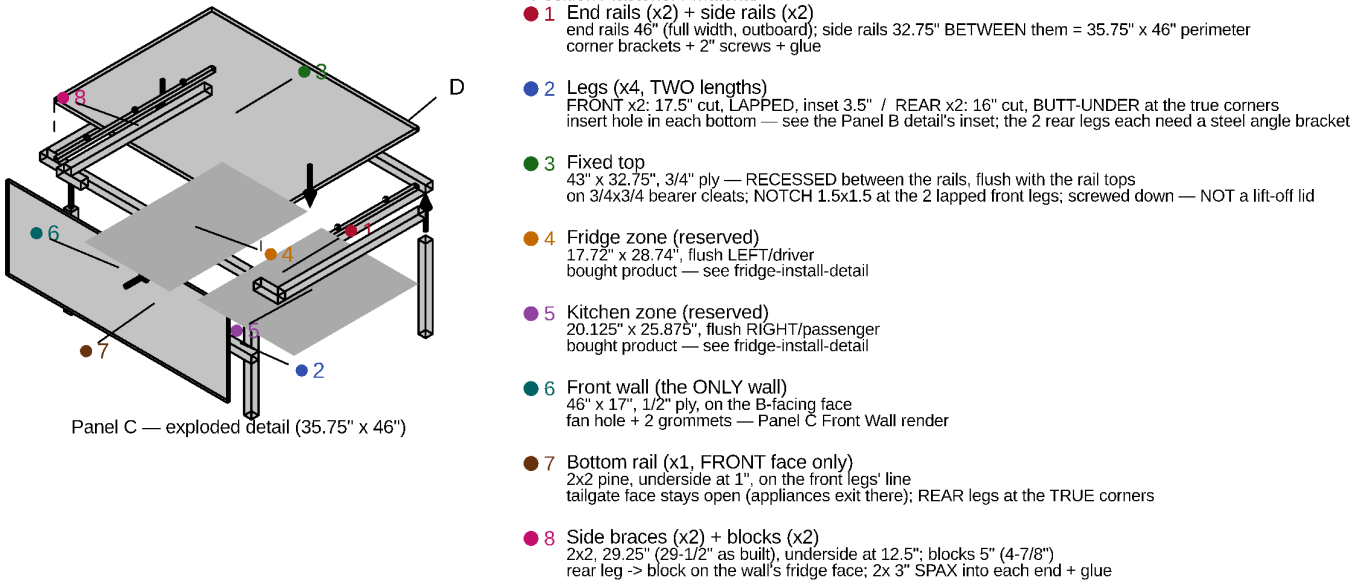
● 7 WAVE 3 hold-down (left)
2x D-ring + 1 low-profile cam strap
no box, no slide — sits on the bay floor, strapped fore-aft

● 8 Bottom rails (x2, END faces only)
2x2 pine, underside at 1"
the SIDE faces stay open — the drawer + WAVE 3 exit there at floor level

*Panel A detail*




*Panel B detail*




*Panel C detail*

***The two hole details in the left margin of each sheet, in full.***

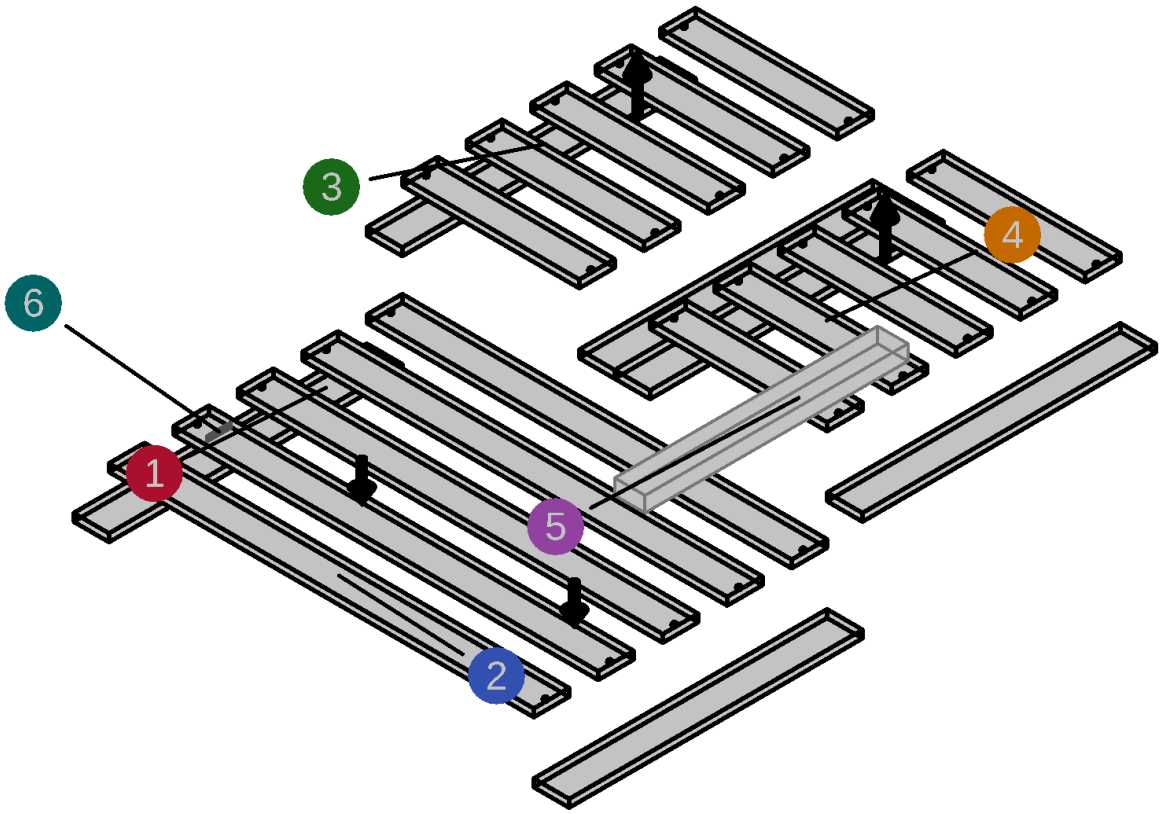
- ***LEG BOTTOM** — the same hole on all 12 legs: 1/2" dia × **1-3/4"** deep, dead center of the 1.5" × 1.5" end grain, taking a 3/8-16 threaded insert driven with a **10mm hex wrench**. The depth is set by the foot's stud, not the insert: bored 7/8" the stud bottoms out on wood and the leg loses half its adjustment. **Drill it before assembly** — see the Leveling Foot render, and the shop photos in Section 3.*
- ***END-RAIL SEAM FACE**, both ends — 2× 3/8" dia × 3/8" deep pin holes, 3" in from each side edge, centered on the rail. Drill the mating panel's face **as a matched pair**: clamp both panels together and drill through a guide block (Component 5).*

***Per-panel differences:***

- ***Panel A** — left (driver-side) bay is WAVE 3 open storage: no drawer box, no slide. The unit is too wide for a boxed drawer.*
- ***Panel B** — no divider, no drawers, no skirts. The side doors don't reach it, so its whole bay is deep storage loaded from above. It is **the full cube**: bottom rails on all 4 faces, since nothing exits Panel B sideways.*
- ***Panel C** — no divider, no drawers; the void stays open for the bought fridge and kitchen unit (Component 7). It carries the ONE wall in the build, on its front (B-facing) face, and two **side braces** (Oct 2026): a 29½" 2×2 each side, 12½" off the floor, from the butt-under rear corner leg's forward face to a 4⅞" block on the wall's fridge face (Panel C sheet, Section F).*

## *Bed Platform Detail*

*Component 2's slatted bed platform (58" x 49" overall, 15 slats glued and pocket-screwed between 1x4 side rails — one flush 3/4" plane, cantilevered 1.5" past the boxes on each side), exploded. **It is three pieces, but only two of them are loose: Panel A's 29" × 49" section is SCREWED DOWN, and Panel B's top is two 29" × 24.5" centerline halves that each lift out on their own.** Per-site leveling is done at the WHEELS with blocks (Block Calculator, Appendix E); the interior leg feet are a one-time set. Both sections rest DIRECTLY on Panel A/B's top rails and the platform ENDS at the B/C seam — Panel C's own fixed deck sits at exactly the same surface height, so the two meet flush and the mattress's last ~20" rides that deck. Leveling happens at the leg feet, down at the floor, and an RV bar bubble level screwed to the driver-side rail's outer edge reads fore-aft pitch while you turn the feet (its twin on the rear-pantry deck edge reads roll):*




# 58" x 49" overall — 15 slats in one 3/4" plane

## 1-6 keyed to the parts table in the text

*Bed frame detail*

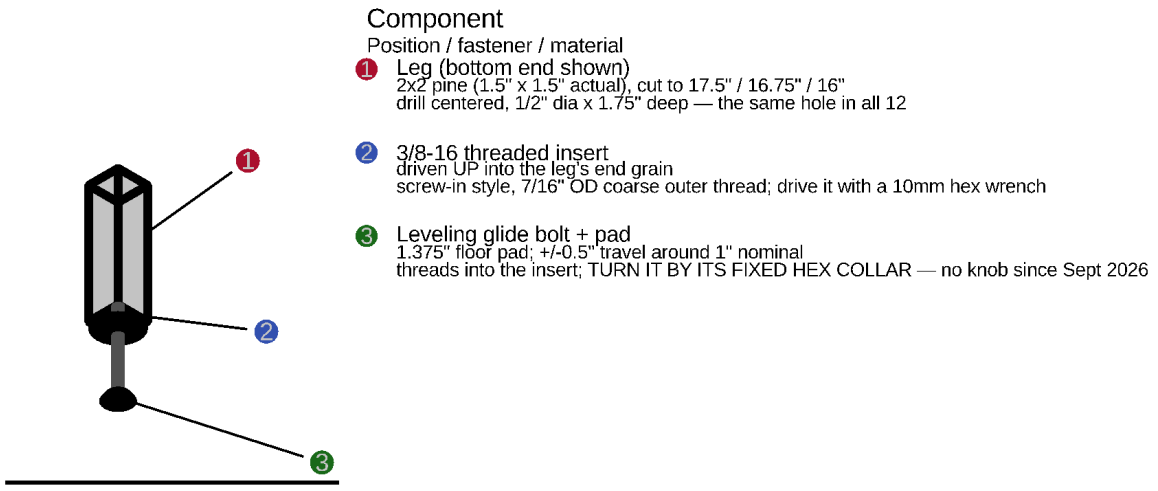
*The numbers on the drawing key to this table.*

| | Part | Size | Notes |
|---|---|---|---|
| **1** | *Side rails (×2 per piece, ×6)* | *6 × 29" × 3.5" × ¾" — 1x4 pine* | *Each piece's spine; its slats pocket-screw into the inner edges* |
| **2** | *Panel A slats (×5)* | *42" × 3.5" × ¾" — 1x4 pine, two per 8ft board* | *~2.9" gaps. This section is **screwed down** — it is also acting as a diaphragm across Panel A's rails* |
| **3** | *Panel B half (×2)* | *29" × 24.5"* | ***Lifts out on its own**, ~6 lb. 24.5" wide clears the 35" side door — a 49" piece never would. No hinge, no lid stay, no opening-angle limit* |
| **4** | *Panel B slats (×10)* | *17.5" × 3.5" × ¾" — 1x4 pine, five per 8ft board* | *5 per half, ~2.9" gaps* |
| **5** | *Center bearer (Panel B's frame)* | *26" × 3" — two 2x2s side by side, top flush with Panel B's long rails* | *1½" of bearing per half, and it **halves the deck's unsupported span** over Panel B, 46" → ~22". Panel B has no center divider where Panel A does. **Drop-in** on 4× 1" corner braces, so it lifts out for the spare* |
| **6** | *Bubble level (PITCH)* | *Larbeti stick-on bar level* | *Self-adhesive — degrease the rail edge first. Its twin (ROLL) mounts on the rear-pantry deck edge; read both while turning the leg feet* |

***Panel B is top-load-only**, which is why its top splits: the measured side-door aperture is 35" × 45" with only 29" ever clear, and it sits over Panel A's footprint, not Panel B's. Lift the mattress clear and take out only the half on the side you're standing at.*

### *Leg Leveling Foot Detail*

*Leveling lives at the floor: each leg is cut 1" short and gets a 3/8-16 insert in its bottom end grain, taking a leveling glide bolt with a floor pad, turned by the stud's own fixed hex collar (12 total: 4 per panel x 3). **No star knob and no jam nut** (a knob takes up too much room under the legs): put a wrench on the collar. Effective leg heights are 18.5" on Panel C's lapped front pair / 17.75" on A/B / 17" on Panel C's butt-under rear corner pair (cuts: 17.5", 16.75", 16" — the deck recess sets the rail height, the lap adds 1.5" of leg above it); leveling at the floor instead of between layers buys 1.25" of headroom, and recessing the deck ply into the rail plane buys another 3/4". To adjust: tip that corner of the box slightly and turn the collar — every leg is exposed at floor level:*




*Leveling foot detail*

*Notes for this sheet:*

- *To level: kneel at the side door, tip that corner of the box slightly, wrench the foot's hex collar. Every leg is exposed at*
- *floor level — nothing boxes it in. The platform above rests DIRECTLY on the box rails (no adjusters up top).*

***Load**: worst case is ~700 lb total (2 people + mattress + platform + boxes + cargo) spread over 12 feet — ~60 lb per foot against the feet's 330 lb rating each, a 5x margin before dynamic factors. The feet are not the weak point. If you want more margin anyway, 1/2"-13 leveling mounts (1,100 lb each) are a drop-in upgrade — same install, just a 5/8" insert hole instead of 1/2".*

***Access**: the practical answer is to stop crawling under the bed at camp entirely — **level the VAN at the wheels** (Andersen/Camco-style curved ramps, standard van-life practice) each site, and set the interior feet ONCE against the van's own floor irregularities. After that first setup you shouldn't need to touch a foot; the two bubble levels tell you at a glance whether the wheels need a ramp.*

***Blocks + calculator (the no-feet workflow)**: the two bubble levels read DEGREES (-7 to 7). Degrees plus the wheelbase are all the math needs, so there's a phone calculator for it — **Sienna Block Calculator** (also saved in the repo as `leveling_calculator.html`): enter the two readings and which end/side is low, and it returns how many Lynx-style leveling blocks to stack under each tire, the total, and the residual tilt after driving up. One 10-pack of blocks covers ~2.9° of correction at 1.5"/block over the 120.5" wheelbase — beyond ~4 blocks per tire, find a flatter spot. This makes the interior feet a ONE-TIME set (against the van's own floor irregularities), not a per-site chore — and if you'd rather delete the feet entirely, replace them with fixed shims set once and let the blocks do all site leveling (saves ~$72, loses the fine-trim option).*

***Installing the calculator on a phone** (two ways):*

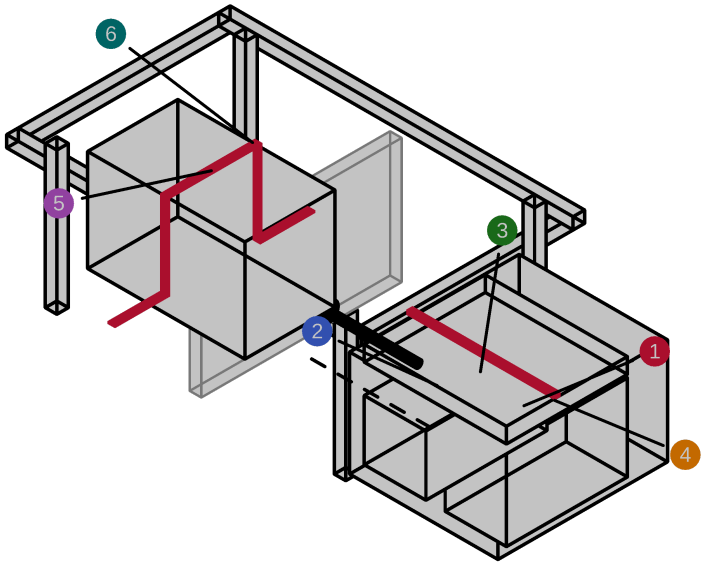
1. *Online: open the calculator link in Chrome on the phone (it needs the claude.ai login), then Chrome menu → **Add to Home screen** — it gets an icon and opens full-screen. Needs signal at launch.*
2. *Offline — the one to rely on at camp: `leveling_calculator.html` is a single self-contained file with zero network dependencies. Send the copy from this repo (or `~/Downloads/sienna_block_calculator.html`) to the phone by email/Drive/Nearby Share, open it once via **Files** → **Open with** → **Chrome**, and bookmark it. It runs with no signal forever and remembers the settings below.*

***Using it at a site:***

1. *Park in the spot you actually want, engine off, **parking brake on**.*
2. *Read the two bed-mounted levels: the fore-aft level on the platform's driver-side rail (pitch) and the side-to-side level on the rear-pantry deck edge (roll). Each reads in degrees; the bubble floats toward the HIGH side, so the opposite end/side is the LOW one.*
3. *Enter both degree readings and tap which end (nose/tail) and side (driver/passenger) are low. One-time settings under "Van & block settings": wheelbase 120.5", track ~68", and the height each block adds — Lynx-style ≈ 1.5", but **measure your own stack once** and enter that.*
4. *The van diagram shows blocks per tire. Lay each stack just ahead of its low tire, **drive up slowly**, and chock a wheel.*
5. *Re-read the levels and refine once if needed — suspension squish and block nesting make a second small pass normal. If any tire calls for more than 4 blocks, the calculator says so: re-park (turning the van around often halves the stack) rather than building a tower.*

### *DELTA 3 / WAVE 3 Stowage Detail*

*Stowage for the EcoFlow gear: the DELTA 3 Plus + Smart Extra Battery, stowed unstacked side by side in Panel A's right drawer, and the WAVE 3, stored as open bay storage (no drawer box, no slide) in Panel A's left bay (its hoses/cord stow separately — item 12 in the fridge installation detail above):*




## DELTA 3 STACK (right drawer) + WAVE 3 (left bay)

Panel A, exploded — Components 2 & 8

DELTA 3 Plus outboard (the WAVE 3 plugs into it), Extra Battery inboard; ~48 lb in a normal drawer — no E-track. The WAVE 3 sits on the raw bay floor, cam-strapped to 2 D-rings.

Found storage: a ~2.6" lift-out tray tops the DELTA stack, The WAVE 3's hoses live on the hook in Panel C's utility bay.

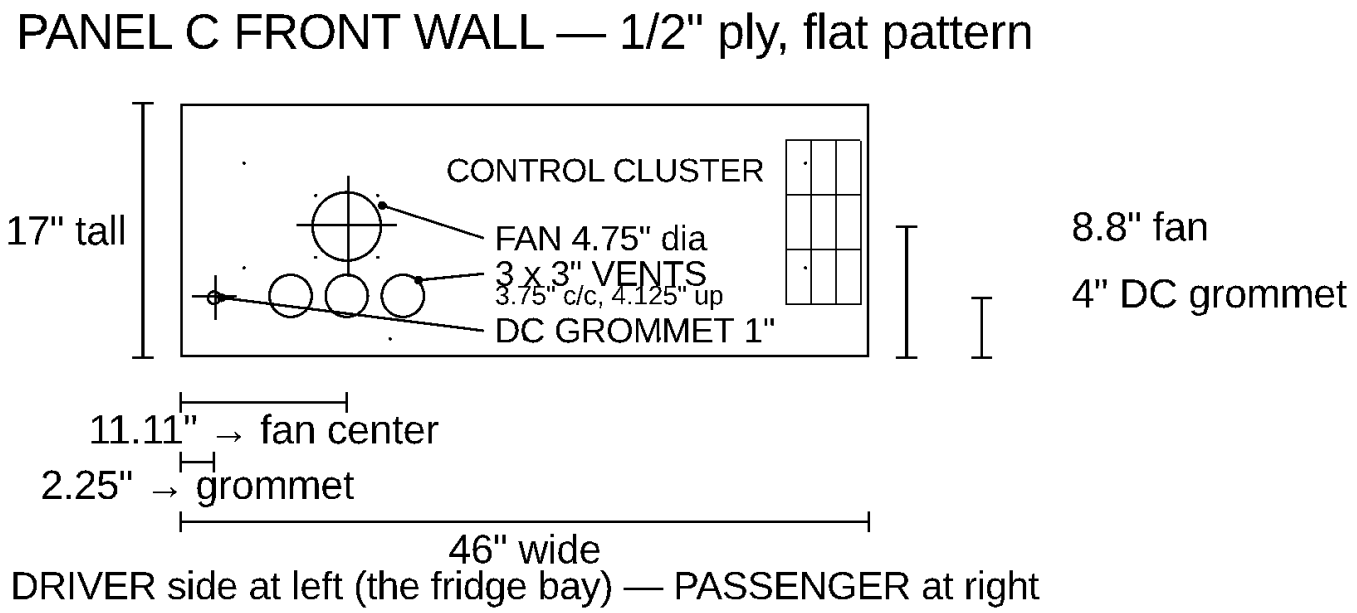
### Component / position

1. DELTA 3 Plus — 7.95"x15.7"x11.16", ~28 lb
   outboard (pull wall); cam-strapped over locating cleats
2. Smart Extra Battery — 8"x15.7", ~20 lb
   inboard — extra runtime only, grabbed less often
3. Lift-out tray — reclaims ~2.6" dead air
   cables, the DELTA's own cords, dongles
4. Cam strap + D-rings + 1" grommet
   strap over both units; WAVE 3 charge cable exits the grommet
5. WAVE 3 — 20.4"x11.7"x13.2", 33.7 lb
   raw bay, 0.35" clear — a boxed drawer would NOT fit
6. Cam strap + 2 D-rings (WAVE 3)
   fore-aft over the top, D-rings on the end bottom rails — no bounce, no walk-out

*DELTA 3 and WAVE 3 stowage detail*

### *Panel C Front Wall*

*The one wall any panel gets, as a flat pattern with **every hole dimensioned**: the 120mm intake-fan hole (with its 105mm screw square) centered on the fridge bay, ONE 1" grommet (fridge DC line low — Power strip 1 plugs into the REAR outlet and never crosses this wall), three 3" intake vent holes under the fan, and the 7 mounting screws (4 into the front legs, 3 into back blocks behind the bottom rail). Panels A and B have no walls or skirts at all, Panel C's sides stay open, and its tailgate face needs no wall — the fridge, the open utility bay between the units, the kitchen unit, and the kitchen drawer face fill it completely:*




*Panel C front wall detail*

*The sheet carries the geometry; here is what each opening on it is for.*

- ***Where it goes.*** *On the front legs' back faces (the fridge side), its bottom edge ½" above the floor. Heights on the sheet run from that bottom edge. This is the ONLY wall on any panel — Panel A has none (both bays face the side doors), Panel B has none (bare-frame deep storage), and Panel C's sides stay open (van wall ~1" away) and its tailgate face is filled by the fridge, open utility bay, kitchen unit and kitchen drawer.*
- ***FAN 4.75" dia*** *— the 120mm intake fan, blowing IN. Cut with a hole saw or jigsaw. Fastened with 4× #8 screws on a 4.13" (105mm) square, the standard 120mm pattern.*
- ***DC CORD GROMMET 1" dia*** *— carries the fridge DC line, in the driver-side cord chase outboard of the front leg. Just the one: Power strip 1 is on the van's rear outlet, so its line never crosses this wall.*
- ***3 × 3" VENTS*** *— passive scoops admitting cool floor-level air, in a row under the fan and centered on its line: centers 3.75" apart, 4.125" up. They take the 3" round snap-in soffit vents on hand; check the hole size on the package. ~21 sq in, which covers the 120mm fan's own 17.7 sq in aperture.*
- ***Opening spacing.*** `params.scad` *asserts a 3/4" minimum web between every pair of openings.*
- ***Fastening*** *— 7× #8 × 1-1/4" screws: 2 into each front leg, and 1 into each of the three 6" back blocks glued behind the bottom rail. The wall sits 1-7/16" behind the rails, so it can't screw to them. Optional: cut two relief notches with its other openings for the backup striker straps (3" wide × 1" up from the bottom edge, at 12¾–15¾" and 30¼–33¼" from the driver end), and drive the block screws at least 1¼" up from the bottom edge, clear of them.*

## 1. Overview

### Vehicle constraints (hard limits — everything below is designed inside them)

*The 2nd row seats are being* ***removed entirely*** *(not just pushed forward), with the 3rd row folded flat.* ***Every figure in this table was measured on the actual vehicle****, with the 2nd row out and the 3rd row folded as it sits in camper mode — see Appendix A for the survey sheet those numbers came off.*

| Constraint | Value | Notes |
|---|---|---|
| *Max interior length* | ***93.75"*** | *Closed rear hatch to the front seatbacks, 2nd row removed — MEASURED V1* |
| *Max floor width* | ***49"*** | *Between the wheel wells (54" forward of them) — MEASURED V2* |
| *Max height* | ***42"*** | *Load floor to headliner, mid-van; 37" back at the gate — MEASURED V3* |
| *Vent intrusion* | ***3.5" per side*** | *Rear lower heat vents, at floor level only — legs must stay clear; the raised deck may overhang — MEASURED V4* |
| *Hatch curvature reserve* | ***0"*** | *Nothing needs reserving: the build never rises high enough to meet the hatch glass/trim curvature — MEASURED V5* |
| ***Usable width at the sleeping plane*** | ***50"*** | *Wall-to-wall at ~18.5" up (49.5" at ~22.5") — MEASURED V7.* ***The walls do NOT flare to ≥53"****, which is what holds the bed platform to 49"* |

***Panel C (35¾") takes up the length*** *— forced by two of the model's own guards: Panels A and B must stay equal (shared step diagrams / cut list), and A + B is* `bed_frame_length`*, so trimming either would change the 58" bed rails for no reason. Panel C has ~7" of fore-aft slack around the fridge. With V5's hatch-curvature reserve at* ***zero****, the pantry drawers can't sit proud of the deck edge, so* `pantry_len` *is 14.5" and the deck carries them fully. All* `assert()` *guards in* `params.scad` *pass with the measured numbers in place.*

*That leaves a **93.75" usable length** (V5 reserves nothing at the hatch) and a **42" floor-level width** (full 49" available at deck height, 50" up at the sleeping plane). The math: Panel A 29" + Panel B 29" + Panel C 35.75" = **93.75" of panels — Panel A sits flush with the front seatbacks (no open floor left over)**, filling usable_length exactly. There's no length lost to a dedicated fridge/kitchen zone — the fridge and kitchen unit both live in the void underneath Panel C's deck, sharing space with the same legs that hold up the mattress. The rear pantry claims the last 14.5" of Panel C's own 35.75" (the tailgate-most strip, right next to the fridge/kitchen void), leaving **79.25" of continuous sleeping run** (Panel A + Panel B + the first 21.25" of Panel C) for the mattress — the HEST Dually Long (78" long) rides here with ~1.25" of spare at the head end. These limits live at the top of `params.scad`: bump any dimension past the envelope and every render fails loudly instead of silently overflowing the van.*

### Rear liftgate opening (MEASURED)

*Every module has to physically pass through the liftgate opening to be lifted in and out — that's the entire point of the modular design, so this matters as much as fitting once installed. Both figures are tape measurements.*

| Constraint | Value | Notes |
|---|---|---|
| Gate opening width | **50"** | Narrowest point — MEASURED V6a |
| Gate opening height | **37"** | Clear height at the center of the opening — MEASURED V6b |
| Height out at the corner radius | **20.5"** | The corners round off hard — MEASURED V6b. Carry modules through the **middle** of the gate, not flat against one side |

***Fit check** (each module carried upright, legs down, the way it sits once installed):*

| Module | Passes through as | Width margin | Height margin |
|---|---|---|---|
| Panel C | 46"W x 18.5"H | **2" per side** | 18.5" to spare |
| Panel A, Panel B | 46"W x 17.75"H | **2" per side** | 19.25" to spare |
| Bed platform (Component 2) | 49"W x 58"L x 3/4"H | enters tilted diagonally (gate diagonal ~62") | — |
| Rear pantry drawer units (prefab, Component 1) | ~12.1"W each | easily | a few lb each — lifts out one at a time |

*The fridge and kitchen unit never need to pass through the gate opening at all — they live inside Panel C's own void as bought products, installed once and left in place, not lifted in and out. The tight spot is not width — the measured 50" opening gives the 46" boxes **2" per side**. It is the **corner radius**: only 20.5" of height survives out at the rounded corner, so an 18.5"-tall box walked in flat against one side will foul long before it runs out of the 37" available in the center. `params.scad` asserts against the measured numbers the same as every other hard limit. The rear pantry's prefab drawer units are separate pieces from Panel C by nature — loosen the cam strap and each ~12.1"-wide unit lifts out and through the gate trivially (Component 1).*

### Side door openings (FULLY MEASURED — aperture 35" × 45", 29" ever clear, opening starts at the front seatback)

*The side doors only matter for **Panel A**: its right drawer (DELTA 3) pulls out the passenger door and its left bay (WAVE 3) is reached through the driver door — the door openings sit over the old 2nd-row footprint, which is exactly where Panel A lives. **Panel B is beyond the door openings entirely** (owner-confirmed), which is why it has no drawers: nothing pulled sideways from it could clear a door, so its bay is top-loaded deep storage instead. Panel C's void houses the fridge and kitchen unit, both pulling out through the open tailgate. Measured V8.*

| Constraint | Value | Notes |
|---|---|---|
| **Usable clear width at the door's real stop** | **29"** | MEASURED V8c. This is the number that governs; `side_door_opening_width` = 29 |
| Side door aperture | **35" × 45"** | MEASURED — the opening itself. The door parks 6" short of its forward edge, which is why only 29" is ever clear |
| Door panel (not the opening) | 50" × 58" | Measured, but these are the door's own outside dimensions — recorded so they aren't mistaken for aperture figures |
| Door opening's fore-aft position | **Y = 0** | MEASURED — the front seatback is essentially **even with the opening** at build height; if anything the seatback intrudes ~¼" into it. `side_door_y0` = 0 is a measurement |

***Reach caveat:** a **29"** clear opening won't expose the full 58" drawer run (Panel A + Panel B) from one position — the reachability check in `params.scad` reports Panel A reachable and **Panel B effectively unreachable from the side.** That is survivable only because Panel B is a top-loaded bay with no drawers. Put your most-used gear in Panel A's drawers, and re-run the check once `side_door_y0` is actually measured.*

### Layout

- ***Rear pantry — tote + microwave** (changed 30 Oct 2026): a **27-gallon Tough Box tote** (bottom 15⅛ × 25", rim 20½ × 30.15", 14.3" tall) sits on the **tailgate end of Panel C's deck**, 25" side across the van, bottom flush with the tailgate edge and ~1½" in from the driver edge. The pots' 11×11 box goes inside it. The **microwave** sits flush with the passenger edge beside it. **One cam strap over both**, hooked to a footman loop on the outside face of each side top rail; a non-slip mat under both; no cleats. The tote's bottom pokes ⅝" past the 14½" pantry end into the foot of the mattress (79¼" run, 78" mattress), and its rim hangs ~2¾" past the deck's tailgate edge ~33" up — **check the closed hatch clears it at the van**. **Power strip 1 and the ROLL bubble level need a new spot** (the old pot bay is now the microwave's) — re-site them at Component 6. See Section 6, Component 1.*

- ***One continuous sleeping deck**, 46" wide x 93.75" long (Panels A + B + C, flush with the front seatbacks — no gap), deck surface sitting **18.5"** above the van floor: **the horizontal ply is recessed INTO the rail plane** (deck recess — Panel C's deck drops between its rails onto 3/4"×3/4" bearer cleats, flush with the rail tops; Panels A/B instead sit their rails at **16.25"**, so the 3/4" bed platform resting ON their rails tops out at the same 18.5" plane). That buys 3/4" of headroom: **19.5" sitting headroom** over the 4" mattress (V3 measured the cabin at 42"). Panel C's `leg_height` stays 17", driven by the fridge's mounted stack (see below), not by the folded 3rd-row well depth — confirm 17" clears your actual well before cutting.*
- ***Panel A and Panel B share the same frame, and neither has a top of its own.** Panel A has a center divider splitting its bay in two, reached through the side doors: the right (passenger) side is the one real drawer (DELTA 3), the left (driver) side is WAVE 3 open storage (the unit is too wide for a boxed drawer — Section 1). **Panel B has NO drawers and no divider** — the sliding-door openings sit over the old 2nd-row footprint (Panel A), not over Panel B, so nothing pulled sideways from Panel B could clear a door. Its whole bay is deep storage instead, loaded from above by lifting out **either half of its two-piece bed top** (long-term/bulky items you don't touch at camp). The slatted bed platform (Component 2, 15 slats between 1x4 side rails, 58" x 49" overall — cantilevered 1.5" past the boxes each side, 3/4" thick; **width set by V7**) spans Panel A + B ONLY and ENDS at the B/C seam: Panel C's own fixed deck is at exactly the same surface height, so the two meet flush and the mattress's last ~20" rides that deck (an 80" platform would have to sit ON that deck — 3/4" too high). It rests DIRECTLY on Panel A/B's top rails as one flush 3/4" plane (slats pocket-screwed between the side rails). Leveling happens at the leg feet down at the floor — each leg is cut 1" short and carries a hand-adjustable leveling foot.*

***SPLIT INTO THREE LIFT-OUT PIECES**, so that reaching Panel B, the only bay with no side access at all, doesn't involve wrestling a 49"×58" frame out and finding somewhere in a packed van to put it:*

- ***Panel A's section** — 29" × 49", one piece, **screwed down and permanent**. It never needs to come out: Panel A's two bays are both reached through the side doors — the DELTA 3 drawer pulls out the passenger side, the WAVE 3 bay is reached by hand from the driver side. Fixing it is a real simplification, not just one less loose part. It becomes a **screwed-down diaphragm** across Panel A's rails, putting back some of the torsional stiffness Panels A and B give up by having no plywood tops (the diagonal corner braces have help rather than carrying that alone); it is the **fore-aft datum** the two loose halves locate against; and it **cannot rattle**, so it needs no anti-rattle pads. The trade: Panel A's bay loses its from-above deep-cleaning route and is side-door-only. Acceptable — the 29" clear side gap exactly spans Panel A's 29" length, and the DELTA drawer (22" fore-aft) comes out through it.*
- ***Panel B's TWO HALVES** — 29" × 24.5" each, **split on the centerline**, each lifting out on its own at ~6 lb. Lift the mattress clear and take out only the half on the side you're standing at.*
- ***A center bearer in Panel B's frame** — 26⅛" of 3"-wide 2×2 (two 2×2s side by side, or one 2×4 laid flat) running fore-aft at the centerline, its top flush with Panel B's long rails. The halves' inner rails land on it, 1½" of bearing each — the same as their outer rails get from the long rails. **Drop-in, not screwed:** it rests on 4× 1" corner braces on the end rails' inner faces and lifts out, cut to the measured gap less ¼" (~25⅞"). Screwed in, it would split the top opening into two 20⅛ × 26⅛" holes, and the 28½" spare can't pass either one edge-on — it could be neither loaded nor pulled out.*

***Why halves rather than one hinged leaf: 24.5" fits back out through the 35" side door, and 49" never could** — a full-width piece could only ever leave via the tailgate, which defeats the point of working from a side door. Halves also mean the mattress only has to be lifted clear of half the width, there's no hinge, lid stay or opening-angle limit (a hinged 29" leaf could only reach ~67° before hitting the headliner — there's just 23.5" above the deck), and **the center bearer halves the deck's unsupported span over Panel B, 46" → ~22"**. The costs: two loose parts to set down instead of one captive lid, and ~1.7 lb over the hinged version.*

⚠️ ***Neither scheme gets you around the mattress.** It's one 78" piece and Panel B is the middle third of the bed — fold the head end back and it lands on Panel B; fold the tail end forward and it lands there too. The mattress comes off, or gets folded clear, first. Halving the width is what makes that manageable rather than a two-person job.*

*Sourcing: five 42" slats (Panel A) and ten 17.5" slats (the halves, five per 8ft board) plus six 29" side rails — **seven 1×4 × 8ft boards, crosscuts only** (a 1x4 is already ¾"×3½", the slat spec, so nothing gets ripped), plus 52" of 2×2 for the center bearer. No bought slat kit: they come stapled to webbing or riveted to metal side frames, neither of which survives being cut down (Section 4 + Component 2). - **Trade-off: no top means no enclosure — answered with CUBE FRAMES.** Removing the plywood tops loses torsional rigidity and a dust barrier. Two fixes are specified: diagonal corner braces up top, and **bottom rails closing each frame into a box** (underside 1" up — at the leg bottoms, just clear of the leveling feet, for the tallest box section and the lowest floor-edge curb) on every face that can take one — Panel A's two END faces (its sides must stay open for the drawer/WAVE 3), ALL 4 faces of Panel B (the full cube), and Panel C's FRONT face (its tailgate face stays open for the appliances; its fixed top + front wall already stiffen it). Panel C's sides can't take a floor-level rail (the fridge riser, the H1 eye bolt and the kitchen strap's footman loop all live in that band), so they get **side braces 12½" up** instead, tying the rear corner legs to the front wall (Oct 2026, after a rear leg broke when the panel was tipped). A closed frame racks far less than rails + brackets alone. The bed frame's slats still have gaps — small items can fall through into the bays below. - **Panel C has no drawers.** Its under-deck void instead holds two bought products side by side across the 46" width: a **BougeRV Rocky 40 (41-quart, dual-zone)** — 17.72" side left-right, reversible lid (manual §4.4), optional detachable B240 battery at the tailgate-facing end — on a heavy-duty slide, against the driver-side REAR CORNER leg (1.5" in from the edge); and a **JAGAHAHA slide-out camp kitchen** (25⅞"L x 20⅛"W x 11.55"H closed, measured, with its own 2-burner stove space and built-in slide), against the passenger-side rear corner leg — its shelves swing out on that side. (Panel C's rear leg pair sits at the TRUE corners — inset legs would stand exactly in the appliances' slide paths, a collision the Rocky 40's extra width exposed. The rear-corner floor vents were checked and do NOT reach the leg area, so those legs land on solid floor.) **Both pull straight out the open tailgate** (not a side door). Neither is something this plan builds from plywood — both are bought, standalone units. Unlike everything else in this build, both are **strapped to the 3rd-row seats' latch hoops** via a shared no-drill anchor board (Section 8 — the owner ruled out drilling the vehicle) — too heavy to rely on the same "rests unbolted" approach as the sleeping panels. The gap between them is an **open utility bay — no door**. The exhaust fan blows the fridge bay's warm air INTO the bay and straight out toward the tailgate, and the control panel (switches + surge protector) mounts at the back of the bay on a backer board hung from the deck underside — everything electrical is reachable by just reaching in. The 2 utility bins that live in the bay's spare volume get hook-and-loop tape under them, since there's no door holding loose items in. - A **HEST Dually Long mattress (78" x 50" x 4", solid foam — no air chambers)** rides on the **49"** cantilevered platform: **25" of width per person** and ~1.25" of spare length parked at the head end beside the rear-pantry cluster. The platform overhangs the 46" boxes by 1.5" per side — the boxes stay at 46" (floor vents + liftgate pass-through), and the mattress lives ~18.5–22.5" up. ⚠️ **V7 measured wall-to-wall up there at 50" / 49.5" — the platform is 49", and the 50"-wide mattress is a ZERO-CLEARANCE fit between the walls that also overhangs its own frame by ~½" per side.** Solid foam tolerates both, but **dry-fit the foam in the van before cutting the platform to final width** — Appendix A, V7. Budget fallback: the DIY 2-layer foam build (Section 4/Component 9), same 50x78 footprint. - Panel legs sit inset **3.5"** from the deck's side edges so they land clear of the floor-level vent intrusion (**MEASURED V4**). The deck itself still overhangs the vents harmlessly at height. - **Power:** the fridge is 12V-native, but runs off the EcoFlow DELTA 3 stack (Panel A) via its own dedicated DC cord rather than the van's rear accessory outlet — see Section 1 for why. **The cooktop, Power strip 1, and the DELTA 3's own AC charging cord each get their own dedicated run to the front console** — see Section 5 for all three routes. - **Fire safety + CO safety (owner-placed):** the fire extinguisher and the low-level CO monitor are **deliberately not located in this plan** — both are owner-supplied and will be positioned manually once the build is in the van. Two reminders: the design has no propane at all (electric induction*

*cooktop), so the only combustion-gas risk is the Sienna's own engine exhaust while idling parked; and a generic 70ppm home-style CO detector is too slow for that job — use a low-level unit (alarms at 9/25ppm within ~60s). -* ***Camp power/cooling (EcoFlow):*** *an EcoFlow DELTA 3 Plus + Smart Extra Battery (~48 lb combined) stows unstacked, side by side, in Panel A's* ***right (passenger-side)*** *drawer — a normal drawer, not strapped down like Panel C's appliances, since 48 lb is well within what the drawer slides already handle and this gear is never used while driving. Within the drawer, the* ***DELTA 3 Plus sits outboard*** *(nearest the pull wall — it's what the WAVE 3 actually plugs into, used whether or not the extra battery is along) and the* ***Smart Extra Battery sits inboard*** *(grabbed less often, for extra runtime only). The* ***WAVE 3 portable AC/heater stores in Panel A's LEFT (driver-side) bay as open storage, not a drawer*** *— it's 20.4" wide, too wide for a boxed drawer's 19" clear interior, but the raw bay (20.75", no box walls to eat into it) fits it with ~0.35" to spare. It sits directly on the bay floor under a cam strap hooked to 2 D-rings on the frame, reached by hand through the driver's side door — Panel A has only one actual drawer (right/DELTA 3). For camp use, the unit gets carried to wherever it actually runs (Panel C's tailgate deck or the front seat — see the sleeping-configurations note below); it's storage-only in Panel A, not used in place there. -* ***Side-door access (V8).*** *The usable clear opening at the door's real stopping point measures* ***29"*** *against Panel A + Panel B's combined 58" drawer run, so barely half that run can land within reach from any one door position. The door's fore-aft position is* ***measured too****: the opening starts level with the front seatback, i.e. at* ***Y = 0****. So the 29" clear gap runs from 0" to 29" —* ***exactly spanning Panel A's 29" length****. Panel A is therefore reachable end to end, and Panel B (29–58") gets* ***zero overlap — blocked outright*** *behind the van's own body structure.* ***Both design decisions rest on measurements:*** *the DELTA 3 stack belongs in* ***Panel A****, and Panel B correctly has no drawers, which is why its bed top is two lift-out halves instead (Component 2). The coincidence is worth noticing — the door runs out precisely at the A/B seam, so there is no margin either way, and the DELTA drawer (22" fore-aft, centered in the 29") clears it with ~3.5" a side. Both units are charged at camp via shore power or solar — this design does not route van power to either (see Section 5). -* ***Why the fridge (and its fan system) run off the DELTA 3 stack, not the van's rear outlet.*** *It costs two things: a long cord run across all 3 module seams, and competing with the WAVE 3 for the DELTA 3's battery budget. This design accepts both costs deliberately, for a concrete reliability win:* ***the fridge keeps running when people are away from the vehicle, regardless of the van's ignition state*** *— many factory 12V outlets cut off with the key out, which would mean the fridge stops cooling every time the van is parked and locked, and putting the fridge on the DELTA 3 makes that risk moot rather than just hoping the outlet turns out to be always-hot.* ***The real numbers make this hold up:*** *the DELTA 3 Plus + Smart Extra Battery combined is 2048Wh; the Rocky 40 is rated 60W max / 45W ECO (dual-zone, compressor duty-cycled) — realistically ~450–1050Wh/day depending on mode, partition use, and ambient temp, so a full charge runs the fridge alone for roughly* ***2 to 4.5 days*** *unattended, worst case to best case. And recharging is fast: the DELTA 3 Plus's AC input maxes out at 1500W, an almost exact match for the Sienna's front console outlet (****confirmed 1500W****) — a full 0-100% charge takes about 56 minutes, so even a short errand run tops it back off. EcoFlow's DELTA 3 line supports charging and discharging at the same time (running the fridge while AC-charging is normal, it just slows the charge rate somewhat), so there's no need to choose one or the other.* ***The fan system runs off the DELTA 3 too, not just the fridge*** *— the fans exist to vent the compressor's heat, so they need to stay available under the exact same conditions the fridge does (parked, people away, ignition off); leaving them on the van's outlet would defeat the point. The rear 12V accessory outlet is unused by this design — free for anything else you'd like to plug in there later.* ***One real trade-off to plan around:*** *the front console circuit has 3 things wanting it (the cooktop, Power strip 1, and the DELTA 3's AC charging, which alone can draw up to the circuit's full 1500W) — see the shared-circuit note in Section 5. -* ***Modular lift-out design****: Panel A, Panel B, and Panel C are each built as an independent, self-supporting module — own frame, own 4 legs (the fridge and kitchen unit are bought products with no frame of their own, and aren't lifted out as part of this; the rear pantry isn't its own module either — see below). All three lift straight out and drop straight back in without touching the others (for deep cleaning or reconfiguring — day-to-day storage access is through the drawers/shelves, not by removing a module). With no tops or skirts, each panel is gripped by its exposed 2x2 top rails (no routed hand-holds needed), and anti-rattle bumpers + alignment pins + hand-released seam draw-latches at the A/B and B/C seams keep things fast to install/remove, quiet in transit, and clamped into one rigid beam (Component 5) — see Section 8. The rear pantry's tote and the microwave just sit on Panel C's deck under one cam strap (Component 1) — unhook the strap and either lifts straight out, contents and all.*

### *WAVE 3 sleeping configurations: tent vs. no tent*

*The WAVE 3 is* ***stored*** *in Panel A's left bay either way (see the DELTA 3/WAVE 3 stowage detail, Renders) — these two configurations are about where it gets carried to and set up for actual camp use, and where its hoses vent. Both use the same unit and DELTA 3 stack.*

***With a tent (tailgate open):*** *carry the WAVE 3 from its Panel A storage bay back toward the tailgate, setting it down on the mattress-covered part of Panel C's deck just forward of the rear-pantry cluster (the pantry's 14.5" occupies the tailgate-most strip of Panel C, so the WAVE 3's spot sits a few inches forward of that) — still blowing toward the open tailgate so the shared van+tent air volume gets conditioned together. A cheap non-slip mat under it keeps it from creeping while it runs. Its intake/exhaust hoses route the extra short distance past the pantry cluster and through the open tailgate gap to true outside air — the open tailgate itself is the path outside, no window seal needed. Power cord runs from the DELTA 3 stack in Panel A's right drawer, the length of Panels A/B/C.*

***No tent (sleeping in the Sienna alone, tailgate closed):*** *with the tailgate shut for security/weather, there's no gap to vent hoses through and nothing to usefully blow into at the tailgate — so the WAVE 3 instead moves to the front of the van:*

- *Move the WAVE 3 forward to the front passenger seat (seatback reclined or folded flat, whichever the unit sits more level on) — the only open seat/floor space once Panels A/B/C fill the rear cargo area. A non-slip mat or a simple bungee strap around the seat frame keeps it from sliding off during setup; it isn't a permanent mount, just an occasional-use position.*
- *Vent the hoses out a front window using the* ***EcoFlow WAVE Series Car Vent Kit*** *— a real EcoFlow accessory,* ***$39, officially compatible with both WAVE 2 and WAVE 3****: a 47.2" x 27.6" waterproof, UV-resistant nylon cover that velcros over the window opening (window cracked, not removed) with sealed duct pass-throughs for both hoses. 15-second install, no permanent modification, fully reversible.*
- *With the front window sealed around the hoses and every other window/door shut, the WAVE 3 conditions the* ***entire sealed cabin****, not just the back — likely more effective in this configuration than the tent setup, since the whole volume it's blowing into is the one you're sleeping in.*
- *Power cord routes back to the DELTA 3 stack in Panel A's drawer — roughly the same length run as the induction cooktop's own forward cord (Section 5), just the opposite direction (front seat instead of tailgate). Budget for an extension or a quick-disconnect if the stock WAVE 3 cord doesn't reach.*
- ***This configuration seals the cabin more tightly than the tent setup, with no open tailgate providing passive fresh-air exchange — the owner-placed low-level CO monitor (see the safety note above; Section 4) matters even more here.*** *Confirm it's active and audible from the sleeping position before using this configuration, same engine-exhaust-while-idling risk as always, just less passive ventilation to dilute it.*

## 2. Dimensions & Layout

| Section | Length | Width | Notes |
|---|---|---|---|
| Rear pantry — tote + microwave (on Panel C's deck, tailgate end) | tote bottom 15⅛" (⅝" past the 14.5" pantry end, into the mattress foot); microwave 14" | tote 25" (rim 30.15") + microwave 17.3" across 46" | **Bought, not built** — a 27-gal Tough Box tote on the driver side + the microwave flush to the passenger edge; one cam strap over both to footman loops on the side rails, mat underneath, no cleats. Power strip 1 + the ROLL bubble level still need a spot. |
| Panel A | 29" | 46" | **No top of its own** — capped by the bed frame's screwed-down section (Component 2); right (DELTA 3) drawer through the passenger door, left bay is WAVE 3 open storage through the driver door (no drawer box) |
| Panel B | 29" | 46" | **No top of its own** — capped by the same bed frame; **no drawers, no divider, no skirts** (the side doors don't reach it): a bare 2x2 frame whose bay is deep storage, loaded from above by lifting the platform + mattress |
| Panel C | 35.75" (21.25" mattress-covered + 14.5" rear pantry) | 46" | **Keeps its fixed top** (recessed flush between its rails on bearer cleats — deck plane 18.5") — the mattress-covered part only reaches ~21.25" into it (the rear pantry claims the last 14.5", Section 1), and the fridge/kitchen void underneath needs the enclosure regardless; no drawers |
| Fridge (BougeRV Rocky 40, in Panel C's void) | 28.74" deep (incl. handles) | 17.72" wide | 15.79" tall — drives `leg_height` via the mounted stack: 0.375" tray hang + 1/2" tray + fridge = 16.665", leaving 0.335" running clearance under the tailgate end rail; on 24" VADANIA slides mounted on the SIDES (nothing under the tray), against Panel C's driver-side rear corner leg (1.5" in from the edge) |
| Kitchen unit (JAGAHAHA, in Panel C's void) | 26" deep (closed) | 20" wide | 11.8" tall; against Panel C's passenger-side rear corner leg (1.5" in from the edge) — shelves swing out that side; own built-in tailgate slide |

**Drawer dimensions** (Panel A's single right-side DELTA 3 drawer — its left bay is WAVE 3 open storage, and Panel B and Panel C have no drawers):

| Drawer dimension | Value | Notes |
|---|---|---|
| Travel (pull-out direction) | 20" | Matches a standard 20" full-extension slide |
| Depth (fore-aft) | 22" | Fits between the panel's front and back legs — the **lapped** legs sit 1.4375" inboard at each end, so the bay's clear fore-aft span is 23¼", not 26". Clear interior 20.5" against the DELTA 3's 15.7" |
| Height | 13.75" | Inside the 16.25" leg-height storage bay (A/B legs, deck recess), under the removable bed platform (Panel A has no fixed deck of its own) |

`leg_height` (17") is driven by the fridge's mounted stack — 0.375" tray hang + 1/2" tray + 15.79" fridge = 16.665", leaving 0.335" of running clearance under the tailgate end rail (the slides mount on the tray's SIDES; nothing sits under it) — not by the folded 3rd-row well depth. **That 17" is Panel C only, and it is the height of the VOID — the rail underside — not the leg: Panels A/B's rails sit at 16.25" (deck recess), so the platform on their rails lands flush with Panel C's recessed deck at the one 18.5" plane. Every leg but Panel C's rear corner pair is lapped past its rail and so is cut 1.5" longer than the void it stands in (Section 3).** Measure your actual well and confirm 17" clears it before cutting — if your well is shallower, the fridge (and therefore every leg on the platform) still needs the full 17" of standing height regardless, so the platform would simply sit a bit higher off the true floor than the well alone would require.

### Walls & skirts, by panel

The panels are open 2x2 frames — plywood skins exist only where something needs one:

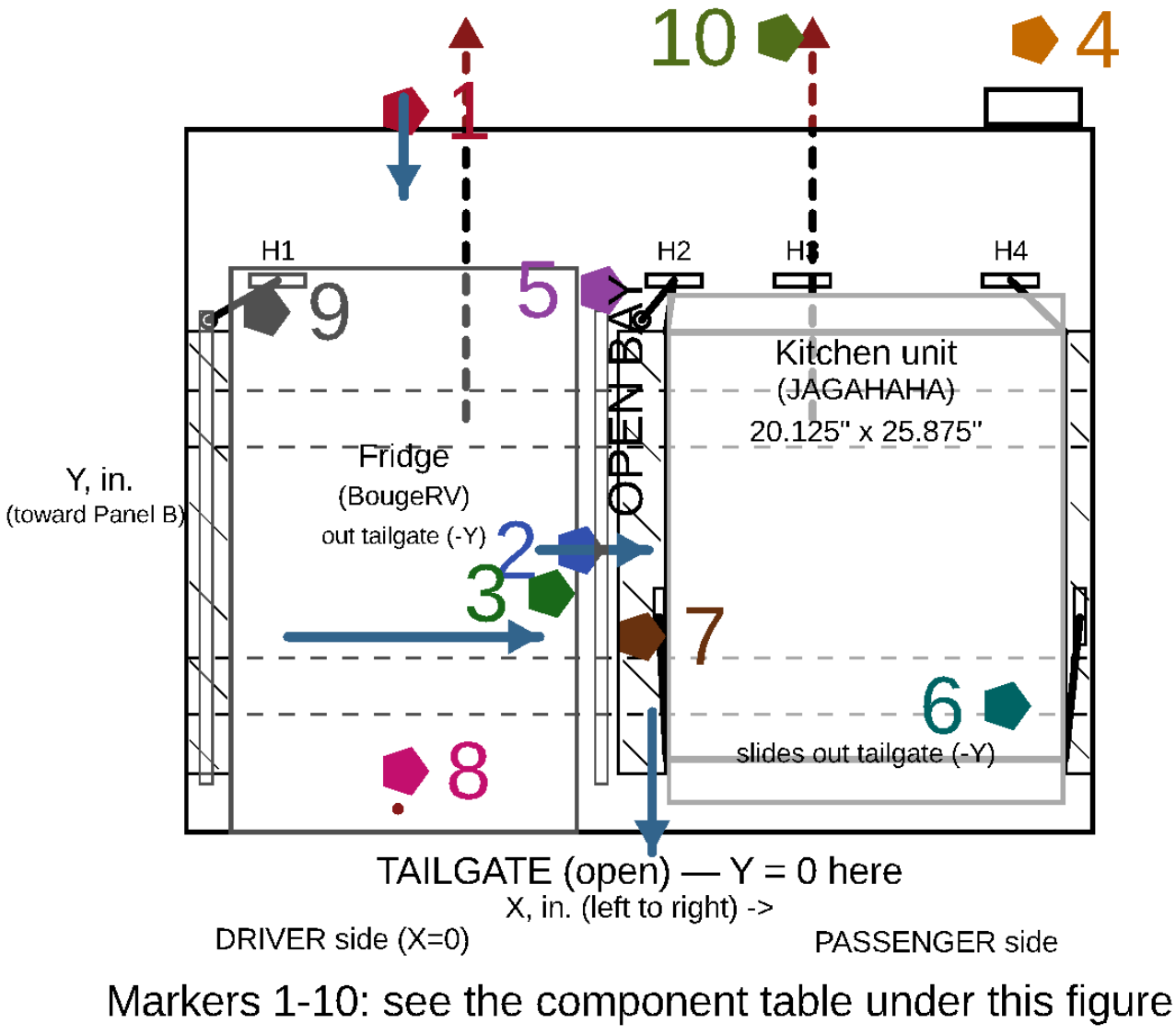
| Panel | Walls/skirts | Why |
|---|---|---|
| Panel A | **None** | Both bays face the side doors — a skirt would block the only access. Passenger side: DELTA 3 drawer. Driver side: WAVE 3 open bay. |
| Panel B | **None** | Nothing to see or reach — it sits under the platform between A and C. Bare frame, deep storage from above. |
| Panel C — front (B-facing) | **The ONE wall**: 1/2" ply, 46" x 17" | Mounts the 120mm intake fan, holds the three low intake vents, and passes the fridge DC line (one 1" grommet — Power strip 1 feeds off the rear outlet, not through this wall) — every opening dimensioned in the Panel C Front Wall render. |
| Panel C — sides | None | The van wall is ~1" away; the exhaust fan still pulls a net flow across the fridge, so side leakage doesn't matter. |
| Panel C — rear (tailgate) | None needed | Fully occupied already: fridge face + the open utility bay + kitchen unit face + kitchen drawer face. |

With no tops and no skirts, every panel's top rails are exposed — **grip those to lift a panel out**.

### Component coordinates

Two coordinate conventions, each picked to match how you'd actually stand at the access point in question — full reasoning and diagrams in `fridge_install_detail.scad` / `rear_view.scad`:

- ***Panel C*** *(fridge + kitchen system, AND the rear pantry — it sits on Panel C's deck, sharing this same coordinate frame): origin (0,0) at Panel C's* ***tailgate-facing DRIVER-side corner****, floor level. X increases toward the passenger side (0-46"), Y increases forward toward the Panel B seam (0-35.75"). Z increases up from the van floor. The rear pantry occupies Y 0-14.5 (the tailgate-most 14.5", above deck level) — the SAME Y range the fridge/kitchen's own tailgate-facing ends sit in, just at a different Z height (Z 18.5+ for the pantry vs. Z 0-16ish for the fridge/kitchen below deck).*
- ***Panel A, Panel B****: origin (0,0) at that panel's* ***own front-left corner*** *(front = toward the front seats). X increases right (0-46"), Y increases toward the tailgate (0 = that panel's own length). Z from the van floor.*




*Fridge install detail*

| Component | Panel | X (in.) | Y (in.) | Z (in.) |
|---|---|---|---|---|
| Kitchen unit (JAGAHAHA) | C | 24.5–44.5 | 0–26 | 0–11.8 |
| Fridge (BougeRV Rocky 40) | C | 2–19.7 (slide rails flank at ~0.6–1.5 and ~20.2–21.2) | 0–28.74 | 0.5–16.67 (hanging tray + fridge) |
| Intake fan (120mm) | C | 11.11 (on the front wall) | ~35 (Panel C's front wall) | 8.8 |
| Exhaust fan (120mm) | C | 19.7 (fridge's kitchen-facing wall) | ~14.4 | 8.8 |
| NTC temp sensor | C | ~18.2 (just inside the fridge's exhaust wall) | ~12.2 | 8.8 |
| Control cluster (switches, surge protector, W1209) | C | 40.4–45.4 | 35.75 (Panel C's FRONT wall) | 3.5–14.5 — on the wall's Panel-B face, standing ~2" off it into Panel B's bay |
| Power strip 2 (cooktop) | C | ~34.5–37.5 | ~26 (kitchen unit's front face) | ~1.5 |
| Right drawer (the only one) | A | 23.75–43.75 | 3.5–25.5 | 0–13.75 |
| Center divider | A | 22.25–23.75 | 1.5–27.5 | 0–16.25 |
| Rear pantry: 27-gal tote (bought; bottom, rim flares 2.6–2.7" each side) | C | driver 1.5–26.5 | 0–15.1 | 18.5–32.8 |
| Microwave (BLACK+DECKER EM720CPN-P) | C | passenger 28.7–46 | 0–14 | 18.5–28.7 |
| Power strip 1 + ROLL bubble level (relocated to the deck edge) | C | ~33 (passenger bay) | on the deck, back edge | ~19.25–20.25 (just above the deck) |

*Coordinates for the fixed/structural items (drawers, panels, fridge, kitchen) come directly from `params.scad` / `platform.scad`'s real geometry. Power strip 1/2 positions are schematic markers in the 2D diagrams (approximate, not separately modeled in the 3D assembly) — confirm exact placement against the real frame once built.*

***Fridge clearance vs. the manual (CR04001, saved user manual):*** *BougeRV asks for* ***200 mm of free air at the compressor end and 100 mm at the sides*** *— free-convection numbers the enclosed bay can't offer (the van wall is ~1" away). The forced cross-flow above is the substitute, and the manual makes the geometry work in its favor: the compressor, louvers, control panel, and power port are* ***all at ONE end*** *(manual-confirmed — this answers measurement row #4), and Component 7 faces that end toward the tailgate. So intake air enters at the front wall, washes down the bay past the fridge, and the exhaust fan on the kitchen-facing wall pulls it out across the louvered end into the open utility bay, where it exits straight toward the tailgate (no door in the way). Two things keep that substitution honest: the louvered end must stay unblocked (the plenum strip in front of the appliances stays empty — see the access notes below), and the NTC probe + controller exist precisely to prove the airflow is doing its job — if the fans can't hold the bay near ambient, the fix is more airflow, not less clearance. Confirm the louver faces on the physical unit when it arrives (measurement row #4).*

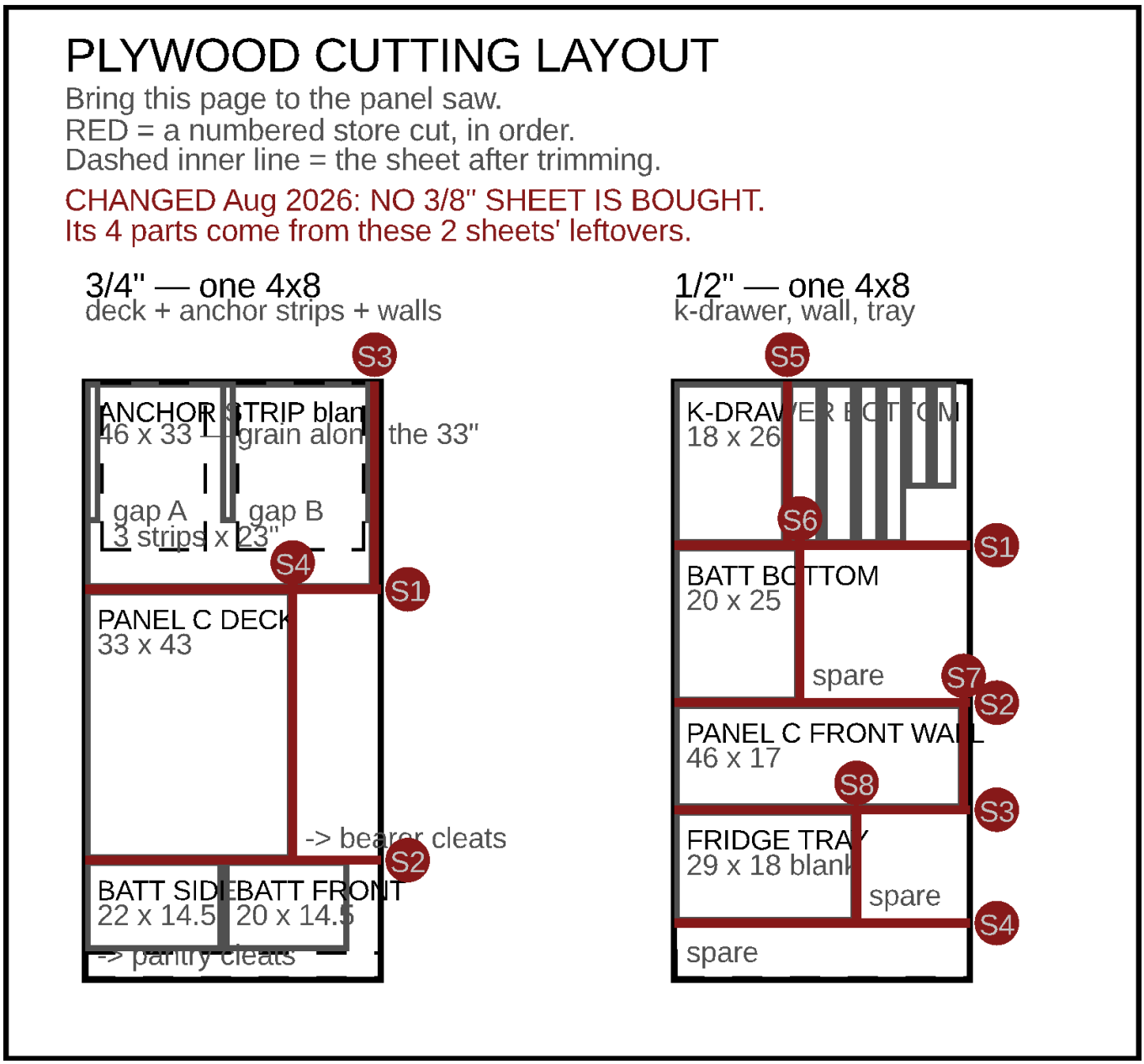
### How under-deck storage is accessed, by panel

- ***Panel A****: a* ***center divider*** *splits its under-deck void in two. The right (passenger) side is the one real drawer on a 20" full-extension slide (the DELTA 3 stack); the left (driver) side is* ***WAVE 3 open storage*** *— no box, no slide, the unit sits on the bay floor under a cam strap, reached by hand through the driver's side door. Both van side doors are used, both for Panel A only.*
- ***Panel B****:* ***no drawers, no divider*** *— the side-door openings don't reach it. Its whole bay is deep storage: lift the platform + mattress off and load from above.* ***The SPARE TIRE stows here*** *(RJ-MODINI kit, flat on 3" cleats at the axle — the best weight placement in the van, Component 3) with* ***2× Sterilite 28-Qt lidded totes restacked on top*** *and the jack kit beside it — you lift out whole labeled totes instead of rummaging a loose pile (plain lidded totes on purpose: with no side access, a slide-drawer front would be a wasted feature). Use it for the long-term/bulky stuff you don't touch at camp.*
- ***Panel C****: no drawers at all. Its entire under-deck void is split between the fridge (left/driver side, on its own slide) and the kitchen unit (right/passenger side, on its own slide) — both bought products, both accessed by pulling them out through the* ***open tailgate****, not a side door. There's no "storage bay" here in the drawer sense; the fridge and kitchen units are the storage, permanently installed rather than removable containers. (There's a ~7–10" strip of under-deck space in front of the stowed appliances, but it's deliberately* ***left empty*** *— it's the intake fan's cool-air plenum, and it's boxed in with poor access, so filling it would choke the fridge cooling. The found-storage additions instead reclaim the dead headroom above the DELTA 3 and WAVE 3, and the open utility bay's spare volume — see Component 8.)*
- *None of the 3 panels needs to be lifted out for routine storage access — lifting a panel (gripping its exposed top rails) is only for deep cleaning or reconfiguring, per Section 8. (Panel B's bay is the exception in spirit: loading it means lifting the platform, though never the panel itself.) The rear pantry's drawer units are the one exception with routine lift-off access: loosen the cam strap and any unit lifts out in seconds — day-to-day restocking just opens the drawers, nothing needs removing.*

### *3. Full Lumber Sizing & Cut List*

*The fridge and kitchen unit are bought products (Section 4) — **neither needs any plywood or lumber**, which is why this cut list is smaller than it would be if they were built.*

#### *Have the store break the sheets down*




*Plywood cutting layout*

*Notes for this sheet:*

- *left to right after S3: 2 cheeks 5.45 x 26, 2 sides 4 x 26,*
- *PLYWOOD CUTTING LAYOUT — STORE CUTS vs SHOP CUTS*

*Every red cut is a straight, full-length or full-width cut a store panel saw can follow, numbered in the order it has to happen. The dashed inner line is the sheet after its factory edges come off. Bring this page to the saw.*

***Two policies, both owner decisions:***

1. ***Trim ~½" off all four factory edges first** (cuts T1–T4 on every sheet). Factory edges are chipped, out of square, and frequently not straight — and every part referenced off one inherits that. Trimming first means every later cut runs off a known-straight edge. **The usable sheet is therefore 47" × 95", not 48" × 96",***

***and every part in the tables below is checked against that.***

2. ***Let the store make the rectangles.*** *Ask for the crosscuts before the rips — a crosscut on a full sheet is far easier to hold straight than a rip, and it makes each following piece small enough to handle alone. What comes home is rectangles; what's left for your saw is only the narrow ¾"×¾" cleats, the anchor board's strips, the deck's leg notches and the holes.*

***Two things to know before you go:***

- ***Store saws hold about ±⅛".*** *For the parts that must end up exact — Panel C's deck, its front wall, the fridge tray —* ***ask for them ~⅛" oversize*** *and trim at home. For everything else, store tolerance is fine.*
- ***Many stores won't rip narrower than ~4".*** *If yours won't, bring the kitchen drawer's 5.45" cheeks and 4" sides home as one wide strip and rip them yourself — easy fence cuts once the sheet is already broken down.*

⚠️ ***No 3/8" ply is bought — these four parts come out of the two sheets you're already buying. Where each part goes:***

| *Part* | *Thickness* | *Cut from* |
|---|---|---|
| *Panel C front wall, 46 × 17* | ***1/2"*** | *the 1/2" sheet's big 47 × 44 leftover* |
| *Fridge tray, 17.72 × 28.74* | ***1/2"*** | *the same leftover,* ***cross-grain*** *(see below)* |
| *Battery drawer, 2 sides + 2 front/back* | ***3/4"*** | *the 3/4" sheet: 1 side + 1 front out of its 47 × 19 strip, the other 2 from the sheet's remaining offcut* |

***Three consequences:***

1. ***It costs ~16 lb*** *over 3/8" parts (build ≈488 lb, Appendix F).* ***12 lb of that is the battery drawer's*** *3/4" walls — a drawer that also carries the 48 lb DELTA 3 stack, so its 20" slide pair sees ~78 lb of a 100 lb rating (margin ~22%). It still passes, and payload has room (Appendix F), but this is the one place the weight is felt rather than just accounted for. If you'd rather not carry it, one 4×4 3/8" panel covers just those four walls.*
2. ***The drawer box's corner biscuits are 2× R3, not R1*** *— R1 is thin-stock sizing and would be lost in 3/4" ply.*
3. ***Build order:*** *nothing on the ¾" sheet depends on another part being cut first.*

***The walls are 13.75" tall.*** *The bay is* `leg_height_ab 16.25 − 1.5 − 1.0 = 13.75"`*. Measure your own frame's bay before cutting — top of the bottom rail up to the underside of the top rail — since it is set by the legs' cut length and is not affected by how far the leveling feet are wound out.*

***Why 3/4" for the walls and not 1/2".*** *Not preference — capacity. The 1/2" sheet's leftover holds the front wall and the tray and then two of the four walls, and stops: every remaining offcut is under 27.5" in its short direction, so each yields exactly one 13.75" strip and no more. The only leftovers that do are 3/4". Mixing (two 1/2" walls, two 3/4") would break the box's 20"/22" arithmetic and its biscuit sizing, so the box is one thickness throughout.*

***And why the tray is cut cross-grain.*** *The only leftover big enough puts the tray's 28.74" across the sheet's width, so its face grain runs along the 17.72" instead. The extra thickness more than pays for it — bending stiffness goes as t³, and (0.5/0.375)³ = 2.4× against roughly a 2× cross-grain penalty — and the two glued 1×3 aprons carry the long span anyway.*

`render.sh` *also generates the top-down/side/rear assembly views; the per-piece tables below stay as tables — they're easier to keep correct as dimensions change, and just as usable at the saw.*

### *Material options & upgrades*

*The base build is* ***Baltic birch plywood + pine framing****, chosen for cost and availability. Two optional upgrades trade a little weight and money for durability — worth it in targeted spots, overkill everywhere:*

- ***Plywood is already Baltic birch*** *for every panel that matters (Panel C's deck, the drawer/kitchen boxes, the fridge tray). No change needed — birch is the dent- and screw-holding win people usually ask for, and it's already specified.*
- ***Poplar (or ash/maple) for the top rails.*** *Pine dents, and the one place that matters is the exposed* ***2×2 top rails you grip to lift each module*** *— those get handled constantly. Swapping just those rails to poplar (a hardwood) resists denting and holds biscuits/screws noticeably better than pine, which suits the biscuit joinery. Cost: poplar is ~20% denser (a few lb of added lift weight across the rails) and pricier per board — so upgrade the* ***grip rails only****, and leave the hidden framing and the under-mattress 1×4 slats as pine (weight and cost win there, and they never show or get bumped). Budget ~$25–40 extra.*
- ***Aluminum angle for the diagonal corner braces.*** *The Panel A/B corner braces can be aluminum L-angle instead of steel/wood — lighter (these modules are carried) and rust-proof. Budget ~$10–15 for the aluminum swap.*

*Neither upgrade changes any dimension in the cut list below — same sizes, different stock. The weight deltas are folded into the note in* ***Appendix F (Weight Budget)****.*

***Weight-reduction swaps.*** *Separately, and pulling the other way, the build is lightened wherever safe — plywood is thin wherever it isn't carrying real load. This is reflected in the cut list and in Appendix F:*

- ***Kitchen-drawer cheeks → 1/2"****;* ***bed platform → 15 (short) slats over 6 rails****, plus Panel B's center bearer (fine under the solid-foam mattress). See Component 2.*
- ***The rear pantry goes further than thinning — it's not plywood at all:*** *it's one bought 27-gallon tote (~7 lb with its strap, Component 1).*
- ***Kept 3/4":*** *Panel C's deck — it carries sitting load.* ***Note the tension with the poplar upgrade above:*** *poplar adds ~20% weight, so if you're chasing weight, keep poplar to the grip rails only or skip it.*

### *Plywood — 1 sheet 3/4" + 1 sheet 1/2", Baltic birch (or shop-grade) — **two sheets, nothing else***

***Thickness note.*** *Two sheets are bought and that's all:* ***one 3/4"*** *and* ***one 1/2"****. Panel C's deck stays 3/4" (it carries sitting load) and so do the battery drawer's four walls (from offcuts); the kitchen boxes, the battery-drawer bottom, Panel C's front wall and the fridge tray are 1/2". Nothing in the build is 3/8" ply; between them the*

*two sheets carry every plywood part.* ***Panel A/B have no tops*** *(Component 2), and* ***Panel B has no drawers.***

***3/4" sheet*** *— Panel C's deck, the Section 8 anchor board's three strips and the battery drawer's walls (Component 1 is prefab drawers). Even so this sheet is still under half used: rip the cleats from the offcut, and keep the rest for the control-panel backer board (Section 6) and repairs:*

| Piece | Qty | Dimensions |
|---|---|---|
| Panel C top (deck) | 1 | 33" x 43" — drops BETWEEN the rails onto bearer cleats, flush with the rail tops (deck recess: 3/4" more headroom) |
| Anchor board strips (Section 8) | 3 | **22½" long**, 2½", ~2½" and 1½" wide, each one piece; tied by steel rungs, not glued (Anchor Board sheet) |
| Deck bearer cleats | 4 | 3/4" x 3/4" strips: 2 x 33" + 2 x 40" (ripped from the offcut; screwed to Panel C's rails' inner faces, tops 3/4" below the rail tops). **Interrupt the FRONT cleat at the 2 lapped front legs** — their tops come up into the deck plane on that face; cut it as 3 short runs rather than one 40" strip |
| **Battery drawer side wall** | 2 | **22" x 13.75"** — full depth, and the slide pair screws to these two. One out of the 47" x 19" strip, one from the sheet's remaining offcut. 3/4": glue + **2× R3** per corner |
| **Battery drawer front/back wall** | 2 | **18.5" x 13.75"** — they fit BETWEEN the side walls, so cut them at 20" with everything else and trim **1.5" (= 2 × 3/4")** off each once the sides exist. One out of the same 47" x 19" strip beside the side wall (22 + 20 = 42 of the 47), one from the sheet's remaining offcut. 3/4", 2× R3 per corner |

***1/2" sheet*** *— the kitchen drawer box + its 2 hanging cheeks and the battery drawer's bottom:*

| Piece | Qty | Dimensions |
|---|---|---|
| Kitchen drawer box | 5 pieces | bottom **18" x 26"**; 2 sides 26" x 4"; front/back **17" x 4"** (4.5" exterior height) |
| Kitchen drawer hanging cheeks | 2 | 26" x 5.45" (1/2" — tied by 3 spreaders into a squared unit, then screwed to Panel C's recessed deck, flanking the drawer) |
| Kitchen drawer spreaders (¾", from the ¾" sheet's offcut) | 3 | 19" x 2" — between the cheek tops, flush, 3/16" over the box |
| Battery drawer bottom (Panel A right) | 1 | **19.25" x 21.25"** (1/2" — the base under the 48 lb stack. **20 x 22 is the box's OUTER footprint**, which is the size for a bottom sitting under the walls; this one is housed in a 3/8" glued rabbet, and that takes 3/4" off each dimension. Cut it last, off the glued-up box, not to the number) |
| **Panel C front wall** | 1 | **46" x 17"** (1/2") — 120mm fan hole + two 1" grommet holes, positions in the Panel C Front Wall render. Cut ~1/8" oversize and trim to the frame |
| **Fridge tray** | 1 | **17⅞" x 28¾"** (1/2" — the fridge's base + ⅛") + two 2½" side aprons, 28¾" each (a 1×4 ripped to 2½"; Fridge Tray sheet), glued+screwed to its edges — the slides' moving members screw to them and their top edges are the fridge's anti-shift lip. Cut from a 29 x 18 blank, **cross-grain** (see the buy note above) |

***No third sheet.*** *Every plywood part is in the two tables above — front wall and fridge tray on the 1/2" sheet, all four battery-drawer walls on the 3/4". The WAVE 3 needs no shelf or glide strips: it sits on the bay floor under a cam strap (Component 8, step 2).*

### *The sheets, as actually cut — from the shop (Sept 9–11 2026)*

***Both sheets are cut, and they broke down as drawn*** *— no dimension, no part and no offcut in the tables above changed. What these add is the method. The division of labour turned out to be two saws —* ***table saw for every long cut, jigsaw for every interior cut*** *(the deck's leg notches and the front wall's holes) — with a drill bit standing between them.*

***Mark off the printed cut plan, on the sheet, before any blade runs.*** *The sheet layout from Section 3 is printed and sitting on the work — not read off a phone and not held in memory. The straightedge here is a* ***4 ft level****, which is the longest true edge most shops own and is what a marked line should be struck against; the face grain direction gets confirmed at this stage too, because for the anchor board it is specified (****grain along the 33" direction****) and it is unrecoverable once the piece is cut. Note what has already happened before this photo: the store broke the sheets into rectangles (Section 3), so what is on the table is a manageable panel rather than a full 4×8.*

***Every long cut went on the table saw.*** *Section 3's cut plan offers a track saw or a circular saw on a straightedge; the saw that was in the shop was a table saw, and for a sheet already broken into rectangles it is the better of the three — a fence gives a repeatable width, where a straightedge has to be re-clamped and re-measured for every cut.* ***What a table saw cannot do is an interior cut****: the deck's leg notches stop part-way across. Those are the jigsaw's, below.*

***Put a reverse-tooth blade in the jigsaw — this is the detail worth stealing.*** *The pack is* ***Bosch T101BR****, 10 TPI: the* ***R is for reverse****, meaning the teeth point down instead of up. An ordinary jigsaw blade cuts on the upstroke and therefore splinters the* ***top*** *face — which on this board is the face you marked, the face you can see, and on veneered ply the face whose thin outer ply tears in long strings. A reverse blade moves that tearout to the underside. Two things come with it: it* ***pushes the saw up off the work****, so hold the shoe down and feed slower than you would, and it wants a* ***flat, fully supported*** *cut — it is not the blade for a plunge. Also set the saw's* ***orbital selector*** *(the SELECTOR dial on this SKIL) to suit the stock: more orbit cuts faster and rougher, and on 3/4" ply the low settings are what keep the edge clean. That dial is the "set your jigsaw for the thickness of the wood" step, and it is the one most people never touch.*

***Both sheets, down to parts — this is what "two sheets, nothing else" looks like.*** *Panel C's deck, the kitchen box pieces, the battery drawer's walls and bottom, Panel C's front wall, the fridge tray, the ripped* ***3/4" × 3/4" bearer cleats*** *(the long thin strips standing at center), and, at the right, the first anchor board, a one-piece comb since replaced by three strips (Section 8). Stand them on edge rather than stacking them flat: ply left lying in a pile in a garage picks up a bow from its own weight and from whatever humidity the slab is holding, and the deck and front wall are the two parts that have to stay flat.* ***Label each piece as it comes off the saw*** *— several of these are near-identical rectangles that differ by an inch, and the cut plan's names stop being obvious about a day later.*

***The WAVE 3 earned its keep before it was ever installed.*** *The same air conditioner this build carries (Component 8, and the cooling appendix) spent the cut days cooling the garage, stacked on its add-on battery exactly as it sits in the van. That is a free rehearsal worth paying attention to: it is where you find out how loud it is, how long the battery really lasts at your ambient, and how much clearance the rear intake grille actually needs behind it — all of which set the WAVE 3 bay's dimensions in Panel A, and none of which a spec sheet tells you.*

### *Frame lumber — 2x2 pine (or 1"x1" aluminum L-channel), sold in 8ft (96") lengths*

*Panel A, Panel B, and Panel C each get their own independent perimeter frame — 2 side rails + 2 end rails sized to that module's own footprint — plus its own 4 legs.* ***Which rail is outboard matters*** *(listing both at full size double-books 1.5" × 1.5" at all four corners): the* ***end rails run the full 46" width*** *and the* ***side rails fit between them****, 3" shorter than the panel's length. End rails are outboard for three reasons — they are the face the lapped legs bear against, Panel C's 46" front wall matches them, and it is the end rail that covers Panel C's true-corner rear legs. No rail is shared between modules, which is what makes each one lift straight out on its own.* ***The rear pantry needs no frame lumber at all*** *— it's bought drawer units sitting on Panel C's already-built deck. The fridge and kitchen unit need no frame lumber either.*

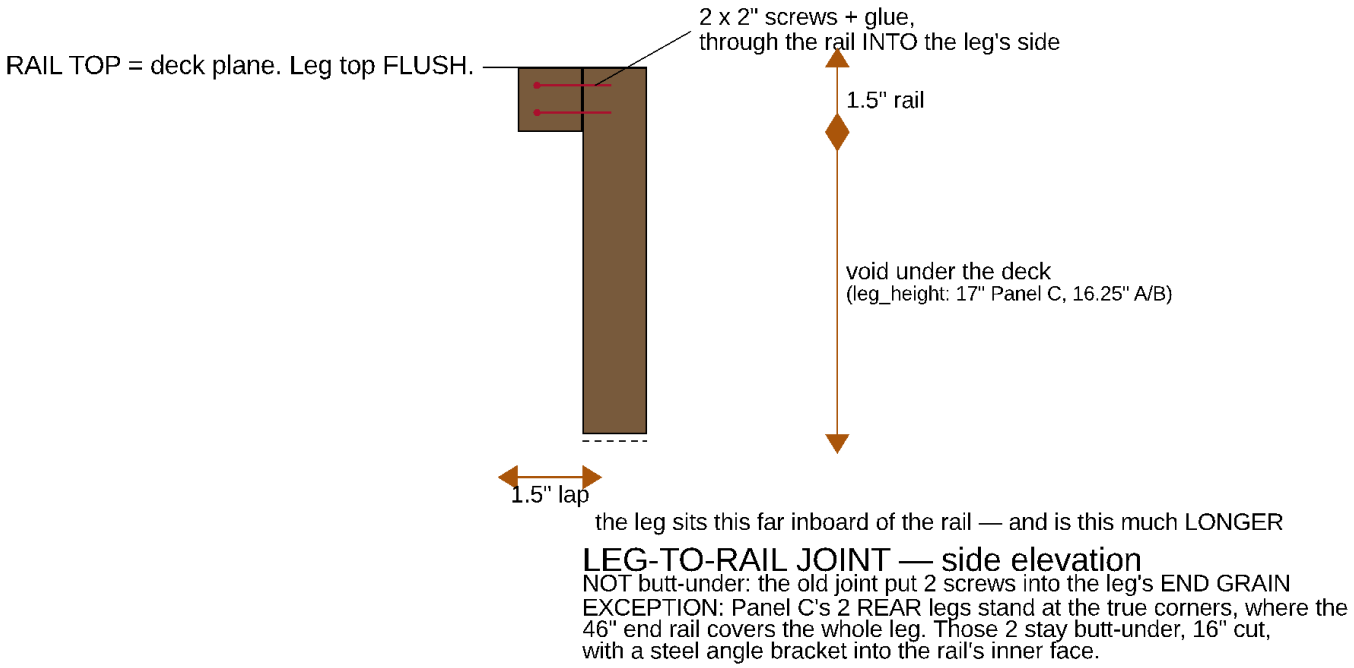
| ***Piece*** | ***Qty*** | ***Cut length*** | ***Notes*** |
|---|---|---|---|
| *End rails (all 3 modules)* | *6* | *46"* | *2 per module.* ***The OUTBOARD pair*** *— full panel width, so they set the 46" and the side rails fit between them* |
| *Panel A side rails* | *2* | ***26⅛"*** | *= 29" panel length − 2 ×* ***1.4375"*** *of end rail (MEASURED stock)* |
| *Panel B side rails* | *2* | ***26⅛"*** | *= 29" panel length − 2 ×* ***1.4375"*** |
| *Panel C side rails* | *2* | ***32⅞"*** | *= 35.75" panel length − 2 ×* ***1.4375"*** |
| *Center divider* | *1* | ***26⅛"*** | *Panel A only — splits its bay into drawer (right) + WAVE 3 (left) runs, see Step 2 diagram. Panel B and C have none.* |
| *Legs — Panel C, FRONT pair* | *2* | *17.5"* | ***lapped****; cut 1" short, the leveling foot makes each an effective 18.5" — floor to rail TOP (Leg Leveling Foot Detail)* |
| *Legs — Panel C, REAR pair* | *2* | *16"* | ***butt-under****, at the true corners — the only 2 legs in the build that are. Effective 17". Each gets its own steel angle bracket into the rail's inner face* |
| *Legs — Panels A/B* | *8* | *16.75"* | ***lapped****; cut 1" short, foot brings each to an effective 17.75" — 3/4" shorter than Panel C's front pair (deck recess: the platform on A/B's rails lands flush with C's recessed deck)* |
| *Panel C side braces + blocks* | *2 + 2* | ***29½"*** *+* ***4⅞"*** | *Underside 12½" off the floor, in the rear legs' outer band under each side rail: rear leg's forward face → a block glued across the front wall's fridge face (wall end → front leg), the block screwed through the leg + wall. Cut each brace to the gap you measure. 2 × 3" SPAX into each brace end + glue (Panel C sheet, Section F)* |
| *Bottom rails (cube frames)* | *5x 46" + 2x* ***23¼"*** | *see notes* | *underside at 1" (at the leg bottoms): Panel A both END faces (2x46), Panel B all 4 faces (2x46 + 2x23), Panel C FRONT face only (1x46) — 2x 2½" screws + glue into each leg.* ***OFFSET those screws ~⅜" either side of the leg's centerline:*** *the leveling-foot bore is ½" dia × 1¾" deep up the leg's end grain, which is DEEPER than the rail band is tall, so a centered screw hits the insert at every height. ~½" of solid wood each side of the bore takes them. Applies to all 14 bottom-rail-to-leg joints. They screw into the legs, so they sit on the legs' line: 1.5" inboard wherever the legs are lapped, which is why Panel B's side pair is 23¼"* |

***The stock measured 1-7/16", not 1½" (MEASURED).*** *The 2×2 bought for this build reads* ***1.4375"*** *across the face (cross-checked at ~36 mm) — neither the 1½" of kiln-dried board nor the 1⅜" of furring, most likely 1½" stock that dried and shrank. Measured governs, so* ***every side rail and Panel B's bottom side pair are sized off 1.4375"****, to hold each panel at its designed 29" / 35.75" length. Size them off 1½" instead and each panel lands ⅛" short — ⅜" across the three, against the*

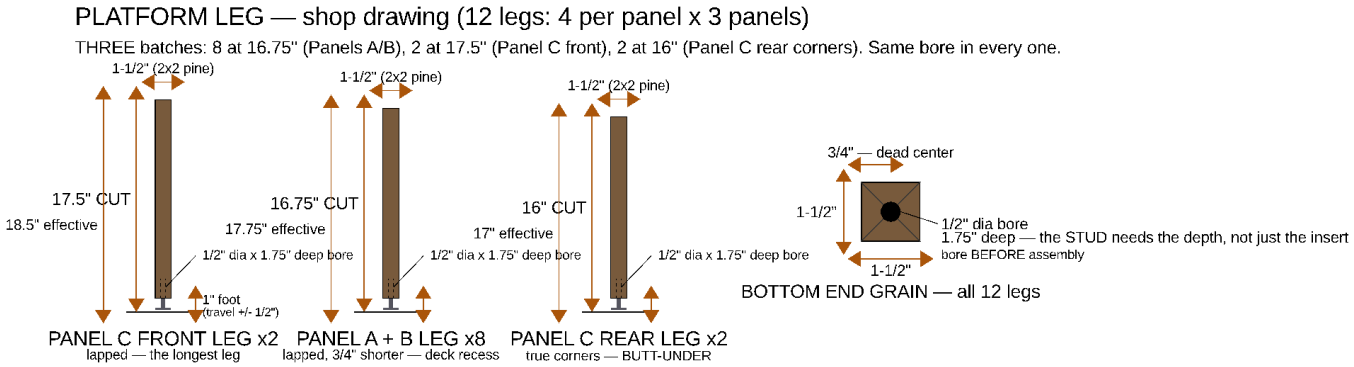
*seatbacks and the tailgate.* ***The 46" end rails, the 46" bottom rails and all twelve legs are unaffected:*** *end rails set the full width regardless, and leg length runs floor-to-rail-top, so stock thickness never entered it.*

***Cut the end rails first, then measure each panel's actual gap.*** *The numbers above are the design intent and your sanity check, not gospel — SPF 2×2 varies board to board and along a single board, which is the same reason H.4 tells you to sort the bundle. So: cut the six 46" end rails, stand each panel's pair up, measure the real opening, and cut that panel's side rails to it. Do the same for Panel B's bottom side pair once its legs are on. Costs nothing but sequence, and it is immune to a board running 1.40" beside one running 1.46". Panel C's deck is already handled this way (H.8 — cut oversize, trimmed to the measured opening).*

***The leg-to-rail joint.*** *A leg is* ***lapped*** *to the inner face of its end rail and runs up* ***flush with the rail top*** *— it is not butted under the rail. The two screws then pass through the rail's face into the leg's side: long grain, loaded in shear, on a real face-to-face glue joint. Butting the leg under the rail instead puts those screws into end grain, where they have almost no withdrawal strength and the glue does nothing. Two consequences run through this whole section: each lapped leg is* ***1.5" longer*** *than the void it stands in, and it sits* ***1.5" further inboard*** *along the panel's length.* ***The two exceptions*** *are Panel C's rear pair — at a true corner the 46" end rail covers the leg's entire footprint, so there is no face to lap to and no room to move inboard (Panel C's width chain has 1.28" of slack). Those two are butt-under, and the steel angle bracket on each is not optional.*




*The lap, in side elevation with the end rail seen in section. The leg's top is* ***flush with the rail top*** *— the 18.5" deck plane — not stopped at the rail's underside. The 1.5" of leg above the void is what makes an A/B leg 16.75" and not 15.25".*

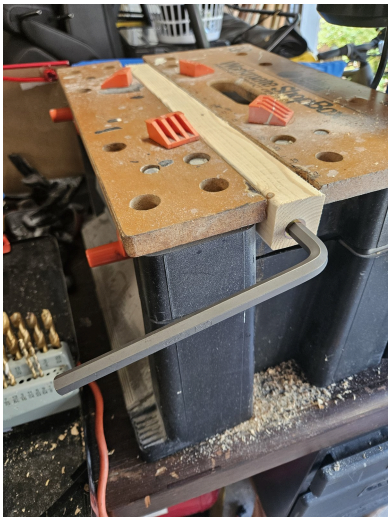



*The leg as a part, all three batches: Panel C front (17.5" cut, lapped), Panel A/B (16.75" cut, lapped), Panel C rear corner (16" cut, butt-under), and the bottom end grain with its 1/2" × 1-3/4" insert bore dead center.*

### *Boring the legs and fitting the feet — from the bench*

*Photographed doing it (Sept 3 2026). Four things here that no drawing was telling you:*

***Two drills, so no bit ever gets swapped.*** *Pilot bit on the right, 1/2" brad point on the left with a* ***blue-tape depth flag*** *wrapped at 1-3/4". Bore the pilot, put that drill down, pick the other one up, bore to the tape. Twelve legs is twelve pilots and twelve bores — one drill means 24 chuck changes. Measure the flag from the bit's* ***full diameter****, not the brad point's tip.*



***The insert takes a 10mm (1 cm) hex wrench*** *— "CRV 10mm" on the shaft here. Not 3/8", not 7/16": metric, and the one tool on this job you are least likely to already own. Clamp the leg so both hands are free to keep the key square; the first two turns set the angle for all of them.*

***Why the bore is 1-3/4" and not 7/8".*** *The foot threaded home — stud, fixed hex collar, 1-3/8" pad. Almost none of the stud is showing, because all of that length is inside the leg. Bored to 7/8" it bottoms out on solid wood and the leg loses the bottom half of its travel. Note also what is not on the face: no stick-on pad, and no star knob on the stud.*

*The feet, for re-ordering: **X002FCXYQT** — "Furniture Leg Leveler … Made in China", 198 g per pack. That is an Amazon-style seller barcode (the `X00…` prefix), **not** a manufacturer UPC: it will find the listing again on Amazon, but it will not scan in a store. Photograph the bag before it goes in the bin.*

***Total linear footage: 1,002⅛" as designed — 801⅛" still to cut** (the twelve legs, 201", are already cut, bored and footed). Remaining: 6@46 + 4@26⅛ + 2@32⅞ + 1@26⅛, plus the cube-frame bottom rails 5@46 + 2@23¼, and Panel B's 2-piece center bearer 2@26⅛. Already cut: 2@17.5 + 8@16.75 + 2@16. Nine 8ft (96") boards hold 864", so the eleven bought leave two spare — see the board-by-board packing plan in **Appendix H**.*

### *Hardware sizing*

| Item | Size needed |
|---|---|
| Drawer slides | 1 pair, 20" full-extension — Panel A's right (DELTA 3) drawer, mounted between the drawer box and the frame's long rail / center divider. Panel B has no drawers (side doors don't reach it) |
| Drawer catches | 1, simple friction catch or small turn latch — Panel A's drawer |
| WAVE 3 hold-down | 2 screw-eye D-rings in Panel A's two END bottom rails, either side of the WAVE 3, + 1 low-profile cam strap fore-aft over the unit — it sits straight on the bay floor (no shelf, no glide strips, no slide hardware), and the strap stops it bouncing into the ~2.8" of headroom or walking out the bay's open driver-side face |
| Wood screws | 1.25" for plywood-to-frame, 2" for frame-to-frame joints |
| Corner brackets | 2" or 3" steel L-brackets, 12 total (4 per module x 3 modules — the rear pantry has no frame of its own, no brackets needed) |
| Cable grommets | 1" diameter, 7 needed (1 cooktop line, 1 Power strip 1 line — both short rear-outlet hops, 3 fridge DC line, 2 DELTA 3 AC charging line — see Section 5) |
| Hand-hold holes | **None** — with no tops or skirts, every panel's 2x2 top rails are exposed; grip those to lift. |
| Anti-rattle bumper strips | 2 strips (one at each lift-out seam: A/B, B/C — the pantry and fridge/kitchen aren't lift-out modules, so no seams of their own). Adhesive-backed felt or closed-cell foam weatherstrip |
| Alignment dowel pins | 4 pins (2 per seam x 2 seams), 3/8" dowel rod, ~3/4" long, friction-fit into a matching hole on each module face |
| Seam draw-latches | 4 over-center draw latches (2 per seam x 2 seams), stainless/zinc, ~3" body. Mounted LOW on the bottom-rail band (~1.75" up), over the inset leg line, one on each side of each seam. Base plate on one module, hooked bail + lever on its neighbour — flip to clamp the two tight against the bumper, flip to release. Alignment pins LOCATE the modules; these CLAMP them into one rigid beam (with the bed platform tying the A/B tops). Hand-released, so the modular lift-out is preserved — see Component 5 |
| Rear pantry — tote + microwave | 1 × 27-gal Tough Box tote (bottom 15⅛ × 25", 14.3" tall) beside the microwave; **1 × 1" cam strap with snap hooks, 8 ft**, over both; **2 stainless footman loops** (from the spares) on the outside face of Panel C's side top rails, 7½" from the tailgate end, ¾" below the deck top, on their #10 × ¾" screws. No cleats: the deck mat grips, the mattress stops them forward |
| Panel B deep-storage totes | 2x Sterilite 28-Qt under-bed lidded totes (23.5" x 16.9" x 5.9"), restacked ON the spare in Panel B's bay — containerized top-load storage (lift the platform, lift out whole totes) |
| Spare tire stowage | RJ-MODINI kit (T155/85R18 on an 18x4 steel wheel, 28.5" dia x ~6.4" stored, ~40 lb w/ 2-ton jack) FLAT in Panel B's bay at the axle: a cradle skid (2 battens 2⅛" × 24" standing on edge, fore-aft between the 2nd-row seat tracks (centered 18½" and 27½" from the driver edge), with 3 cleats 3" × 30" glued and screwed flat across them, all ¾" ply — 2⅞" tall, so the cleats clear the seat tracks and caps and the spare still clears the 2-7/16" bottom-rail curb (cut 14 Oct)) over a non-slip liner offcut + 1" cam strap to 2 footman loops on the bottom rails + tool case nested in the wheel — install/access diagrams on the Spare Tire Stowage render |
| Panel A/B diagonal corner braces | 4 used (2 per panel, Panels A + B) from the purchased Sumnacon 4" 8-pack (4 spares), steel flat-strap — recovers some of the racking rigidity lost by removing each panel's own top |
| Leg leveling feet | 12 total (4 per panel x 3 panels), 3/8-16 screw-in threaded insert (7/16" OD coarse outer thread — the Anwenk kit's "T-nut") + leveling stud with a 1-3/8" pad and a fixed hex collar (the collar is the adjustment — no star knob or jam nut, which take up too much room under the legs). Legs are CUT to 17.5" (Panel C front pair) / 16.75" (Panels A/B) / 16" (Panel C rear corner pair) and drilled in the bottom end grain (1/2" dia x **1-3/4"** deep — deep enough for the whole stud, not just the insert; test-fit in offcut first) — the foot brings each back to an effective 18.5" / 17.75" / 17" (Leg Leveling Foot Detail render). Drive the insert with a **10mm hex wrench**, pilot-drill before the 1/2" (two drills saves 24 bit changes over 12 legs), and flag the depth with tape on the bit. Skip the kit's stick-on pads — the bare nylon grips better and doesn't shed. Pack barcode **X002FCXYQT** ("Furniture Leg Leveler", 198 g) — an Amazon seller code, not a retail UPC. Tip a corner slightly and wrench the collar to level |
| Bed platform side rails | 6, all **29" x 3.5" x 3/4"** (1x4 pine) — one pair per piece: Panel A's screwed-down section (set 49" apart outside-to-outside, 42" slats between them) and Panel B's two halves (each set 24.5" apart, 17.5" slats between them). One flush 3/4" plane cantilevered 1.5" past the boxes per side (Bed Platform Detail render). **Width set by measurement V7** |
| Bed platform slats | 15: **5 × 42"** (Panel A section, ~2.9" gaps) + **10 × 17.5"** (five per Panel B half, ~2.9" gaps), all 3.5" x 3/4" 1x4 pine — two 42" or five 17.5" per 8ft board |
| Panel B center bearer | 1, **26" x 3" x 1.5"** — two 2x2s side by side (or one 2x4 laid flat), fore-aft at Panel B's centerline between its end rails, top flush with its long rails. Carries the inner edge of both bed halves (1.5" bearing each) and halves the deck's unsupported span, 46" → ~22". **Drop-in**: rests on 4× 1" corner braces, cut to the measured gap less ¼", so it lifts out and the spare can pass |
| Ryobi R-series biscuits | R1 (thin stock / 1/2" ply), R2, R3 (3/4" ply) — buy an R1/R2/R3 assortment for the DBJ50. Wood glue (already in the BOM) goes in every biscuit slot + mating face |

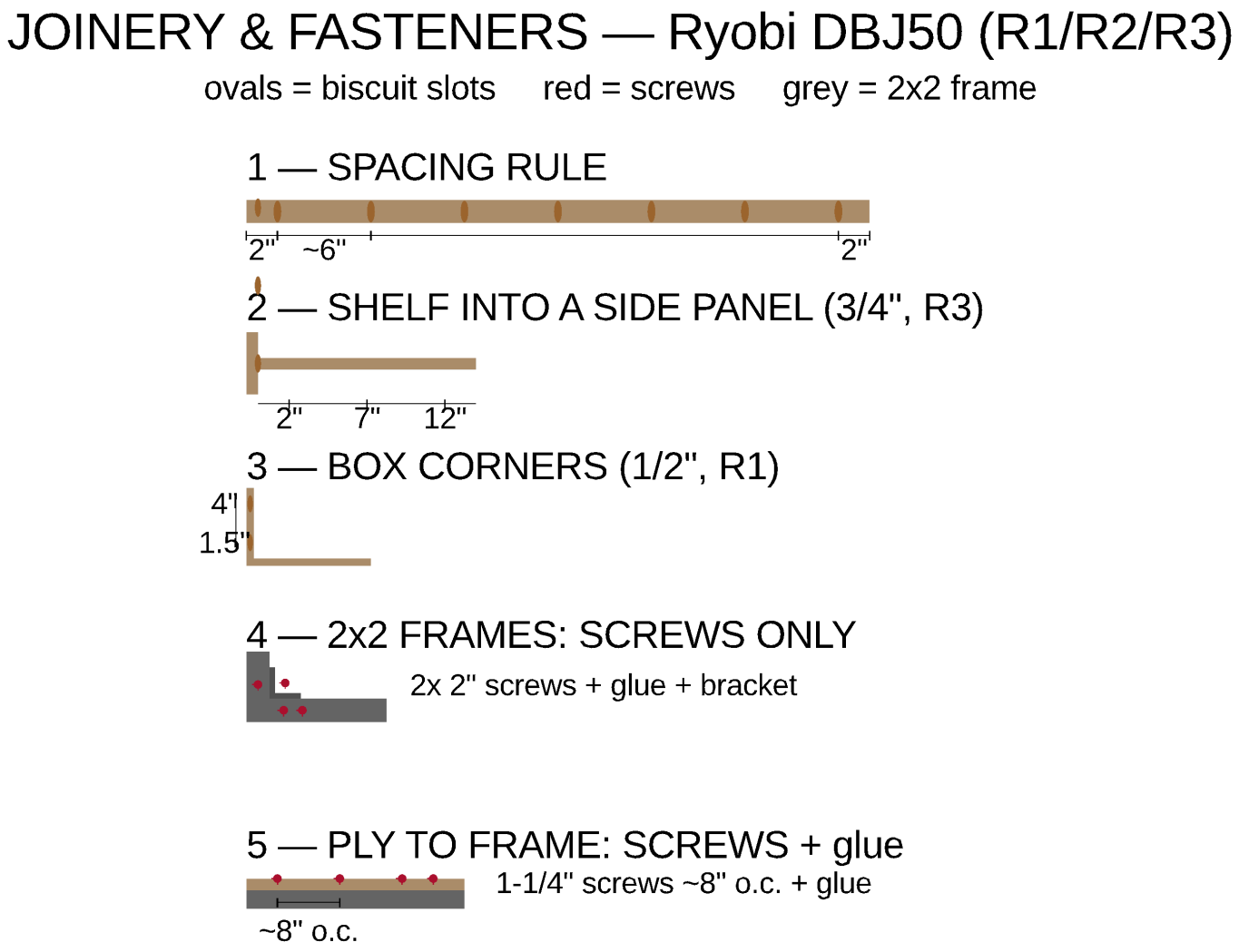
### Joinery & Fasteners (Ryobi DBJ50 biscuit guide)

Every wooden joint's method — biscuit vs. screw — with the exact biscuit spacing is on the **Joinery & Fastener Guide** render below. The short version, using the Ryobi DBJ50 detail biscuit joiner (R1/R2/R3 biscuits, 0–9/32" depth, 1-1/2" 6-tooth blade, 90°/45° fence):

- **Biscuits where clean plywood faces meet** (hidden strength + self-alignment, no screw heads on show faces): every drawer/tray box corner — the kitchen drawer (**2× R1 per corner, centered on the ½"**) and the battery drawer (**2× R3 per corner** — its walls are ¾", and R1 would be lost in that thickness). **Rule**

***for any biscuited joint: first biscuit 2" from each end, then ~6" on center, always centered on the stock thickness; glue the slot and the mating face, then clamp.***

- ***Screws (+ glue) for everything into the 2×2 frame or hardware:*** *all frame corners get* ***2× 2½" screws + a steel corner bracket*** *(a biscuit would blow out the 1.5" stock — the one place to skip biscuits); Panel C's fixed top and front wall, the base cleats, and the kitchen-drawer cheeks are screwed to the frame; slides and threaded inserts use their own hardware.*
- ***PAINT AFTER ASSEMBLY, never before (matte black).*** *The frames get BEHR flat black (Section 4), but* ***glue does not bond to painted wood*** *— and every frame corner, all 12 leg laps and all 14 bottom-rail joints are glue plus screws, with the lap joints relying on a real long-grain glue face. So build all three frames complete, then paint. Pre-painting and masking ~38 joint faces is the alternative, and one missed patch is a dead joint you will not discover for a year. Painting last also blacks out the steel corner brackets, the diagonal braces and every screw head in the same pass, which reads as intentional.* ***Trade to accept:*** *flat black burnishes, and the exposed top rails are the grip surfaces you lift each module by — they will show hand wear. Touch-up on flat is trivial; keep the leftover quart.*
- ***No fasteners:*** *the adjustable shelf rests on pins, and the DELTA 3 tray + utility-bay bins just drop in (the bay bins get hook-and-loop tape — the bay is open to the tailgate).*




*Joinery and fastener guide*

*The five numbered vignettes on the sheet give the geometry and spacing; what each one means is below.*

| | Vignette | What it specifies |
|---|---|---|
| 1 | **Spacing rule** — any biscuited joint, edge-on | First biscuit **2" from each end**, then **~6" on center**, always **centered on the stock thickness**. Glue the slot and the face; clamp until set |
| 2 | **Shelf into a side panel** (¾" ply → R3) | **3 × R3 per end**, at **2" / 7" / 12"** from the front, centered on the ¾". Glue + clamp |
| 3 | **Box corners** (½" ply → R1; ¾" → R3) | **2 × R1 per corner** + glue, all four corners — the kitchen drawer and the fridge-tray apron. **Panel A's battery drawer is ¾" → 2 × R3** |
| 4 | **2×2 frames** (rails / legs / bottom rails / dividers) | **Screws only.** 2 × 2½" screws + glue per corner + a steel corner bracket. **Length is 2½", not 2"** — the screw crosses a full 1.5" rail before it bites, so a 2" screw leaves only ½" of engagement; 2½" gives a full 1" with ½" of wood still beyond the tip. A biscuit would blow out the 1.5" stock — the one place not to use one. **Rail-to-leg is a LAP**: the screws go through the rail's face into the leg's side, with the leg's top clamped flush to the rail's top before drilling |
| 5 | **Ply to frame** (Panel C's top, front wall, cleats, cheeks) | **Screws + glue.** 1¼" screws ~8" o.c. into the rail. Front wall: 8 screws, positions on the Panel C Front Wall render. Cheeks and cleats: 2" screws ~6" o.c. |

### Method by component

| Component | Joint | Method (biscuit sizes are Ryobi R1/R2/R3) |
|---|---|---|
| Kitchen drawer box (½") | box corners | 2 × R1 per corner + glue; bottom glued and screwed on UNDER the walls (the 18" × 26" bottom is the box's outer footprint, so there is no rabbet) |
| Panel A battery drawer box (¾" walls, ½" bottom) | box corners | **2 × R3** per corner + glue; bottom in a glued rabbet |
| Fridge tray + slide apron (½" + pine) | edge / face | R1 biscuits **or** 1¼" screws + glue |
| Kitchen-drawer hanging cheeks ↔ deck (¾") | face-to-face | 2" screws every ~6" down through the deck into the cheeks' top edges, plus 2 into each spreader; the cheeks and spreaders are glued and screwed (#8 × 1¼", 2 per end) into one squared unit first |
| Panel C fixed top ↔ 2×2 rails | ply-to-frame | 1¼" screws every ~8" + glue (R2 optional at the corners, to align) |
| Panel C front wall ↔ frame | ply-to-frame | 8 × #8 × 1¼" screws (exact positions on the Front Wall render) |
| **All 2×2 frames** (rails / legs / bottom rails / dividers) | 2×2 corners | 2 × 2½" screws + glue + a steel corner bracket — **no biscuit** |
| **Leg ↔ end rail** (10 of 12 legs) | **lap**, leg top flush with the rail top | 2 × 2½" screws through the rail's FACE into the leg's side + glue. Clamp flush first — the leg top is the deck-plane reference |
| **Leg ↔ end rail** (Panel C's 2 rear corner legs) | butt-under | 2 × 2½" screws down into the leg's end grain + **a steel angle bracket into the rail's inner face** — the bracket is the joint here, the screws are alignment |
| Bed platform slats ↔ 1x4 side rails | end-to-edge | Glue + 2 × 1¼" Kreg SML-C125 pocket screws per end (Kreg 320 jig set for ¾"), slats crown-up |
| DELTA 3 tray / utility-bay bins / adjustable shelf | none | Drop in or rest on pins — no fasteners |

## *4. Bill of Materials (BOM)*

| Item | Qty | Est. Unit Cost | Est. Total |
|---|---|---|---|
| 3/4" Baltic birch plywood, 4x8 sheet (Panel C deck + the anchor board's strips + **all 4 battery-drawer walls** + the bearer cleats ripped from the offcuts — see Section 3) — **PURCHASED 2026-09-06** | 1 | $65 | $65 |
| Rear pantry — 27-gal Tough Box tote with lid, black/yellow (CPC27GTBY) — **BOUGHT**; replaces the IRIS drawer cluster (30 Oct 2026; the 2 IRIS 3-packs already bought are spares or returns) | 1 | — | — |
| Tote + microwave hold-down — 1" cam strap with snap hooks, 8 ft (to buy) + 2 stainless footman loops from the spares (on hand). The old pantry kit's recessed deck D-rings, felt and Velcro ties are spare | 1 | $10 | $10 |
| Panel B totes — Sterilite 28-Qt under-bed lidded (23.5" x 16.9" x 5.9"), restacked on the spare | 2 | $12 | $24 |
| RJ-MODINI spare kit — 2011-2026 Sienna incl. Hybrid (T155/85R18 steel, 28.5" dia + 2-ton jack + wrenches/sockets + cases) — **$349.95 on Amazon** | 1 | $350 | $350 |
| Spare hold-down hardware: 2x 1" footman loops (from a 4-pack, screws included) + 1" x 6 ft cam-buckle strap (Masterwise multi-pack, purchased — a ~5.5 ft over-stack run, so 6 ft reaches; spares cover the pantry + DELTA 3 drawer straps); skid + liner from offcuts | 1 set | $24 | $24 |
| 1/2" Baltic birch or shop-grade plywood, 4x8 sheet (kitchen drawer box & cheeks + battery-drawer bottom + **Panel C front wall + the fridge tray**) — **PURCHASED 2026-09-06** | 1 | $50 | $50 |
| 2x2 pine (8ft lengths) or aluminum L-channel — **PURCHASED 2026-09-06**. ✅ **MEASURED 2026-09-06: 1-7/16" (1.4375") actual**, not 1½". Every side rail and Panel B's bottom side pair are cut to suit — see the measured-stock note in Section 3. Eleven boards bought, **nine used, two spare** (H.9) | 12 | $8–15 | $96–180 |
| Panel drawer slides, 20" full-extension (1 pair — Panel A's right/DELTA 3 drawer; Panel B has no drawers, Panel A's left bay has none either) | 1 pair | $16 | $16 |
| Drawer catches (friction catch or small turn latch) — **PURCHASED 2026-07-20: WOOCH 30 lb magnetic door catch, $6.48. VERIFY THE COUNT** — the build needs 2 (Panel A drawer + kitchen drawer) and the order was one line item | 1 | $3 | $6.48 |
| WAVE 3 hold-down: 2 screw-eye D-rings (Panel A's end bottom rails) + 1 low-profile cam strap over the unit | 1 set | $11 | $11 |
| Corner brackets (frame joints, 4 per module x 3 modules — the rear pantry has no frame, no brackets needed) — **PURCHASED 2026-07-20: 2× "8 Pack L Bracket Corner Braces 2 inch stainless" = 16, $20.54** (12 used, 4 spare) | 12 | $1 | $20.54 |
| **SPAX #8 × 2½" Interior Flat Head, Torx T-Star Plus, 133 ct / 1 lb, bit included** — every 2×2 frame joint: corners, leg laps, bottom-rail-to-leg, divider and center bearer (~88 used of 133) — **PURCHASED 2026-09-06** | 1 box | $12.63 | $12.63 |
| **Everbilt #8 × 1¼" Phillips Flat Head, 100 ct** — the 12 corner brackets (48), the 8 diagonal braces (32), and ply-to-frame — **PURCHASED 2026-09-06**. ⚠️ A 2nd box is likely: Panel C's deck and front wall are not covered by the first 100 | 1 box | $8.98 | $8.98 |
| Anti-rattle bumper strip (adhesive felt/foam weatherstrip roll) | 1 | $10 | $10 |
| Alignment dowel pins (3/8" dowel rod, cut to 4 pins) | 1 | $3 | $3 |
| Seam draw-latches (over-center, 4 total — both sides of the A/B and B/C seams) | 4 | $4 | $16 |
| Utensil bins + non-slip liner (loose, inside the drawers/bins as desired) | 1 set | $12 | $12 |
| Wood glue | 1 | $8 | $8 |
| Ryobi R-series biscuit assortment (R1/R2/R3) for the DBJ50 — carcass, shelf, drawer & box joints (see the Joinery & Fastener Guide) | 1 | $12 | $12 |
| Panel A/B diagonal corner braces (2 per panel, A + B = 4 used — Sumnacon 4" 8-pack, 4 spares, **PURCHASED**) | 1 pack | $9 | $9 |
| 1" zinc corner braces, 4-pack — Panel B's **drop-in** center bearer rests on them. Not the 2" stainless corner brackets: their 2" upright is taller than the 1-7/16" rail | 1 pack | $4 | $4 |
| VELCRO Extreme Outdoor, 1" × 10 ft — holds Panel C's non-slip pantry mat down so it can be replaced (loop on the deck, hook on the mat) | 2 rolls | $11 | $22 |
| Leg leveling feet — Heavy Duty 3/8-16 furniture levelers w/ T-nut kit, 4-pack (330 lb per foot; buy 3 packs = 12) | 3 packs | $12 | $36 |
| Bed platform lumber — 1x4 x 8ft pine boards (five 42" slats + ten 17.5" slats + six 29" side rails; crosscuts only) — **PURCHASED 2026-09-06** | 7 | $5 | $35 |
| Kreg Pocket Hole Jig 320 + stepped bit — the bed slats' pocket holes — **PURCHASED** | 1 | $40 | $40 |
| Kreg SML-C125-100 pocket screws, #8 × 1¼" coarse thread, 100-pack — 2 per slat end, 60 used — **PURCHASED** | 1 | $8 | $8 |
| Kitchen drawer slide pair — 24" side-mount — **PURCHASED 2026-07-20: Liberty 942405 soft-close ball-bearing 24", 2-pack, $37.63** | 1 pair | $37.63 | $37.63 |
| Kitchen drawer magnetic catch — see the drawer-catch line above: **one WOOCH catch is purchased, and two are needed** (box + cheeks come from the plywood sheets' spare area) | 1 | $3 | $3 |
| RV bubble levels — Larbeti stick-on 2+2 pack, bar + T-shape (PITCH on the platform's driver-side rail edge, ROLL on the rear-pantry deck edge) — **PURCHASED July 2026** | 1 | $10 | $10 |
| RV leveling blocks, Lynx-style 10-pack + 1 wheel chock (per-site leveling at the WHEELS — see the Block Calculator note, Appendix E) | 1 | $40 | $40 |

| Item | Qty | Est. Unit Cost | Est. Total |
|---|---|---|---|
| **HEST Dually Long mattress** (hest.com/products/dually — 78" x 50" x 4" solid foam, memory-foam top, washable waterproof cover INCLUDED; no air chambers) | 1 | $530–590 | $530–590 |
| — Budget fallback: DIY 2-layer foam (queen 4" HD base + 1.5" memory topper, both cut to 50" x 78" with a serrated/electric knife + spray adhesive + a separate waterproof cover) — swaps in for the HEST at ~$185–255 total | (alt) | ($185–255) | ($0 in total below) |
| Cable grommets (1 cooktop line, 1 Power strip 1's line — both short hops to the rear outlet now, 3 fridge DC line, 2 DELTA 3 AC charging line) — EASYEAH 20-pack covers all 7, **PURCHASED** | 1 pack | $10 | $10 |
| 16 AWG extension cord (household AC-rated — cooktop to the REAR outlet, short + slack loop for the slide) | 1 | $8 | $8 |
| 16 AWG extension cord (household AC-rated — Power strip 1 to the REAR outlet beside it, short) | 1 | $8 | $8 |
| **HDX 30 ft. 16/3 Extension Cord Reel — 4 grounded outlets + surge protector, gray** — **PURCHASED 2026-09-06, $29.98. This adds SHORE POWER:** without it, every AC path in this plan runs off the van's 1500W inverter (engine on) or the DELTA 3's battery, with no way to take an external outlet. With the reel the DELTA 3 charges from a campground pedestal or a house outlet **without idling the van**, and its surge protection guards the DELTA 3 and the fridge electronics against notoriously rough pedestal power. ⚠️ **Two hard limits — see the shore-power note in Section 5** | 1 | $29.98 | $29.98 |
| Fridge DC line — OYMSAE 12-ft 16AWG SAE-to-SAE 2-pack: the cord AND its 2 seam quick-disconnects in one part, **PURCHASED July 2026** (don't buy cord and connectors separately) | 1 pack | $22 | $22 |
| 16 AWG cord (household AC-rated — DELTA 3 AC charging line, front console to Panel A, ~3-4ft) | 1 | $8 | $8 |
| Per-module surface wire raceway — a CordMate II kit for the 3 AC lines (~$28) + a 123" open-slot raceway kit for the DC line (~$18, screws included), AC and DC in separate channels + screw-mount cord clips | 1 set | $46 | $46 |
| SAE quick-disconnect connectors (2 for the fridge DC line's 2 seams — DC-rated, matches the fridge's actual 12V circuit) | 2 | $8 | $16 |
| Fused (15A) male car-plug → SAE adapter — DELTA 3 end of the fridge line (**P1**; its fuse is the line's upstream protection). Any 16AWG fused SAE adapter kit works | 1 | $9 | $9 |
| SAE → female 12V socket pigtail — fridge end (**P7**; the fridge's own cord plugs into it). Often sold in the same kit as the P1 adapter | 1 | $9 | $9 |
| 18AWG red/black hookup wire, ~10 ft (in-bay branch runs, P6–P12) | 1 | $8 | $8 |
| Crimp terminal assortment — #10 heat-shrink rings (fuse-block studs) + 0.25" female spades (switch tabs) | 1 kit | $12 | $12 |
| WAGO 221-413 lever nuts (fan feed + fan return splits — **P12**) | 1 pack | $10 | $10 |
| 2-way AC outlet tap (REAR outlet — Power strip 1 + the cooktop share it) | 1 | $5 | $5 |
| **BougeRV Rocky 40 (CR04001), 41QT dual-zone** (bougerv.com Rocky V3.0 — specs verified against the saved user manual: 28.74" x 17.72" x 15.79", 40.6 lb (18.4 kg), 60W max / 45W ECO, reversible lid, optional B240 battery) — **PURCHASED July 2026 direct from bougerv.com (New Rocky V3.0 41QT, JULY7 coupon)** | 1 | $400–500 (verify invoice) | $400–500 |
| Heavy-duty 24" full-extension drawer slide pair — VADANIA #D76 locking, 379 lb, **PURCHASED July 2026** (locks closed AND extended; loaded fridge can realistically hit 60–90 lb). Rails mount VERTICALLY on the tray's sides — never under it (see the fridge slide detail) | 1 pair | $79 | $79 |
| Steel riser angles for the fridge slides, ~2"x2"x3/16" x 24" (1 per fixed rail, drilled to match — bolted atop the anchor board's rail-line strips with 1/4-20 screws into T-nuts; the board raises the rails ~0.85", and the moving member screws lower on the tray apron to compensate, Section 8). **As cut (14 Oct):** 24" each, flanges pointing outboard, away from the tray; the driver riser stays 2" (it overhangs the board edge ¼" in clear air), the kitchen riser's flange is ripped to 1" so it clears the kitchen unit | 2 | $12 | $24 |
| 1x3 pine, 6 ft (rip the fridge tray's two 28¾" side aprons from it) — **PURCHASED 2026-09-06** | 1 | $6 | $6 |
| **JAGAHAHA wooden slide-out camp kitchen**, right-side variant (kitchen faces out the passenger side), built for a 2-burner stove (confirmed product link — **no listed price found in research; treat as an UNVERIFIED estimate** until you check the live listing) | 1 | $300–350 | $300–350 |
| COOKTRON Portable Induction Cooktop 2 Burner w/ Removable Iron Cast Griddle Pan — confirmed exact product, 18.1"x9.1"x4.3" (fits the JAGAHAHA's 23"x15.7"x5.7" stove tray with margin on every axis); see Section 7 for why the Duxtop 9620LS was dropped | 1 | $189.99 | $189.99 |
| BLACK+DECKER EM720CPN-P microwave, 0.7 cu ft / **700W output** — Target TCIN 50568401, pickup San Antonio North — **PURCHASED 2026-09-06**. ⚠️ **Two things still open, one settled — see the power note in Section 7.** (a) 700W is the output rating; input draw for this class is **~1200W, UNVERIFIED** — read the nameplate. (b) ✅ **SETTLED: never run at the same time as the cooktop.** The van has ONE 1500W inverter feeding BOTH outlets, so the remaining watch item is pausing DELTA 3 AC charging before use. (c) **Home: Panel C's deck, flush with the passenger edge at the tailgate end, directly above the kitchen drawer** — beside the pantry tote, door facing the tailgate so it's used from outside with the hatch up. It is 17.3" W × 14" D × 10.2" H (~22 lb, sibling-model figure). **Its vents are on the back**, ~½" from the mattress foot: pull the mattress foot back or slide the microwave toward the tailgate before running it, and run it only with the hatch open. For travel it shares the tote's strap (Component 1) | 1 | $54.99 | $54.99 |
| 120mm 12V fan + PWM temperature controller w/ NTC probe (W1209-style, fridge-bay cooling) | 1 | $15–20 | $15–20 |
| 3" round snap-in soffit vents, 4-pack — 3 in the front wall's low intake row under the fan, 1 spare (**PURCHASED**) | 1 pack | $12 | $12 |

| Item | Qty | Est. Unit Cost | Est. Total |
|---|---|---|---|
| *12V fused distribution block (control panel) — **PURCHASED July 2026: Nilight 6-way blade fuse block w/ negative bus, waterproof cover, label stickers, fuse assortment included** (no separate ATO fuse pack needed)* | *1* | *$17.98* | *$17.98* |
| *Illuminated toggle switches (SW1 fridge, SW2 fans, SW3 spare — the cooktop is AC, switched at Power strip 2)* | *3* | *$5* | *$15* |
| ***Windmill Smart Fan 2-in-1 air circulator** (personal comfort, Power Strip 1) — **PURCHASED July 2026*** | *1* | *$69* | *$69* |
| *Anchor board — three ¾" ply strips, one piece each (cut from the ¾" sheet's offcut) + ¼-20 T-nuts and 1" button heads for the 2 risers (Anchor Board sheet)* | *1* | *$10* | *$10* |
| *Steel flat bar 3" x 48" x 3/16" (Steelworks 11679, Lowe's) — the 2 rungs under the anchor board, cut to 46"* | *2* | *$20* | *$40* |
| *Forged ¼-20 × 2" shoulder eye bolts with nuts, hot-dip galvanized (Hardware Essentials 851885), 5-pack, 500 lb WLL — one down through each fridge riser's front end (+ 2 ¼-20 jam nuts)* | *1* | *$18* | *$18* |
| *¼" grade 30 galvanized chain, by the foot (RELIABILT at Lowe's; Everbilt 806666 at Home Depot), 1,250–1,300 lb WLL — two pieces, eye bolt to hoop plus ~3" each to wrap the ⅝" hoop bar* | *2 ft* | *$3.67* | *$7.34* |
| *¼" stainless quick links (Everbilt 43394), 1,200 lb WLL — 2 per chain: to the eye bolt, and closing the loop around the hoop* | *4* | *$4* | *$16* |
| *#8 x ⅝" flat-head wood screws, 12-pack (Everbilt 825381) — rungs up into the strips, chamfered into the steel* | *1* | *$1.38* | *$1.38* |
| *DEWALT 4-1/2" cut-off wheels, 5-pack — aluminum oxide, 7/8" arbor, angle grinder. Cuts the 2 rungs from the flat bar and the 2 riser angles from the angle stock (H.8). Cut **dry** — never put fluid on an abrasive wheel — **PURCHASED 2026-09-05*** | *1 pack* | *$6.46* | *$6.46* |
| *Tap Magic 30004P ProTap cutting fluid, 4 oz — for the **drilling** that follows the cuts: the 2 rungs' screw holes, the 2 riser angles' 1/4-20 holes, and the anchor board's steel work generally. Nothing to do with the cut-off wheels — **PURCHASED 2026-09-06*** | *1* | *$4.98* | *$4.98* |
| *Neoprene rubber sheet, ⅛" × 16" × 60" roll — cut into strips under the anchor board's strips* | *1* | *$20* | *$20* |
| *Blue medium-strength threadlocker (Loctite 242 or equivalent — on the anchor board's machine screws)* | *1 bottle* | *$8* | *$8* |
| *Erickson ratchet straps (01415), 400 lb WLL / 1,200 lb break, 1" x 10' (2: kitchen unit, front to H2/H4 and rear to the footman loops; 3 optional backups to the striker loops; 3 spare)* | *2 packs of 4* | *$32* | *$64* |
| *Fridge hold-down — 1 low-profile cam strap + 2 screw-eye D-rings (hooks to the fridge's 2 end handles, down to the tray's side apron — keeps the fridge from lifting off the tray in a hard stop)* | *1 set* | *$10* | *$10* |
| *Cable clips for the fridge DC line's slide-travel slack (3, screwed to the slide's fixed outer rail — same VIPMOON-type clip as the cooktop run, buy links below)* | *3* | *$1* | *$3* |
| *Tension rod + blackout curtain fabric* | *1* | *$40* | *$40* |
| ***BEHR Premium Plus 1 qt. Black Flat Low Odor Interior Paint & Primer** (130004) — **matte black on all three frames**. ~52 sq ft of frame surface vs ~100 sq ft/coat, so one quart covers two coats with margin; paint-and-primer in one because raw pine drinks the first coat. Low-odor / GREENGUARD Gold, which matters in a vehicle you sleep in. **Apply AFTER assembly — see the paint note in Section 3** — **PURCHASED 2026-09-06*** | *1 qt* | *$17.48* | *$17.48* |
| *Interior satin polyurethane for Panel C's deck and the drawer/tray surfaces things slide across (NOT spar urethane — H.5 note 5)* | *1 qt* | *$20* | *$20* |
| *Sandpaper, misc finishing supplies* | *—* | *$15* | *$15* |
| *6-outlet power strip, individual switches (Power strip 1 + Power strip 2) — CRST metal, **PURCHASED July 2026*** | *2* | *$25* | *$50* |
| *UMETRE 9-Piece Detachable-Handle Induction Cookware Set — nests flat, its 11x11 box goes in the pantry tote (Component 1); induction-compatible, dishwasher-safe — **PURCHASED July 2026*** | *1 set* | *$45* | *$45* |
| *PeaceOut Toyota Sienna Sliding Doors Bug Nets — Sienna-specific, sold as a pair (driver + passenger sliding doors), no-see-um mesh, magnetic + hook-and-loop attachment (temporary, no screws), carry bag included — lets both side doors stay open for ventilation while sleeping without bugs getting in. Real listed price is in CAD; $ figure here is an approximate USD conversion — confirm current exchange rate before ordering.* | *1 pair* | *~$197 (CAD $269.99)* | *~$197* |
| *DELTA 3 drawer hardware: rubber non-slip mat on the drawer floor (**no locating cleats** — owner, 30 Oct 2026), 2 D-rings on #8 × ⅝" screws ~7½" down from the box's top edge + cam strap, turn button on the frame's top rail (e.g. Wright Products V20A, or a 3" × ¾" ply tab), 1" cable grommet (Panel A right drawer — Power Drawer Tie-Down sheet)* | *1 set* | *$15* | *$15* |
| *WAVE 3 hose/cord storage hook, in the open utility bay (screwed up into the deck underside at the bay's kitchen side; bundle the hoses with their strap so nothing swings — see Renders)* | *1* | *$4* | *$4* |
| *Non-slip mat, WAVE 3's point-of-use surface (Panel C's deck or the front seat — not the Panel A storage bay, see Renders)* | *1* | *$12* | *$12* |
| *FOUND STORAGE — DELTA 3 drawer top tray (reclaims ~2.1" of dead headroom above the DELTA 3 stack — 13.75" box − 1/2" bottom − 11.16" stack: cables, the DELTA 3's own cords, dongles) — shallow lift-out bin, **build it 1.75" or shallower*** | *1* | *$8* | *$8* |
| *FOUND STORAGE — utility-bay bins (the open bay's spare volume beside the control panel: dish soap, sponges, utility odds) + hook-and-loop tape under each (no door holds them in)* | *2* | *$4* | *$8* |

| Item | Qty | Est. Unit Cost | Est. Total |
|---|---|---|---|
| EcoFlow WAVE Series Car Vent Kit — official accessory, WAVE 2/3 compatible, velcro window seal + duct pass-throughs, no permanent modification (no-tent sleeping configuration, Section 1) — **PURCHASED July 2026** (order EFUS-426448, $40) | 1 | $40 | $40 |
| SRS airbag emulators, Sienna 2nd row — **PURCHASED July 2026**: 0651 Round x2 + 5522 "Square" x2 (shipped). Click into the van-side yellow floor plugs when the seats are out so the airbag system reads normally (Section 9). **VERIFY before seat-removal day that the 5522 mates with the Sienna's rectangular floor plug** | 1 set | ~$60 | ~$60 |

***EcoFlow DELTA 3 Plus + Smart Extra Battery and WAVE 3 (already owned, not priced here)*** *— see the DELTA 3/WAVE 3 stowage detail in Renders and Section 1.*

### *Recommended purchase links — slides + electrical (checked July 2026)*

- ***Rear pantry****: the 27-gallon Tough Box tote (bought) + a 1" × 8 ft cam strap with snap hooks; the anchors are 2 of the stainless footman loops already on hand.*
- ***Panel B deep-storage totes****: Sterilite 28-Qt under-bed lidded tote ×2 — plain lidded totes on purpose (Panel B has no side access, so slide-drawer versions would waste money).*
- ***Spare tire****: RJ-MODINI kit, 2011-2026 Sienna incl. Hybrid ($349.95) — T155/85R18 on an 18x4 steel wheel with the factory 60.1mm hub-centric bore; at **28.5" rolling diameter it's the closest match to the ~29.1" OE tires** (−2%, vs −4.8% for the T145 kits — gentlest on the AWD systems). Includes a model-specific 2-ton jack, lug + ratchet wrench, 4 sockets, gloves, and cases (the tool case nests inside the wheel barrel). Skid + liner come from offcuts already in the build. (Alternatives compared: ValueSpare ~$270-385 steel/27.7", Modern Spare ~$350-460 alloy/27.7".)*
- ***Spare hold-down hardware****: NVAAV 1" stainless footman loops, 4-pack w/ screws (~$10, need 2) + Cartman 1" x 12 ft cam-buckle lashing straps, 6-pack — **PURCHASED July 2026**: footman loops (2 packs) + a Masterwise 6-ft x 1" cam-strap multi-pack (the spare's over-stack run is ~5.5 ft, so 6 ft reaches; spares cover the pantry + DELTA 3 drawer straps).*
- ***Leveling blocks****: Lynx Levelers 10-pack — the "legos"; pair with the Sienna Block Calculator.*
- ***Bubble levels****: **PURCHASED July 2026** — Larbeti stick-on 2+2 pack (bar + T-shape, self-adhesive): PITCH bar on the platform rail edge, ROLL on the rear-pantry deck edge, 2 spares. (The adhesive mount skips two screw holes in the poplar rail; degrease the surfaces first.)*

*Specific products verified against this plan's specs, full URLs shown for copy-paste. Quantities refer to what THIS build needs; several listings are multi-packs. The low-level CO monitor is owner-supplied (battery-powered, see the BOM note above) and deliberately absent here.*

| Item (plan spec) | Product | Link |
|---|---|---|
| Panel drawer slides — only 1 pair needed (Panel A's right/DELTA 3 drawer; Panel B has no drawers — the side doors don't reach it — and Panel A's left bay is WAVE 3 open storage, Section 1), 20" full-extension, 100 lb class — **PURCHASED** (July 2026) | GlideRite 20" 100 lb ball-bearing slides, 5-pair pack ($46.99; 4 spares — one could serve as the kitchen drawer's pair if you go 20" instead of 24" there) | https://www.amazon.com/20-inch-Extension-Ball-Bearing-Drawer-Over-Travel/dp/B07KFLS2C9 |
| Fridge slide — 1 pair, 24" — **PURCHASED** (July 2026) | VADANIA #D76 (VD2576) 24" industrial heavy-duty WITH LOCK — 379 lb, 3" (76mm) rail, 3-fold full extension, locks both closed (transit) and extended (loading), 1 pair ($79.20). INSTALL NOTE: the 76mm rail wants ~3" of flat vertical face — the rails stand VERTICALLY, one flanking each side of the tray: the moving member screws to a 1x3 apron glued+screwed along each tray edge, and the fixed member screws to a steel riser angle bolted to the no-drill anchor board (Section 8; the board raises the rails ~0.85" — the moving member screws lower on the apron so the tray keeps its height). NEVER mount them flat under the tray — undermount adds ~1.2" of height the 17" void doesn't have, and the fridge would hit Panel C's end rail (the mounted stack clears it by 0.33"). | https://www.amazon.com/VADANIA-Industrial-Extension-Bearing-Widening/dp/B08C56Z6GF |
| Fridge slide, premium alternate | Hettich 24" full-extension, 500 lb (~$70+) | https://www.amazon.com/Drawer-Slide-Extension-Capacity-Hettich/dp/B0016LEBBG |
| 120mm 12V fans — need 2 (intake + exhaust) | GDSTIME 120mm x 25mm 12V dual-ball-bearing fan, ~$9 — sold singly, ORDER QTY 2. (Verify 12V / 120x120x25mm on the spec table before buying any fan listing.) | https://www.amazon.com/GDSTIME-Bearings-Brushless-Cooling-Exhaust/dp/B00N1Y4BMA |
| W1209 temp controller + NTC probe (need 1; 2nd is a spare) | HiLetgo 2pcs W1209 12V with case + waterproof probe (~$10) | https://www.amazon.com/HiLetgo-Temperature-Controller-Thermostat-One-Channel/dp/B07VDWRZKB |
| Fused 12V distribution block — **PURCHASED** (July 2026) | Nilight 6-way blade fuse block w/ negative bus, waterproof cover, label stickers + fuse assortment included ($17.98) | https://www.amazon.com/Nilight-Negative-Standard-Waterproof-Automotive/dp/B09NPQBXCG |
| Illuminated toggle switches — need 3 — **PURCHASED** (July 2026) | Ampper 12V 20A illuminated round rockers, 10-pack ($8.99) | https://www.amazon.com/Illuminated-Rocker-Switches-Ampper-Terminals/dp/B0BZPY5D9L |
| Control panel enclosure | **None.** The switches, surge strip and W1209 screw straight to the backer board: the utility bay is 2.78" wide, and a box that holds those parts is wider than the parts. | — |
| 16 AWG AC cords — cooktop → REAR outlet + Power strip 1 → REAR outlet (both short, no seams) + DELTA 3 charging → front console | DEWENWILS 3ft 16AWG SJTW indoor/outdoor extension cord, ETL listed, 2-pack (~$13) — 1 pack covers the two short rear-outlet hops; add a 2nd cord for the DELTA 3 charging line (or size one to your console distance) | https://www.amazon.com/DEWENWILS-Extension-Weatherproof-Outdoor-Grounded/dp/B0891VDLV5 |
| 16 AWG DC cord run + 2 seam disconnects (fridge line — this one really is 12V DC, DELTA 3 to fridge, so the SAE connector IS the right part here) | OYMSAE 12 ft 16AWG SAE-to-SAE quick-disconnect cables, 2-pack — the SAE plugs ARE the seam disconnects (~$14) | https://www.amazon.com/OYMSAE-Extension-Disconnect-Connector-Automotive/dp/B0B2JLGJCR |
| 16 AWG cord, ~3-4ft (DELTA 3 AC charging line, front console to Panel A — household AC-rated, no seam crossing) | DEWENWILS 3ft 16AWG SJTW indoor/outdoor extension cord, ETL listed, 2-pack (~$13) — same product as the cooktop line above, 1 pack covers this with a spare | https://www.amazon.com/DEWENWILS-Extension-Weatherproof-Outdoor-Grounded/dp/B0891VDLV5 |
| 2-way AC outlet tap (REAR outlet — Power strip 1 + the cooktop share it; the front | GE Outlet Extender Wall Tap, UL listed, 125VAC (~$5-13) | https://www.amazon.com/GE-Adap |

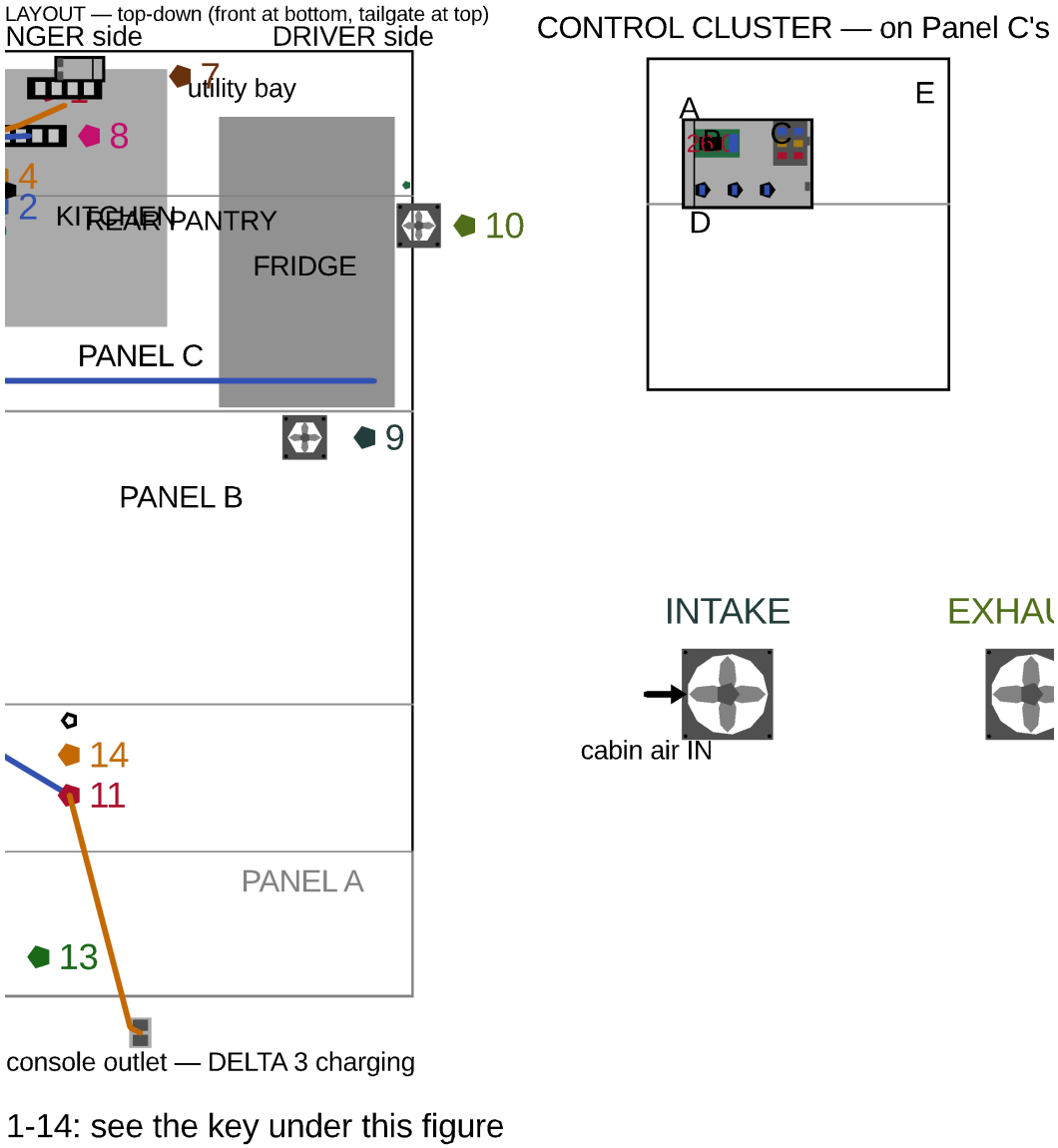
| | | |
|---|---|---|
| *console outlet is the DELTA 3's alone)* | | *ter-Grounded-Outlets-52203/dp/B00006IBFC* |
| *Leg leveling feet — 12 needed (buy 3 four-packs); 3/8-16 leveling feet WITH T-nut kit, 330 lb per foot* | *Heavy Duty Furniture Levelers w/ T-Nut Kit, 4-pack (~$12/pack)* | *https://www.amazon.com/Furniture-Levelers-Adjustable-Leveling-Cabinets-Sofa/dp/B081ZT4Q4G* |
| *1" cable grommets — need 7 (the 20-pack covers all pass-throughs) — **PURCHASED** (July 2026)* | *EASYEAH 1" rubber grommets, 20-pack ($9.99)* | *https://www.amazon.com/Diameter-Hole%EF%BC%8CRubber-Synthetic-Protection-Double-Sided/dp/B085S6S6KH* |
| *6-outlet power strips, individual switches — need 2* | *CRST 6-outlet metal strip, individual switches, 1200J (~$25 each)* | *https://www.amazon.com/CRST-Individual-Protectors-ECO-Friendly-Protector/dp/B097JH3W3W* |
| *Cord clips — ~10-12 along the cooktop cord run* | *VIPMOON 100x 1/4" R-type nylon screw-mount clamps w/ screws (~$9) — screw-mount, NOT adhesive: stick-on clips let go under vehicle vibration and temperature swings. Size up to 3/8" if the SAE cable measures over ~0.28"* | *https://www.amazon.com/VIPMOON-Mounting-Fastener-Electrical-Management/dp/B07HGZ83JB* |

*Slides + electrical together run **~$290-305**. These links satisfy the corresponding BOM rows above — don't double-count.*

***Estimated total (summing the BOM table above): ~$3,569–3,939 with the fridge + kitchen included, or ~$2,869–3,089 if you already own equivalents** — the owner-supplied CO monitor and fire extinguisher aren't priced here. The itemized BOM table is the authoritative figure. It includes the bought HEST Dually Long mattress (~$530–590, cover included — the DIY 2-layer fallback, ~$185–255, stays in the BOM as the budget swap), a real BougeRV Rocky 40 + JAGAHAHA kitchen (~$700–850 combined, Rocky price UNVERIFIED), the fridge cooling/control electronics (2 fans, controller, sensor, surge protector, ~$85–120 combined), the no-drill securing hardware — rated straps, the anchor board's steel rungs, chain hardware and neoprene (~$140, Section 8) — and the RJ-MODINI spare kit (~$350, Appendix F). Two optional material upgrades are documented but NOT in the total above — poplar top-rails (~$25–40) and aluminum corner braces (~$10–15), see Section 3's "Material options & upgrades."*

## 5. Power Cord Routing

*Every purchased electrical component at its real position — icon-based top-down layout, the control-cluster mounting elevation, and the fan installation detail (screw pattern + airflow direction):*




*Electrical layout*

***Marker key.*** *Markers 1–14 on the sheet index this table.*

| # | Component |
|---|---|
| 1 | Power strip 1 — on the rear-pantry deck edge (phone / light / Windmill fan) |
| 2 | Power strip 1's cord — a SHORT hop to the rear outlet beside the pantry (no seams) |
| 3 | REAR AC outlet — passenger rear quarter trim, **22.5" up, 10" in** from the sidewall (MEASURED V9b). Shares the one 1500W inverter |
| 4 | Cooktop cord — SHORT run to the rear outlet, with a slack loop for the kitchen's slide (no seams) |
| 5 | 2-way outlet tap at the rear outlet — Power strip 1 and the cooktop share it |
| 6 | 1" grommets — deck-edge exits for the two short AC cords |
| 7 | Power strip 2 — ON the slide-out kitchen unit (cooktop); cord slack for the slide, fed from the rear outlet |
| 8 | Control cluster (no enclosure) — on Panel C's FRONT wall, Panel-B face (Section 2 below) |
| 9 | Intake fan, 120mm — on Panel C's FRONT wall, over the fridge's B-facing end |
| 10 | Exhaust fan, 120mm, into the open bay + NTC probe inside the bay at that wall, in the hot exhaust |
| 11 | Fridge DC line — DELTA 3 (Panel A) forward to the fridge (Panel C), its own dedicated run |
| 12 | SAE quick-disconnects — on the fridge DC line, one at each seam it crosses |
| 13 | DELTA 3 AC charging cord — the front console's ONE outlet (1500W, verified) to Panel A's drawer |
| 14 | DELTA 3 drawer grommet — WAVE 3 charge cable exit (see the stowage detail) |

*Notes for this sheet:*

- *backer board at the back of the open bay (no door — reach in)*
- *A backer board (3/4" offcut) — 4x #8 x 1" screws up into the deck underside; the switches, surge strip and W1209 mount to its face (no enclosure, see Section 2)*
- *B W1209 controller (in its case) — NTC probe wire exits the bottom grommet,*
- *runs to the fridge's kitchen-facing wall (Section 3)*
- *C Nilight 6-way fuse block — feed from the DELTA 3's DC output (Panel A); fans + W1209 each*
- *get their own fused circuit (5A is plenty)*
- *D 3x Ampper switches (fridge / cooktop strip / fans) — 20mm holes in the*
- *backer board's face (no enclosure)*
- *FAN MOUNTING — face-on, both fans identical hardware*
- *NTC probe: zip-tie 1-2" from the exhaust cutout*
- *Cut a 4.5" round hole (or 4.5" square) in the wall; fan mounts over it with*
- *4x #8 x 1-1/4" screws through the corner holes. Grill goes on the AIR-ENTRY side.*
- *Arrow on the fan's side frame shows its blow direction — point it per the labels above.*
- *Wiring: both fans in parallel off the W1209 relay (Section 5 wiring schematic).*

***The fridge (and its fan system) now runs off the DELTA 3 stack, not the van's rear outlet** — see the "Why the fridge and fan system now run off the DELTA 3 stack" note in Section 1 for the full reasoning (reliability while people are away from the vehicle, backed by real runtime/recharge numbers). This means the fridge needs a cord — a DC line from the DELTA 3 (Panel A's right drawer) forward to the fridge (Panel C), crossing both module seams (A/B and B/C). It is the only cord in the build that crosses them; the cooktop and Power strip 1 both run short cords to the van's rear outlet and cross none. The Sienna's rear 12V accessory outlet is simply unused now — nothing in this design plugs into it, so it's free for anything else later.*

***Cord management — run each module's cords in a surface raceway, not loose clips.** Instead of loose zip-tied/clip-covered runs, route the cords through a **segmented plastic surface raceway (wire duct / Wiremold-style)** — a short length screwed to the inside of each module's frame, one channel per panel, meeting at the existing seam disconnects. Two rules keep it correct: (1) the raceway must stay **segmented per module** (a separate length in A, B, and C) — NOT one continuous conduit spanning the seams, or a panel can't be lifted out; the inline AC connectors / SAE DC disconnects still live at each seam. (2) Keep the **120V AC and 12V DC lines in separate channels** (a divided duct, or two small ducts side by side) — don't bundle them together. This protects the cords from abrasion, makes every run traceable, and lets you pull or add a wire later without fishing it through the frame — a real serviceability win over clipped cords for a small added cost (see the raceway line in Section 4's BOM). Where a run has to cross into a module's interior bay, a snap-cover raceway lifts open without unscrewing.*

***The van has exactly 2 AC outlets (VERIFIED): 1 in the front center console (1500W) + 1 in the back passenger area.** That rear outlet transforms the routing: the two tailgate-end AC loads (the cooktop and Power strip 1) plug into it with **short cords and ZERO seam crossings**. The front console outlet is dedicated to the DELTA 3's charging cord. **Four cords need routing in total, each on its own independent line:***

### *Cooktop line (Power strip 2 → REAR outlet) — short*

***Power strip 2 rides the slide-out kitchen unit** so it travels to wherever you're actually cooking — the JAGAHAHA extends to ~70" out the tailgate in use. Its cord just hops to the **rear AC outlet in the back passenger area**, a few feet away. This is a full 120V AC household circuit, so use a household-rated extension cord.*

1. *Drill one 1" grommet hole in the frame rail at the kitchen unit's corner post nearest the rear outlet.*
2. *Run the short cord from the strip, through the grommet, to the rear outlet's 2-way tap (shared with Power strip 1 — below).*
3. *Leave a **slack loop** sized to the kitchen's full ~70" slide extension, coiled near the stowed position, so the cooktop keeps power at the cook position without strain.*

### *Power strip 1 line (rear pantry deck edge → REAR outlet) — short*

*Power strip 1 (phone/device charging, reading light, the Windmill fan) mounts at the **rear pantry's deck edge** in the open bay (Component 1) — which sits directly beside the rear outlet. One short 120V AC cord:*

1. *Drill a 1" grommet at the deck edge where the cord drops off the pantry bay.*
2. *Run it to the rear outlet's 2-way tap. Leave ~1 ft of slack so lifting the pantry's drawer units never requires unplugging anything.*

### *Fridge DC line (DELTA 3 → Panel C fridge bay)*

*The fridge's DC line runs from the DELTA 3 Plus's DC output (Panel A's right drawer) back toward the tailgate, through Panel B, into Panel C's fridge bay. Unlike the cooktop and Power strip 1's lines, this one really is 12V DC, so an automotive/solar SAE quick-disconnect is the right connector type here. Like the cooktop's line, it crosses the same 2 remaining module seams (Panel A → Panel B, Panel B → Panel C), so it needs its own quick-disconnects too. This is the trade-off Section 1 already flagged: a long cord run in exchange for the fridge no longer depending on the van's ignition state.*

1. *Drill a 1" grommet hole in Panel A's frame rail near the DELTA 3 drawer, and one more at each of the 2 seams it crosses (Panel A → B, Panel B → C) — 3 total, reusing the same rail-channel approach as the cooktop line (a dado groove or a zip-tied cord under a clip-cover strip both work).*
2. *Install a quick-disconnect connector at each of the 2 seams so Panel A and Panel B still lift out independently.*
3. *Terminate the cord at the fridge's DC input inside Panel C's void, near the control cluster (Component 7).*
4. *Leave a few inches of slack at the DELTA 3 end so opening/closing that drawer doesn't strain the connection.*
5. ***At the fridge end, clip the slide-travel slack loop to the slide's FIXED outer rail — not the tray, and not left loose** — 2-3 screw-mount cable clips (same VIPMOON-type clips as the cooktop line, Section 4 BOM) along the rail's inside face. The outer rail never moves, so the clipped loop stays out of the pinch point between the tray and the rail as the tray slides home; a loose coil left to dangle can get caught under the tray or crushed against the rail on the way in.*

#### *Every connection point, P1–P12 (matches the numbered markers on the wiring diagram, Renders)*

*The ONE 16AWG SAE line runs DELTA 3 → fuse block; **both** circuits branch at the Nilight block, and every black (−) return lands on its negative bus:*

| # | Where | What lands there — terminal, connector, crimp |
|---|---|---|
| P1 | DELTA 3 Plus 12V car-power outlet (Panel A right drawer) | Fused (15A) male car-plug → SAE adapter — hand-plug, no tools. That fuse is the whole line's upstream protection |
| P2 | Panel A/B seam | SAE quick-disconnect #1 — the OYMSAE cord ends mate; zip-tie a strain-relief loop each side of the seam |
| P3 | Panel B/C seam | SAE quick-disconnect #2 — same part; both disconnects hang slack, never taut |
| P4 | Panel C front wall, low driver side | 1" grommet — the cord just passes through into the fridge bay, no termination |
| P5 | Nilight 6-way fuse block, main studs (in the utility bay) | Red → **+IN** stud, black → **−IN** stud: strip 3/8", crimp #10 heat-shrink ring terminals, M5 nuts snug. F1 = 10A ATO (fridge), F2 = 3A ATO (fans), F3–F6 spare |
| P6 | SW1 "FRIDGE" — Ampper rocker, 3 spade tabs | F1's 16AWG red → **PWR** tab; **LOAD** tab → pigtail red; **GND** tab → neg bus (powers the glow ring). 0.25" female spade crimps on all 3 |
| P7 | SAE → female 12V socket pigtail | SAE end mates toward SW1's LOAD side; the socket zip-ties in the utility bay near the fridge's kitchen-facing wall |
| P8 | Fridge DC input jack | The fridge's OWN 12V cord: plug → socket, plug → fridge. Slack loop for the 24" slide, clipped to the FIXED rail (step 5 above) |
| P9 | SW2 "FANS" — Ampper rocker | F2's 18AWG red → **PWR** tab; **LOAD** tab → W1209 "+12V"; **GND** tab → neg bus |
| P10 | W1209 controller — 4 screw terminals | **+12V** ← SW2 LOAD; **GND** ← neg bus (18AWG black); **K0** ← short jumper from +12V; **K1** → fan feed wire. Strip 1/4", screws snug. Settings: P0 = H mode, fans ON ~95°F / OFF ~85°F |
| P11 | W1209 2-pin probe header | NTC probe plug clicks on; the probe TIP zip-ties inside the bay at the fridge's kitchen-facing wall, next to the exhaust fan |
| P12 | WAGO 221-413 lever nuts (×2) | Nut A: K1 feed → both fan REDS. Nut B: both fan BLACKS → one 18AWG black → neg bus. Levers fully closed, tug-test every wire |

***Wire + termination spec:** main run = the OYMSAE cord's own 16AWG; in-bay branches = 18AWG red/black hookup wire. Crimps: heat-shrink #10 rings (block studs) and 0.25" female spades (switch tabs) — crimp, tug-test, then shrink. SW3 is a spare on the panel (the cooktop is 120V AC, switched at Power strip 2's own outlet switch).*

*The fan system's own wiring (fuse/switch/PWM controller/fans, Section 1) taps off this same DELTA 3 source rather than a separate line — see the fridge wiring schematic in Renders. The rail-channel routing is illustrated in the Step 8 diagram — it carries only this DC line, since both AC loads plug into the rear outlet.*

### *DELTA 3 AC charging line (front console → Panel A drawer)*

*A second, independent cord charges the DELTA 3 stack itself from the front console's AC outlet while driving — this is what makes the fridge's long runtime (Section 1) actually sustainable trip after trip, instead of draining down with no way to recharge until you're back home. Like the cooktop and Power strip 1's lines, this is a 120V AC household cord (charging the DELTA 3's AC input), not a DC connector. Panel A is the module closest to the front seats (the pantry sits at the opposite end, on Panel C — Section 1), so this cord **crosses zero panel seams**, just a short run through the open floor gap in front of Panel A.*

1. *Drill a 1" grommet hole in Panel A's front end rail, near the DELTA 3 drawer.*
2. *Run the cord forward, through the open floor gap (nothing built there — Section 1), to a grommet hole near the front of the platform.*
3. *Continue along the van floor track to the front console AC outlet — same routing as the other two forward-running lines.*

4. *Leave enough slack at the Panel A end that pulling the DELTA 3 drawer open doesn't strain the connection.*

***Outlet facts (VERIFIED): 2 AC outlets total, both fed by the van's single 1500W inverter.*** *The rear outlet sits on the* ***passenger-side rear quarter trim: 22.5" above the cargo floor, ~10" forward of the liftgate scuff plate****, socket center* ***10" laterally in from the right sidewall*** *(MEASURED V9b, owner Aug 1 2026), directly above the 12V battery access panel and below the rear cup holders.* ***That is 4" ABOVE deck level (18.5"), not below it — which makes the cord runs easier:*** *both cords reach the socket from deck level instead of having to drop down the gap between the kitchen unit's outer face and the van wall. Plug access doesn't depend on sliding the kitchen out.* ***Serviceability note: the 12V battery access panel sits under/behind the kitchen's passenger rear corner — keep it clear: the anchor board's passenger strip starts 3" in from the hatch, so check it clears the panel (Section 8), and battery service = unstrap + slide the kitchen out (Component 7).***

***Shared-inverter wattage: every AC load shares one 1500W pool (confirmed — both outlets ride the same inverter).*** *Plugging the cooktop + Power strip 1 into the rear outlet and the DELTA 3 charger into the front one cleans up the cords, but not the budget: the DELTA 3's AC input alone (up to 1500W at full X-Stream speed) can consume the entire rated capacity. Running the induction cooktop while fast-charging the DELTA 3 could oversubscribe the inverter. In practice this rarely collides — cooktop use happens at camp with the engine off, DELTA 3 charging while driving — but if you ever want both at once (e.g., cooking at a rest stop with the engine idling), cap the DELTA 3's AC charge rate in its app (EcoFlow lets you limit input wattage) to leave headroom.* ***The cooktop's own half of that budget is now measured, not guessed:*** *both burners at P6 draw ~1460W and a single burner at max ~950W (bench-tested Sept 2026 — Section 7), so dual-burner cooking uses essentially the whole inverter on its own.*

***The WAVE 3 is the one thing still fully self-contained.*** *It draws from the DELTA 3 Plus (in Panel A's drawer) via its own charge cable, and isn't wired to any van circuit at all.*

## 6. Illustrated Build Manual — By Component

*Everything from Sections 3-4 reorganized around **10 self-contained components**, in the order you'd actually build them. Each component lists its parts, then walks through numbered steps in instruction-manual format: **the new parts for that step on the left, an exploded assembly view showing how they go in on the right** — drawn as woodworking-plan line art (white parts, black edges, letter labels keyed to the parts list, arrows showing insertion direction). Every piece dimension traces back to the cut list (Section 3) and every hardware item to the BOM (Section 4). Panels A and B share identical frame/divider/drawer construction, so those steps intentionally reuse the same diagrams in both components.*

*Cut every piece for every component up front, per Section 3's cut list, before starting Component 1 — sand all edges once cut.*

### Component 1: Rear Pantry — Tote + Microwave

***Changed 30 Oct 2026: the IRIS drawer cluster is out.** The rear pantry is one **27-gallon Tough Box tote** (black/yellow, CPC27GTBY; measured: bottom 15⅛ × 25", rim 20½ × 30.15", 14.3" tall) on the tailgate end of Panel C's deck, beside the **BLACK+DECKER microwave** (17.3" W × 14" D × 10.2" H). The pots' 11×11 box goes inside the tote, so there's no pot crate. **One cam strap holds both, and there are no cleats:** the rubber mat grips, the mattress stops them going forward, and the closed hatch stops them going back. Shop drawing: the **Tote + Microwave Strap Anchors** sheet at the end of this PDF.*

***Parts needed:***

- *27-gallon Tough Box tote with lid (bought)*
- *The black non-slip mat on the deck's tailgate end (Panel C sheet)*
- *2 stainless footman loops from the spares, on their own #10 × ¾" screws (or 2 surface clip D-rings on #8 × 1" screws). **Not** the recessed flush D-rings from the old pantry kit: they need a pocket cut, and the anchors go on the 1½" rail face*
- *1 × 1" cam strap with snap hooks, 8 ft*

***Layout** (looking forward from the tailgate):*

- ***Tote:** 25" side across the van, bottom flush with the deck's tailgate edge, ~1½" in from the driver edge. The bottom pokes ⅝" past the 14½" pantry end, and the foot of the mattress takes it (a 79¼" run for the 78" mattress). Its rim hangs ~2¾" past the deck's tailgate edge, ~33" off the floor. **Check at the van that the closed hatch clears it:** V5 was taken at deck height only. If it doesn't, the tote moves ~2" forward and the mattress no longer fits its run.*
- ***Microwave:** flush with the passenger edge, door facing the tailgate. The gap to the tote closes from ~2¼" at the deck to ~⅜" at its top corner; that side has no vents. **Its vents are on the back**, ~½" from the mattress foot: pull the mattress foot back or slide the microwave out toward the tailgate before running it, and run it only with the hatch up.*

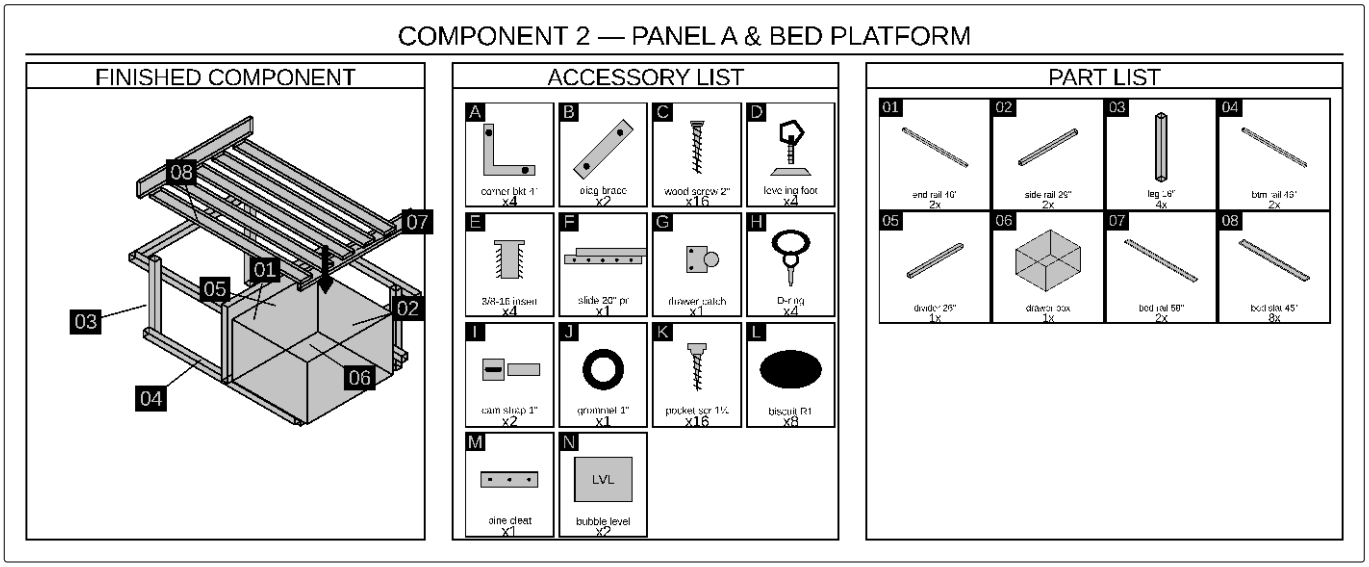
***Build it:***

1. ***Mark the anchors.** 7½" from the tailgate end, on the outside face of both side top rails. That's the middle of the tote's bottom and close to the microwave's middle, so the strap crosses square, and it's clear of the tailgate legs.*
2. ***Fit the anchors.** A footman loop on each mark, centered on the rail's 1½" face (¾" below the deck top), screws in a level line along the rail; pilot first. They sit below the deck, so they take none of its width, and stand ~¾" into the ~2" to the van wall.*
3. ***Mat down, tote and microwave in place.***
4. ***Strap.** Hook one end, run it up over the tote lid and down across the microwave's top, hook the other end, and cinch the buckle on top of the tote lid. It presses the whole lid but only the microwave's outer top corner; with the mat, that's enough for ~22 lb. Snug, not cranked: the microwave's case is thin sheet metal.*
5. ***Load it.** Heavy things low in the tote. Re-cinch at the start of each drive.*

***Still open:** Power strip 1 and the ROLL bubble level were to go on the deck edge in the old pot bay, which the microwave now fills. Re-site them with the control cluster at Component 6.*

***Retired:** the 4 IRIS drawer units (6 bought — spares or returns), the pot crate (not bought), the pantry cleats, and the 2 recessed deck D-rings from the pantry kit (spares).*

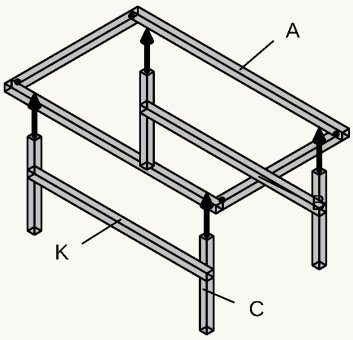
### Component 2: Panel A & Bed Platform



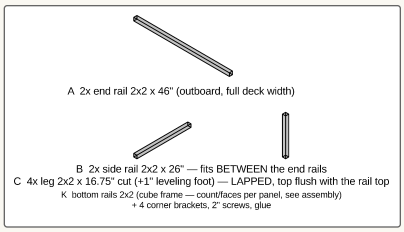

*Panel A and Panel B share the same frame construction — neither has a top of its own; the three-piece slatted bed frame caps both (see Steps 6-7 below, and the Bed Frame Detail render): Panel A's section is* ***screwed down*** *(it is also acting as a diaphragm — see Section 1), and Panel B's top is two centerline halves that lift out individually.* ***Panel B also has a center bearer*** *for those halves' inner edges — the one frame difference from Panel A's divider, which splits its bay instead. They diverge at the drawers:* ***Panel A's left (driver-side) bay is WAVE 3 open storage, not a drawer*** *— the WAVE 3 (20.4" wide) is too wide for a boxed drawer's 19" clear interior, so it rests directly on the bay floor instead, reached by hand through the driver's side door. Panel A ends up with only ONE actual drawer (right side, DELTA 3).*

***Parts needed:***

- *Side rails — 2,* ***26⅛"*** *cut, 2x2 pine (they fit BETWEEN the end rails: 29" panel − 2 × 1.4375" measured stock)*
- *End rails — 2, 46" cut, 2x2 pine (the outboard pair, full deck width)*
- *Legs — 4, cut to* ***16.75"****, 2x2 pine (1.4375" square measured — plenty for this load); the leveling foot brings each back to an effective 17.75", which is the floor-to-rail-TOP height because these legs are* ***lapped*** *to the end rails' inner faces, not butted under them (Section 3)*
- *Leg leveling feet — 4 (3/8-16 screw-in insert in the leg's bottom end grain + leveling stud; the stud's own fixed hex collar is the adjustment — no knob), one per leg (Leg Leveling Foot Detail render)*
- *Bottom rails — 2, 46" cut, 2x2 pine (END faces only — the side faces stay open for the drawer/WAVE 3)*
- *Center divider — 1,* ***26⅛"*** *cut, 2x2 pine*
- *Corner brackets — 4*
- *Diagonal corner braces — 2 (recovers racking rigidity lost without a top).* ***They mount on the frame's UNDERSIDE****, not the rail tops — the top face is the deck plane and the bed section is screwed down flat onto it, so anything proud of it is a rock you can never shim out. Same face as the 4 corner brackets. Note the frame is built top-face-DOWN, so that face is the one pointing up at you on the bench*
- *Drawer box — 1 (right/DELTA 3 side only), 5 pieces: bottom 19.25"x21.25" (1/2" ply, into a 3/8" glued rabbet), 2 side walls 22"x13.75" + 2 front/back walls 18.5"x13.75" (cut at 20", trimmed 1.5" each) (****3/4" ply*** *— these are the parts no other leftover could hold, Section 3). Glue +* ***2× R3*** *per corner since it carries the 48 lb DELTA stack.*
- *Drawer slide — 1 pair, 20" full-extension*
- *Drawer catch — 1, friction catch or small turn latch*
- *DELTA 3 drawer hardware (right drawer) — rubber non-slip mat (no locating cleats, as built), 2 D-rings + cam strap, a turn button on the frame's top rail, 1" cable grommet*
- *WAVE 3 hold-down (left bay) — 2 screw-eye D-rings in the end bottom rails + 1 low-profile cam strap over the unit (no glide strips, no shelf)*
- *Bed platform lumber — seven 1x4 x 8ft pine boards: five 42" slats + ten 17.5" slats + six 29" side rails — crosscuts only, no hardware (Bed Platform Detail render)*
- *Panel B center bearer — 52¼" of 2x2 (two* ***26⅛"*** *pieces side by side, rough), out of the 2x2 frame stock; glued up, then trimmed to the measured gap less ¼" (~25⅞") — it is* ***drop-in*** *on 4× 1" corner braces*
- *RV bubble levels — Larbeti stick-on 2+2 pack (PURCHASED): PITCH bar mounts here, ROLL one goes to the rear-pantry deck edge (Component 1)*




**1**




K: bottom rails (2x2, underside at 1") close the frame into a box — both end faces only (drawer + WAVE 3 exit the sides)

*Build the frame: 2 side rails (B, **26⅛"**) dropped BETWEEN 2 end rails (A, 46" — the outboard pair, full deck width), joined with corner brackets and 2½" screws, then the 4 legs (C) — CUT to 16.75" (Panels A/B run 3/4" shorter than Panel C's front pair: deck recess), drilled in the bottom end grain (1/2" dia x 1-3/4" deep, centered — deep enough for the foot's whole stud; test-fit in offcut) for the 3/8-16 screw-in threaded insert, so the leveling foot brings each leg back to an effective 17.75" (Leg Leveling Foot Detail render). Much easier to drill before assembly. Each leg is **LAPPED** to its end rail — 1.5" inboard, against the rail's inner face, top clamped **flush with the rail top** before you drive the 2 screws through the rail's face into the leg's side (Section 3). Legs inset **3.5"** from the deck's side edges to clear the floor-level vent intrusion (V4). Then close the bottom: 2 END-face bottom rails (K, 46", underside 1" up at the leg bottoms — Panel A's SIDE faces stay open so the drawer and WAVE 3 can exit) plus 2 diagonal corner braces — **on the frame's UNDERSIDE, never the rail tops** (the top face is the deck plane; a strap on it holds the bed section up off the wood and it rocks). Set each strap so its screws clear the corner bracket's: a 4" strap lands ~2¾" along each rail, the bracket's screws stay inside 2". up top — the part-cube racks far less than rails + brackets alone. (Panel B's frame differs only at the bottom rails: it closes all 4 faces.)*

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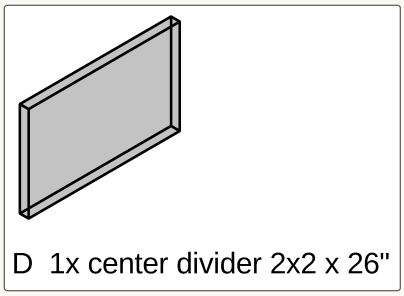
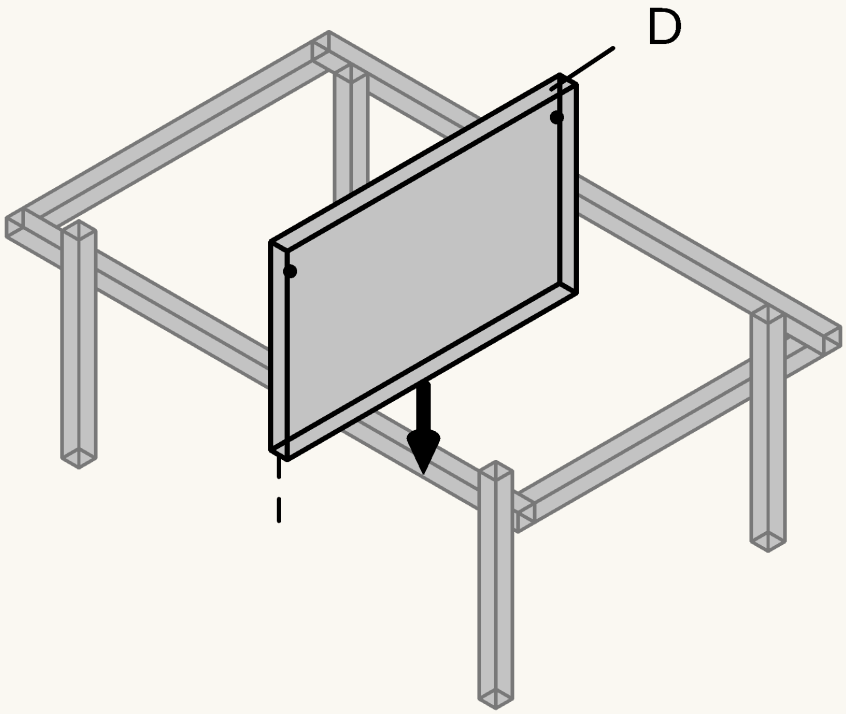
**2**




*Drop the center divider (D) into the bay, centered on the panel's width — it splits the void into a left and right bay and carries one side of the right bay's drawer slide.*

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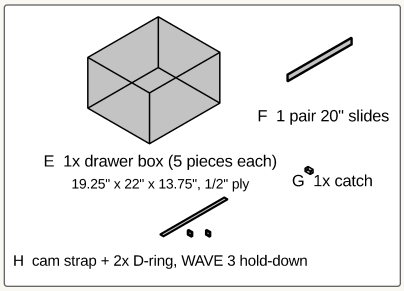
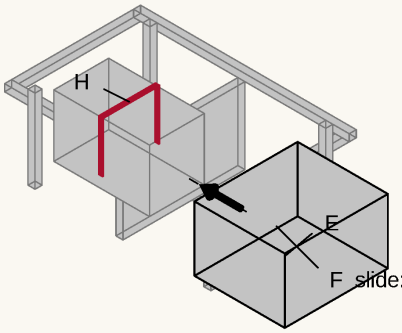
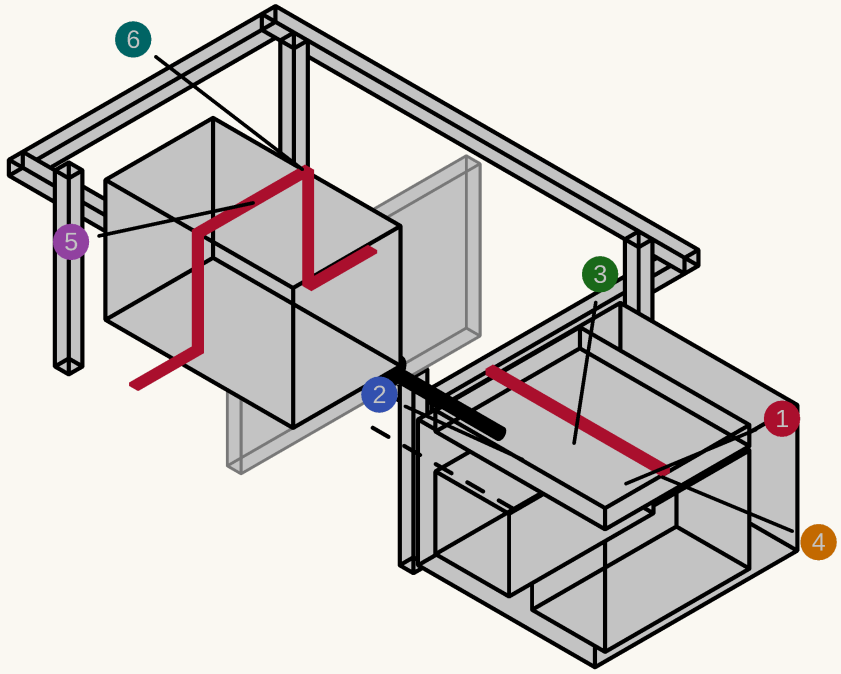
**3**




Left bay: WAVE 3 open storage (shown in place for reference) — no box, no slide

*Build the right (DELTA 3) drawer box (E, 5 pieces, open top) — biscuit the corners with **2x R3** biscuits each + glue, bottom in a glued rabbet (**R3, not R1** — the walls are ¾" and an R1 is thin-stock sizing, lost in that thickness; Joinery guide, vignette 3) (Joinery & Fastener Guide) — and slide it onto its pair of 20" full-extension slides (F) — one slide side on the frame's outer rail, the other on the center divider. Add a catch (G) so it doesn't slide open in transit. On the LEFT bay, skip the drawer entirely: the WAVE 3 sits straight on the bay floor — screw the 2 D-rings (H) into the end bottom rails either side of it for its cam strap (Component 8).* ***The full shop drawing for this box — plan, section, corner detail, fit checks, cut list and assembly order — is the DELTA 3 Power Drawer sheet appended at the end of this PDF.***




**4**

*DELTA 3 drawer hardware (parts list above): pine cleats, 2 D-rings + cam strap, 1" grommet*

# DELTA 3 STACK (right drawer) + WAVE 3 (left bay)

Panel A, exploded — Components 2 & 8

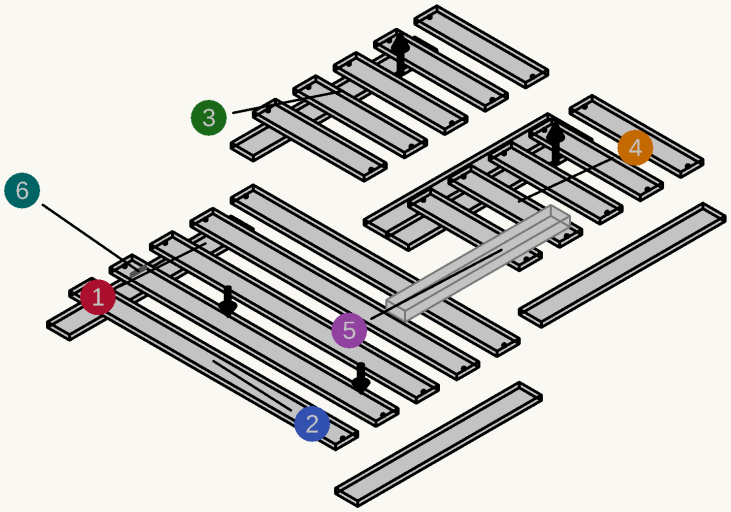
DELTA 3 Plus outboard (the WAVE 3 plugs into it), Extra Battery inboard; ~48 lb in a normal drawer — no E-track. The WAVE 3 sits on the raw bay floor, cam-strapped to 2 D-rings.

Found storage: a ~2.6" lift-out tray tops the DELTA stack, The WAVE 3's hoses live on the hook in Panel C's utility bay.

## Component / position

1. DELTA 3 Plus — 7.95"x15.7"x11.16", ~28 lb
   outboard (pull wall); cam-strapped over locating cleats
2. Smart Extra Battery — 8"x15.7", ~20 lb
   inboard — extra runtime only, grabbed less often
3. Lift-out tray — reclaims ~2.6" dead air
   cables, the DELTA's own cords, dongles
4. Cam strap + D-rings + 1" grommet
   strap over both units; WAVE 3 charge cable exits the grommet
5. WAVE 3 — 20.4"x11.7"x13.2", 33.7 lb
   raw bay, 0.35" clear — a boxed drawer would NOT fit
6. Cam strap + 2 D-rings (WAVE 3)
   fore-aft over the top, D-rings on the end bottom rails — no bounce, no walk-out

*In the right drawer, add the DELTA 3 fixtures: a rubber non-slip mat on the drawer floor (as built — no locating cleats), D-rings ~7½" down from the top edge + cam strap to hold the stack down in transit, and a cable grommet for the WAVE 3's charge cable. DELTA 3 Plus outboard (pull wall), Smart Extra Battery inboard — see Component 8.*




**5**

*seven 1x4 boards: 6 rails + 15 slats + glue and 1¼" pocket screws, plus 52" of 2x2 for Panel B's center bearer (parts list above)*

## 58" x 49" overall — 15 slats in one 3/4" plane

### 1-6 keyed to the parts table in the text

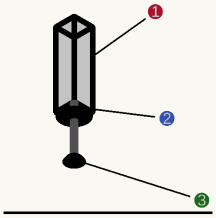
*Crosscut everything from seven 1x4 x 8ft pine boards — nothing gets ripped, a 1x4 is already 3/4" x 3-1/2": six 29" side rails, five 42" slats, and ten 17.5" slats (two 42" or five 17.5" per board).* ***Build it as THREE lift-out pieces****: Panel A's section is one pair of rails on edge-guides 49" apart outside-to-outside (measurement V7) with its five 42" slats between them; each of Panel B's two halves is a pair of rails 24.5" apart with five 17.5" slats. Every gap lands at ~2.9". Glue and pocket-screw each slat end into the rails' inner edges: two pocket holes per end drilled with a Kreg 320 jig set for 3/4" stock, glue on the end grain, then 2× Kreg SML-C125 1-1/4" coarse-thread screws. Set every slat crown-up, gapped with 13" ripped-plywood spacers (Bed Platform Assembly sheet). Then add Panel B's center bearer — two 2x2s glued side by side, fore-aft at its centerline, cut to the gap between the end rails less ¼" (~25⅞") and dropped onto four 1" corner braces so it lifts out for the spare, top flush with its long rails: it carries the inner rail of both halves (1-1/2" of bearing each) and halves the deck's unsupported span over Panel B. Last,* ***screw Panel A's section down*** *to its rails (2" screws every ~8", countersunk so nothing sits proud under the foam): it is permanent, it stiffens Panel A against racking, and it is the datum the two loose halves register against. Panel B's halves get no fasteners at all — they just lift, and at 24.5" wide either one will pass back out through the 35" side door. Everything sits in one flush 3/4" plane that ENDS at the B/C seam — Panel C's own deck is at the same height, so the sleeping surface stays flush and the mattress's last ~20" rides that deck. The platform adds only 3/4" to the stack. Last, stick one Larbeti bar level (self-adhesive — degrease the surface first) on the driver-side rail's outer edge at mid-span (the 3/4" face fits it exactly) — it reads fore-aft PITCH from the slider door while you turn the leg feet; its twin reads ROLL from the rear-pantry deck edge (Component 1).*

**6**

*no new parts — the assembled platform from Step 5 (leveling feet already on the legs, Step 1)*

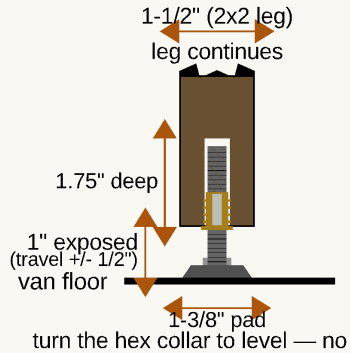
Component
Position / fastener / material

1. Leg (bottom end shown)
   2x2 pine (1.5" x 1.5" actual), cut to 17.5" / 16.75" / 16"
   drill centered, 1/2" dia x 1.75" deep — the same hole in all 12
2. 3/8-16 threaded insert
   driven UP into the leg's end grain
   screw-in style, 7/16" OD coarse outer thread; drive it with a 10mm hex wrench
3. Leveling glide bolt + pad
   1.375" floor pad; +/-0.5" travel around 1" nominal
   threads into the insert; TURN IT BY ITS FIXED HEX COLLAR — no knob since Sept 2026



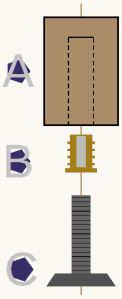

LEG LEVELING FOOT — shown 3x actual size (12 total: 4 per panel x 3 panels)

Legs are CUT to 17.5" (Panel C front, lapped) / 16.75" (A/B, lapped) / 16" (Panel C rear corners, butt-under) — the foot makes up the last 1".




turn the hex collar to level — no knob

EXPLODED — assembly order




SECTION A-A — installed, mid-travel (6x)

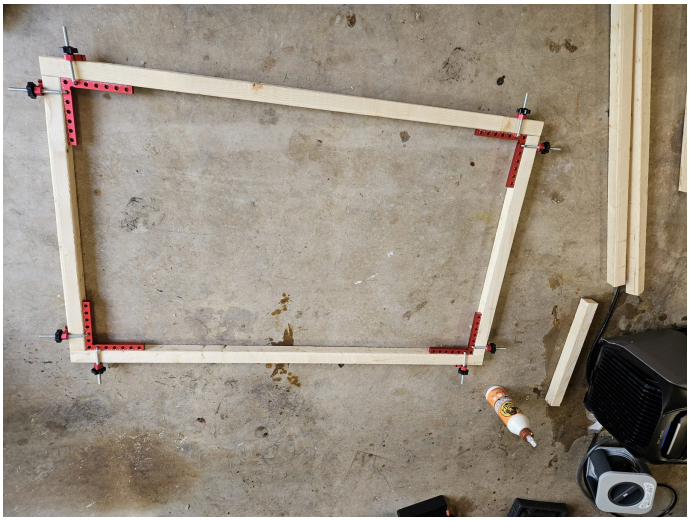
***Parts, keyed to A–C on the exploded view**: **A** the leg — drill its end grain ½" dia × 1¾" deep, centered. The depth is for the whole stud, not just the insert; test-fit in an offcut first. **B** screw-in insert, 7/16" OD with a 3/8-16 bore — drive it in flush with a **10mm hex wrench** in its own hex socket. **C** the stud + its fixed hex collar — screw the stud into the insert to 1" exposure, and **turn it by that collar** to level. **Three parts per foot**: no star knob and no jam nut — a hand wheel on the bottom of all 12 legs takes up too much room down there. Leveling is a wrench on the collar. Measure your collar before you pack a wrench; it is not 9/16". **12 feet total** (4 per panel × 3 panels): 3× Anwenk leveler 4-packs. **Rating 1,320 lb per foot** (Anwenk) — 4 under the heaviest panel is ~5,280 lb of capacity against a ~90 lb module. **Skip the kit's stick-on pads** — the bare nylon pad grips the van floor better. Effective leg height = cut length + the 1" foot.*

*Set the finished platform down directly on Panel A + B's top rails and Panel C's mattress zone, centered so it overhangs the boxes 1.5" per side — nothing fastens it, it rests there and lifts straight off for drawer-bay access (it enters/exits the van tilted diagonally through the liftgate). Then LEVEL the boxes at the floor: with a level on the platform, tip each box corner slightly and **wrench that foot's hex collar** until the bubble centers along both axes. Re-check after the van is parked on level ground. **Bring the wrench that fits the collar — there is no hand grip on these feet.** With no knob there is nothing to lock; the collar is simply the thing you turn. **From the drawing:** SECTION A-A — installed, mid-travel (6x actual size) LEG LEVELING FOOT — engineering drawing. 12 feet total (4 per panel x 3 panels): 3x Anwenk leveler 4-packs. Rating: 1,320 lb per foot (Anwenk) — 4 feet under the heaviest panel = ~5,280 lb capacity vs ~450 lb actual load. NOTE: skip the kit's stick-on pads — the bare nylon pad grips the van floor better and doesn't shed.*

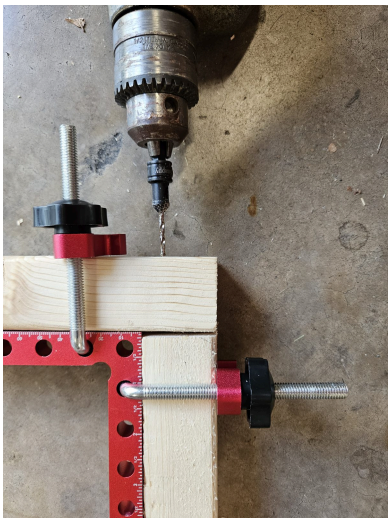
### *Panel A, as actually built — from the bench (Sept 7 2026)*

***Panel A is assembled, and it went together as drawn** — nothing below changes a dimension, a part or a fastener. What these add is the method, which no exploded view shows: how the corners were squared, drilled, driven and braced, and the two throwaway jigs that made it repeatable.*

***Three drills, so no bit ever gets swapped.*** *Two cordless and one corded: the corded one carries the* ***pilot/countersink*** *bit (it does the drilling, so it never runs flat), one cordless carries the* ***Torx*** *driver for the SPAX screws, and one carries the* ***Philips*** *for the corner-bracket screws. Same trick as the two-drill setup for boring the legs (Section 3), and it pays better here, because every corner runs pilot/countersink → Torx → Philips: one drill would mean three chuck changes per corner, four corners per panel.* ***Note the drives differ*** *— SPAX are Torx, the brackets came with Philips. That is the thing you find out at the bench if you didn't check it at the store.*



***Glue up the whole frame flat, in four right-angle clamps, before a single screw.*** *Each clamp holds its corner at 90° with two thumbscrews, so the frame sets to its shape rather than to whatever the first screw pulls it into. Check the diagonals while everything is still adjustable: a 46" × 29" frame that is square on the floor is what lets the A/B seam close flat and the alignment pins register later (Component 5). Note there is no bench in this photo — the garage floor is the flattest reference surface in most shops, and it is what this panel was built on.*

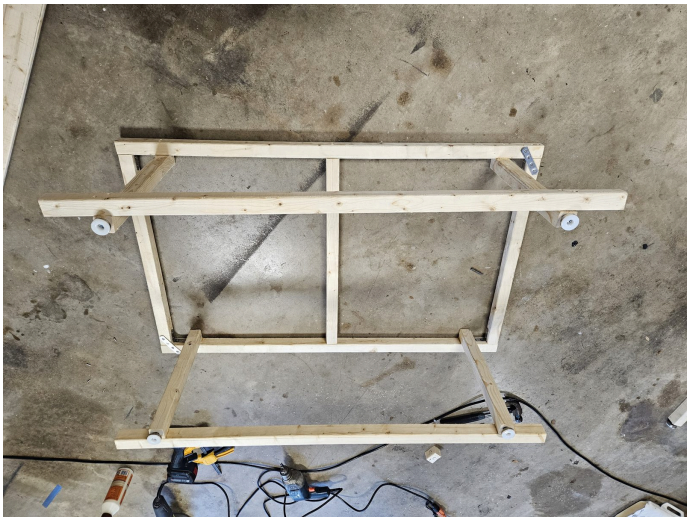


***Pilot and countersink in one pass, drilled through the clamp.*** *2×2 pine this close to an end will split if you drive a 2½" screw into it dry, so the pilot is not optional. The countersink half matters for a second reason: the head has to pull flush, because a corner bracket sits flat on that same rail face afterwards and will rock on a proud screw. Drill it with the corner still clamped — that is the whole point of clamping first.*

***Two screws per corner, and they are 2½", not 2".*** *Driven through the outboard end rail's face into the end of the side rail behind it, both pulled below flush. Through 1-7/16" of end rail, a 2" screw leaves only about ½" of bite in the side rail — not a joint to trust on a module that gets lifted in and out of a van by its rails. The same measured 1-7/16" stock that sets every side rail at 26½" is what drives this (Section 3).*

***The bracket goes on after the screws, not instead of them.*** *Four per panel, flat steel, on the inside of the corner. The glued and screwed joint is what carries load; the bracket's job is to hold the angle — to keep the corner from working loose over a few hundred miles of road input. Its screws are the short Philips ones, which is the third drill's only job on this panel.*

***Clamp the leg before you drill it — and cut a block so you only measure once.*** *A* ***lapped*** *leg has to sit flush with the rail top and stay exactly there while the pilot and both SPAX screws go through the rail's face into the leg's side (Section 3). Unclamped, it walks under the bit, and it ends up proud of the rail — which is the one surface the bed platform lands on. Two bar clamps hold it; the offcut marked* ***2"*** *sets the leg's position across the panel.* ***Be exact about what that 2" spans, because it is not the leg inset:*** *it is the* ***gap between the side rail's inner face and the leg's outer face****. The side rail sits flush with the deck's edge, so 1-7/16" of measured rail plus the 2" block puts the leg's outer face* ***3-7/16"*** *in from the edge — the 3.5"* `leg_inset` *the model carries, to within 1/16", and still 3/16" clear of the 3¼" vent intrusion that set that number (V4). Read the 2" as an inset instead and you would move every leg 1½" outboard, straight into the vent.* ***It is a gauge, not a shim:*** *cut it once, register the leg against it at every corner, and the position never gets measured again. Cheapest jig in the build.*



***Panel A complete, upside down — legs, feet, divider, bottom rails and braces all on.*** *Shown inverted, which is how you work on it: the four white* ***leveling feet*** *are already threaded into the legs' end grain (Section 3), the* ***center divider*** *is in, splitting the bay into the DELTA 3 drawer run and the WAVE 3 bay, and the two* ***END-face bottom rails*** *are closing the frame. The* ***metal straps at two diagonally opposite corners*** *are the diagonal corner braces from the parts list — 4" steel flat-strap, and* ***two per panel is the spec, not a shortfall****. They recover part of the racking stiffness Panel A lacks without a plywood top; Panel A gets the rest back later from its bed section, which is screwed down and acts as a diaphragm across the rails (Step 5, and Section 1).*

### *The battery drawer, as actually built — from the bench*

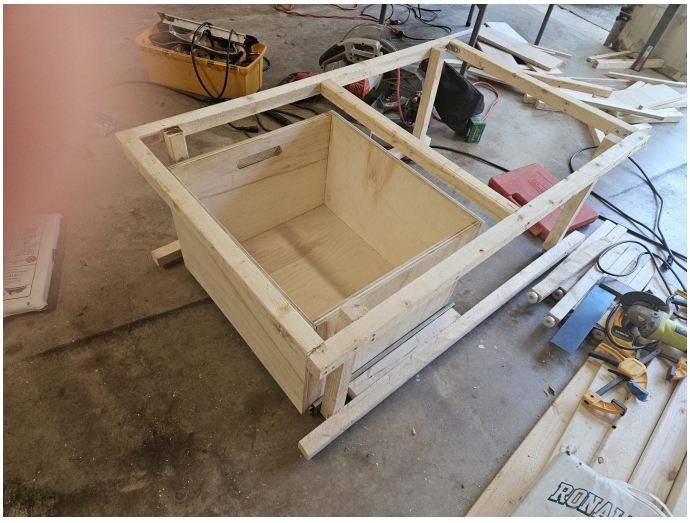
***The DELTA 3 drawer is built and running on its slides in Panel A's right-hand bay.*** *The sizes are on the DELTA 3 Power Drawer sheet at the end of this PDF. These photos add the order of operations.*

***Cut the bottom's rabbets first, on the router table, while the walls are still loose boards.*** *⅜" deep × ½" wide along the bottom inside edge of all four walls, so the ½" bottom lands flush with the box's underside. A fence clamped to the table and one pass per wall keeps all four identical.*

***All four walls cut and rabbeted:*** *two side walls at 22 × 13¾" that run full depth and carry the slides, and two front/back walls at 18½ × 13¾" that fit between them. Label each one as it comes off the saw. Two of them come out of the offcut rotated, and the grain direction is hard to read later.*

***Biscuits dry-fitted and ready for glue.*** *Two R3 biscuits per corner, 2" in from each end of the joint. The slot near the end of this wall is the cord slot for the DELTA 3's charging line. Cut it before glue-up, while the wall still lies flat.*

***Glued and clamped in one go:*** *every biscuit slot, the mating faces and the bottom's rabbet all the way round. Check the diagonals before the glue grabs, then leave it. The bottom is structural here: it's what stops a 48 lb load racking the box on the slides.*



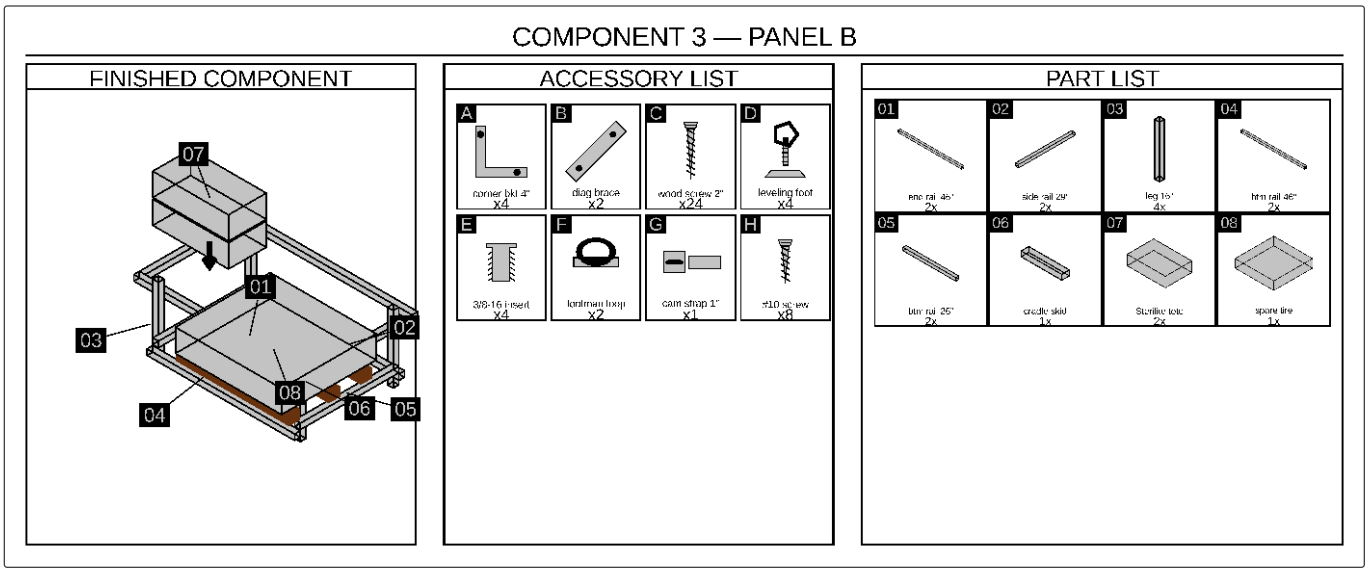
***In Panel A's frame, closed.*** *The slides mount to the frame's outer rail on one side and the center divider on the other. Note the cord slot in the box's wall, so the charging line can stay plugged in.*

***At full extension.*** *The whole box clears the frame, so both units lift straight out from above. Leave enough slack in the charging line for this travel.*

***From the side door, closed.*** *The pull wall sits just inside the frame, under the top rail, with a routed hand slot instead of a handle, so nothing sticks out into the door opening.*

***Loaded:*** *the DELTA 3 Plus and the Smart Extra Battery side by side, with the charging cable leaving through the cord slot. The cam strap goes across both units, D-ring to D-ring, before the van moves.* ***As built (30 Oct): no locating cleats.*** *The rubber mat on the drawer floor holds the units, the strap pins them down, and a turn button on the frame's top rail keeps the drawer shut. Power Drawer Tie-Down sheet.*

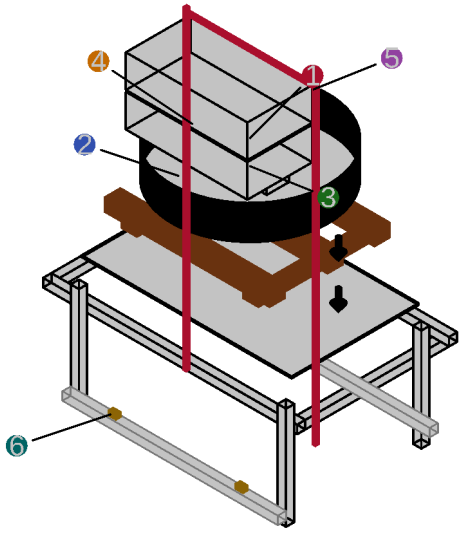
### *Component 3: Panel B*



*Same frame construction as Panel A — and that's the whole build.* ***Panel B has no drawers, no divider, and no skirts****: the sliding-door openings sit over Panel A's footprint, not Panel B's (owner-confirmed), so nothing pulled sideways from this panel could ever clear a door. Its bay is deep storage instead — lift the platform + mattress and load it from above.* ***The SPARE TIRE lives here*** *(RJ-MODINI kit, flat on 3" cleats at the axle — see the Spare Tire Stowage render), with* ***2× Sterilite 28-Qt lidded totes restacked on top*** *and the jack kit strapped beside it, so whole labeled boxes lift out instead of loose items. The Panel B detail render dimensions* ***every hole in the panel****: the 4 leg-bottom leveling-insert holes (1/2" dia x 1-3/4" deep, dead center of each leg's end grain) and the alignment-pin holes on both seam faces (2x 3/8" dia x 3/8" deep, 3" in from each side edge).*

***Parts needed:***

- *Side rails — 2,* ***26⅛"*** *cut, 2x2 pine (BETWEEN the end rails)*
- *End rails — 2, 46" cut, 2x2 pine (outboard, full deck width)*
- *Legs — 4,* ***16.75"*** *cut, 2x2 pine (leveling feet make up the inch; lapped to the end rails, tops flush with the rail tops)*
- *Bottom rails — 4: 2x 46" + 2x* ***23¼"*** *(all four faces — the full cube; the side pair is 23¼" and not 26⅛" because the lapped legs sit 1.4375" inboard at each end) — the 46" end rail that meets Panel C: optional: 3" × ½" relief notches at 12¾–15¾" and 30¼–33¼" from the driver end for the backup striker straps*
- *Corner brackets — 4*
- *Diagonal corner braces — 2 (****underside****, like Panel A's — see Component 2)*
- *Deep-storage totes — 2x Sterilite 28-Qt under-bed lidded (23.5" x 16.9" x 5.9"), restacked ON the spare (plain lidded, NOT slide-drawer — no side access to use a drawer front)*
- *SPARE TIRE — RJ-MODINI kit for the 2011-2026 Sienna incl. Hybrid (T155/85R18 steel wheel, 28.5" dia x ~6.4" stored, ~40 lb incl. 2-ton jack/wrenches/cases — VERIFY weight + dims on arrival). Stows FLAT in this bay at the AXLE, on a* ***cradle skid****: 2 battens 2⅛" × 24" standing on edge, fore-aft between the 2nd-row seat tracks (centered 18½" and 27½" from the driver edge), with 3 cleats 3" × 30" glued and screwed flat across them, all ¾" ply — 2⅞" tall, so the cleats clear the seat tracks and caps and the spare still clears the 2-7/16" bottom-rail curb (cut 14 Oct). One piece, so the cleats can't wander. NOTHING screws into the van floor: the skid just sits on a non-slip liner offcut (same roll as the drawer liner)*
- *SPARE INSTALL (once — see the exploded view on the Spare Tire Stowage render): (A) lay the liner offcut on the van floor mid-bay, (B) set the skid on it, (C) spare FLAT on the skid,* ***valve stem UP*** *(pressure checks without unloading), tool case nested in the wheel barrel, (D) 2 Sterilite totes restacked on top, (E) 1" cam-buckle strap OVER the whole stack, hooked to* ***2x 1" footman loops screwed (#10 x 3/4") to the inside faces of Panel B's bottom rails***
- *SPARE ACCESS (emergency-only, no tools — 4 steps on the render): 1. lift the mattress + platform slats off; 2. lift out the 2 totes, still packed; 3. pop the cam buckle, lift out the tool case; 4. tilt the spare upright and roll it over Panel C's deck, out the gate*
- *NOTE: the spare does NOT fit above the kitchen (28.5" disc vs ~24" before the fridge blocks it) — Panel B is the only interior spot, and it's the best one: ~40 lb at the axle beats any hitch basket (which would lever ~130+ lb onto the rear axle)*




Exploded stack (top -> bottom)

1. Spare, FLAT, valve stem UP (pressure checks in place)
2. Cradle skid: 3x 3" cleats on 2 battens (offcut), one piece
3. Tool case (2-ton jack + wrenches) nested in the wheel barrel
4. 2x Sterilite totes restacked on top
5. 1" cam strap over the whole stack
6. 2x footman loops, #10x3/4" into the bottom rails (strap anchors)

Access (emergency-only, no tools)

1. Lift the mattress + platform slats off
2. Lift out the 2 totes (they stay packed)
3. Pop the cam buckle, lift out the tool case
4. Tilt the spare upright, roll it over Panel C, out the gate

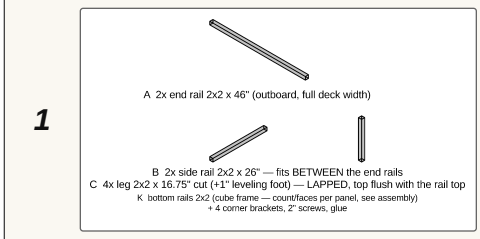
SPARE TIRE — flat in Panel B's bay at the AXLE (Component 3), exploded stack

Stack: 3" cleats + 6.4" spare + 5.9" tote, under the 17.75" deck

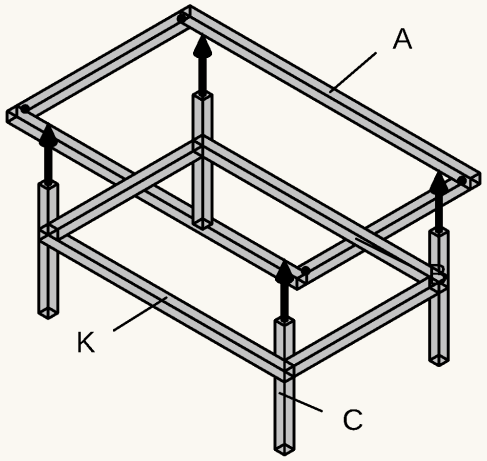
*Spare tire stowage*

***Does it fit?*** *Height stack, floor upward: 3" cleats (clearing the 2.5" floor-edge curb) + 6.4" spare + 5.9" tote = 15.3", under Panel B's 17.75" deck. Nothing screws to the van floor — the skid sits on the liner and the strap ties to footman loops on the bottom rails, which is also the best weight placement in the van (at the axle), so no hitch basket is needed. The kitchen-drawer slot does* ***not*** *fit the 28.5" disc, so that drawer stays where it is.*

- *Best weight placement in the van — no hitch basket needed. NOTHING screws to the van floor: the skid sits on the liner, the strap ties to footman loops on the bottom rails.*



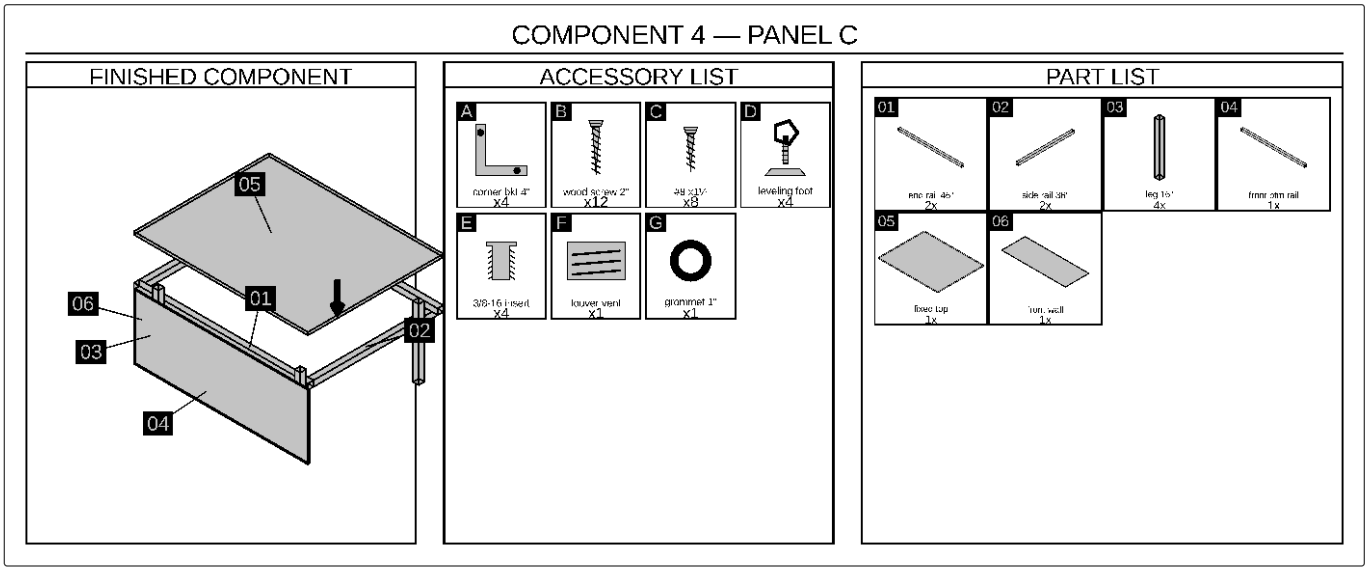

**1**




K: bottom rails (2x2, underside at 1") close the frame into a box — all 4 faces (full cube)

*Build the frame — like Panel A's step 1: side rails (B, **26⅛"**) between the end rails (A, 46"), corner brackets, 4 legs (C) inset 3.5" from the deck edges and LAPPED to the end rails with their tops flush (V4; drilled for their leveling inserts BEFORE assembly), the 2 diagonal corner braces, and bottom rails on ALL FOUR faces (2x 46" at the leg bottoms, underside 1" up, + 2x **23¼"** side rails set **1¾" higher**, just above the leveling-insert bore, so their screws clear the end rails' and the bore) — **the full cube**: nothing exits Panel B sideways, so every face can close, and this is the stiffest frame of the three. That's the entire panel — no divider, no drawers, nothing else. Drill the alignment-pin holes when mating it to its neighbors (Component 5).* ***The full shop drawing for this frame — plan, elevation, the bottom-rail screw order, the drop-in center bearer, cut list and build order — is the Panel B Frame sheet appended at the end of this PDF.***

### *Component 4: Panel C*



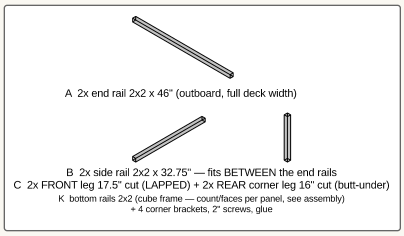

*Build this one before Component 1 (Rear Pantry) — the pantry tote and the microwave sit on this panel's finished deck at the tailgate end (one cam strap to footman loops on the side rails — no fasteners into either), with no frame or top of its own.*

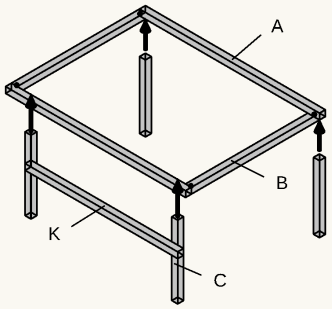
***Parts needed:***

- *Panel C top — 1, 33" x 43", 3/4" ply (drops BETWEEN the rails;* ***notch 1.5" x 1.5" out of its front edge at 3.5" in from each side****, for the 2 lapped front legs) + 4 bearer cleats (3/4" x 3/4": 2 x 33" + 2 x 40", from the offcut — the front one cut as 3 runs around those 2 legs)*
- *Side rails — 2,* ***32⅞"*** *cut, 2x2 pine (BETWEEN the end rails: 35.75" panel − 2 × 1.4375" measured stock)*
- *End rails — 2, 46" cut, 2x2 pine (outboard, full deck width)*
- *Legs — 4 in* ***two lengths****, 2x2 pine (leveling feet make up the inch): FRONT pair* ***17.5"****, lapped to the front rail's inner face, tops flush with the rail top; REAR pair* ***16"****, butt-under at the TRUE corners — the one pair in the build that cannot be lapped (Section 3)*

- *Bottom rail — 1, 46" cut, 2x2 pine (FRONT face only, on the front legs' line — 1.5" inboard of the front face). Optional: 3" × ½" relief notches at 12¾–15¾" and 30¼–33¼" from the driver end (and 1" in the front wall) if you run the backup straps to the striker loops under it*
- *Steel angle brackets — 2 extra, one per REAR corner leg (that bracket is the joint; the screws into end grain are alignment only)*
- *Corner brackets — 4*
- *Front wall — 1, 46" x 17", 1/2" ply, with the 120mm fan hole + one 1" grommet hole + three 3" low intake vent holes pre-cut (Panel C Front Wall render — every opening dimensioned)*
- *3" snap-in soffit vents — 3 (the low intake row in the front wall), from the 4-pack on hand*
- *Non-slip mat — 1, ~14¼" × 43", black rubber, on the deck's pantry end (the ~14¼" of deck the mattress doesn't cover), with no cutouts (the strap anchors are on the side rails, not the deck) — **held with Velcro so it can be replaced**: loop side on the deck (stapled at each end), hook side on the mat, a strip all round ~1" in from the edge and none across the middle, where a ridge would rock the drawer stack. Test the adhesive on the mat's back first; contact cement if it won't hold. Full method: the Panel C Frame sheet, last step*
- *VELCRO Extreme Outdoor 1" × 10 ft — 2 rolls (heat-rated adhesive), for the mat*
- *No divider, no drawers — the void stays fully open for Component 7 (Fridge & Kitchen Install)*

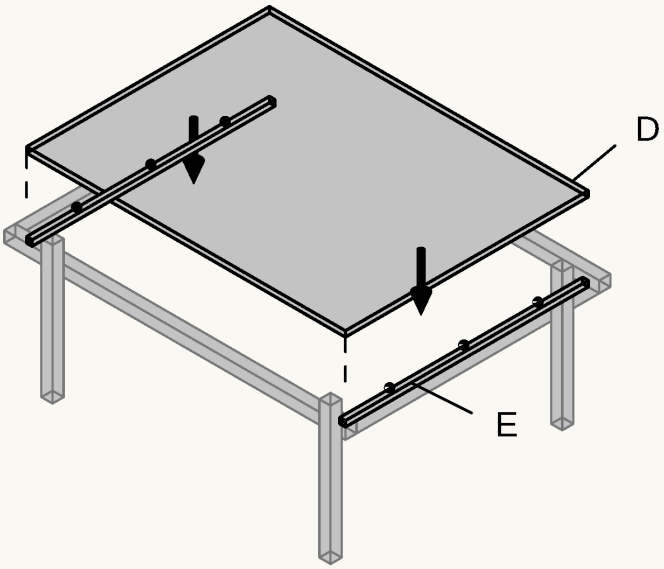
***1***



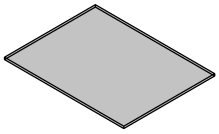




K: bottom rails (2x2, underside at 1") close the frame into a box — front face only (appliances exit the tailgate; rear legs at the TRUE corners)

*Build the frame — same construction as Panels A/B, with **32⅞"** side rails (B) between the 46" end rails (A): corner brackets, 4 legs (C) in TWO lengths — FRONT pair **17.5"**, inset 3.5" and LAPPED to the front rail's inner face with their tops flush; REAR pair **16"**, BUTT-UNDER at the TRUE corners (the fridge/kitchen slide paths pass exactly where inset rear legs would stand, and at a true corner the 46" end rail covers the whole leg so there is nothing to lap to — give each of those two a steel angle bracket; the rear-corner floor vents were checked and do NOT reach the leg area, so those legs land on solid floor). Add the FRONT-face bottom rail (46", underside 1" up at the leg bottoms, on the front legs' line) — the tailgate face stays open for the appliances.*




**2**



D 1x top 32.75" x 43" (3/4" ply — drops BETWEEN the rails)
E 3/4 x 3/4 bearer cleats on the rails' inner faces, tops 3/4" down + 1-1/4" screws down into the cleats + toe-screws into the rails

## no divider — void stays open for the fridge + kitchen unit

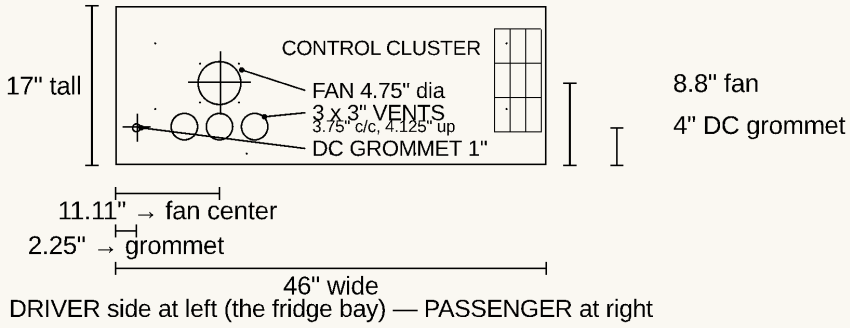
*Screw the bearer cleats (E) to the rails' inner faces — cleat tops exactly 3/4" below the rail tops — then drop the top (D) BETWEEN the rails onto them, flush with the rail tops (the deck recess: 3/4" more headroom), and screw it down into the cleats + toe-screw into the rails. **Notch the deck for the 2 front legs** — they are lapped, so their tops come up into the deck plane 1.5" inboard of the front rail: two 1.5" x 1.5" bites out of the deck's front edge, at 3.5" in from each side. Interrupt the front bearer cleat at the same two places. (The rear corner legs are butt-under and need no notch.) NO divider, since the fridge and kitchen unit need the full width of the void underneath.*

---

**3**

*front wall: 46" x 17" 1/2" ply, holes pre-cut per the Panel C Front Wall render*

## PANEL C FRONT WALL — 1/2" ply, flat pattern



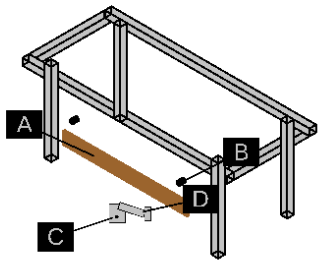

DRIVER side at left (the fridge bay) — PASSENGER at right

*Cut the front wall and its openings to the render's dimensions — the 4.75" (120mm) fan hole centered on the fridge bay (11.11" from the driver edge, 8.8" up), one 1" grommet hole at **2¼"** in, 4" up (3" puts its flange on the leg) (fridge DC line — Power strip 1 feeds off the rear outlet, so this wall doesn't pass its line), and **three 3" intake vent holes** in a row under the fan, centered on its line (3.75" apart, 4.125" up) — passive cool-air scoops that feed the fridge the coolest floor-level cabin air, each screened with a 3" snap-in soffit vent — then screw it to the front legs' BACK faces (the fridge side), its bottom edge ½" below the leg bottoms so it never touches the floor: 2x #8 x 1-1/4" into each front leg + 1 into each of **three 6" 2×2 back blocks** glued behind the bottom rail between the legs (7 total). *The wall sits 1-7/16" behind the rails, so screws into the top and bottom rails can't reach them.* **The full shop drawing for this panel — plan, the front-wall section, the wall's openings, the rear corner, the deck notch, cut list and build order — is the Panel C Frame sheet appended at the end of this PDF.** This is the ONLY wall on any panel; the intake fan bolts over the fan hole in Component 7. Panel C's sides stay open and its tailgate face is fully occupied by the fridge, the open utility bay, the kitchen unit, and the kitchen drawer.*

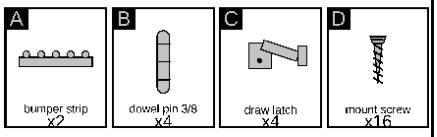
### *Component 5: Anti-Rattle Bumpers, Alignment Pins & Seam Draw-Latches*

## COMPONENT 5 — BUMPERS, ALIGNMENT PINS & SEAM DRAW-LATCHES

### FINISHED COMPONENT




### ACCESSORY LIST



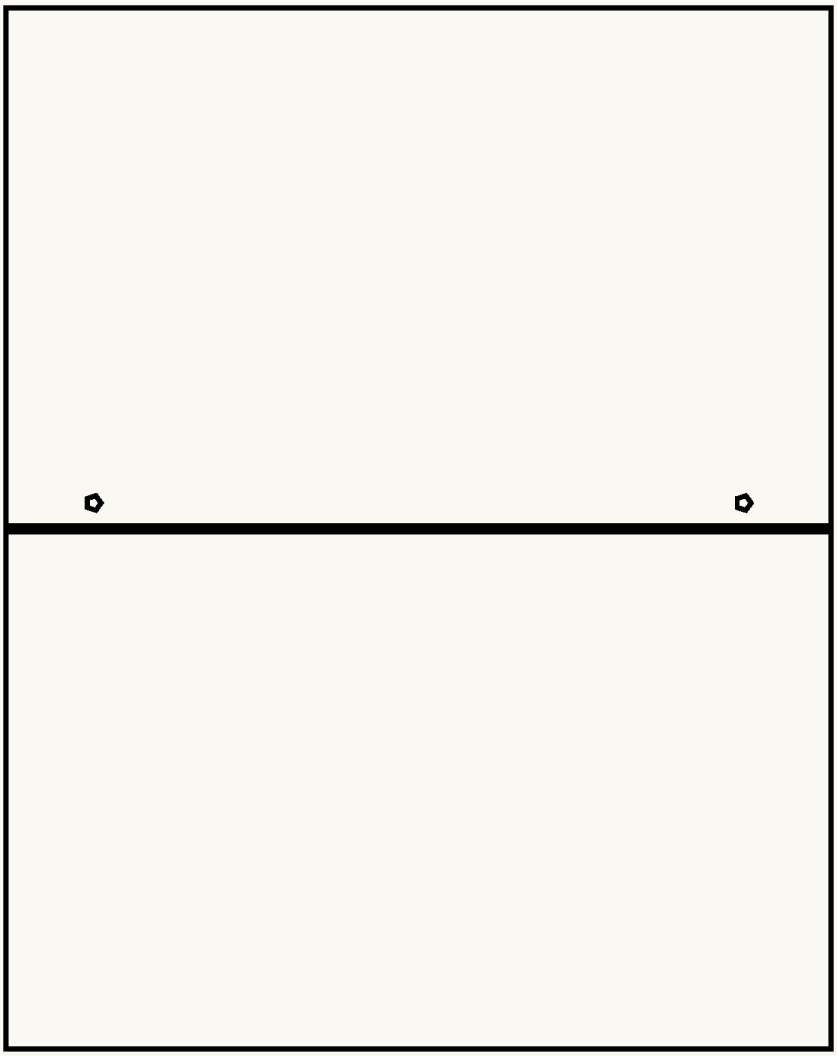

### PART LIST

(no cut parts)

hardware mounts to existing panels

***Parts needed:***

- *Anti-rattle bumper strips — 2 (adhesive-backed felt or closed-cell foam weatherstrip), one per seam*
- *Alignment dowel pins — 4 (2 per seam), 3/8" dowel rod, ~3/4" long*
- *Over-center draw latches — 4 (2 per seam × 2 seams), stainless/zinc, with their keepers and mounting screws*



**1**

*1 bumper strip + 2 dowel pins per seam (A/B seam and B/C seam)*

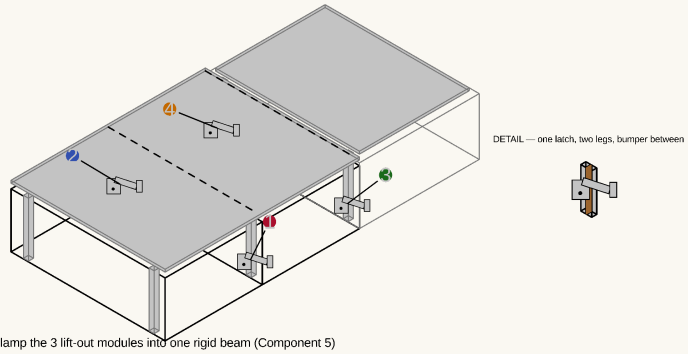
# Bumper strip + 2 alignment pins

## at every module seam (A/B shown)

*At each seam, apply an adhesive bumper strip across the full seam face on one side of the joint, then drill matching 3/8" holes and friction-fit 2 alignment dowel pins — the panels register in the same spot every time they're lifted out and reinstalled, and don't rub or squeak in transit.*

**2**

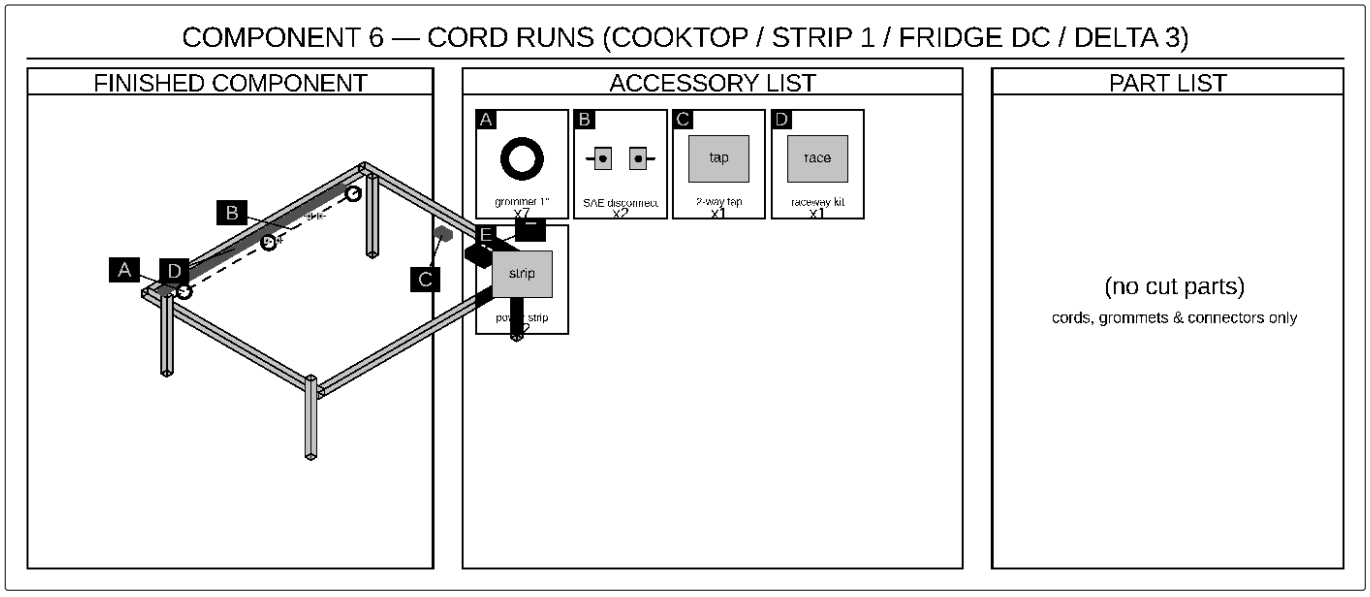
*2 over-center draw latches per seam (one each side), mounted low on the bottom-rail band*




SEAM DRAW-LATCHES — clamp the 3 lift-out modules into one rigid beam (Component 5)

*The alignment pins only locate the modules; the draw latches clamp them together. Mount one over-center draw latch on each SIDE of each seam (4 total: both sides of A/B and B/C),* ***low on the bottom-rail band (~1.75" up), over the inset leg line*** *— hand-reachable from the side door / tailgate. The base plate screws to one module's leg/rail, the hooked bail + lever to its neighbour's; flip the handle to pull the two modules tight against the bumper strip, flip to release. This ties the three boxes into one rigid beam: the bed platform already couples the Panel A + B tops, and a low latch completes a top-and-bottom couple that kills sway and rattle. It stays fully modular — the latches are hand-released, so any panel still lifts out in seconds.* ***Honest scope:*** *this buys rigidity and quiet, not a higher weight rating — the cube-framed boxes are already strong (5× margin at the feet). Don't rely on the latches to carry more load; their job is to stop relative motion between modules.* ***From the drawing:*** *SEAM DRAW-LATCHES — clamp the 3 lift-out modules into one rigid beam (Component 5) 4 latches total: one each side of BOTH the A/B and B/C seams, low on the bottom-rail band (~1.75" up, hand-reach). Alignment pins LOCATE the modules; the latches PULL them tight against the bumper strip. Hand-released — panels still lift out in seconds. (front seats upper-left, tailgate lower-right; the bed platform already ties the Panel A + B tops) base plate on one leg, hooked bail + lever on the other;.*

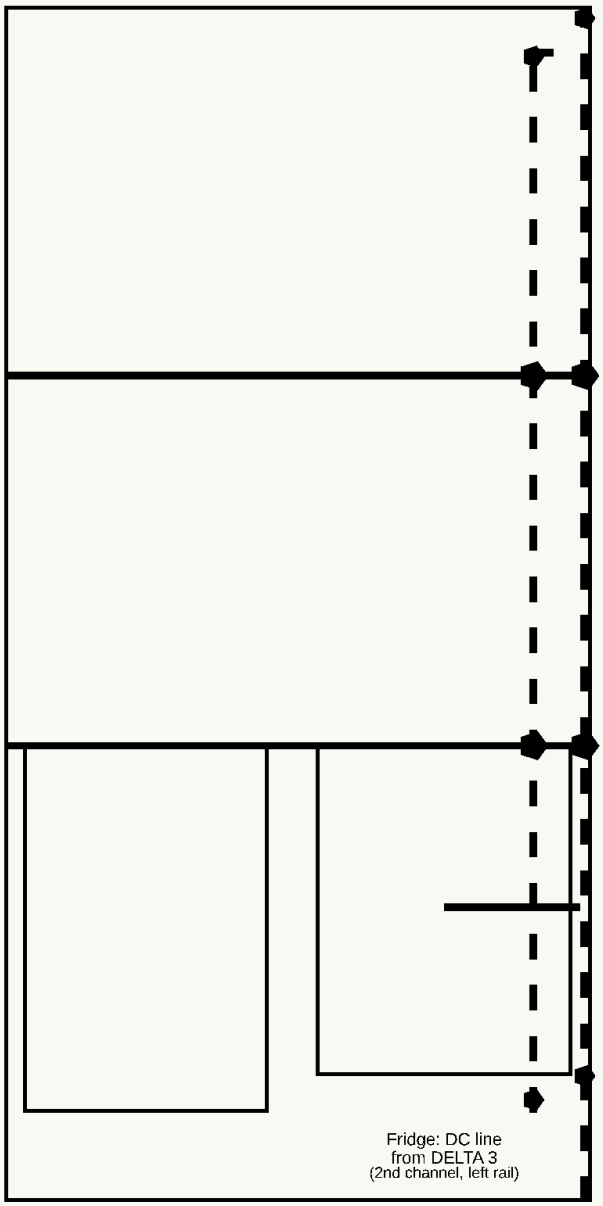
### *Component 6: Cord Runs — Cooktop, Power Strip 1, Fridge DC & DELTA 3 Charging*




***Parts needed:***

- *16 AWG extension cord (household AC-rated), short (cooktop → REAR outlet, + slack loop for the slide)*
- *16 AWG extension cord (household AC-rated), short (Power strip 1 → REAR outlet beside the pantry)*
- *16 AWG DC cord (automotive/solar SAE-rated), ~8ft (fridge line — DELTA 3 in Panel A back to the fridge in Panel C)*
- *16 AWG cord (household AC-rated), ~3-4ft (DELTA 3 AC charging line — front console to Panel A)*
- *Cable grommets — 7, 1" diameter (1 cooktop line, 1 Power strip 1 line, 3 fridge DC line, 2 DELTA 3 charging line)*
- *SAE quick-disconnect connectors — 2 (DC-rated, for the fridge DC line's 2 seams — the ONLY line that still crosses seams; both AC tailgate loads plug into the rear outlet)*
- *Fused (15A) car-plug → SAE adapter (P1, DELTA 3 end) + SAE → female 12V socket pigtail (P7, fridge end)*
- *18AWG red/black hookup wire ~10 ft, crimp assortment (#10 heat-shrink rings + 0.25" female spades), 2× WAGO 221-413 lever nuts*
- *Per-module surface wire raceway (AC + DC separated) + screw-mount cord clips — 1 set*
- *2-way AC outlet tap — 1 (REAR outlet: Power strip 1 + the cooktop share it — see Section 5)*

**1** *cooktop's short cord + 1 grommet + slack loop (parts list above)*



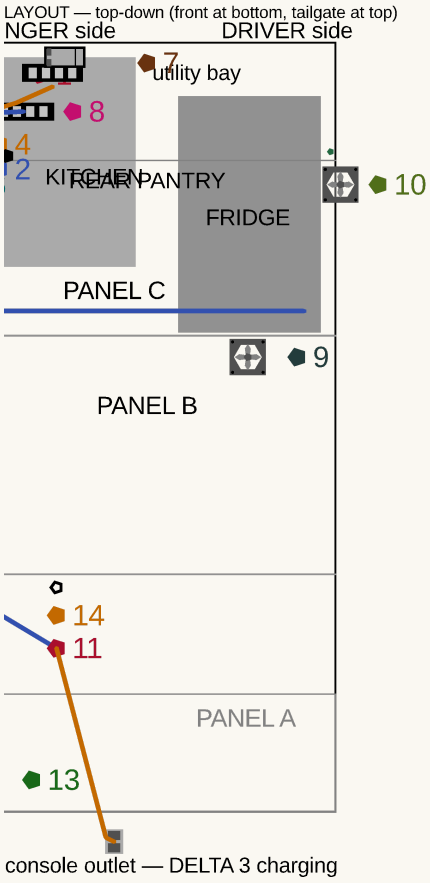

Power channels: cooktop (right rail) + fridge DC line (left rail)

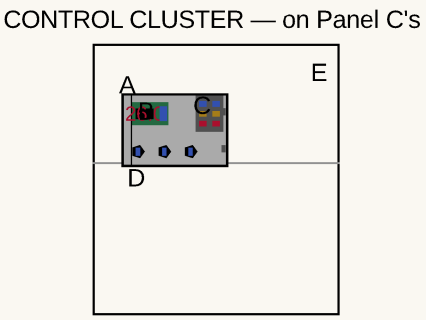
cooktop: kitchen unit -> Panel C -> B -> A -> console outlet

fridge: DELTA 3 (Panel A) -> B -> C -> fridge DC input

(large circles = quick-disconnect at each panel seam)

*Drill the single grommet at the kitchen unit's corner post and run the cooktop's short cord to the REAR AC outlet in the back passenger area (a few feet — no seams, no inline connectors). Coil a slack loop sized to the kitchen's full ~70" slide extension near the stowed position so the cooktop keeps power at the cook position. Power strip 2 rides ON the slide-out unit. The outlet is at **22.5" up** (i.e. just above the deck plane), ~10" forward of the liftgate scuff on the passenger quarter trim, socket center 10" in from the sidewall (MEASURED V9b) — position the grommet to line up with it.*

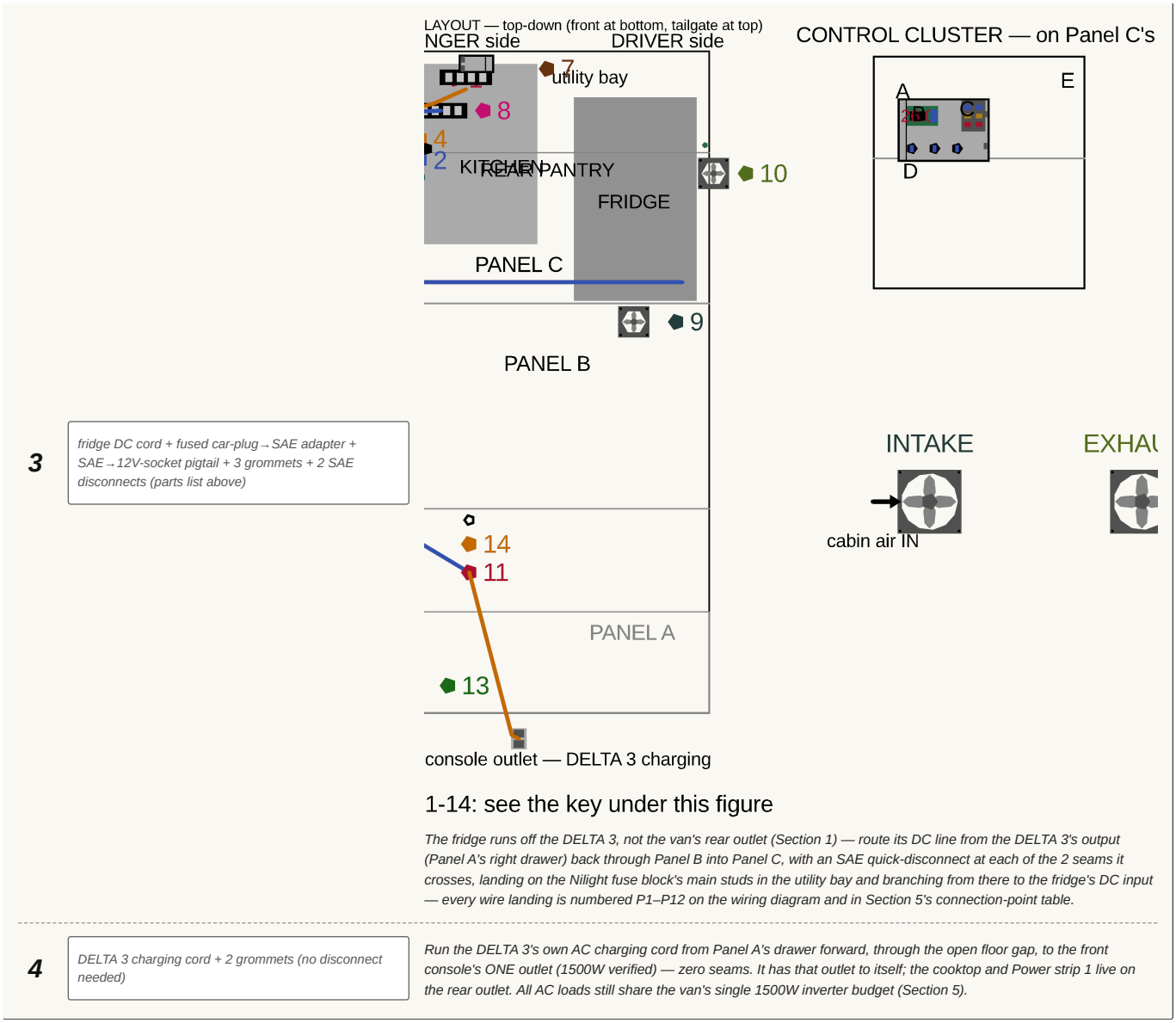



**2** *Power strip 1's short cord + 1 grommet + the rear outlet's 2-way tap (parts list above)*

1-14: see the key under this figure

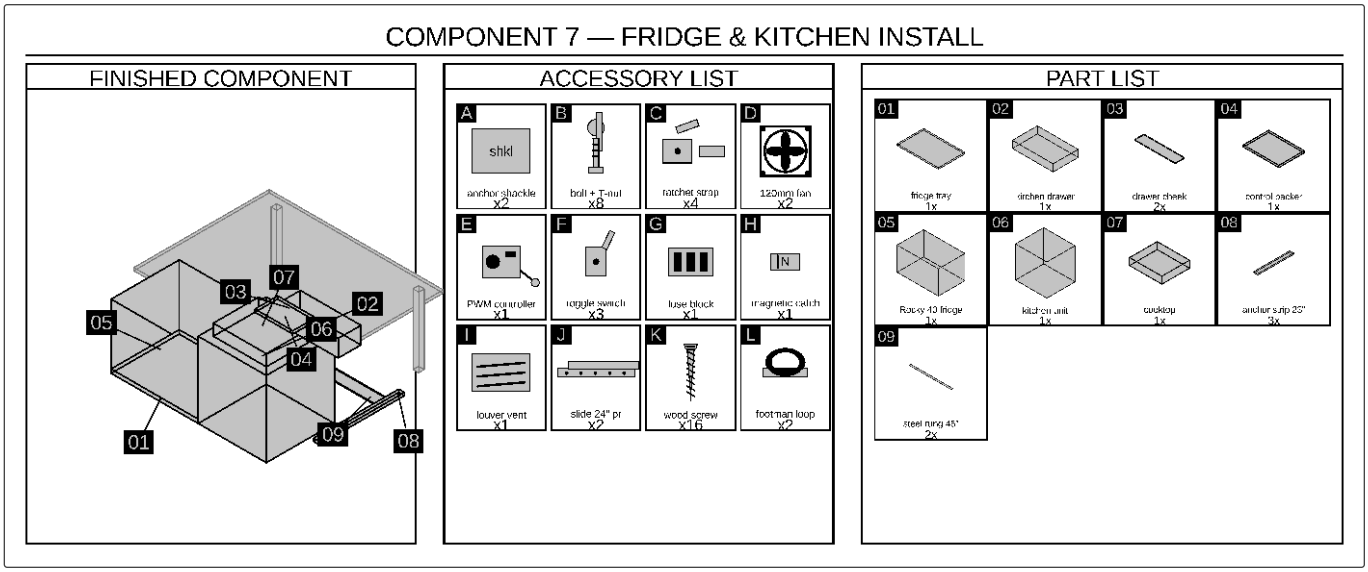
*Power strip 1 mounts on the rear pantry's deck edge in the open bay (Component 1) — directly beside the REAR AC outlet. One grommet at the deck edge, one short cord to the outlet's 2-way tap (shared with the cooktop), ~1 ft of slack so lifting the pantry's drawer units never needs an unplug.*




*The fridge runs off the DELTA 3, not the van's rear outlet (Section 1) — route its DC line from the DELTA 3's output (Panel A's right drawer) back through Panel B into Panel C, with an SAE quick-disconnect at each of the 2 seams it crosses, landing on the Nilight fuse block's main studs in the utility bay and branching from there to the fridge's DC input — every wire landing is numbered P1–P12 on the wiring diagram and in Section 5's connection-point table.*

| | | |
|---|---|---|
| **3** | *fridge DC cord + fused car-plug→SAE adapter + SAE→12V-socket pigtail + 3 grommets + 2 SAE disconnects (parts list above)* | |
| **4** | *DELTA 3 charging cord + 2 grommets (no disconnect needed)* | *Run the DELTA 3's own AC charging cord from Panel A's drawer forward, through the open floor gap, to the front console's ONE outlet (1500W verified) — zero seams. It has that outlet to itself; the cooktop and Power strip 1 live on the rear outlet. All AC loads still share the van's single 1500W inverter budget (Section 5).* |

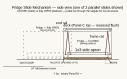
### *Component 7: Fridge & Kitchen Install*



***Parts needed:***

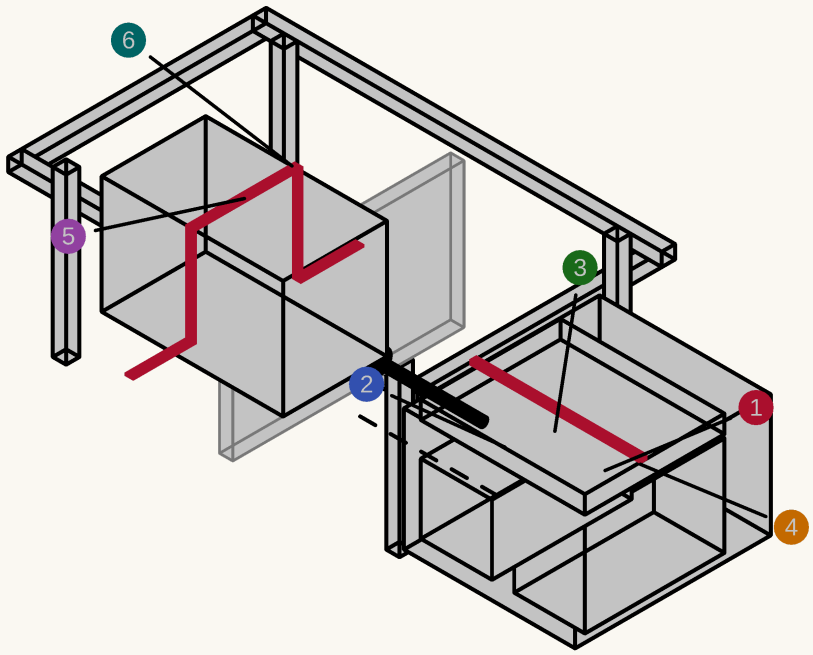
- *BougeRV Rocky 40 (41QT dual-zone) + fridge tray (1/2" ply, 17⅞" x 28¾", + two 2½" aprons glued and screwed to its long edges)*
- *Heavy-duty 24" full-extension drawer slide pair — VADANIA #D76 locking, 379 lb (purchased)*
- *JAGAHAHA slide-out camp kitchen + COOKTRON induction cooktop*
- *120mm 12V fan + PWM temperature controller w/ NTC probe*
- *12V surge protector/fused distribution block, 3 illuminated toggle switches, ¾" backer board (no enclosure)*
- *Anchor board: three ¾" ply strips tied by two 3" × 3/16" steel rungs, on ⅛" neoprene (Section 8, Anchor Board sheet)*
- *2 forged ¼-20 eye bolts + 2 short ¼" chains + 4 quick links (the risers to seat-latch hoops H1/H2) + a 3rd eye bolt, cut short, in the kitchen riser + 1 stainless footman loop*
- *2 Erickson 400lb-WLL ratchet straps (the kitchen unit, in its notches) + up to 3 optional backups to the striker loops*
- *1 low-profile cam strap + 2 stainless footman loops (fridge hold-down, through its 2 end handles)*
- *3 screw-mount cable clips (fridge DC line slack, clipped to the slide's fixed outer rail)*
- *Utility bay fit-out — 2 low-profile bins + hook-and-loop tape (no door: the bay between the fridge and kitchen stays open)*
- *Kitchen drawer — box (1/2" ply, 18" x 26" x 4.5") + 2 hanging cheeks (1/2" ply, 26" x 5.45") + 3 ¾" spreaders (19" x 2") + 24" full-extension slide pair (100lb) + magnetic catch*

| | | |
|---|---|---|
| **1** | *no parts — inspection only* | ***The October 2026 van survey sets this component**: the folded-seat surface is solid to 24¾" from the hatch; from 26" forward the side trim closes the van to 42"; and the 3rd-row seats' four latch hoops sit in the headrest trough 28" from the hatch, their bars 1⅞" below the loaded seat surface, at 4¾", 24¾", 31¼" and 41¾" from* |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | Panel C's driver line. Those hoops are the anchors. | | | | | |
| | **2** | anchor board (3 strips + 2 steel rungs) + neoprene + 3 eye bolts, chains and quick links + 1 footman loop + 2 ratchet straps | Bench-build the anchor board (Anchor Board sheet): three one-piece plywood strips, two steel rungs set flush into grooves in their undersides and screwed up into each strip, T-nuts for the two risers, an eye-bolt hole in each riser's front end, footman loops for the kitchen's rear strap, and neoprene cut to each strip. In the van: neoprene, then the board, back ends 3" from the hatch; wrap H1's chain around its bar, close it with a quick link and lay it forward; bolt the fridge tray assembly down; wrap the second chain around H2, quick-link both chains to their eye bolts and tighten the eye bolts' nuts. Re-check after the first drive. | **3** | fridge + 1/2" ply tray (edge frame) + 24" 200lb slide pair |  | Mount the Rocky 40 on its plywood tray hanging BETWEEN the slide pair — rails vertical against the tray's 1x3 side aprons (side-mount, nothing under the tray); each fixed rail's steel riser angle is already bolted to the anchor board (step 2 — the board raises the rails ~0.85"; screw the moving members to the aprons lower in their overlap so the tray keeps its 0.5" hang), against Panel C's driver-side rear corner leg (1.5" in from the edge), pulling straight out the open tailgate between the corner legs. Face the compressor/battery end toward the tailgate (B240 swaps without a full slide-out), re-hinge the reversible lid to open toward the van wall, and let the unit stand 12 hours before first power-up (manual). Screw 2 D-rings into the tray's side apron level with the fridge's 2 end handles, and hook a low-profile cam strap over the top from handle to handle — snug enough to stop the fridge lifting off the tray, loose enough that it doesn't bind the lid at full slide extension (test this before loading the fridge). Connect its DC input to the fridge line routed in from the DELTA 3 (Component 6, step 3) — leave enough slack for | **4** | 2 fans + controller/NTC probe + control cluster on its backer board | | Bolt the intake fan over the front wall's fan hole (Component 4 step 3 — blowing IN) and snap the three 3" soffit vents into the wall's vent holes below it (cool floor-level air in), install the exhaust fan + NTC probe (NTC just INSIDE the bay at the fridge's kitchen-facing wall, in the hot exhaust airflow, feeding the controller — it blows the warm air INTO the open utility bay and straight out toward the tailgate), and the control panel enclosure (switches + surge protector) at the back of the open bay, screwed to a backer board (the 3/4" sheet's offcut) hung from the deck underside — every switch reachable by just reaching into the bay. Wire everything per the schematic's numbered connection points (P1–P12 — the exact terminal, connector, and crimp for every wire is in |

| | | |
|---|---|---|
| | *the slide's full travel, and clip that slack loop to the FIXED outer rail (Component 6, step 5) so it can't get pinched when the tray slides home.* | *Section 5's connection-point table). The CO monitor and fire extinguisher are owner-placed — position them yourself once the build is in the van (the CO monitor stays battery-powered and unwired either way).* |

*### Component 8: EcoFlow Stowage (DELTA 3 + WAVE 3) ![Component 8 — finished component, accessory list & part list](renders/steps/comp-8-header.png) **Parts needed:** - EcoFlow DELTA 3 Plus + Smart Extra Battery, EcoFlow WAVE 3 (already owned, not priced) - WAVE 3 hose/cord storage hook - WAVE 3 hold-down: 2 screw-eye D-rings + 1 low-profile cam strap (Component 2's left bay) - Non-slip mat (camp-use point of use — Panel C's deck or the front seat, not the storage bay) - EcoFlow WAVE Series Car Vent Kit (no-tent sleeping configuration only) - FOUND STORAGE: DELTA 3 drawer top tray (shallow lift-out bin) + 2 utility-bay bins (velcroed — Component 7 step 5) **Battery placement — low and counterbalanced.** Heavy batteries want to sit **low and centered**, and this design already gets both most of the way there: the DELTA 3 stack (~48 lb) rides in Panel A's floor-level under-deck drawer — about as **low** as it can go. On the **lateral** axis it looks off to one side (passenger), but the ~34 lb WAVE 3 sits in the **opposite** (driver) bay of the same Panel A, so the two heaviest movable items counterbalance each other — the whole build's lateral center of mass ends up only **~1.1" off centerline** (Appendix F), which is trivial on a ~4,500 lb van. So the "get it centered" concern is effectively already answered. Don't use a longer drawer to slide the DELTA to the true centerline: a 48 lb load cantilevered at the end of a long slide is tippy when open and hard on the hardware, it fouls the bed-platform cantilever near Panel B, and it wouldn't free clean storage (the drawer sweeps its own travel path). If you still want to nudge it inboard for free, shift Panel A's center divider a couple inches toward the driver side to re-center the DELTA bay. See **Appendix F** for the full weight-distribution analysis.*




**1**

*uses Panel A's right-drawer fixtures (Component 2, step 4)*

## DELTA 3 STACK (right drawer) + WAVE 3 (left bay)

Panel A, exploded — Components 2 & 8

DELTA 3 Plus outboard (the WAVE 3 plugs into it), Extra Battery inboard; ~48 lb in a normal drawer — no E-track. The WAVE 3 sits on the raw bay floor, cam-strapped to 2 D-rings.

Found storage: a ~2.6" lift-out tray tops the DELTA stack, The WAVE 3's hoses live on the hook in Panel C's utility bay.

### Component / position

1. DELTA 3 Plus — 7.95"x15.7"x11.16", ~28 lb
   outboard (pull wall); cam-strapped over locating cleats
2. Smart Extra Battery — 8"x15.7", ~20 lb
   inboard — extra runtime only, grabbed less often
3. Lift-out tray — reclaims ~2.6" dead air
   cables, the DELTA's own cords, dongles
4. Cam strap + D-rings + 1" grommet
   strap over both units; WAVE 3 charge cable exits the grommet
5. WAVE 3 — 20.4"x11.7"x13.2", 33.7 lb
   raw bay, 0.35" clear — a boxed drawer would NOT fit
6. Cam strap + 2 D-rings (WAVE 3)
   fore-aft over the top, D-rings on the end bottom rails — no bounce, no walk-out

*Stow the DELTA 3 Plus outboard (pull wall — it's what the WAVE 3 plugs into) and the Smart Extra Battery inboard, side by side on the rubber mat (no locating cleats — owner, 30 Oct 2026), cam strap over both into D-rings ~7½" down from the box's top edge (#8 × ⅝" screws: the slides are on the other face of the ¾" wall). Turn the turn button down over the pull wall before the van moves (Power Drawer Tie-Down sheet). Connect the fridge DC line (via its fused car-plug→SAE adapter — connection point P1) and the AC charging cord (both routed in Component 6) to the DELTA 3 Plus's 12V car-power outlet and AC input port — it powers the fridge/fans continuously and gets charged from the front console while driving (Section 1).*

---

**2**

*uses Panel A's left-bay D-rings + cam strap (Component 2, step 3) + storage hook*

*Slide the WAVE 3 into Panel A's left bay, straight onto the bay floor — no box, no slide hardware, no glide strips, just reach in through the driver's side door — and cinch the cam strap fore-aft over its top between the 2 D-rings on the end bottom rails. The strap is not optional: the unit has ~2.8" of headroom to bounce into and nothing else stops it walking out the bay's open driver-side face on a hard turn (same pattern as the DELTA 3's strap). Its hoses don't stay with it — mount the hose/cord hook in the open utility bay (screwed up into the deck underside at the bay's kitchen side; bundle the hoses with their own strap so nothing swings out of the open bay) for when the unit is carried off to run at camp.* ***Found storage*** *(DELTA 3/WAVE 3 detail render, side sections): the DELTA 3 stack is only 11.16" tall in a 13.75" box — 13.25" clear over the 1/2" bottom — so drop a shallow lift-out tray, **1.75" or less**, in the ~2.1" of dead air on top for cables/cords (the cam strap runs under the tray). It doesn't touch the frame or the units' function. (No shelf over the WAVE 3 — the headroom there is left to the strap.)*

---

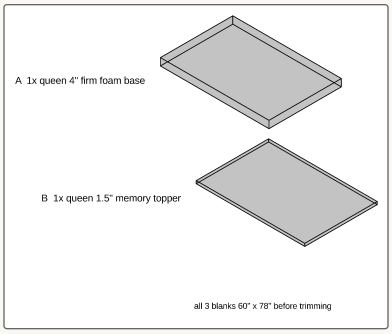
**3**

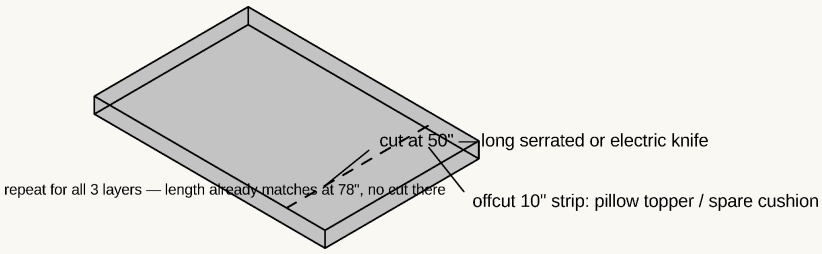
tent config: non-slip mat only · no-tent config: non-slip mat + WAVE Car Vent Kit

***Tent*** *(tailgate open): carry the WAVE 3 from Panel A's bay to Panel C's deck, just forward of the rear-pantry cluster (which occupies the tailgate-most 14"), on the non-slip mat, and route its hoses past the pantry and through the open tailgate gap — no window seal needed.* ***No tent*** *(tailgate closed): carry it instead to the front passenger seat, on the same mat, and install the Car Vent Kit over a cracked front window for the hoses.*

*### Component 9: Mattress ![Component 9 — finished component, accessory list & part list](renders/steps/comp-9-header.png) **Primary path — BUY, no build:** the HEST Dually Long (78" x 50" x 4") drops onto the platform as-is: solid foam, no inflation, washable waterproof cover included. Heads go at the rear-pantry end. Done. **Budget fallback — DIY 2-layer build** (the steps below): same 50" x 78" footprint, ~$185–255 all-in, 5.5" thick (costs ~1.5" of the headroom the HEST keeps). **Fallback parts needed:** - Queen 4" firm high-density foam base + queen 1"-2" memory foam topper - Spray adhesive (or large fabric strips) - Waterproof mattress cover*

**1**

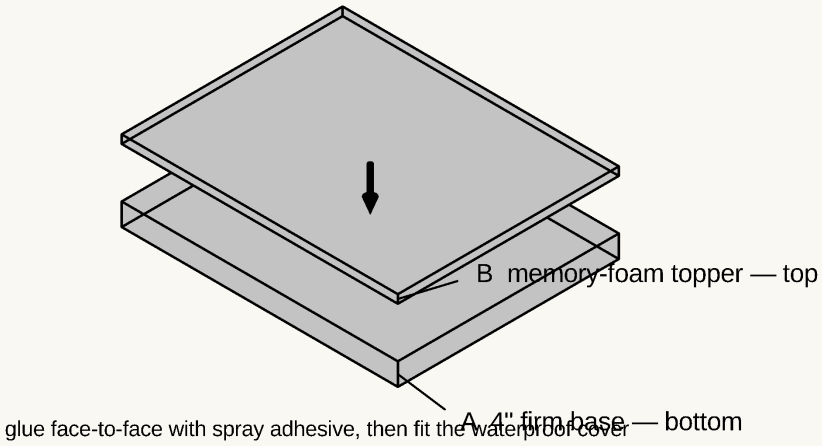



*FALLBACK ONLY (skip if you bought the HEST). Buy both foam layers at queen size (60" x 80") and trim to 50" wide x 78" long with a long serrated or electric knife; the offcut strips work as a pillow topper or spare cushion.*

**2**

the 2 trimmed layers from step 1
+ spray adhesive (or wide fabric strips)
+ waterproof cover, fitted after gluing




*FALLBACK ONLY. Stack the firm base (A) with the memory-foam topper (B) on top; bond face-to-face with spray adhesive, fit the waterproof cover, and lay it on the platform.*

*### Component 10: Final Assembly, Curtain & Test Fit ![Component 10 — finished component, accessory list & part list](renders/steps/comp-10-header.png) **Parts needed:** - Tension rod + blackout curtain fabric - Wood sealant/varnish, sandpaper/misc finishing supplies **Steps:** 1. Sand and apply 2 coats of wood sealant/varnish to all plywood surfaces (especially edges) to prevent moisture damage. 2. Install the curtain rod (tension rod + blackout fabric) between the front seats and the platform area for privacy. 3. Test fit everything: confirm the cooktop's and Power strip 1's cords reach the REAR outlet, the DELTA 3's charging cord reaches the front console outlet, confirm the fridge's DC line reaches the DELTA 3 in Panel A with its disconnects seated at both seams, and confirm the tailgate closes with everything loaded. 4. Do a full lift-out/reinstall dry run on Panel A, Panel B, and Panel C — gripping each by its exposed top rails, each should come free and drop back in without binding, rubbing, or forcing a bumper/pin out of place. For **Panel A and Panel B specifically**, also test lifting the one-piece bed platform off the box rails first (they have no top or lid of their own — the platform is the only thing capping them), then confirm each panel's drawers pull smoothly once it's clear. For **the rear pantry specifically**, unhook the cam strap and confirm the tote and the microwave lift off Panel C cleanly, and that the strap re-cinches square over both. *No diagram — finishing and test steps, not new construction.* --- ## 7. Induction Cooktop & Power Strip Design ### Recommended cooktop: [COOKTRON Portable Induction Cooktop 2 Burner w/ Removable Iron Cast Griddle Pan] (https://www.amazon.com/dp/B09MCR1SDT) - **Real measurements confirmed from the JAGAHAHA's own listing photos**: the stove tray is **23"L x 15.7"W, with 5.7" of clearance** when pulled out. - The **JAGAHAHA kitchen unit is built for a 2-burner stove** (per its own listing) — a single-burner cooktop would leave half its designed stove space unused. - **Confirmed exact product** (real listing photos with dimension callouts): **18.1"L x 9.1"W x 4.3"H** (briefcase-style unit with a fold-down handle) — fits the tray with real margin: ~4.9" of length, ~6.6" of width, ~1.4" of height to spare. Two 6.3"-diameter induction zones, **1800W nameplate, which is a COMBINED cap across both zones — not 1800W per burner** (the manual, model FS-RC121G, rates each side at **1100W max, at P9**, and caps the pair at 1800W together — see the bench test below), plus an included removable non-stick cast-iron griddle pan that lays across both zones — a direct match for "two burner or griddle" in one unit. - **Price: $189.99** (confirmed current listing price). - Requires magnetic (induction-compatible) cookware — test with a magnet before buying pots/pans if you don't already own compatible ones (the included griddle pan is already induction-ready). See the cookware recommendation below. **Power check — confirmed:** the front console AC outlet is rated at **1500W (verified against the real van, Appendix A)**, and a second AC outlet exists in the **back passenger area** (the cooktop + Power strip 1 plug in there — Section 5). Both outlets are CONFIRMED to share the van's single 1500W inverter — treat 1500W as the TOTAL AC budget across everything at once. ### Bench test — measured draw*

*(TESTED, not estimated) The COOKTRON was run **off the EcoFlow DELTA 3 Plus on the bench, base unit only (no Smart Extra Battery), with the van not involved at all**. That is a valid stand-in for the real install: the DELTA 3's inverter and the van's are **both 1500W**, so every number below transfers straight to the rear outlet. Method: one burner stepped through P1–P9 watching the power station's output display, then the second burner brought in on top. | Setting | Measured draw | Verdict | |---|---|---| | 1 burner, P1 | ~380W | Ramp from P1 to P9 was clean — no faults, no voltage instability | | 1 burner, P5–P6 | ~650–700W | Stable continuous draw — the setting for long or unattended simmering | | 1 burner, P9 (max) | ~934–1000W, mostly under 950W | Fine on its own; ~55–60 min of runtime from 74% charge | | **2 burners, P6 + P6** | **~1460W combined, sustained without fault** | **The confirmed maximum for continuous dual-burner cooking** | | 2 burners, one pushed to P9 | ~1600W combined | Handled briefly, but at/over the inverter ceiling — NOT safe as a sustained setting | | Grill button (middle button, pan icon) | Runs BOTH zones synchronously at one temperature, defaulting to **C5 (410°F)** — the manual's own description | **Tripped the overload (EO/EU) in testing — leave it alone in the van** | **What the manual says, and why the cooktop is NOT the problem (model FS-RC121G).** The manual's own power table rates **each side identically**, and the ladder is modest: | P level | P1 | P2 | P3 | P4 | P5 | P6 | P7 | P8 | P9 | |---|---|---|---|---|---|---|---|---|---| | Watts, one side working alone | 400 (intermittent) | 400 | 500 | 600 | 700 | 800 | 900 | 1000 | **1100** | So **a single burner tops out at 1100W** — the 1800W on the box is the pair's combined ceiling, nothing more. Against that, the bench numbers are not a shortfall at all: ~934–1000W measured at P9 against 1100W rated, ~650–700W at P5–P6 against 700/800W rated, ~1460W for P6+P6 against 1600W rated. Every reading sits **~8–10% under nominal**, consistently, which is what a power station's own display and a duty-cycled induction load look like — not a defect, not bad pans, and not a wiring fault. **This cooktop is simply a 1100W-per-burner appliance**, well under a household burner's 2000–3000W, and slow boils are the honest consequence rather than a symptom. **The derate rule is the trap, and it is in the manual (item 3).** With both stoves running, total power *"should not exceed 1800W… once total power [goes] over 1800W after setting of one stove, the power of another one will decrease accordingly."* **Turning one burner up silently turns the other one down.** Also worth knowing: entering power mode **defaults each stove to P7 (900W)** — so simply switching both on lands you at 1800W combined without touching a knob, which is over the van's inverter before you have cooked anything. **Cook accordingly:** **boil on ONE zone with the other OFF** (that is the full 1100W and the fastest this unit boils), lid on, and only the water you need. Expect the griddle — which lays across BOTH zones — to share the 1800W rather than double it. **Working limits that come out of this:** - **Both burners at P6 is the tested ceiling for continuous cooking** — ~1460W measured, on the DELTA 3 or the van's rear outlet, same 1500W either way. **Read the margin honestly:** P6+P6 is *nominally* 1600W, over the inverter, and it only fit because real draw runs ~10% under nominal. For everyday cooking use **P6 + P5** (~1500W nominal, ~1360W real) and keep the headroom. - Pushing either burner past P6 while both are running walks into the inverter's limit; testing faulted (EO/EU) there. One burner alone can go to P9 freely — and that single-zone P9 is also the **fastest boil the unit can give you**, since the 1800W rating is shared, not per-burner (above). - **Never use the sync/boost button in the van.** It takes combined wattage out of your hands — that button is what caused the overload trip. - **The microwave is never run with the cooktop — which closes the only hard conflict.** The BLACK+DECKER EM720CPN-P is 700W *output*; a microwave of this class draws roughly **1200W at the wall (UNVERIFIED — read the nameplate)**. Because **one 1500W inverter feeds BOTH van outlets** (Section 5), microwave + cooktop at the everyday P6+P5 setting would have been ~1200 + ~1360 = **~2560W against 1500W available**. The owner has confirmed the two are **never used at the same time**, so no design change is needed: the microwave plugs into the rear outlet alongside the cooktop's 2-way tap, and the discipline is operational, not electrical. **The one case left to watch is not a cooking case.** The DELTA 3 AC-charges from the front console *while driving*, and that charge alone can pull the full 1500W — so **pause the charge before reheating at a rest stop**, or you are back over the inverter with the cooktop untouched. If that ever gets annoying, run the microwave off the **DELTA 3's own AC output** instead: a separate 1500W budget, at the cost of a cord from Panel A and ~60Wh a go (3 min at 1200W). Cheap simultaneity, if you want it. - **Runtime, DELTA 3 Plus base unit alone:** ~35–40 min at P6/P6 (~1460W), or ~55–60 min at single-burner P9 (~950W) from ~74% charge. Enough for a normal meal; a longer cook session wants the Smart Extra Battery in the loop (2048Wh combined). - **A useful side finding: the cooktop runs fine straight off the DELTA 3.** The routing in Section 5 feeds it from the van's rear outlet, but this test proves the fallback works — you can cook with the van's own inverter off or unavailable. - In the van, that ~1460W sits against the ONE 1500W inverter both outlets share, so dual-burner cooking means nothing else meaningful on AC at the same time, and the DELTA 3's own charging cord has to be capped in the app or unplugged (Section 5). **The COOKTRON stays** — the test confirmed the unit itself, so the plan keeps it rather than looking for a lower-draw replacement. ### Cookware: UMETRE 9-Piece Detachable-Handle Induction Set (PURCHASED July 2026) The detachable-handle design is what won: the whole set nests flat and ships in an **11" x 11" box that goes straight into the rear pantry's tote** (Component 1). Induction-compatible (verified against the COOKTRON), dishwasher-safe. Trade-off, noted honestly: it's a nonstick coating rather than bare stainless, so pack silicone/wood utensils and expect to re-buy pans in a few seasons of camp abuse. ### Kitchen unit: bought product, no slide-out tray to build The JAGAHAHA kitchen unit is a standalone product with its **own** slide-out mechanism and stove space already built in — there's no cooktop tray to build. Set the COOKTRON cooktop into that built-in space and run its cord through a grommet at the back of the unit, then a SHORT hop to the REAR outlet right beside the tailgate (Section 5 — no seam crossings; the front console outlet is the DELTA 3's alone). No plumbing, no water lines, no sink. ### Multiport AC power strips (2 total) | Location | Purpose | |---|---| | Power strip 1 — mounted on the rear pantry's deck edge (Component 1/6) | Phone/device charging, reading light, and the **Windmill Smart fan** — easy reach from the sleeping area, right by where your head/torso would be | | Power strip 2 — mounted ON the slide-out kitchen unit (travels to the cook position; cord has a slack loop for the slide) | Powers the induction cooktop and any other small kitchen appliance (kettle, blender) | **Shore power — the HDX 30 ft 16/3 reel.** This is the design's only path to an *external* AC source, and it is worth having: it charges the DELTA 3 stack at a campsite or in the driveway with the van switched off, which nothing else here could do. Two limits govern it, and both are about the cord, not the van: - ⚠️ **UNWIND THE REEL COMPLETELY BEFORE DRAWING REAL CURRENT.** A coiled cord reel cannot shed heat — the wraps insulate each other — and reels are commonly derated to roughly a third of their rating while still spooled. At the ~12.5A a DELTA 3 charge pulls, a coiled reel is a genuine fire risk. Pull all 30 ft off, every time, even if you only need six of them. - **16/3 is about 13A / ~1560W for the whole reel, all 4 outlets combined.** A DELTA 3 charging at its full 1500W is 12.5A — about 96% of the cord, with no margin left. So the same one-at-a-time discipline that governs the van's inverter applies here for a different reason: **DELTA 3 charging OR the microwave OR the cooktop, not two of them.** Voltage drop is fine (~3V over 30 ft at 12.5A, ~2.5%); it is the current rating that binds. **Each strip runs on its own short cord to the REAR outlet's 2-way tap (Section 5)**. They're not unlimited: a 6-8 outlet power strip is for convenient access to the inverter's one 1500W budget, not for adding capacity — don't run the induction cooktop and multiple other high-draw devices at the same time, and remember the DELTA 3's charging cord (front console) draws from the SAME inverter pool. **Recommended power strip:** a compact 6-outlet strip with individual switches (e.g., a basic surge-protected strip, ~$15-20 each) — the individual switches let you kill power to specific outlets without unplugging everything, useful for lights left on overnight. All cooktop and power strip hardware costs are already folded into the master BOM in Section 4. ## 8. Modular Lift-Out Design & Weight Panel A, Panel B, and Panel C each rest unbolted on the van floor — nothing is fastened down, and no module shares a frame rail with its neighbor. Pull the bumper strips free, lift any of the three out by their exposed 2x2 top rails (no tops or skirts in the way), and the sleeping deck is gone for normal minivan use; drop them back in and the alignment pins register everything in the same spot every time. The rear pantry is not one of these three lift-out modules — its tote and the microwave just sit on Panel C's deck under one strap (unhook it, no tools, Component 1) — lift them out first and its deck is clear. **The fridge and kitchen unit are the one deliberate exception to "nothing is fastened down"** — see below. ### Securing heavy components (fridge + kitchen unit) **The "rests unbolted, lifts out" design that works for the sleeping panels is not safe for the fridge (up to ~90 lb loaded) or the kitchen unit (45 lb).** In a hard stop they become projectiles, so both are tied to the van through the anchor board, without drilling. **The Sienna's factory cargo hooks are not adequate for this**: Toyota rates them for a cargo net. The build uses parts of the van that are built to hold seats in a crash instead. **What carries*

*the load: the 3rd-row seats' latch hoops.** With the 3rd row folded into its well, the four steel hoops its seat backs latch to sit in the headrest trough, 28" from the hatch, their ⅝" bars 1⅞" below the folded-seat surface with weight on it (2⅜" bare). They are anchored to the floor pan to hold the seats in a crash. **The anchor board** (overview at the start of this plan; shop drawing on the Anchor Board sheet) gathers the fridge slides and the kitchen unit's straps onto one chassis: three one-piece plywood strips tied by two steel rungs set into their undersides, lying on neoprene on the folded-seat surface. **Three load paths:** 1. **Forward and lift — two chains to the hoops.** A forged ¼-20 shoulder eye bolt hangs down through the front end of each fridge riser, just ahead of its strip; a ¼" quick link and a short ¼" chain, wrapped once around the hoop's ⅝" bar and closed by a second quick link, take it forward and down to hoop H1 (driver riser) and H2 (kitchen riser). The run is only about 5", too short for any ratchet strap's mechanism, so the eye bolt's nut is the tensioner: turn it down to draw the chain tight, then jam a second nut on top. The load runs hoop → chain loop → quick link → chain → eye bolt → riser → four bolts → strip → rungs → the whole board. Wrap H1's chain on before the fridge tray goes in: the tray covers it. 2. **The kitchen unit** — a strap across its top in its front notches down to H2 and H4, and one in its rear notches down to the kitchen riser's eye bolt and a footman loop on the passenger strip. The notches stop it sliding along the straps. 3. **Lateral** — the strips' edges, the van walls and trim close on both sides, the neoprene's grip, and the straps' angles. **Optional backup:** ratchet straps from the board forward to the three 3rd-row striker loops on the 2nd-row floor. They pass under Panel C's front edge and Panel B's end rail through the 3" relief notches there. **The anchor board secures the tray to the van; it doesn't secure the fridge to the tray.** The fridge's own hold-down is the cam strap through its end handles to the footman loops on the tray's kitchen-side apron (Fridge Tray sheet). **Install and upkeep:** 1. **Bench-build the board** (Anchor Board sheet): strips, rungs, T-nuts, eye-bolt holes, loops, neoprene. 2. **In the van:** neoprene, then the board, back ends 3" from the hatch. Hook the H1 strap in and lay it forward. 3. **Bolt the fridge tray assembly down**, then chain both eye bolts to H1 and H2 and tighten them. 4. **Kitchen unit in**, both straps seated in its notches and tensioned. 5. **Re-tension every strap after the first drive**, then check at the start of each trip. 6. **To return the van to stock:** release the straps and lift everything out. No holes were made. **Approximate weight per module** (plywood + frame + legs only, rough estimates — excludes mattress and small hardware): | Module | Est. weight | |---|---| | Rear pantry — 27-gal tote + strap (lifts out in one piece) | ~7 lb empty | | Panel A (incl. divider + 1 drawer box/slide + corner braces + 2 bottom rails + WAVE 3 D-rings/strap, no top) | ~37 lb (pull the DELTA 3 drawer box off its slides first and it drops to ~15 lb) | | Panel B (bare cube frame — bottom rails on all 4 faces + corner braces; no top, no divider, no drawers) | ~15 lb | | Panel C (frame + front bottom rail + deck + front wall + kitchen drawer & cheeks + control cluster backer — the fridge TRAY + slides stay bolted to the anchor board (strapped to the van, no holes), and the fridge/kitchen units are tracked separately below) | ~61 lb — the heavy one (unclip the kitchen drawer box from its slides first: ~48 lb) | | BougeRV Rocky 40 fridge (empty weight, manual spec) | 40.6 lb / 18.4 kg (realistically 60-90 lb loaded — why it's on a 379 lb locking slide, not a lift-out module) | | JAGAHAHA kitchen unit (real product weight) | 45 lb | | EcoFlow DELTA 3 Plus + Smart Extra Battery (Panel A right drawer, included in that drawer's own weight above) | ~48 lb | | EcoFlow WAVE 3 (Panel A left bay, stored in place — it doesn't need removing for driving) | 33.7 lb | **Panel A (~37 lb) and Panel B (~15 lb) are comfortable one-person lifts; Panel C (~61 lb) is the outlier** — unclip its kitchen drawer box from the slides first (2 minutes, drops it to ~48 lb) or make it a two-person lift. A and B are well under the ~50 lb rule of thumb for a solo carry (weights in the table above), and Panel C gets under it once its kitchen drawer box is off. The rear pantry's tote (~7 lb empty) and the microwave (~22 lb) ride on top of Panel C when installed and lift out separately once the strap is unhooked. **Fit and quiet-in-transit checklist:** - Each panel is lifted by gripping its exposed 2x2 top rails (no tops or skirts in the way); a two-hand, centered grip keeps the lift balanced. - Bumper strips + alignment pins + seam draw-latches (hardware in Section 3, installed in Section 6, Component 5) go at the Panel A/B and Panel B/C seams. The bumpers stop the panels rubbing/rattling, the pins register them in the same spot each time, and the 4 over-center draw latches (both sides of each seam, low on the bottom-rail band) clamp the three modules into one rigid beam so they can't creep or rock relative to each other — with the bed platform tying the A/B tops, the low latches complete a top-and-bottom couple. All hand-released, so any panel still lifts out in seconds. - Do the full lift-out/reinstall dry run in Section 6, Component 10 before you consider the build done — that's the real test of whether the modular fit is tight enough to stay quiet but loose enough to lift easily (with a second person, given the weight note above). Test the well depth, fridge dimensions, and drawer weight against your actual van and materials before cutting — the numbers above are solid starting estimates but real-world results vary by trim year and exactly what plywood you source. --- ## 9. Seat Removal & Reinstall (Camper ↔ Passenger Mode) The 2nd-row seats come out entirely for camper mode (Section 1) and go back in for passenger use. The Sienna's 2nd-row seats carry SRS (airbag/occupancy) wiring, so pulling them without the right procedure leaves airbag warning lights on the dash — this build uses **SRS airbag emulators (0651 Round ×2 + 5522 "Square" ×2 — purchased July 2026; VERIFY the 5522 mates with the rectangular plug)** plugged into the vacated van-side floor connectors so the airbag system reads normally with the seats out. **The 30-minute battery wait is a safety step, not a convenience step** — the airbag computer holds backup electrical power in capacitors that must fully drain before any SRS connector is touched. Do not skip or shorten it in either direction of the transition. ### Transitioning to Camper Mode (Removing Seats) 1. **Turn off the vehicle** completely and open the rear trunk hatch. 2. Locate the 12V battery in the right rear trunk wall panel and **disconnect the negative terminal**. 3. **Wait exactly 30 minutes.** Do not skip this; the airbag computer holds backup electrical power in capacitors that must drain completely. 4. Unbolt the seats and **unplug the floor harnesses**. 5. Firmly click the **0651 Round** and **5522 "Square"** emulators (purchased July 2026 — verify the 5522 mates with the rectangular plug) into the van-side yellow floor plugs. Give them a light tug to verify they are locked. 6. Reconnect the negative 12V battery terminal and tighten it. Start the van; the dashboard lights will clear normally. ### Transitioning to Passenger Mode (Reinstalling Seats) 1. Turn off the vehicle and **disconnect the negative 12V battery terminal** again. 2. **Wait 30 minutes** to de-energize the SRS system. 3. Unplug the Ling Labs dongles and store them safely in your glovebox or center console for the next trip. 4. Bolt the captain's chairs back down securely and **plug the factory seat harnesses back into the floor**. 5. Reconnect the negative 12V battery terminal. ### Storage Tip When in camper mode, the vehicle-side plugs sit flat on the floor. Use a small piece of plastic wrap or a ziplock bag secured with a rubber band around the emulators to prevent dirt, dust, or spilled camping liquids from getting into the electrical contact points. --- ## Appendices — Working Sheets These references consolidate the plan into sheets you actually work from. Four are **also a live web page** (interactive check-off boxes, or a calculator; all printable); the weight budget is also a spreadsheet file: - **Shopping List** — https://claude.ai/code/artifact/24d09bf4-2f51-42d5-879d-003ab1c65326 - **Cut List** — https://claude.ai/code/artifact/fc8fa178-40f9-40dc-9ecf-f8dc181ad479 - **Build Sequence** — https://claude.ai/code/artifact/eff00cab-1e69-4a54-8eec-2d9346072e92 - **Leveling Block Calculator** — https://claude.ai/code/artifact/149333c6-8f02-47a2-915f-52d26d9059d9 - **Weight Budget** — spreadsheet in the repo as `weight_budget.csv` Appendices B and C consolidate Sections 4 and 3; Appendix B (build order), Appendix E (leveling calculator), and Appendix F (weight budget) are new here. **Appendix H is the one you hand to somebody else** — the Home Depot order and the cut-desk instructions, written out so an associate can work from it without reading the rest of this plan. ### Appendix A — Vehicle Measurements & Rear-Floor Survey (COMPLETE — measured Aug 1 2026) **DONE for the van — surveyed Aug 1 2026 (owner), 2nd row out and 3rd row folded as it sits in camper mode.** Every V and F line below carries a measured finding, and `params.scad` matches. The fridge/kitchen table further down is still open (those are bought products, measured on arrival). **Read the four findings that drive the design before building anything: V1 (interior length 93¾"), V3 (height 42"), V4 (vents 3.5" per side) and V7 — the walls do NOT flare out at the sleeping plane, which holds the bed platform to 49".** All of them feed directly into [`params.scad`](params.scad) — every diagram, cut list, and BOM total in this plan is drawn from the measured numbers. This appendix is **the record of what was measured and what is still open**, so any figure in the plan can be traced back to a tape reading. The `assert()` guards in `params.scad` force the design to adjust rather than silently overflow the van. #### The van (once the 2nd row is physically out) **Reading these drawings.** Same format as the rear-floor survey below: each measurement gets its own plan of the whole Sienna — **front at the left, tailgate at the right**, driver side along the bottom — with **only that measurement drawn in red** and tagged (V6a, V6b, ...). Those tags match the blank lines under each drawing one-for-one. Several of these are not plan dimensions at all — interior height, the gate opening, wall-to-wall width up*

*at platform height, floor slope — so those sections carry an **elevation or section inset**, and the plan just shows where it is taken. Fore-aft distances all work from the **closed hatch**, the same datum the survey uses. ##### V1 — Interior length — closed hatch to front seatbacks ![Van measurement V1 — interior length](renders/vanmeas-v1.png) - **Measure:** The interior length along the floor, from the closed hatch to the front seatbacks. - **Feeds:** `van_interior_length` — the number the whole panel train is sized against: Panel A + B + C fill it exactly, so if it comes back short, every panel shrinks. - **V1** — hatch to seatbacks: **93¾"** (MEASURED — owner, Aug 1 2026). `van_interior_length` = 93.75. Panel C absorbs the length (**35¾"**) — Panels A and B are asserted to stay the *same* length (they share one set of step diagrams and one cut-list entry), and their combined length *is* the bed platform's 58" side rails, so shaving either one would ripple through the lumber list for nothing. Panel C has ~7" of fore-aft slack around the 28.74" fridge. ##### V2 — Interior width — between the wheel wells ![Van measurement V2 — interior width](renders/vanmeas-v2.png) - **Measure:** The width at the floor, at its narrowest point between the two wheel wells — not at a wider spot fore or aft of them. - **Feeds:** `van_interior_width` — every 46"-wide panel and the floor envelope come off this. - **V2** — floor pinch, wheel well to wheel well: **49"** (MEASURED — owner, Aug 1 2026). The floor opens out to **54" forward of the wheel wells**, but the pinch is what gates the 46" boxes. `van_interior_width` = 49. ##### V3 — Interior height — floor to headliner ![Van measurement V3 — interior height](renders/vanmeas-v3.png) - **Measure:** Cargo floor up to the headliner, taken at the sleeping run rather than at the tailgate where the roof drops away. - **Feeds:** `van_interior_height` — caps the whole stack: deck + mattress + sitting headroom. - **V3** — floor to headliner: **42"** mid-van (MEASURED — owner, Aug 1 2026), and only **37" at the gate** where the roof drops (that is the V6 gate height). `van_interior_height` = 42. Sitting headroom over the mattress is **19.5"**, and the rear-pantry cluster's roof clearance is ~6.7" — a third drawer tier would foul the headliner by 1.7", so the 2×2 cluster is the ceiling. ##### V4 — Vent intrusion width, each side, at floor level ![Van measurement V4 — vent intrusion](renders/vanmeas-v4.png) - **Measure:** How far the floor vent / trim kick eats into the usable width down at floor level. One measurement per side. The deck may overhang it, but a **leg may not stand on it**. - **Feeds:** `vent_intrusion_width` — drives `leg_inset`. - **V4a** — driver side: **3.5"** (MEASURED — owner, Aug 1 2026) - **V4b** — passenger side: **3.5"** (MEASURED — owner, Aug 1 2026) - `vent_intrusion_width` = 3.5, so `leg_inset` is 3.5". Sketched on the sheet as ~3¼" over a ~15" fore-aft run; **3.5" is the number to build to.** Floor-level usable width is 42" (49 − 2×3.5), and the leg line at 39" still sits inside it. ##### V5 — Hatch curvature clearance ![Van measurement V5 — hatch curvature](renders/vanmeas-v5.png) - **Measure:** How far forward of the closed hatch you have to stop building, because the glass and trim curve inward above the floor. - **Feeds:** `hatch_curvature_clearance` — it is why the panels total 94", not 96". - **V5** — hatch to the usable boundary: **0"** (MEASURED — owner, Aug 1 2026) — *"the build isn't high enough to hit curvature."* Nothing has to be reserved at the tailgate end: `hatch_curvature_clearance` = 0. With no reserve for the rear-pantry drawer units (14.3" deep) to hang into, **`pantry_len` = 14.5"** and the deck carries the whole drawer (costs ½" of sleeping run). ##### V6 — The gate (tailgate) opening — width and height ![Van measurement V6 — gate opening](renders/vanmeas-v6.png) - **Measure:** Standing at the open hatch: the opening's width at its **narrowest** point, and its height **where the rounded corners start cutting in** — not the widest or tallest point of either. - **Feeds:** `gate_opening_width` / `gate_opening_height` — these gate what can physically pass into the van at all. - **V6a** — narrowest width: **50"** (MEASURED — owner, Aug 1 2026) — the 46" boxes have **2" of clearance per side**. - **V6b** — clear height: **37"** at the center of the opening (MEASURED — owner, Aug 1 2026), but only **20.5" out where the corner radius starts cutting in**. `gate_opening_width` = 50, `gate_opening_height` = 37. **Carry the 18.5"-tall boxes through the middle of the gate** — a module walked in flat against one side will foul the rounded corner well before it runs out of height in the center. ##### V7 — Wall-to-wall width up at platform height ![Van measurement V7 — wall-to-wall at platform height](renders/vanmeas-v7.png) - **Measure:** Interior wall-to-wall width at **~18.5"** and **~22.5"** above the floor, taken at several points along the sleeping run. - **Feeds:** Gates `bed_frame_width`. - **V7a** — at ~18.5" above the floor: **50"** (MEASURED — owner, Aug 1 2026) - **V7b** — at ~22.5" above the floor: **49.5"** (MEASURED — owner, Aug 1 2026) - ⚠️ **The walls barely flare over the wheel wells: 50" is the usable width** at the sleeping plane (owner). So the cantilevered platform is **49"** (`bed_frame_width`), i.e. **1½" of overhang per side**, keeping ~½" of insertion clearance per side. The slats are **42"**. Param `van_platform_width` = 50 records the limit and asserts against it. - ✅ **Dry fit: the 50" HEST goes in, with a little play to spare.** It is still the widest thing in the van — the usable width is also 50", and the frame under it is 49", so the foam overhangs its own frame by ~½" per side. The walls **narrow** with height here (50" at 18.5", 49.5" at 22.5"), so what little pinch there is lands on the mattress's top outer edges rather than its base, and solid foam absorbs it (no coils to distort). No trimming, chamfering or replacement needed — build the platform at 49" as specified. ##### V8 — The side sliding door opening ![Van measurement V8 — side door opening](renders/vanmeas-v8.png) - **Measure:** The door opening's fore-aft width and its height — and the one that actually matters, the **usable clear** width and height at the door's real stopping point, not the rough opening. - **Feeds:** `side_door_opening_width` / `side_door_opening_height` — decides whether the DELTA 3 stack stays in Panel A or has to move (see the reachability note in Section 1). - **V8a** — fore-aft width: **35"** (MEASURED — owner, Aug 2026). - **V8b** — height: **45"** (MEASURED — owner, Aug 2026). - **V8c** — usable clear at the door's actual stop: **29"** (MEASURED — owner, Aug 1 2026) — **this is the number that governs.** The sliding door parks 6" short of the aperture's forward edge, so 6" of the 35" is never open. `side_door_opening_width` = 35 / `_height` = 45 record the aperture; `side_door_clear_width` = 29 is what the reach check runs off. With the door's fore-aft position still a placeholder, the check reports Panel A reachable and **Panel B blocked outright** — fine, because Panel B is a top-loaded bay with no drawers, and it is why its bed top is two lift-out halves (Component 2). Re-run the check once `side_door_y0` is measured. ##### V9 — The two AC outlets ![Van measurement V9 — AC outlets](renders/vanmeas-v9.png) - **Measure:** **VERIFIED: 2 outlets total, both fed by the one 1500W inverter**, so they share a single power budget. Front: center console. Rear: passenger-side rear quarter trim — **22.5" above the cargo floor, ~10" forward of the liftgate scuff plate, socket center 10" laterally in from the right sidewall** (MEASURED — owner, Aug 1 2026), directly above the 12V battery access panel and below the rear cup holders. - **Feeds:** Section 5/7 — the cooktop and Power strip 1 tap the rear outlet; DELTA 3 charging has the front one. Section 1/7 uses the 1500W rating for the DELTA 3's charge rate and the cooktop check. - **V9a** — front console outlet — wattage rating: **1500W** (already verified). - **V9b** — rear outlet position: **22.5" up, 10" inboard of the right sidewall** (MEASURED — owner, Aug 1 2026). At 22.5" the socket sits **4" ABOVE the 18.5" deck plane** — the cooktop and Power strip 1 cords reach it from deck level instead of having to drop down the gap between the kitchen unit and the van wall. See Section 5's routing note. - **Check** — does the quarter trim / cup holder intrude inboard of the 46" deck width at deck height near that corner?: **the trim sits ~10" in from the sidewall at outlet height**, so the corner is where the deck and the trim come closest — keep the anchor board's passenger strip clear of it (it starts 3" in from the hatch). ##### V10 — Cargo floor level — flat, or sloped front-to-back? ![Van measurement V10 — cargo floor level](renders/vanmeas-v10.png) - **Measure:** Lay a long level along the run and check at several stations. Nothing in the model accounts for a slope: every module's four legs are one length, which only gives a level deck if the floor itself is level. - **Feeds:** Not currently a param — a slope means per-leg shimming rather than one `leg_height`. - **V10** — level, or slope over what run?: **mixed — mostly level; level at the ends, with minor change in between** (MEASURED — owner, Aug 1 2026). No slope big enough to force per-leg shimming, so the one-length-legs assumption holds. The leveling feet already on every leg absorb the "minor change" — set them once on a flat surface and adjust only if the deck rocks. ##### Also worth noting while you are in there — no drawing needed - **Folded 3rd-row well depth.** Already covered by **F3b** in the rear-floor survey below, which measures the same well. It confirms `leg_height` (17" on Panel C, fridge-driven; Panel A/B legs are 16.25" with the deck recess — that clears it). - **Rear 12V accessory outlet — exact position.** Informational only, **not a blocker**: this design doesn't use it. The fridge and fan system run off the DELTA 3 stack instead (Section 1). - **Factory sunroof — PRESENT** (MEASURED — owner, Aug 1 2026), opening size not measured. Only relevant if you later pursue a roof vent — measure it then. #### The fridge (BougeRV) and kitchen unit (JAGAHAHA), once purchased ![Measurement guide: fridge and kitchen](renders/measurement-fridge-kitchen.png) *The sheet carries only the numbered dimension tags. Each number is a row of the table below, which is where the question, the listing spec, and the blank for your measurement live.* | # | Measure |*

*Goes in `params.scad` as | Listing spec (unverified) | Your measurement | |---|---|---|---|---| | 1 | Fridge length as installed (left-right, the 450mm side) | `fridge_ext_length` | 17.72" | 17¾" (owner, July 2026 — matches) | | 2 | Fridge depth, front to back incl. handles | `fridge_ext_width` | 28.74" (body 28.03") | 27¾" (owner, July 2026 — ~1" under; hard to tape a rounded-corner case square, and easy to undercount how far the handle protrudes. Spec used, since the tray is already cut to it.) | | 3 | Fridge height | `fridge_ext_height` | 15.79" | 15⅝" (owner, July 2026 — matches within rounded-corner tolerance) | | 4 | Which side is the compressor/vent on? | (confirms the intake/exhaust fan placement in Section 2 still makes sense) | Manual (CR04001): compressor, louvers, control panel & power port ALL at one end; free-air clearances 200mm that end / 100mm sides | Manual answers the "which end" question — that end faces the tailgate (Component 7). Still confirm on the physical unit, and see Section 2's clearance note for how the fan system substitutes for the manual's free-air numbers | | 5 | Fridge empty weight | (Section 8 weight table) | 40.6 lb (18.4 kg, manual) | | | 5b | Floor vents at Panel C's REAR corners | Panel C's rear legs sit at the TRUE corners now | must be vent-free for ~4" | ✅ **CLEAR — the rear-corner floor vents do not reach the leg area** (owner, Aug 2026). Panel C's rear legs land on solid floor | | — | Fridge cord length | (confirms it reaches the new DC line's connection point inside Panel C's void — Section 5) | — | | | — | Fridge slide clearance — how far it must come out before the lid opens fully | `fridge_slide_length` (confirms 24" is enough) | 24" slide | | | 6 | Kitchen unit width, closed | `kitchen_box_width` | 20" | **20⅛" (20.125")** (RE-MEASURED — owner, Oct 18 2026; it read 20.5" in August, and the open issue below was written against that, so the bay is ⅜" wider than it says). At x 24⅜"–44½" from Panel C's driver line | | 7 | Kitchen unit length, closed | `kitchen_box_length` | 26" | **25⅞" (25.875")** (RE-MEASURED — owner, Oct 18 2026; 25 13/16" in August). Its hatch end sits **1½" in** from Panel C's tailgate edge (`kitchen_hatch_inset`) | | 8 | Kitchen unit height, closed | `kitchen_box_height` | 11.8" | **11.55"** (MEASURED — owner, Aug 2 2026). A quarter-inch shorter, so the kitchen drawer above it gains that much clearance | | 9 | Stove tray length x width x clearance height | (Section 7 — the COOKTRON cooktop's fit depends on this; the current numbers came from listing photos, not a spec sheet) | 23" x 15.7" x 5.7" | | | 10 | Kitchen unit empty weight | (Section 8 weight table) | ~45 lb | | | — | Cord pass-through location for the cooktop's power cord | (Section 5/6 cord routing) | — | | | — | Confirm it still extends to ~70" open | (sanity check — this happens outside the vehicle, doesn't affect interior length) | ~70" | | **The kitchen unit is designed to be strapped down, and it tells you where.** The owner's photos of the physical unit (Aug 2 2026) show **notches routed into the box's top edges**, and the maker's own assembly video shows a cam strap seated **in** such a notch, running around the box. That is the unit's designed hold-down load path: a strap over the top sits down in the notches and cannot slide along the box. There are **4 notches and 4 D-rings — 2 of each per face**, so it is **2 straps**, one across each face, seated in that face's notch pair. The unit also carries folding stainless handles on its ends and two oval hand-holds, so lifting points are built in. > **OPEN ISSUE — the measured width costs the utility bay 0.5".** At 20.5" > instead of 20", the kitchen leaves the open utility bay **2.78" wide** where > the control cluster needs **2.8" + 0.4" working clearance = 3.2"** — short > by **0.42"**. The hard assert is an `echo`, so every render logs the > shortfall rather than refusing to build. **RESOLVED: the control cluster mounts outside the bay.** The full width chain comes to **44.72" of 46"**, leaving the bay **1.28"** — too narrow to mount anything. There is no room inside Panel C's void — over the fridge stack is 1.09", over the kitchen under the drawer 0.5", over the drawer under the deck 1.2". It mounts on **Panel C's front wall, on the wall's Panel-B face** — the one surface in the build with real space: 46" × 17" carrying only a 4.75" fan hole, a 9"×2" louver and a single 1" grommet. **Where exactly.** The owner confirmed that **the spare is centered in Panel B**, and that fixes what stands against this wall: a 28.5" spare centered in 46" spans **x 8.75–37.25**, and the two 16.9" totes stacked on it span **6.1–39.9**. So the wall's clear area isn't the middle at all — it's a **~6" full-height column at each end**. The cluster is therefore tall and narrow, **X 40.4–45.4", Z 3.5–14.5"**, in the passenger column. It lands over the passenger front leg (x 41–42.5), which is better screw purchase than the wall itself. `params.scad` asserts it clears the fan, the louver, the grommet, the spare and the totes, and stays on the wall. **Build order note:** fit the wall and drive its perimeter screws *before* mounting the cluster — it covers the two leg screws at x 41.75. Panel A was ruled out for a physical reason rather than a spatial one — its drawer has excellent access, but the W1209's thermistor lead is about 39" and its probe must sit in the **fridge** bay, roughly 59" away through two seams. **Access:** lift one of Panel B's two lift-out bed-top halves and reach down the front wall. Panel B is top-loaded by design — the side doors don't reach it — so this is the access route the panel already has, not a new one. The spare and totes are accounted for in `params.scad`. **What the bay is.** Not a mounting location: an **air path and a finger gap**. The exhaust fan blows the fridge bay's warm air through it and out the open tailgate face, and at 1.28" × 17" it presents **21.8 sq in** of cross section against the 120mm fan's own **17.7 sq in** aperture, so it does not choke the flow — `params.scad` asserts exactly that. #### The rear floor under Panel C — what does the anchor board actually sit on and strap to? (SURVEYED Aug 1 2026 — findings below) > **Superseded in part (October 2026).** The anchor board no longer uses steel tongues to the 2nd-row rails. A second survey found the 3rd-row seats' latch hoops in the headrest trough (28" from the hatch, 1⅞" below the loaded seat surface) and the side trim closing the van to 42" from 26" forward; the board now straps to those hoops (Section 8, Anchor Board sheet). The F1–F8 findings below are kept as the record of the van. **HEADLINE FINDING (owner, Aug 1 2026):** **The stowage well is full of folded seats, not open**, and the folded seatbacks make a load surface solid enough to bear on across the whole Panel C footprint (F1/F2/F3) — so no filler blocking, and the well's edges are not a constraint. But **there is no hard step face at the striker row** for the steel tongues to butt against (F7 — it is carpet over a soft ~1" step), and **the 3rd row's seat-mount bolts are occupied by the seats** and unavailable as hardpoints (F6). That leaves Section 8's load path: **the 2nd-row rail rear ends in tension/compression, plus the 3 striker straps.** The rail ends measure **42" from the hatch** and the striker row **44.5"**, so the steel tongues are **16" long** — 4" lapped on the bridge, 9" of span, 3" on the cap. **The problem statement (from a photo of the actual cargo area):** the rearmost stretch of the Sienna — exactly where Panel C, the fridge slide, and the kitchen unit live — is not plain sheet-metal floor. It's the 3rd-row stowage area: a **deep cargo well** the folded seatbacks drop into, so much of the "floor" back there is actually **folded seatback over a void**, plus the seats' own latch/hinge hardware and (passenger rear corner) the hybrid's 12V battery access panel. Section 8's no-drill securing design — an anchor board bearing on this surface, steel tongues bolted to the rear ends of the 2nd-row floor rails, straps into the 3 striker loops — needs the same geography mapped before its outline can be cut: where the surface is solid, where the well and its edges are, exactly where the strikers and the rail ends sit, and what Panel C's legs land on. Do the whole survey with the 3rd row folded exactly as it will sit in camper mode and the 2nd-row carriages parked fully forward. **Reading these drawings.** Each measurement below gets its own drawing of the whole vehicle in plan, so you can see where in the van you are working. All eight are the same base plan — **front at the left, tailgate at the right**, driver side along the bottom — showing the body outline, the front seats, the empty 2nd-row bay with its long-slide floor rails, the folded 3rd row over its stowage well, and Panel C's footprint. **Only that section's measurement is drawn in red**, tagged with lettered items (F4a, F4b, ...) that match the blank finding lines under each drawing one-for-one — so there is a line to write on for every thing you are asked to measure. F3 and F7 add a small section inset, because a depth and a step height cannot be shown in plan; F8 adds three detail bubbles for the rail-end cases you have to tell apart. The body outline is illustrative, and every dashed interior feature is drawn at its ASSUMED position — confirming those is the entire point of this survey. ##### F1 — Map the load surface into its three zones ![Survey F1 — load-surface map](renders/survey-f1.png) - **Measure / check:** With the 3rd row folded flat, map the load surface: press and knock along the whole Panel C footprint and mark with masking tape where it is **folded SEATBACK**, where it is the **well/tub** (hollow underneath), and where it is **carpet over solid pan**. - **Feeds:** Where the anchor board's strips can bear solidly, and where they would need filler blocking. - **Finding** (MEASURED — owner, Aug 1 2026): **"Seats in all areas. Can handle load."** There is no three-zone map to draw — the folded seatbacks cover the whole Panel C footprint and bear the load. **No filler blocking needed.** The trade-off is that the bearing surface is upholstered rather than steel, so the anchor board's rubber mat is doing real work; keep the straps tight. ##### F2 — Fore-aft position of the stowage well ![Survey F2 — well edges, fore-aft](renders/survey-f2.png) - **Measure / check:** Where the stowage well's rear and forward edges sit, both measured from the closed hatch (the same fore-aft datum as V1). - **Feeds:** The board's strip lengths, and where filler blocking starts and stops. - **F2a / F2b** (MEASURED — owner, Aug 1 2026): **not applicable — "doesn't matter, seats fill it in."** With the 3rd row folded into it, the well is not*

*an open cavity with edges to work around, so there is nothing for the strip lengths or filler blocking to be positioned against. ##### F3 — The well's width, and its depth ![Survey F3 — well width and depth](renders/survey-f3.png) - **Measure / check:** The well's width across the van, and its depth below the surrounding load floor. - **Feeds:** Sizing the filler blocking that carries a board strip across the well, down to its structural bottom. - **F3a / F3b** (MEASURED — owner, Aug 1 2026): **not applicable — "well is filled with seats, doesn't matter."** No depth to blocking-out, and nothing here challenges `leg_height` (17" on Panel C, 16.25" on A/B). ##### F4 — The 3rd-row striker loops ![Survey F4 — striker loops](renders/survey-f4.png) - **Measure / check:** For each of the 3 loops: its fore-aft distance from the hatch, its lateral position, and its inside clearance (a strap hook has to seat in it). Also whether the FOLDED seatbacks leave all 3 loops exposed and hookable. - **Feeds:** Section 8's striker straps plus the steel tongues' length. - **F4a** — hatch → striker row: **44.5"** (MEASURED — owner, Aug 1 2026). - **F4b** — loop-to-loop spacing: **~12.5–13"**, with the outer two loops **~11" in from each sidewall** (scaled off the tape in the Aug 2026 striker-row photos; the finding line itself says "see pictures"). Three loops across the row, roughly 11" / 24" / 37" on a tape zeroed at the driver sidewall. - **F4c** — inside clearance / exposure: **loops are 1.5" tall, and all 3 are exposed and hookable with the seats folded — "yes"** (MEASURED — owner, Aug 1 2026). 1.5" of clear height is enough for a cam-strap S-hook or a small carabiner; check the hook you buy against it. ##### F5 — What Panel C's legs and the kitchen bear on ![Survey F5 — leg and kitchen bearing points](renders/survey-f5.png) - **Measure / check:** Where Panel C's 4 legs and the kitchen unit's footprint actually bear: solid floor, or folded seatback? A deck leg standing on a seat cushion is springy — it will not sit level or solid. - **Feeds:** `leg_height`, deck levelness, and the kitchen unit's strap-down. - **F5a** — what each of the 4 leg pads bears on (MEASURED — owner, Aug 1 2026): **the REAR legs land on folded seats; the FRONT legs land on the floor, but beside the vent intrusions** (see the Aug 2026 photos). So two legs bear on upholstery — springy, exactly the case this section warned about. **Set the leveling feet with the deck loaded**, not empty, so the foam-and-fabric compression is already taken up, and re-check after the first night out. - **F5b** — what the kitchen unit's footprint bears on: same mixed surface as the rear legs (folded seat). It is strapped down rather than standing on levelers, so compression matters less — but keep the strap tight enough that settling doesn't loosen it. ##### F6 — Fold the 3rd row, or remove it? ![Survey F6 — fold vs remove the 3rd row](renders/survey-f6.png) - **Measure / check:** Does a solid bearing surface require REMOVING the 3rd row rather than folding it? If so, does the 3rd row carry its own SRS / seatbelt-pretensioner wiring? - **Feeds:** Section 9 covers 2nd-row removal only — a 3rd-row removal procedure (and possibly more emulators) would be NEW scope. Removal would also expose the 3rd row's own seat-mount bolts as bolt-in anchor-plate hardpoints (no new holes) — the strongest points in the zone if the striker plan needs help. - **F6a** — does the row have to come OUT? **No — "folded is good."** (MEASURED — owner, Aug 1 2026) **Section 9 stays as written: 2nd-row removal only.** No 3rd-row removal procedure, no extra SRS emulators, no new scope. - **F6b** (MEASURED — owner, Aug 1 2026): **no SRS / pretensioner wiring on the 3rd row** — but **the seat-mount bolts are in use by the seats and are NOT available** as anchor-plate hardpoints, so the rail ends + striker straps carry the load. ##### F7 — The step/riser at the striker row ![Survey F7 — step/riser at the striker row](renders/survey-f7.png) - **Measure / check:** Is there a square, hard vertical face at the step by the striker row (metal under the trim?) for Section 8's two steel tongues to butt against — and how tall is it? - **Feeds:** The FALLBACK forward load path, in compression, if F8 rules the rail ends out. - **F7a** — is the step face square and hard? **No — "covered in carpet"** (MEASURED — owner, Aug 1 2026). Not a square metal face, so **the compression fallback is out**: do not size the steel tongues expecting to butt them against this step. - **F7b** — step HEIGHT: **1"** (MEASURED — owner, Aug 1 2026) — low and soft. It will stop a board from creeping rearward under gentle load, but it is not a crash-path member. ##### F8 — The 2nd-row rails' rear ends ![Survey F8 — 2nd-row rail rear ends](renders/survey-f8.png) - **Measure / check:** With the seat carriages parked fully forward: where do the rails' REAR ends sit (distance from the hatch, lateral spacing), and what is at each end — an end cap that unclips, exposed track bolt heads, an open track slot, a lip a clamp plate could grab? Photograph the end hardware close-up. - **Feeds:** Section 8's PRIMARY forward connection: the anchor board's 2 steel tongues bolt or clamp to these rail ends (no new holes). This measurement decides the bracket detail and the tongue length. - **F8a** — hatch → rail REAR ends: **42"** (MEASURED — owner, Aug 1 2026). **The rail ends reach — with room to spare.** They sit 2.5" NEARER the hatch than the striker row (44.5", F4a), so the tongues are **16" long**. This is the PRIMARY forward connection, which matters because F6 and F7 both eliminated fallbacks. - **F8b** — rail lateral spacing: **17.5"** center to center (MEASURED — owner, Aug 2026). This is what the anchor board's bridge gets drilled to. - **What is AT each end** (MEASURED Aug 2026): a **moulded plastic end cap** held by **one flush slotted fastener**, over an **open-top track channel with a 0.75" slot**. The metal track sits **0.5"** above the floor pan and the plastic housing over it reaches **1.0"**. Under the plastic cap is a **steel saddle cap** with a ¼" hole in its flat top face — see Section 8's connection detail (short version: a vertical pin into that hole carries the forward load). **Photos to take while you're in there** (each one answers a specific question later, at the workbench, without crawling back in): 1. **Wide shot from the open tailgate**, 3rd row folded as it will sit in camper mode, with a tape measure laid fore-aft along the floor from the hatch — establishes the Y datum every other photo hangs off. 2. **Carpet/liner pulled back at the tailgate end** — what's actually under it: painted steel, a plastic tub, or the seat mechanism. 3. **Straight-down shot into the stowage well** (seats up if needed) with the tape showing its depth and its front edge's distance from the hatch. 4. **Close-ups of the seat latch/hinge hardware** anywhere it falls inside Panel C's 35.75" footprint — this is what a board strip or a deck leg would be fighting for space with. 5. **The passenger rear corner** showing the 12V battery access panel — the one spot the anchor board must stop clear of (Section 8). 6. **The striker row straight-on**, tape held across it — all 3 loops in frame, with the step's face visible below them. This pairs with F4 and F7. 7. **Each 2nd-row floor rail's rear end, close up** (carriages parked forward) — the end cap/bolts/slot the tongue bracket has to mate with. This pairs with F8 and is what the bracket gets fabricated against, at the workbench. **If the survey comes back ugly** — rail ends short of the tongues' reach or with nothing to grab (F8), loops buried under the folded seatbacks, no hard face at the step, nothing solid for the strips to bear on — the Section 8 design flexes rather than dies: the striker-row step as the tongues' compression stop (F7), filler blocking down to the well's structural bottom (F3), the well's forward wall as a bearing target (F2/F3), or the 3rd row's own seat-mount bolts with anchor plates (F6 — likely means removing the 3rd row, which is new Section 9-style scope). Every one of those changes geometry that `params.scad` currently takes on faith, so settle this survey before cutting Panel C or the anchor board. --- ### Appendix B — Build Sequence & Assembly Order The order to build in — **dependency-first, not the Component numbering.** Work top to bottom; the "needs" is what must be finished before each phase. **Phase 0 — MEASURE & PREP, before any cutting** (Appendix A, 9) — *critical* 1. Measure the real van and update `params.scad` — **DONE** (Appendix A): interior 93.75", liftgate 50×37", side-door usable clear 29", wall-to-wall at platform height **50"** — under the ≥53" gate, so the platform is 49". *Still open: the side-door aperture height and its fore-aft position (`side_door_y0`).* 2. Confirm the rear-corner floor vents don't reach the last ~4" (Panel C's rear legs sit there). **DONE — they do not reach the leg area; Panel C's rear legs land on solid floor.** 3. Remove the 2nd-row seats; install the Ling Labs SRS emulators (Section 9). 4. Buy & verify materials (Appendix C); verify the Rocky 40 + JAGAHAHA dimensions on arrival (both purchased July 2026). **Phase 1 — Cut all plywood & lumber** (Section 3 / Appendix D) 1. Cut both sheets + all boards. The ½" sheet nests to ~95% — lay it out before cutting. 2. Drill the leg leveling-insert holes now (far easier before assembly). 3. Pre-cut the Panel C front-wall openings (fan hole, 2 grommets, louver — Front Wall render). **Phase 2 — Build the three frames** (Components 2/3/4, step 1) 1. Panel A frame: rails + 4 legs + divider + 2 END-face bottom rails (underside 1") + 2 diagonal braces (**on the frame's underside**, same face as the corner brackets). Corners: 2× 2½" screws + bracket + glue. 2. Panel B frame — the full cube: bottom rails on ALL 4 faces, no divider/drawers. 3. Panel C frame: front pair inset 3.5", REAR pair at the true corners; FRONT-face bottom rail only. 4. Install the 12 leveling feet (no knobs — the collar is the adjustment). **Phase 3 — Panel C build-out (fridge / kitchen / cooling)** (Components 4 + 7) 1. Set the fixed top between the rails on its bearer cleats (flush with the rail tops — deck recess) and screw it down; add the front wall (mount the low intake louver). 2. Bench-build the no-drill anchor board (strips + steel rungs + risers, Anchor Board sheet), set it in the van on its neoprene, and chain the risers' eye bolts to the 3rd-row seat-latch hoops H1 and H2; nothing is drilled into the van. *Re-check the chains and straps after the first drive.* **The hybrid's 12V battery access panel is at the passenger rear corner (under the rear outlet) and stays reachable by*

*unstrapping/sliding the kitchen out.** 3. Fridge on its tray + locking slide (driver corner); kitchen unit strapped down (passenger corner). Let the fridge stand 12h before first power-up. 4. Hang the kitchen drawer; velcro the 2 utility bins into the open bay (no door). 5. Cooling + control cluster: intake fan (blows IN), exhaust fan + NTC on the fridge's kitchen-facing wall, W1209 + fuse block + switches on the backer board in the open utility bay. **Phase 4 — Rear pantry (tote + microwave)** (Component 1 — *needs Panel C's finished deck; nothing to build*) 1. Screw a footman loop to the outside face of each side top rail, 7½" from the tailgate end. 2. Mat down; set the tote (driver side) and the microwave (flush to the passenger edge). 3. One cam strap over both, hooked to the two loops, buckle on the tote lid. 4. Re-site Power strip 1 + the ROLL bubble level with the control cluster (Component 6). **Phase 5 — Panel A build-out (EcoFlow bays)** (Components 2 + 8) 1. Right bay: R1-biscuited drawer box on a 20" slide + catch; DELTA 3 cleats + D-rings + a cable tray. 2. Left bay: no shelf, no glide strips — the WAVE 3 sits on the bay floor. 2 D-rings in the end bottom rails + a cam strap fore-aft over the unit. **Phase 6 — Bed platform** (Component 2 — *needs Panel A & B built*) 1. Assemble the three pieces — Panel A: 2× 29" rails 49" apart + 5× 42" slats; each Panel B half: 2× 29" rails 24.5" apart + 5× 17.5" slats (glue + 2 pocket screws per slat end, crown-up; Bed Platform Assembly sheet). Drop Panel B's center bearer onto its corner braces first, so both halves have something to land on (it lifts out again for the spare), and **screw Panel A's section down** — it is permanent. 2. Screw the PITCH bubble level to the driver-side rail edge. **Phase 7 — Seam hardware** (Component 5): bumper strips + 2 alignment pins at the A/B and B/C seams. **Phase 8 — Cord runs** (Component 6): cooktop + Power strip 1 → SHORT cords to the REAR AC outlet (2-way tap, no seams); DELTA 3 charging → front console outlet (no seams); fridge DC → the one long run, Panel A → C with 2 SAE seam disconnects. **Phase 9 — Install in the van** (Components 10 + 8) 1. Set Panel A → B → C; register the alignment pins. 2. Connect power + stow the DELTA 3 stack and WAVE 3; load Panel B — RJ-MODINI spare flat on its skid (liner under), valve up, tool case in the wheel, 2 totes on top, cam strap to the footman loops. 3. Lay the platform → HEST mattress → tension-rod curtain. **Phase 10 — Commission & test** (Component 10) — *critical* 1. Lift-out / reinstall dry run (each panel by its top rails; pantry drawer units by loosening the cam strap). 2. Level the van at the WHEELS (Appendix E) + set the interior feet once. 3. Power-up + cooling test (fridge stood 12h; fans ramp off the NTC); pull every drawer/slide. 4. Owner-place the low-level CO monitor + fire extinguisher. ### Appendix C — Materials Shopping List Consolidates Section 4's BOM into a shopping order with links. Prices are estimates; items marked **UNVERIFIED** need a live-listing check. **Purchased** = already bought (July 2026); **owned** = existing gear, not priced. **ASK THE STORE TO CUT THE SHEETS** — hand them the cutting layout in Section 3: trim ~½" off all four factory edges of every sheet, then the numbered crosscuts before the rips, exact-fit parts ~⅛" oversize. You come home with rectangles; only the anchor board's three strips, the ¾"×¾" cleats and the holes are left for your saw. **Lumber & Plywood** — 3/4" Baltic birch sheet ($65), 1/2" Baltic birch sheet ($50), **no 3/8" material at all (those four parts come out of these two sheets' leftovers, ~–$50 and ~+16 lb; Section 3)**, 12× 2×2 pine 8ft ($96–180), 7× 1×4 pine 8ft ($35), 1× 1×3 pine 6ft ($6). **Frame / fasteners** — rear pantry: 27-gal Tough Box tote (*bought*) + a 1" × 8 ft cam strap with snap hooks ($10) — the 2 IRIS 3-packs ($61) and the old hold-down kit ($26) were bought before the switch + 2× Sterilite 28-Qt Panel B totes ($24) + spare hold-down hardware: footman loops + cam-strap 6-pack ($24) + the RJ-MODINI spare kit ($350 — stows in Panel B at the axle); corner brackets ×12 (*purchased* — 16 came in 2 packs, $20.54); diagonal corner braces ×8 (*purchased*, $24); seam draw-latches ×4 (*purchased* — houya toggle latches, $16); wood screws ($20); wood glue ($8); [Ryobi R-series biscuit assortment](https://www.amazon.com/) ($12); alignment dowel pins ($3); anti-rattle bumper strip (*purchased* — Shappy felt strips, 2× 120", $9.73). **Drawers & slides** — Panel A drawer slides (GlideRite 20" 100lb 5-pack, *purchased*, $47); drawer catch ($3); kitchen-drawer slide 24" (*purchased* — Liberty 942405 soft-close, $37.63) + catch (*purchased*, WOOCH — but 2 are needed, verify); fridge slide 24" heavy-duty locking (VADANIA, *purchased*, $79). **Leveling** — leg leveling feet (3 four-packs, $36); Larbeti stick-on bubble levels 2+2 ($10, PURCHASED); [Lynx leveling blocks 10-pack](https://www.amazon.com/Lynx-Levelers-00015-Leveling-Blocks/dp/B000BUV1RK) + chock ($40). **Fridge & kitchen** — [BougeRV Rocky 40](https://www.bougerv.com/products/rocky-12v-camp-fridge) ($400–500 less JULY7 coupon, **PURCHASED**); [JAGAHAHA slide-out kitchen](https://www.amazon.com/dp/B0FLDCNYZX) ($300–350, **PURCHASED**); [COOKTRON induction cooktop](https://www.amazon.com/dp/B09MCR1SDT) ($190); UMETRE 9-piece detachable-handle cookware set (purchased) ($65–90). **Electrical & cooling** — 120mm 12V fans ×2 ($18–20); W1209 controller + probe ($10); Nilight fuse block (*purchased*, $18); Ampper switches 10-pack (*purchased*, $9); 3" snap-in soffit vents ×3 (*purchased*, 4-pack); grommets (EASYEAH 20-pack, *purchased*, $10); power strips ×2 ($50); AC cords ($24 — two short rear-outlet hops + the DELTA 3 charging line); fridge DC cord + SAE seam disconnects (one OYMSAE 2-pack does both, purchased, $22); 2-way rear-outlet tap ($5); cord clips ($9); fused car-plug→SAE adapter + SAE→12V-socket pigtail ($18); 18AWG hookup wire, crimp assortment + WAGO lever nuts ($30). **Anchoring (no-drill, Section 8)** — Erickson ratchet straps, 2 4-packs ($64); eye bolts, ¼" chain and quick links for the hoop connections ($30); two 3" × 3/16" × 48" steel flat bars for the rungs ($40); 2 more quick links for the hoop loops ($8); anchor-board T-nuts and screws ($15); ⅛" neoprene ($20); fridge-slide steel riser angles ×2 ($24). **Comfort & interior** — [HEST Dually Long mattress](https://hest.com/products/dually) ($530–590); Windmill Smart fan ($69, PURCHASED); tension rod + blackout curtain ($40); [PeaceOut Sienna bug nets](https://peaceout.ca/en/products/toyota-sienna-front-sliding-doors-bug-nets) (~$197); [EcoFlow WAVE vent kit](https://us.ecoflow.com/products/wave-car-vent-kit) ($39); non-slip mat ($12); WAVE 3 hose hook ($4); WAVE 3 D-rings + cam strap ($11); DELTA 3 drawer hardware ($15). **Found storage** — DELTA 3 drawer tray ($8); utility-bay bins ×2 + hook-and-loop tape ($8). **Finishing / vehicle** — utensil bins + non-slip liner ($12); wood sealant ($20); sandpaper/misc ($15); SRS airbag emulators — 0651 Round x2 + 5522 Square x2 (~$60, **PURCHASED**; verify the 5522 fits the rectangular plug). **Already owned (not priced):** EcoFlow DELTA 3 Plus + Smart Extra Battery, EcoFlow WAVE 3. **Total: ~$3,370–3,740 with the fridge + kitchen, or ~$2,670–2,890 if you already own equivalents** (see Section 4 for the full itemized BOM and every purchase link). ### Appendix D — Cut List (consolidated) Finished sizes; add saw kerf. Full context in Section 3. **3/4" Baltic birch plywood — 1 sheet (4×8):** Panel C deck 1× **33×43"** (it drops BETWEEN the rails onto bearer cleats); anchor board strips 3× 22½" long (2½", ~2½", 1½" wide; Section 8 — rip at home); deck bearer cleats 4× ¾×¾ (2× 33" + 2× 40"); rear-pantry hold-down cleats ~4× 1×1×12–14" (both ripped from the offcuts) — the rest is spare stock. **1/2" Baltic birch plywood — 1 sheet (4×8):** kitchen-drawer box (bottom 1× **18×26"**; sides 2× 26×4"; front/back 2× **17×4"**) + hanging cheeks 2× **26×5.45"**; battery-drawer bottom 1× **19.25×21.25"**; **Panel C front wall 1× 46×17"**; **fridge tray 1× 17.72×28.74"** (from a 29×18 blank, cross-grain). **No 3/8" sheet.** **Battery-drawer walls — 3/4", from the 3/4" sheet's offcuts:** 2× **22×13.75"** sides + 2× **18.5×13.75"** front/back. One of each comes out of the 47×19" strip; the other two come from the sheet's remaining offcut. 2× R3 biscuits per corner. **2×2 pine — 11 boards bought, 9 used, 801⅛" of parts still to cut:** end rails 6× 46" (the outboard pair on every panel, full deck width); side rails 2× **26⅛"** (Panel A) + 2× **26⅛"** (Panel B) + 2× **32⅞"** (Panel C) — **each ~2⅞" shorter than its panel, because it fits BETWEEN the end rails, sized for the stock as measured (1-7/16" rather than 1½")**; center divider (Panel A) 1× **26⅛"**; legs **2× 17.5" (Panel C front, lapped) + 2× 16" (Panel C rear, butt-under) + 8× 16.75" (Panels A/B, lapped)** — **all twelve already cut, bored and footed**, each cut 1" short with the foot making up the rest (*the legs are LAPPED to the rails*); bottom rails 5× 46" + 2× **23¼"** (Panel B's side pair — the lapped legs sit 1.4375" inboard at each end); Panel B center bearer 2× **26⅛"**. Section 3's remaining total is 801⅛" and matches; 9 boards (864") covers it, leaving 2 of the 11 spare — see the board-by-board packing plan in **Appendix H**. **1×4 pine — 7 boards (8ft), bed platform:** side rails 6× 29"; slats 5× 42" + 10× 17.5". **Trim / cleats / scrap:** none for the pantry or the DELTA 3 drawer (both held by rubber mats and straps, as built). ### Appendix E — Leveling: Block Calculator Per-site leveling is done at the **wheels** with the [Lynx blocks](https://www.amazon.com/Lynx-Levelers-00015-Leveling-Blocks/dp/B000BUV1RK), driven by the phone **[Sienna Block Calculator](https://claude.ai/code/artifact/149333c6-8f02-47a2-915f-52d26d9059d9)** (open it in Chrome and *Add to Home screen*, or use the offline copy `sienna_block_calculator.html`). The interior leg feet are then a **one-time** set against the van's own floor. **Using it:** park (engine off, brake on) → read the two bed-mounted bubble levels (pitch on the platform rail, roll on the rear-pantry deck edge) in degrees → enter both readings + which end/side is low → it returns blocks-per-tire. Set the one-time van settings once (wheelbase 120.5", track ~68", and *your measured* block height, Lynx ≈ 1.5"). Stack ahead of the low tires, drive up slowly, chock, re-check; if any tire needs >4 blocks, re-park. **The math** (so it's not a black box): a

*low END needs `tan(pitch°) × wheelbase` of lift; a low SIDE needs `tan(roll°) × track`. Blocks per tire = `round(that deficit ÷ block height)`. One 10-pack (1.5"/block over the 120.5" wheelbase) covers about 2.9° of pitch correction. ### Appendix F — Weight Budget & Weight-Distribution Analysis Every component's weight, computed from the real material dimensions (Baltic birch at 42.5 lb/ft³, pine at 28 lb/ft³) plus manufacturer specs for the appliances/power gear. The live spreadsheet is in the repo as **`weight_budget.csv`** (open it in any spreadsheet app; regenerate with `weight_budget.py` if you change materials). Figures marked *est.* are estimates to confirm. | Component | Category | Weight (lb) | Zone | Notes | |---|---|--:|:--:|---| | Panel A frame (2×2 pine) | Structure | 12.1 | front | rails+legs+divider+bottom rails (16.75" legs — deck recess + lap) | | Panel A drawer box + slides (**¾" birch walls**, ½" bottom) | Structure | **34.2** | front | holds the DELTA 3 stack; walls cut from ¾" offcuts (no 3/8" sheet) | | Panel B frame (2×2 pine, full cube) | Structure | 12.8 | mid | bare-frame deep-storage box (16.75" legs — deck recess + lap; 23" side bottom rails) | | Panel C frame (2×2 pine) | Structure | 10.1 | rear | rails+legs+front bottom rail (front legs 17.5" lapped, rear pair 16" butt-under) | | Panel C deck (¾" birch, 33×43 recessed + cleats) | Structure | 27.2 | rear | drops between the rails onto bearer cleats, flush (deck recess) | | Panel C front wall (**½" birch**) | Structure | **9.6** | mid | intake fan + grommets; from the ½" sheet's leftover | | Fridge tray (**½" birch** + 2 1×3 side aprons) | Structure | **8.0** | rear | hangs between the side-mount slide rails; from the same leftover | | Kitchen drawer + ½" cheeks (birch) | Structure | 13.9 | rear | hung over the kitchen unit; 18" box | | Rear pantry: 27-gal tote + strap | Structure | 7.0 *est.* | rear | was 15.0 for the IRIS cluster; the subtotals below still carry 15.0 | | Panel B: 2× Sterilite 28-Qt under-bed totes | Hardware/misc | 4.0 | mid | stacked ON the spare | | RJ-MODINI spare kit + jack (Panel B, flat at the axle) | Hardware/misc | 40.0 | mid | T155/85R18 steel-wheel kit + cleats/strap — VERIFY kit weight | | Bed platform (1×4 pine, 6 rails + 15 slats) | Structure | 23.8 | front | Panel A's fixed section + Panel B's 2 lift-out halves | | Panel B center bearer (2× 2x2 pine, 26") | Structure | 1.9 | mid | halves the deck's unsupported span over Panel B, 46" → ~22" | | Fridge (BougeRV Rocky 40, empty) | Appliances | 40.6 | rear | manual spec; +20–40 lb loaded | | Kitchen unit (JAGAHAHA) | Appliances | 45.0 | rear | listing spec | | EcoFlow DELTA 3 Plus + Extra Battery | Power/climate | 48.0 | front | Panel A passenger drawer | | EcoFlow WAVE 3 A/C | Power/climate | 33.7 | front | Panel A driver bay | | Mattress (HEST Dually Long) | Bedding | 35.0 *est.* | mid | solid foam + waterproof cover | | Fridge + kitchen-drawer slides (heavy) | Hardware/misc | 8.0 | rear | 24" heavy-duty pairs | | Fridge-slide steel riser angles (2) | Hardware/misc | 9.8 | rear | 2×2×3/16" × 24" steel, fixed rails → anchor board (aluminum angle saves ~6 lb) | | 12 leveling feet + inserts | Hardware/misc | 4.3 | mid | 4 per panel; no star knobs or jam nuts | | Anchor board (3 ¾" ply strips + 2 3" steel rungs) + straps + neoprene | Hardware/misc | 22.8 | rear | no-drill tie-down chassis at Panel C, strapped to the seat-latch hoops (Section 8) | | Corner brackets, braces, screws, glue | Hardware/misc | 6.0 | mid | spread across all modules | | Seam draw-latches (4) + pins + bumpers | Hardware/misc | 1.5 | mid | at the A/B + B/C seams | | Electrical (2 fans, controller, fuse, 2 strips, cords) | Hardware/misc | 6.0 | mid | cooling + power runs | | Cooktop + cookware (stowed) | Hardware/misc | 12.0 | rear | induction top + pots in the kitchen | *30 Oct 2026: the pantry tote is ~8 lb lighter than the IRIS cluster, and the ~22 lb microwave is not in this table — together about +14 lb on the totals below, not yet rolled in.* **Subtotals:** Structure **171.0** · Appliances **85.6** · Power/climate **81.7** · Bedding **35.0** · Hardware/misc **114.4** **Build total (added): ≈ 488 lb** — including Panel B's 2 storage totes (~4 lb), the **RJ-MODINI spare kit stowed at the axle (~40 lb)** — carried inside deliberately instead of on a hitch basket, which would have added its own 30–60 lb of carrier plus a ~1.4× lever on the rear axle — and the no-drill anchor board (~23 lb). **But the 2nd-row seats come OUT** (Section 9), so the conversion *removes* weight too: | | Weight | |---|--:| | Build (added) | +488 lb | | **2nd-row seats removed** (2 × ~48–70 lb) *est.* | **–120 lb** | | **NET permanent weight added vs. stock** | **≈ 368 lb** | So the van, converted and empty, weighs only about **368 lb more than a stock seatless curb** — and because the seats were part of the factory curb weight, pulling them **frees ~120 lb of payload** for your gear. **Weigh your actual seats to confirm** — captain's chairs vary; 120 lb is an estimate. **Loaded scenarios** (payload = everything you add to the van — build + people + cargo): | Scenario | Weight | |---|--:| | Net build (added − seats removed) | ≈ 368 lb | | + provisions (food, water, cookware, bedding ~120 lb) | ≈ 488 lb | | + 2 occupants (~340 lb, in the front seats while driving) | ≈ 828 lb | #### Weight distribution - **Fore–aft:** the build's own center of mass sits **≈ 51.3" back** from the front seatbacks — aft of the 47" geometric center (the prefab pantry and the axle-stowed spare pull it forward; the anchor board pulls it back). Breaking it into thirds: **front 31%** (152 lb — Panel A + the DELTA 3/WAVE 3 stack + the bed platform), **middle 25%** (121 lb — Panel B carries the spare + totes at the axle), **rear 44%** (215 lb). That rear share is the **tailgate cluster** — the prefab drawer pantry (~15 lb), the fridge (41 lb), the kitchen unit (45 lb), the anchor board (~23 lb), and Panel C's own structure — most of which has to live at the back for slide-out access. - **Lateral:** **≈ 1.0" toward the passenger side** — trivial (the fridge-slide steel risers on the driver side pulled it in a bit more). This is by design: the DELTA 3 stack (48 lb, passenger) and the WAVE 3 (34 lb, driver) sit in opposite bays of Panel A, and the fridge (driver) and kitchen (passenger) also cross-balance in Panel C. The reviewer's "heavy battery off to one side" concern is effectively cancelled by the WAVE 3 on the other side. - **Height:** everything is low (floor-level bays) except the 16.8" pantry cluster at the rear and, when parked, the sleeping load on the ~20–27" platform. Overall CG stays low — no rollover concern beyond a stock minivan. #### Effects on drivability - **Payload headroom:** loaded for travel (net build + provisions + 2 people, with the seats out) is **≈ 828 lb** — and that already includes the spare tire + jack. A 4th-gen Sienna's payload is roughly **1,100–1,300 lb** (the AWD Woodland is at the lower end), so there's a **~272–472 lb margin** — the removed seats buy back ~120 lb of that. Comfortable, but not unlimited once you add lots of water and gear. **VERIFY the exact figure on the driver door-jamb sticker** — that number, and the **rear GAWR**, are the hard limits. - **Rear bias:** with 44% of the build mass in the rear third (much of it at/behind the rear axle), the fixed build alone loads the rear axle and slightly lightens the front — which reads as lighter steering, a bit more rear squat, and a raised headlight aim. **But while driving, the two occupants (340 lb) sit up front and more than offset that**, so the *loaded* vehicle is reasonably balanced. The rear bias matters most when driving **solo with a full rear and no front passenger**. - **Lateral / height:** negligible effect — the ~1.0" lateral offset is unnoticeable, and the low CG keeps handling stock-like. #### Recommendations 1. **Verify payload + rear GAWR** on the door-jamb sticker before loading, and stay under both. The Woodland AWD has the least payload — treat ~1,100 lb as the working assumption until you read the sticker. 2. **Stow heavy movable cargo forward/mid, not at the tailgate.** Water, canned food, and tools belong in **Panel B** (middle, near the axle) — that offsets the fixed rear cluster. Water is the big one: **8.3 lb/gal**, so a 6-gal jug is ~50 lb — keep it low and mid/forward. 3. **Re-aim the headlights** after the build is in and loaded — rear squat raises the beam. 4. **Set tire pressures to the door placard** (consider the upper end for the rear given the bias) and re-check with the van loaded. 5. **Driving solo?** Expect a slightly lighter front end with a full rear — nothing unsafe at these weights, just drive to it. 6. **Leg-foot load is not a concern:** ~488 lb + occupants + cargo (~1,048 lb worst case) over 12 feet is well under the feet's 330 lb-each rating (~87 lb/foot worst case, about a quarter of capacity). 7. **If you add heavy items later** (second battery, water tank, awning), re-run `weight_budget.py` and re-check the rear axle before a trip. ### Appendix G — Nighttime Van + Tent Cooling Setup A **night-only** cooling arrangement for hot-weather campsites (worked out for a Texas-summer trip, but applies anywhere overnight lows still run warm): condition the van and an attached tailgate tent as one shared air volume with the WAVE 3 alone, and skip the factory AC overnight entirely. This builds directly on the WAVE 3 "with a tent" configuration already described in Section 1 — this appendix just adds the setup diagram and the night-specific hints. ![Nighttime van + tent cooling setup](renders/night-cooling-setup-detail.png) *The tent is drawn broken.* At true scale its 127" footprint runs back from the tailgate against a 93.75" van, so more than half the sheet would be an empty dashed rectangle — and since sheet height sets printed text size, that emptiness would cost every label on the drawing about half its size. **Marker key.** | # | What | Where | |---|---|---| | 1 | WAVE 3 + non-slip mat | Panel C's deck, a few inches forward of the pantry, facing the open tailgate | | 2 | Intake/exhaust hoses | Past the pantry and out the open tailgate gap to true outside air | | 3 | DELTA 3 Plus + Smart Extra Battery | Panel A's right (passenger) drawer — cord runs the length of the van | | 4 | Tent connection sleeve | Elastic, wraps the open tailgate/liftgate. Not part of this build — the tent's own accessory | | 5 | Tent footprint (example) | 127" × 96", a VEVOR SUV tent standing in for yours | | 6 | Conditioned airflow | Fans from the tailgate into the tent — one shared van+tent air volume, NIGHT ONLY | Two night-specific points are expanded below: why the factory AC is skipped here (it needs the van sealed, which the tent sleeve prevents), and sealing the tent's own mesh vents plus adding a small circulation fan to carry the WAVE 3's output to the tent's far*

*end. **Why no factory AC at night:** the Sienna Hybrid's electric AC compressor cools the cabin well, but it needs the van **sealed** (tailgate/doors shut) to work efficiently, and it cycles the gas engine on and off to keep the hybrid battery fed — noisy, intermittent, and it burns fuel right through the park's quiet hours. A tailgate-attached tent needs the opposite: the tailgate **open** so the connection sleeve has something to wrap around. Trying to run both at once means either the AC fights a leaky open tailgate (worse cooling, more engine cycling) or the tent's connection collapses — there's no clean way to combine them. The WAVE 3 has neither problem: it's silent-ish, battery-only, and its whole design point (Section 1) is blowing straight through the open tailgate into the shared van+tent space. **Setup steps (matches the numbered callouts in the diagram):** 1. Carry the WAVE 3 from its Panel A storage bay back to Panel C's deck, just forward of the rear-pantry cluster, on a non-slip mat — facing the open tailgate. 2. Route its intake/exhaust hoses past the pantry and out through the open tailgate gap to true outside air. No window seal needed — the open tailgate is the vent path. 3. Leave the DELTA 3 Plus + Smart Extra Battery in Panel A's right (passenger) drawer — that's the WAVE 3's power source. Run its cord the length of the van (Panels A → B → C) to reach the unit. 4. Attach the tent's elastic connection sleeve around the open tailgate/liftgate opening. 5. Pitch the tent in its normal footprint just beyond the tailgate (swap in your own tent's dimensions — the diagram uses an example ~127" x 96" SUV tent as a stand-in). 6. Zip every one of the tent's own mesh windows/vents shut except the one facing the van — every extra opening is cool air leaking out into the Texas night air. 7. Turn the WAVE 3 on and let it condition the shared van+tent volume; hoses keep dumping heat/moisture out the tailgate gap the whole time. **Hints:** - **Add one small low-power circulation fan inside the tent**, aimed to push the WAVE 3's output toward the tent's far end — its own airflow may not reach that far unassisted, and a clip fan costs only a few watts against the 2048Wh bank. - **If you want the factory AC's extra cooling power too**, use it for a short **pre-bed knockdown** only: seal the van fully (tailgate shut, tent not yet attached) and run the AC 10-15 minutes before quiet hours to clear the day's heat soak, then switch to the WAVE 3 + tent setup above for the actual overnight run. - **Battery budget:** expect roughly 7-10 hours of overnight runtime from a full DELTA 3 stack in this configuration (nighttime ambient is far easier to hold than a daytime load) — plenty for one night; plan a solar/shore-power top-up if running it multiple nights back to back. - **If you need two independently-cooled sleeping zones instead of one shared space** (e.g., people sleeping in both the van and the tent), the WAVE 3 can't do both — the alternative is to seal the van and run the factory AC there while pitching the tent as a **fully stand-alone** shelter (its own door, sleeve not attached to the tailgate) with its own separate cooling. That trades the shared-air convenience for two independently sealed zones. - **Safety note carries over from Section 1/4:** this configuration doesn't run the engine overnight, so it doesn't add to the CO-monitor concern that a factory-AC-all-night approach would — but keep the low-level CO monitor active regardless if you ever do run the engine near the sleeping area. ### Appendix H — Home Depot Order: Owner Notes **This appendix is in two halves, on purpose.** *This* half is for you — what's already bought, what each number means, what to expect at the cut desk, and the decisions behind the order. The *other* half is **Appendix H-S — Store Copy**: three self-contained pages, no design history and no owner commentary, that you print and hand across the counter. **Print PDF pages 127–129** (their own footers read *Page 126–128 of 163* — the offset is the cover sheet, which the printer counts and the footer does not). Everything on the order is already in Section 4's BOM and Section 3's cut list. What this appendix adds is the filter — **only the building materials Home Depot sells, minus everything already sitting in the garage** — and the cut list rewritten as saw-desk instructions with a diagram per sheet. #### H.1 — Already bought on Amazon: do NOT buy these again **Cross-checked Aug 3 2026 against the Amazon order export** (`Your Orders.zip`, 3,113 orders, filtered to this build). These four stay off the order because you already own them: | Not on the order | What already covers it | Ordered | Order total | |---|---|---|---| | **Steel corner brackets ×12** | **2 × "8 Pack L Bracket Corner Braces, 2 inch stainless"** = **16 brackets** — 12 needed (4 per frame × 3), 4 spare | 2026-07-20 | $20.54 | | **Kitchen drawer slide pair, 24"** | **Liberty 942405 soft-close ball-bearing 24", 2-pack** = the one pair the kitchen drawer needs | 2026-07-20 | $37.63 | | **Adhesive felt / anti-rattle weatherstrip** | **Shappy felt strips, 2 packs × 120"** = 240"; the two seams need ~92" and the pantry tape comes off the same roll | 2026-07-19 | $9.73 | | **Magnetic / friction catch ×2** | **WOOCH magnetic door catch, 30 lb** — ⚠️ **count this one before you go.** The build needs **two** (Panel A's drawer + the kitchen drawer) and the export shows a single line item; if it was a 1-pack, buy one more | 2026-07-20 | $6.48 | **Also confirmed bought, so they stay off the sheet:** the 20" Panel A drawer slides (5 pairs), the VADANIA 24" locking fridge slides, all 12 leveling feet, the Nilight fuse block, the illuminated switches, both 120mm fans, the W1209 controller, the EASYEAH grommets, both CRST power strips, the CordMate II + open-slot raceway, the OYMSAE SAE cords, the 3-outlet tap, the VIPMOON cord clips, the D-rings, the footman loops, the Erickson straps, the cam straps, the Lynx blocks, the bubble levels, the toggle draw-latches, the Sumnacon flat-strap braces, the louver vent (3" round soffit vents, 4-pack), the cookware, the Windmill fan, the blackout curtains, the jig saw + blades, the countersink set, the 23/32" dado bit, the corner clamps, the DEWALT 4-1/2" cut-off wheels and the Tap Magic ProTap cutting fluid. **What's still to buy.** **All lumber is bought (2026-09-06)** — both plywood sheets, the 2×2s, the 1×4s and the 1×3 (Section 4). The store copy's *materials* half is spent; only its **cut-desk instructions** (H.6–H.8) still apply. Nothing in the export covered: **wood glue** in a current bottle — the last was an 8 oz Titebond II in **March 2025**, so check the shelf before buying another — **spar urethane** (last one 2017), **#8 screws** in either length, **¼-20 machine screws or T-nuts**, **biscuits**, **the 3/8" dowel**, and **the steel flat bar and angle** — the *stock* only: the cut-off wheels and the cutting fluid for working it are both bought (Section 4). All of that is yours to pick up, and four of them aren't settled yet — **H.5**. #### H.2 — What the export also turned up (not Home Depot, but it changes the BOM) Actual paid amounts, so Section 4's estimates can stop being estimates. **These are order totals including tax**, which is why they run a little over list: | Item | Section 4 says | Actually paid | Ordered | |---|---|---|---| | JAGAHAHA slide-out kitchen | $300–350, **UNVERIFIED** | **$410.27** | 2026-07-16 | | COOKTRON induction cooktop | $189.99 | **$174.81** | 2026-07-16 | | RJ-MODINI spare tire kit | $350 | **$378.82** | 2026-07-19 | | VADANIA 24" locking fridge slides | $79 | **$85.73** | 2026-07-17 | | Leveling feet | 3 packs / $36 | **4 packs / $60.26** — 16 feet for 12 legs | 2026-07-20 + 07-22 | **Two things worth looking at before you build:** 1. ⚠️ **The 30-Qt BougeRV shows as a completed order, not a cancelled one.** Section 4 says the mistakenly-ordered 30-Qt Amazon unit "is cancelled" and the Rocky 40 was bought direct from bougerv.com. The export lists the **30-Qt at $224.69, ordered 2026-07-16, status Closed**, and there is **no refund or return record for it** in the returns files. Either the cancellation didn't take or the refund happened outside this export — worth confirming you aren't holding two fridges. Every dimension in this plan is built around the **Rocky 40** (28.74 × 17.72 × 15.79"), so the 30-Qt does not drop into Panel C's bay. 2. **You own E-track anchors the design doesn't use.** A SNAP-LOC E-track tie-down 10-pack ($43.29, 2026-07-20) is in the export; the anchor board straps to the 3rd-row seat-latch hoops instead, and nothing is drilled into the van. **Don't let the E-track talk you into drilling.** *Minor:* two SUV tents are in the export — VEVOR ($172.01) and JOYTUTUS inflatable ($324.74). Appendix G's cooling diagram draws the **VEVOR** 127" × 96" footprint as the stand-in; if the JOYTUTUS is the keeper, its footprint is the one that should be on that sheet. #### H.3 — Why there's no 3/8" line on the order **There is no 3/8" purchase** — the four parts come out of the two sheets you're already buying.** Panel C's front wall and the fridge tray off the ½" sheet's leftover; the battery drawer's four walls off the ¾" sheet, one side and one front from its 47 × 19 strip and the other two from the remaining offcut. **What it costs, so it isn't a surprise later:** the parts are thicker, which puts **~16 lb** on the build (Appendix F). **12 lb of that is the battery drawer**, whose 20" slide pair carries ~78 lb of its 100 lb rating. Nothing fails, and it saves ~$50 and a third sheet to haul. **The one part worth reconsidering:** a single 3/8" 4×4 panel would cover just those four drawer walls and hand the 12 lb straight back. **Two knock-ons:** the drawer's corner biscuits are **R3** at ¾", and two of its walls come out of the ¾" sheet's remaining offcut rather than the 47 × 19 strip. Full reasoning — including why the walls can't be ½" and why the tray ends up cross-grain — is in **Section 3**. #### H.4 — On the sheet goods, before you pick them off the rack - **"Baltic birch" is not what Home Depot sells.** True Baltic birch (void-free, all-birch core, metric thicknesses) is a hardwood-dealer product; HD stocks birch-*faced* ply like PureBond. Section 3 already allows "or shop-grade," so the HD sheet is the plan-compliant buy. - **Pull the sheets yourself and sight down them.** Reject a crowned face or a delaminated edge: Panel C's deck and the anchor board both come off the ¾" sheet, and they're the two parts where flat and square actually matter. - **The model/SKU numbers on the store copy and in H.5 are from a July 2026 lookup** and are there to help an associate find the rack, not as a guarantee. HD rotates models and prices constantly — treat every*

*one as *verify at the desk*. - **Sight down the 2×2s one at a time.** Twelve boards from a furring bundle will include two you can't use; the store copy asks for the straightest on the rack, but you're the one who has to sort them. #### H.5 — What I pick up myself, and the four specs still open **None of this goes on the store copy.** An associate can hand you a box of screws, but they can't tell you which length engages a T-nut, or whether your joiner takes R-series biscuits — and a store page full of items with question marks on it just gets you the wrong ones. So the store copy is the cut desk only. This is the rest of the trip, in aisle order, with what's actually undecided called out: | Qty | Item | Model / SKU (from the July 2026 lookup) | Settled? | |---|---|---|---| | 1 box | **#8 × 1¼" flat-head wood screws** (100+) | Everbilt Model 801832 / SKU 204275496 | ✅ **BOUGHT 2026-09-06, $8.98.** Brackets, braces, plywood-to-frame. A 2nd box is likely — 100 barely covers the 12 brackets + 8 braces | | 1 box | **#8 × 2½" screws** for every 2×2 frame joint — see the note at Joinery rule 4 | **SPAX Torx T-Star Plus, 133 ct / 1 lb, bit included** | ✅ **BOUGHT 2026-09-06, $12.63.** ~88 used of 133. *A separate #8 × 2" box is still wanted for the cleats, cheeks and Panel A's bed-section screw-down — those do NOT take 2½"* | | 1 | **Wood glue, 16 oz** | Titebond II Model 5004 / SKU 100144210 | ✅ — but check the shelf: an 8 oz bottle was bought March 2025 | | ~30 ea | **¼"-20 machine screws + T-nuts**, + 4× ¼-20 bolts/washers/nylocks | fastener drawers | ⚠️ **Length is open — see 1 below** | | 1 | **Biscuits for the Ryobi DBJ50** | HD stocks #0/#10/#20 (A05BIS01/11/21) | ⚠️ **Sizing is open — see 2 below** | | 1 | **Steel flat bar, 2" × 3/16" × 36"** | metal rack | ⚠️ **May not be stocked in 3/16" — see 3 below** | | 2 | **Steel angle, 2" × 2" × 3/16" × 24"** (or one 48") | metal rack | ⚠️ **Same — see 3 below** | | 1 | **⅜" hardwood dowel, 36"** | Waddell 6506U / SKU 100572181 | ⚠️ **Fit is open — see 4 below** | | 1 | **Clear satin spar urethane, 1 qt** | Varathane 9341H / SKU 100141869 | ⚠️ **Probably the wrong product — see 5 below.** Note this line is only about the *deck and slide surfaces*; the **frames are matte black** (BEHR flat, bought 2026-09-06) | **1. The ¼-20 screws are 1" button heads.** They pass down through a riser's 3/16" flange and the ¾" strip into a T-nut flush in the strip's underside, so 1" engages the barrel with nothing sticking out below. **2. The biscuits may not exist in the size this plan names.** The joinery guide specifies Ryobi **R1/R2/R3**, which is the DBJ50's own sizing; Home Depot stocks standard **#0/#10/#20**. Before buying: open the DBJ50's chip guide and see which family it actually cuts. Note the plan needs the **larger** size in one place — the battery drawer has ¾" walls, so those corners want **R3 (or #20)**. Get the assortment if there is one; a joiner that takes both is common. **3. The steel is the one line likely to fail at Home Depot.** Both pieces are specified at **3/16"** wall, and HD's metal rack is usually ⅛" and ¼" with 3/16" hit-or-miss. What substitutes: - **Flat bar for the 2 anchor-board rungs:** 3/16" × 3" (Steelworks 11679 at Lowe's, 48"; Home Depot's 2" Everbilt 801897 was unavailable). ¼" also works if you groove the strips ¼" deep instead; ⅛" is too flexible. - **Angle for the 2 fridge-slide risers:** the model reserves **0.25"** for this (`fridge_riser_t`), described as "3/16" angle + fit allowance", so **either 3/16" or ¼" lands inside the allowance.** ⅛" does not — it drops the fixed rails 1/8" and changes the moving-member overlap on the tray apron. - If neither thickness is on the rack, this is a metal-supply errand, not a Home Depot one. Don't let an associate talk you into ⅛". **4. The dowel needs a test fit, not a purchase decision.** The 4 alignment pins are ¾" long out of ⅜" dowel, friction-fit into ⅜" holes. Nominal ⅜" hardwood dowel usually runs a few thousandths over or under, and both failure modes are bad here (won't seat / won't hold). Buy the stick, then **drill a ⅜" test hole in an offcut before you cut the pins** and sand the dowel to a firm push fit. **5. Spar urethane is the wrong finish for this build — decide before you buy.** Spar varnish is formulated for exterior UV and stays deliberately soft/flexible; every surface here is interior, and softness means it stays tacky against foam and gear. A **satin interior polyurethane** (or a wipe-on poly) is the better match for plywood that things slide across. The SKU above is from the July list; it's the one line in this appendix I'd change on principle rather than on a measurement. #### H.6 — What to expect at the cut desk - **The cut service is a courtesy and the policy is per-store** — some do a couple of cuts free and charge a little after, some are more generous, and any store can decline if the saw is down or the desk is slammed. **Call ahead**, go on a weekday morning, and bring a little cash. - **Expect about ±⅛", and no guarantee of square.** That's exactly why the deck, the front wall and the fridge tray are ordered oversize on the store copy. - **They will not rip narrow strips** (the limit is commonly ~4") and won't cut small offcuts. Every part under 4" in this build is already a shop cut — **H.8**. - **Mark your good face** with a pencil arrow before the first cut, and **write each piece's name on it as it comes off the saw.** Four rectangles of the same thickness on a cart stop being identifiable within a minute, and the panel saw leaves tearout on one face — you want that face down/hidden in the build. - **Don't rotate parts to "save" material.** The cut order on the store copy already puts the grain where the build needs it — in particular the anchor board's face grain has to run along its **33"** dimension. #### H.7 — If they cap the number of cuts The store copy asks for **8 edge trims + 12 part cuts** across two sheets. If you're told there's a limit, spend it in this order and take the rest home: 1. **Sheet 1's S1–S2 and Sheet 2's S1–S4** (6 cuts) — the crosscuts that turn two 4×8 sheets into pieces that fit in the van. This is the part you can't do without. 2. **Sheet 1's T1–T4** (4 cuts) — the deck and the anchor board are both referenced off those edges. 3. **Sheet 1's S3–S4 and Sheet 2's S5–S8** (6 cuts) — the rips, hardest to hold straight with a circular saw. If you can only have two, take **Sheet 1's S4** (the deck) and **Sheet 2's S7** (the front wall). 4. **Sheet 2's trims — skip if you must.** Nothing off that sheet has to be dead square except the front wall, and you're trimming that at home anyway. #### H.8 — Do NOT ask the store to cut these Yours, at home — each is either narrower than a store saw will rip, shaped rather than straight, or has to be fitted to something you'll measure first: - **The anchor board's three strips** — 22½" long, 2½", ~2½" and 1½" wide, ripped and crosscut from the ¾" offcut. Each is one piece; the two steel rungs tie them together (Anchor Board sheet). - **The other two battery-drawer walls** — one **22 × 13.75** side and one **18.5 × 13.75** front, side by side out of the ¾" sheet's 47 × 19 strip (22 + 18.5 = 40.5 of the 47, with 6.5" to spare). Rip the strip to 13.75" wide first, then two crosscuts; the 47 × 5.25 band left over is pantry-cleat stock. All four walls take **2 × R3** biscuits per corner, not R1. - **Panel C deck's two leg notches** — 1.5" × 1.5" out of the front edge, 3.5" in from each side, for the lapped front legs. Jigsaw them after the deck is fitted to its opening, not before. - **Every strip under 4" wide:** the ¾×¾ deck bearer cleats (2× 33" + 2× 40"), the ~1×1 pantry hold-down cleats, the DELTA 3 locating cleats. - **All the ½" narrow parts out of piece E (29 × 26):** rip the 2 hanging cheeks (5.45 × 26) and the 2 drawer sides (4 × 26), then crosscut 17" off what's left and rip the 2 front/back pieces (4 × 17). - **The final trim on the three oversize parts** — deck, front wall, fridge tray — each fitted to its own measured opening or appliance. - **Every hole:** the front wall's 120mm fan and 2 grommet holes, the leg-bottom ½" × 1¾" leveler-insert bores, the biscuit slots, the anchor board's ~30 T-nut holes. - **The steel** — yours, not the store's: two 24" risers from the angle and two 46" rungs from the flat bar. The **DEWALT 4-1/2" cut-off wheels** do the cutting on an angle grinder, **dry**. The **Tap Magic ProTap fluid** is for the *drilling* that follows. Go slow, keep the bit wet, and **deburr every cut edge** before it goes anywhere near a strap or a wire. #### H.9 — Lumber: take it home full length, and cut to this plan **8-ft boards fit in the van** with the rows out (they run forward between the front seats), so **don't** have the store crosscut them: the cut list needs 34 pieces out of the 2×2s at tighter tolerance than a store crosscut holds. The plywood is the opposite — **uncut 4×8 sheets do not go in a Sienna**, which is why the cut desk isn't optional. After the store cuts, the biggest piece coming home is the **46 × 33** anchor board blank and nothing exceeds 47" in any direction, against a **93.75"** measured interior (V1). The 2×2 cut list is **22 pieces totalling 801⅛"** — the twelve legs (12 pieces, 201") are already cut, bored and footed. Nine 8-ft boards hold 864", so the **eleven bought leave two spare**. Allow ~⅛" kerf per cut. Side rails fit *between* the end rails, and every length is sized for **the stock as measured, 1-7/16" rather than 1½"**. | Board | Cut into | Used | |---|---|---| | 1–5 | 46" + 46" each | 92" | | 6 | 46" + 23¼" + 23¼" | 92½" | | 7 | 32⅞" + 32⅞" + 26⅛" | 91⅞" | | 8 | 26⅛" + 26⅛" + 26⅛" | 78⅜" | | 9 | 26⅛" + 26⅛" + 26⅛" | 78⅜" | | 10–11 | — | **two spare boards, untouched** | *Which gives exactly: 11× 46" (6 end rails + 5 bottom rails), 2× 32⅞" (Panel C side rails), 7× 26⅛" (4 Panel A/B side rails + Panel A divider + Panel B's 2-piece center bearer), and 2× 23¼" (Panel B's side bottom rails). All twelve legs are already cut.* **Cut the eleven 46" pieces first, then work to the real openings.** Every 46" piece is a fixed dimension you can cut off a stop block with confidence. Everything else on this list is a *fitted* length: stand each panel's end-rail pair up, measure the actual gap, and cut that panel's side rails to it rather than to the table above (Section 3). Two boards are spare precisely so a mis-measure is not a trip back to the store. **1×4 — 7 boards bought, 6 for the bed (all bed parts cut 14 Oct):** one 1×4 was ripped to 2½" for the fridge-tray aprons instead of the 1×3, so the bed comes out of 6 boards plus the 1×3. Board 1: 42 + 42. Board 2: 42 + 42. Board 3: 42 + 29 + 17½. Board 4: 29 + 29 + 29. Board 5: 29 + 29 + 17½ + 17½ (tight — square one end only). Board 6: 17½ × 5. **1×3 — 1 board:** the last two 17½" slats, 2½" wide, one in the middle of*

*each Panel B half (its gaps open from ~2.9" to ~3.15"). ### Appendix H-S — Store Copy (PDF pages 127–129) The next three pages are written to be handed to someone who has never seen this plan: the order, then one page per sheet with the cuts in the order they must happen and a diagram of where each one falls. **Nothing on them refers to anything else in this document.** Owner-side notes, prices, weights and the reasoning all stay in Appendix H.*

#### STORE COPY — page 1 of 3: the order **Project S'mores — Sienna camper build. Two sheets to cut, plus lumber.** **This is the whole order.** Two plywood sheets to be cut on the panel saw, and lumber I'm taking uncut. I'm picking up hardware, glue and finish myself — nothing on that side needs your help. | Qty | Item | Model / SKU if it helps | Notes | |---|---|---|---| | 1 | **¾" (23/32") birch plywood, 4×8 sheet** | PureBond birch, Model 165921 / SKU 100077837 | **To be cut — page 2.** Please let me pick the sheet off the rack | | 1 | **½" (15/32") birch plywood, 4×8 sheet** | PureBond birch, Model 833185 / SKU 100020218 | **To be cut — page 3.** Please let me pick the sheet off the rack | | 12 | **2×2 × 8 ft pine** | Select pine, Model 532352 / SKU 205031916 | **Uncut** — I'll sort for the straightest | | 7 | **1×4 × 8 ft pine** | Select pine, Model 922484 / SKU 202535808 | **Uncut** | | 1 | **1×3 × 6 ft pine** | same rack as the 1×4 | **Uncut** | **What I'm asking for at the saw — please read this first:** > **First**, before any part cuts, take about **½" off all four factory edges** of each sheet. Every measurement on the next two pages is taken from those trimmed edges, so the trims come first. > > **Then** the numbered cuts on that sheet's page, **in the order numbered.** The finished size of each piece is written next to it, and the diagram shows where each cut falls. > > **Three pieces have to end up exact** — they're marked **CUT ⅛" OVER**. Please cut those about an eighth of an inch big; I'll trim them at home. > > **Nothing narrower than 4"** — I'll rip those strips myself. Same for the small offcuts marked **KEEP**: those are material I'm taking home whole, not scrap. **Two things that would help:** I'll mark the good face of each sheet with a pencil arrow — if the saw tears out one face, please keep the tearout on the unmarked side. And I'll label pieces as they come off, so a moment to set each one aside separately is appreciated.

#### STORE COPY — page 2 of 3: the ¾" sheet **One ¾" (23/32") birch 4×8 sheet.** Usable size after the four edge trims: **47 × 95**. | # | Cut | What it makes | |---|---|---| | T1–T4 | Trim ~½" off all four factory edges | A square **47 × 95** blank. Everything below is measured from these edges | | S1 | Crosscut **33"** off one end | piece **A** = 47 × 33 · piece **B** = 47 × 62 | | S2 | Crosscut **43"** off piece B | piece **C** = 47 × 43 · piece **D** = 47 × 19 → **KEEP** | | S3 | Rip piece **A** to **46"** wide | **ANCHOR BOARD BLANK, 46 × 33** | | S4 | Rip piece **C** to **33"** wide | ⚠️ **PANEL C DECK, 33 × 43 — CUT ~⅛" OVER** (33⅛ × 43⅛) · plus 14 × 43 → **KEEP** | *33 + 43 + 19 = 95 exactly, so the saw kerf comes out of piece D, which is offcut stock — that's fine. Please take S1 and S2 from the same end, in that order.* ![Sheet 1 cut diagram](renders/sheet-cut-1.png)

#### STORE COPY — page 3 of 3: the ½" sheet **One ½" (15/32") birch 4×8 sheet.** Usable size after the four edge trims: **47 × 95**. Four crosscuts first, then four rips. | # | Cut | What it makes | |---|---|---| | T1–T4 | Trim ~½" off all four factory edges | A square **47 × 95** blank | | S1 | Crosscut **26"** off one end | piece **A** = 47 × 26 | | S2 | Crosscut **25"** off what's left | piece **C** = 47 × 25 | | S3 | Crosscut **17"** off what's left | piece **F** = 47 × 17 | | S4 | Crosscut **18"** off what's left | piece **G** = 47 × 18 · plus a 47 × 9 strip → **KEEP** | | S5 | Rip piece **A** to **18"** wide | **KITCHEN DRAWER BOTTOM, 18 × 26** · plus **29 × 26 → KEEP, please don't cut it further** | | S6 | Rip piece **C** to **19¼"** wide, then crosscut to **21¼"** | **BATTERY DRAWER BOTTOM, 19¼ × 21¼** · plus **27¾ × 25 → KEEP** (the lift-out tray comes out of this) and a 19¼ × 3¾ drop | | S7 | Rip piece **F** to **46"** wide | ⚠️ **FRONT WALL, 46 × 17 — CUT ~⅛" OVER** | | S8 | Rip piece **G** to **29"** wide | ⚠️ **FRIDGE TRAY BLANK, 29 × 18 — CUT ~⅛" OVER** (it gets trimmed to final size at home) · plus 18 × 18 → **KEEP** | *26 + 25 + 17 + 18 = 86 of the 95, so the leftover 47 × 9 strip absorbs all four kerfs.* ![Sheet 2 cut diagram](renders/sheet-cut-2.png) ### Appendix I — Hitch Bike Rack: Hollywood Racks Sport Rider (HR1500) **Status: EVALUATED, NOT YET BOUGHT — and it has one hard prerequisite the plan has never established (see "The prerequisite" below).** This appendix exists because e-bikes are the one piece of gear this build cannot swallow. Everything else is inside: the spare at the axle, the pantry prefab, the appliances under Panel C. Two fat-tire e-bikes are ~60–80 lb each and roughly the size of Panel B — there is no interior scheme that carries them without deleting the bed. So they go outside, on the hitch, and this appendix is the honest accounting of what that costs. **Product:** [Hollywood Racks Sport Rider for Electric Bikes, SKU HR1500](https://hollywoodracks.com/products/sport-rider-fat-tire-electric-bike-hitch-rack) — **$529.99**. #### What the rack is rated for | Spec | Figure | What it means for us | |---|---|---| | Bikes carried | **2** | Add-on 3rd/4th bike trays are a different Hollywood model — this one is a 2-bike platform, period | | Weight per bike | **80 lb max** | Comfortably above a typical fat-tire e-bike; **weigh yours** — some cargo/moped-style e-bikes exceed 80 lb and are disqualified outright | | Total bike weight | **160 lb max** | The rack's own limit, separate from the hitch's limit | | Rack weight | **58 lb** | Two-person lift onto the hitch, and 58 lb of dead payload even with no bikes on it | | Receiver | **2" only, Class 2 or 3** | **No 1¼" version.** This is the prerequisite — see below | | Max tire width | **5"** | Covers essentially all fat tires | | Wheel size | **20" – 32"** | Two small-wheel adapters are included for 20" fat-tire bikes | | Wheelbase | **36" – 60"** | Long-tail cargo e-bikes can exceed 60" — measure axle-to-axle before ordering | | Spacing between bikes | **10"** | Wide bars or big panniers may interfere; stagger the bikes front-to-back | | Security | Keyed-alike **locking hitch pin** + **8 ft cable** with locking frame hooks | Locks bikes to rack and rack to van on one key | | Anti-wobble | Tool-free hitch tightening system | Meaningful — a loose receiver is what turns 200 lb of cantilever into a hammer | | Motion | **Folds flat** when empty · **tilts down** for cargo access | The tilt is load-bearing for this build — see the Panel C conflict | | Warranty | Limited lifetime | | **Manufacturer restrictions that apply to us:** no front-hitch mounting · **max 8" of hitch extension** (so no long extender to clear the tailgate — see the conflict below) · not compatible with bike covers · prohibited on trailers or towed vehicles · **step-through frames need the BA-Pro Bike Adapter, sold separately** (most step-through e-bikes do — budget another ~$40–60 and confirm against your frames). #### The prerequisite — does the van have a hitch? **This plan has never established that the van has a receiver, and nothing in it should be read as saying it does.** The Sport Rider needs a **2" Class 2 or 3 receiver**. Before any of the rest of this appendix matters, answer one of two questions: *is there already a receiver under the bumper*, and if so, *what size is the opening and what does its sticker rate it at?* Take a photo of the receiver and its rating label and this appendix can be closed out. If there is no hitch, the common aftermarket options for a 4th-gen Sienna are all Class 3 / 2" and all install without drilling the vehicle frame (fascia and underbody panels come temporarily loose): | Hitch | Class / opening | GTW | **Tongue weight** | Price | |---|---|---|---|---| | Curt C22UR | III / 2" | 3,500 lb | **350 lb** | ~$211 | | etrailer e56RR | III / 2" | 3,500 lb | **350 lb** | ~$236 | | TowSmart TS82VR | III / 2" | 4,500 lb | **675 lb** | ~$232 | Tongue weight is the number that governs here, not gross trailer weight — a bike rack is pure tongue load. **UNVERIFIED and important:** hitch manufacturers rate tongue weight for a *trailer*, where the load sits at the ball a few inches off the receiver. A loaded bike rack puts its mass a foot or two farther back, so the bending moment on the hitch is higher for the same pounds. Read the specific hitch's own instructions for a cargo-carrier / accessory derate before assuming the full 350 lb is available. Also confirm the Sienna is a **Hybrid** (all 4th-gen are) and whether that changes the exhaust/underbody routing on the hitch you pick. #### The weight math — this is the part that bites Two loads, both real: - **Case A — rack at its ceiling:** 2 × 80 lb bikes + 58 lb rack = **218 lb** - **Case B — realistic fat-tire e-bikes:** 2 × 65 lb bikes + 58 lb rack = **188 lb** **Against payload**, the full number counts, every pound, no leverage discount. Appendix F puts the loaded-for-travel van (net build + provisions + 2 occupants, seats out) at **≈ 828 lb** against a working payload assumption of **~1,100 lb**: | | Weight | |---|---:| | Appendix F, loaded for travel | ≈ 828 lb | | + Sport Rider, Case B (188 lb) | ≈ **1,016 lb** | | + Sport Rider, Case A (218 lb) | ≈ **1,046 lb** | | Working payload assumption (VERIFY on the door-jamb sticker) | ~1,100 lb | **That leaves roughly 54–84 lb of margin, down from ~272 lb.** The rack does not break the payload budget — it consumes most of what was left of it. Practical consequence: with the rack loaded, **water becomes the constraint**. A 6-gallon jug is ~50 lb (Appendix F, recommendation 2), so two jugs and a cooler of food put you over. If you carry bikes, carry less water and refill more often, or leave the DELTA 3's Smart Extra Battery (24 lb of the 48 lb stack) at home for bike trips. **Against the rear axle**, leverage applies. A load hung a distance *d* behind the rear axle on a wheelbase *L* adds **(1 + d/L)** of itself to the rear axle and **removes (d/L)** from the front. The Sienna's wheelbase is **120.5"** (the figure Appendix E's leveling calculator already uses). **`d` is UNVERIFIED — measure it:** from the rear axle centerline to the receiver pin is roughly 45" on a 4th-gen Sienna, and the *loaded* rack's center of mass sits farther back still, call it ~60". Both cases: | *d* = 45" (at the pin) | *d* = 60" (loaded CG) | |---|---:|---:| | Lever factor onto rear axle | **1.37×** | **1.50×** | | Case A (218 lb) → rear axle | +299 lb | **+327 lb** | | Case A (218 lb) → front axle | −81 lb | **−109 lb** | | Case B (188 lb) → rear axle | +258 lb | +282 lb | | Case B (188 lb) → front axle | −70 lb | −94 lb | So a loaded rack puts roughly **260–330 lb onto the rear axle and lifts 70–110 lb off the front.** Appendix F already flags this build as rear-biased (44% of build mass in the rear third) and notes that the two front occupants offset it while driving. **The bike rack lands on the same side of the scale as the build's existing bias, and it is the single largest thing that has been added to that side.** Check the **rear GAWR** on the door-jamb sticker — that is the number this can actually violate, before payload does. **Reconciling with Appendix F.** Appendix F deliberately rejected a hitch cargo basket for the spare tire, on exactly this reasoning: "30–60 lb of carrier plus a ~1.4× lever on the rear axle." That call still stands, and the 1.4× figure there is the same lever computed above — this appendix does not overturn it. The difference is that the spare tire *had* an interior home at the axle, which is the best place in the van for it, and the bikes have no interior home at all. The lever cost is accepted here because the alternative is not carrying bikes; it was rejected there because the alternative was strictly better. **The two decisions are consistent, and stacking them is not** — do not put the spare on a hitch basket *and* the bikes on the hitch rack. #### The one real design conflict: Panel C's tailgate access **This is the thing to resolve before buying, and it cannot be resolved from a catalog — it needs a tape measure on the actual van.** This build's entire rear third opens out the back. Both Panel C appliances — the BougeRV Rocky 40 fridge and the JAGAHAHA camp kitchen — **pull straight out the open tailgate on their slides** (Section 6, Component 7). Appendix G's tent cooling setup runs with the tailgate open and the WAVE 3 blowing out the back. A loaded bike rack sits directly in all of that. The Sport Rider tilts down for cargo access, which is the feature that makes this survivable, but three things have to be checked with bikes actually on it: 1. **Does the power liftgate clear the tilted rack?** Tilt-down clearance is quoted for a *trunk*, not a minivan liftgate that swings up through a long arc. If the liftgate strikes the tilted bikes, the power mechanism will fight it. **Measure with the rack tilted and bikes mounted, and open the liftgate by hand the first time.** 2. **Can the fridge and kitchen still slide out over the tilted rack?** They extend rearward on 24" slides at roughly deck height. The tilted rack and bikes occupy the ground space immediately behind the bumper. This may work fine, or it may mean the bikes have to come off before dinner — which is a real usability cost worth knowing about in advance, not in a campground at dusk. 3. **The 8" extension limit closes the easy escape.** The obvious fix — a hitch extender to push the rack back until everything clears — is capped by Hollywood at **8" of extension**, and every inch of extension multiplies the bending moment on both the hitch and the rack. 8" is not enough to solve a liftgate interference, and going past it voids the rack's rating. Two secondary items: **the bikes will obscure the license plate and rear lights**, which is a ticketable offense in many states — plan on a light board or a relocated plate; and **the rear parking sensors / backup

*camera** will see the rack permanently, so expect nuisance alerts and a partly blocked camera view. #### Cost | Item | Cost | |---|--:| | Hollywood Sport Rider HR1500 | $529.99 | | BA-Pro Bike Adapter (if either bike is step-through) | ~$40–60 | | Class 3 2" hitch, **if the van does not already have one** — parts only | ~$210–236 | | Hitch installation (if not DIY) | ~$100–200 | | Light board / plate relocation (if bikes obscure the plate) | ~$30–60 | | **Total, worst case (no existing hitch, step-through frames, installed)** | **~$910–1,090** | #### Open items before ordering 1. **Photograph the rear bumper area and any existing receiver, plus its rating label.** Nothing else in this appendix is actionable until this is answered. (2" opening required — a 1¼" receiver rules this rack out entirely.) 2. **Weigh both bikes** and measure each one's **wheelbase (36–60")** and **tire width (≤5")**. Confirm whether either frame is step-through. 3. **Read the door-jamb sticker** for actual payload *and* **rear GAWR** — this appendix's ~1,100 lb is Appendix F's working assumption, not a measurement. 4. **Measure `d`** — rear axle centerline to receiver pin — and redo the lever table with the real number. 5. **Test the tailgate and both Panel C slides with the rack tilted and loaded**, before the first trip rather than on it. 6. **Re-aim the headlights** after the rack is on and loaded — this adds to the rear squat Appendix F already calls for re-aiming after. **Not rolled into Appendix F's totals on purpose.** The 218 lb is a bolt-on carried some trips and not others, so folding it into the ≈488 lb build total would misstate the van's permanent weight. If you decide to make the rack permanent, add it to `weight_budget.py` as a `rear` zone line and re-run — the fore-aft center of mass and the rear-third share will both move noticeably.*

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# DELTA 3 Power Drawer

Panel A, passenger-side bay. The ¾" plywood box that carries the EcoFlow DELTA 3 Plus + Smart Extra Battery on a pair of 20" full-extension slides.

| ASSEMBLY | BOX, OUTER | BOX, CLEAR | LOAD |
|---|---|---|---|
| **Comp. 2 · step 3–4**<br>with Component 8, step 1 | **20 × 22 × 13¾"**<br>W (travel) × D (fore-aft) × H | **18½ × 20½ × 13¼"**<br>inside the ¾" walls | **~78 lb**<br>48 lb stack + box + contents |
| **SLIDES** | **JOINERY** | **DRAWN FROM** | **STATUS** |
| **20" full-ext.**<br>100 lb class pair, ~22% margin | **2 × R3 + glue**<br>per corner; bottom in a rabbet | **Build plan**<br>Sections 3, 6 · params.scad | **Built**<br>bay set by the legs' wood length |

## A What sets the sizes

2 rules · 1 adjustment

AS BUILT **No locating cleats, and the drawer runs along the 22" walls**

The units sit on a rubber non-slip mat instead of cleats (owner, 30 Oct 2026), held down by the cam strap into D-rings ~7½" down from the box's top edge, and a turn button on the frame's top rail keeps the drawer shut. Fig 1 below is drawn turned 90°: as built, the slides are on the 22" side walls and the drawer pulls out along them, with the two units side by side across the 18½" width. **The Power Drawer Tie-Down sheet** draws it as built.

RULE **The walls are as tall as the bay: 13¾"**

Panel A's bay runs from the top of the bottom rail to the underside of the top rail. The bottom rail hangs off the leg's wood bottom and rises with it, so the bay is set by the leg's wood length L alone:

```
bay = L − 1.5 (top rail) − 1.5 (bottom rail) = L − 3 · L = 16¾" → 13¾"
```

**If the legs ever measure 16½", the bay is 13½": walls 13½", 1.84" of air over the stack, and the lift-out tray 1½" or less. Measure the bay opening itself; the feet don't change it.**

RULE

### The 22" side walls run full depth; the front and back fit between them

On a 20 × 22 outer box only one pair can run the whole way. The side walls do, because the slides screw to their outer faces and a slide wants one continuous wall under it, not one interrupted by end grain at both ends.

```
22 − 2(¾) = 20½ clear depth · 20 − 2(¾) = 18½ clear width · front/back = 20 − 1½ = 18½"
```

**Sides 2 @ 22 × 13¾, front/back 2 @ 18½ × 13¾. Cut the side walls first and trim the front/back pair to the box they actually make: ¾" ply runs over and under its nominal thickness.**

ADJUST

### Set Panel A's deck plane with the feet

As built, the frame reads 16" from the floor to the rail bottom, so the rail top sits at 17½" and the bed platform lands at 18¼", ¼" below Panel C's fixed 18½" deck. **Turn all four feet out ¼"** to remove the step at the B/C seam. This only moves the deck plane: the bay, and so the drawer, are unaffected.

## B Plan view

looking down into the open box · 1:1 proportions

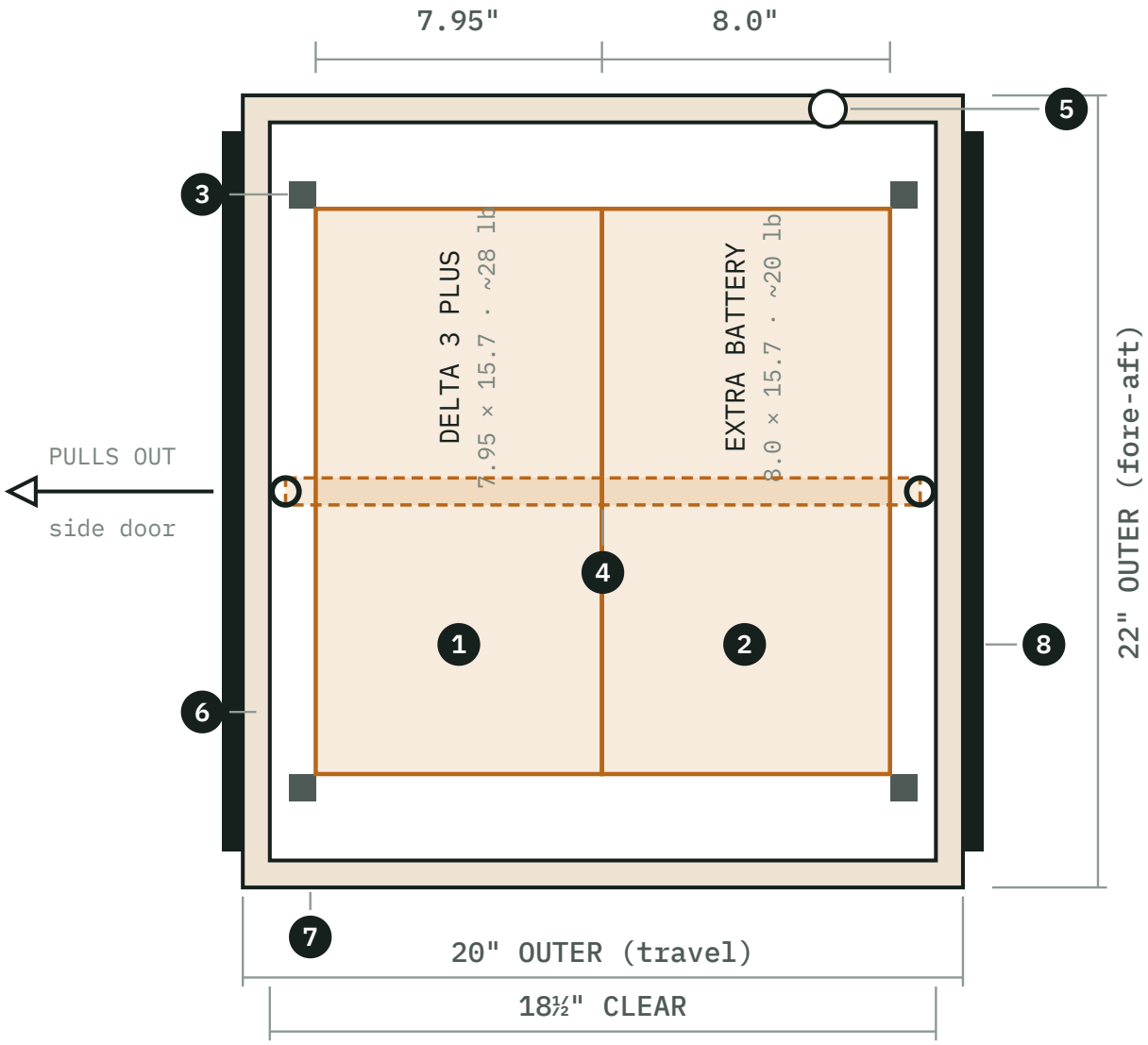



**FIG 1 — PLAN, BOX OPEN**

DELTA 3 Plus outboard against the pull wall (it is what the WAVE 3 and the fridge line plug into), Smart Extra Battery inboard. 2.55" of spare width and 4.8" of spare depth around the pair.

1. **EcoFlow DELTA 3 Plus** — 7.95 × 15.7 × 11.16", ~28 lb. Outboard, at the pull wall: it carries the 12V outlet for the fridge line (P1) and the AC input.
2. **Smart Extra Battery** — 8.0 × 15.7 × 7.8", ~20 lb. Inboard — runtime only, reached far less often.
3. **Locating cleats ×4** — 1×1 pine blocks, glue + 1¼" screws. Bracket the 15.95 × 15.7" footprint at its corners so the stack cannot walk on the slide.
4. **Cam strap + 2 screw-eye D-rings** — strap runs across the 20" width, over both units, into a D-ring at each side wall.

| | |
|---|---|
| 5 | **1" cable grommet** — in a *fore-aft* wall, not a side wall: the side pair's outer faces are where the slides land. Leave slack for full extension. |
| 6 | **Side wall ×2 — 22 × 13¾"**, ¾" ply. Full depth, continuous under the slide (Flag 2). |
| 7 | **Front/back wall ×2 — 18½ × 13¾"**, ¾" ply. Between the side walls (Flag 2). |
| 8 | **20" full-extension slide pair**, 100 lb class. One side to the frame's outer rail, the other to Panel A's center divider. |

## c Section & corner detail

where the height goes

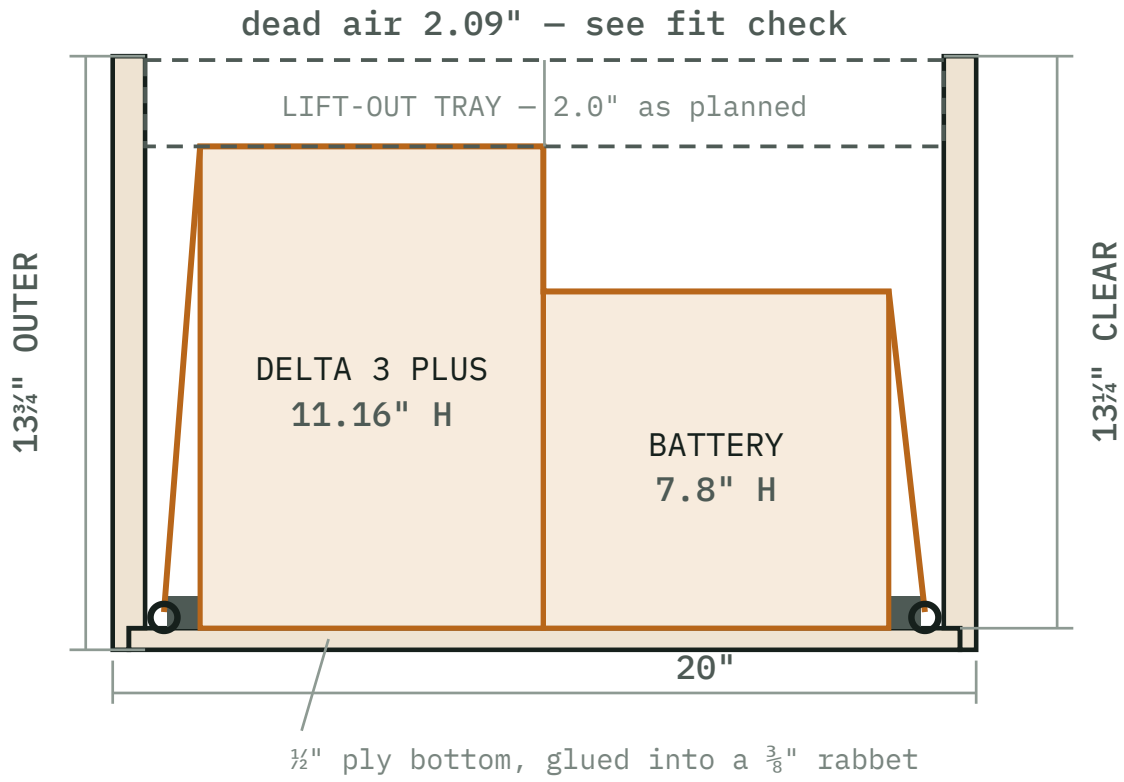



**FIG 2 — SECTION, LOOKING FORE-AFT**

The whole height budget: ½" of it goes to the bottom, 11.16" to the DELTA 3 Plus, and what's left is the tray's. The strap drapes over both units, *under* the tray — so it eats into the same 2.09".

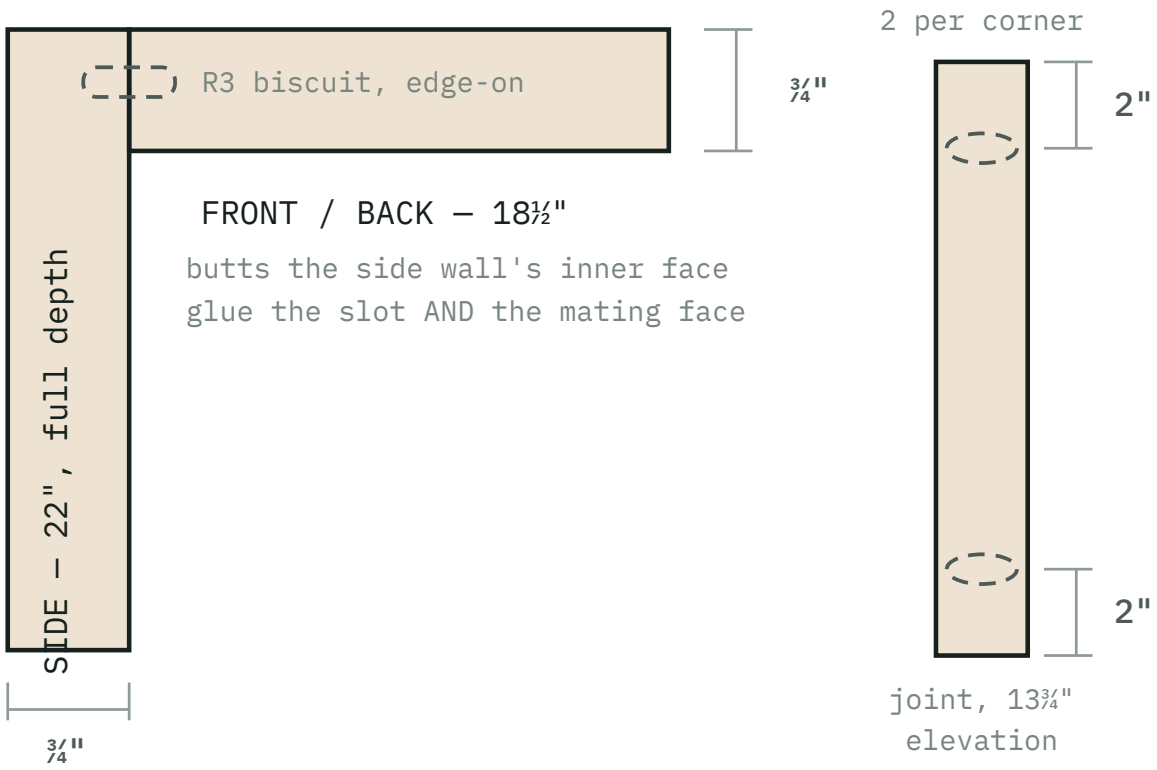



**FIG 3 — CORNER JOINT, ×8**

Butt corner, **2 × R3 biscuits + glue** — R3, not R1: R1 is thin-stock sizing and is lost in ¾" ply. Biscuit heights follow the plan's own rule, 2" in from each end of the joint.

## D Where it sits

Panel A, looking forward

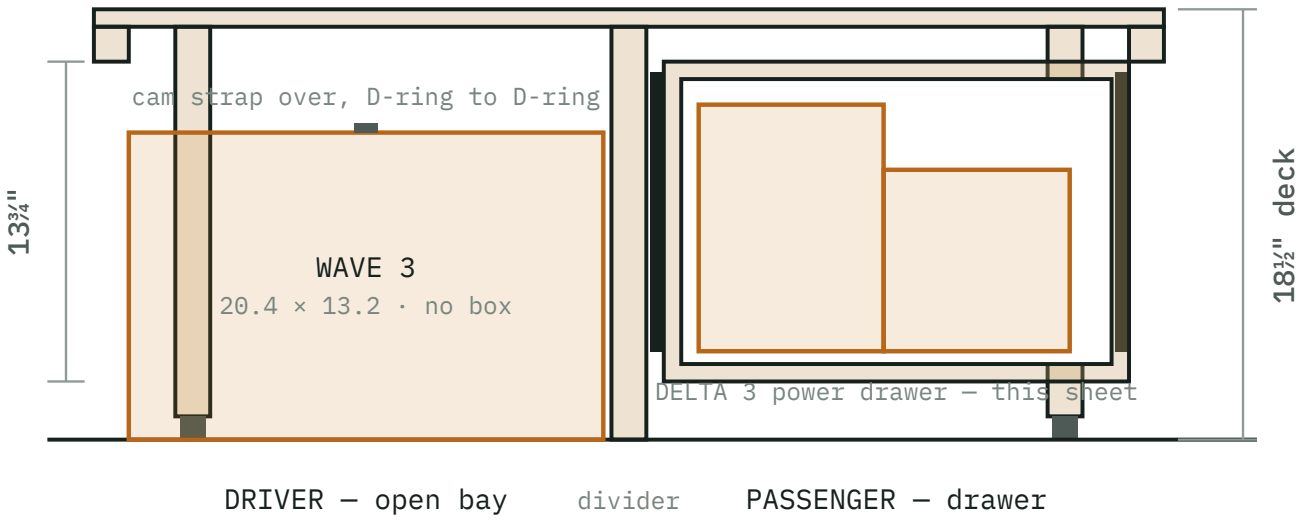



**FIG 4 — PANEL A, FRONT ELEVATION**

Schematic, to scale in both axes. The drawer is the only actual drawer in Panel A: the driver bay has no box at all, because the WAVE 3's 20.4" width will not go inside an 18½" clear one. The two heavy items sit on opposite sides and counterbalance each other.

## E Cut list

5 parts · walls from the ¾" sheet

| PIECE | QTY | CUT TO | STOCK & SOURCE |
|---|---|---|---|
| **Side wall** full depth, carries the slide | 2 | 22 × 13¾ | ¾" Baltic birch — one from the 47 × 19 strip, one from the sheet's remaining offcut (mind the grain) |
| **Front / back wall** between the side walls — trim 1½" off each | 2 | 18½ × 13¾ | ¾" Baltic birch — one from the same 47 × 19 strip, one from the sheet's remaining offcut. Cut at 20" with the rest, then take 1½" off each in one cut once the side walls are made |
| **Bottom** into a glued ⅜" rabbet | 1 | 19¼ × 21¼ | ½" Baltic birch — **piece C** of the ½" sheet (S2 crosscuts 25" off the trimmed 47 × 95 blank; rip it to width at S6). It sits in a ⅜" rabbet, which takes ¾" off the 20 × 22 outer footprint each way. Cut it last, off the glued-up box. |
| **Lift-out tray, if built** same sheet, same piece | 1 | 18¼ × 20¼ base | ½" birch — out of piece C's own 27¾ × 25 keep, right beside the bottom. Walls ≤1¾" tall. A bought shallow bin does the same job. |
| **Locating cleats** | 4 | small blocks | 1×1 pine, off the frame stock |

**Order of operations:** one side wall and one front/back wall come from the 47 × 19 strip, the other two from the ¾" sheet's remaining offcut. Rip the 47 × 19 strip to the wall height first, then crosscut; the leftover band is pantry-cleat stock.

## F Fit check

every number traced to its source

| AXIS | AVAILABLE | NEEDED | SPARE | WHERE IT COMES FROM |
|---|---|---|---|---|
| Width | 18½" clear | 15.95" | **2.55"** | DELTA 3 Plus 7.95 + Extra Battery 8.0, side by side |
| Depth | 20½" clear | 15.7" | **4.80"** | Both units share the same 15.7" depth |
| Height, to the stack | 13¼" clear | 11.16" | **2.09"** | 13¾ box − ½" bottom; the taller unit is the Plus |
| Height, with the tray | 2.09" dead air | 2.0" | **0.09"** | The cam strap runs under the tray, so a 2" tray leaves almost nothing. **Build the tray 1¾" or shallower.** |
| Load | 100 lb pair | ~78 lb | **~22%** | 48 lb of EcoFlow + ~12 lb of ¾" box + contents |

## G Assembly order

Component 2 step 3 → Component 8 step 1

| | |
|---|---|
| 01 | **Cut the five pieces**<br>Walls to the bay height — **13¾"**, or 13½" if the bay measures that (Flag 1). Side walls 22", front/back **trimmed 1½" to 18½"** (Flag 2). Bottom 19¼ × 21¼ out of the ½" sheet, cut last to fit the rabbets. Label each piece the moment it comes off the saw — the two from the offcut may come off rotated and the grain direction is not recoverable by eye later. |
| 02 | **Rabbet the four walls for the bottom**<br>⅜" deep × ½" wide along the bottom inside edge of all four walls, so the ½" bottom lands flush with the box's underside. Cut these before the biscuit slots — it is far easier on a loose board. |
| 03 | **Slot the corners — 2 × R3 each**<br>Eight slots in total, centered on the ¾" thickness, 2" in from each end of the joint. **R3, not R1:** R1 is thin-stock sizing and disappears in ¾" ply. Set the DBJ50's fence to 90° and check the depth on an offcut first. |
| 04 | **Dry-fit the whole box**<br>All four walls plus the bottom, no glue. Measure both diagonals — they should match. This is the last cheap moment to find out that the wall height or the corner convention is off. |
| 05 | **Glue up**<br>Glue every biscuit slot *and* the mating faces, and the bottom's rabbet all the way round. Clamp, re-check the diagonals before the glue grabs, and leave it. The bottom is structural here — it is what stops a 48 lb load racking the box on the slide. |
| 06 | **Hang it on the slides**<br>20" full-extension pair: one slide to the frame's outer rail, the other to Panel A's center divider, both on the *side* walls' outer faces. Run it out fully and back a few times loaded with a sandbag before you trust it with the batteries. |
| 07 | **Add the catch**<br>Friction catch or a small turn latch. Not optional — 48 lb rolling itself open on a corner is the failure that takes the slide with it. |
| 08 | **Fit the drawer's fixtures**<br>Four 1×1 cleats glued and screwed to the floor, bracketing the 15.95 × 15.7" footprint. Two screw-eye D-rings at the side walls on the strap's line. The 1" grommet goes in a **fore-aft** wall — the side walls' outer faces belong to the slides — with enough cable slack for full extension. |

| | |
|---|---|
| 09 | **Load and strap**<br>DELTA 3 Plus outboard at the pull wall, Smart Extra Battery inboard, both down over the cleats. Cam strap across the width, through both D-rings, cinched. Shallow tray on top last, if you build one. |
| 10 | **Connect it**<br>Fridge DC line to the Plus's 12V car-power outlet via its fused car-plug→SAE adapter (connection point P1), and the AC charging cord to the AC input. From here the drawer runs the fridge and fans continuously and recharges off the front console while you drive. |

---

Drawn from `sienna_camper_build_plan.md` (Sections 3 and 6, Components 2 and 8) and `params.scad`. Figures are to scale in both axes; they are assembly drawings, not templates — do not scale off a print.

**Model not yet matched:** `params.scad` carries `delta3_tray_h = 2`, and `renders/sheet-cut-2.svg` draws a 20 × 22 bottom on the ½" sheet. Where they differ from this sheet, this sheet is right.

# Power Drawer Tie-Down

The DELTA 3 drawer drawn from three directions, showing what holds the batteries in the box and what holds the box in the frame. Drawn as built from the bench photos: the slides are on the 22" side walls, and the drawer pulls out along them.

| | | | |
|---|---|---|---|
| BOX, OUTER<br>**20 × 22 × 13¾"**<br>across × travel × height | UNITS<br>**side by side**<br>both 15.7" long, in the direction of travel | GRIP<br>**rubber mat**<br>no cleats (owner, 30 Oct) | HOLD-DOWN<br>**1 cam strap**<br>2 D-rings, side wall to side wall |
| KEEP SHUT<br>**turn latch**<br>on the frame's top rail, over the pull wall | SCALE<br>**16 px = 1"**<br>all three views | | |

## A Top view

looking down into the open box · side door at the bottom

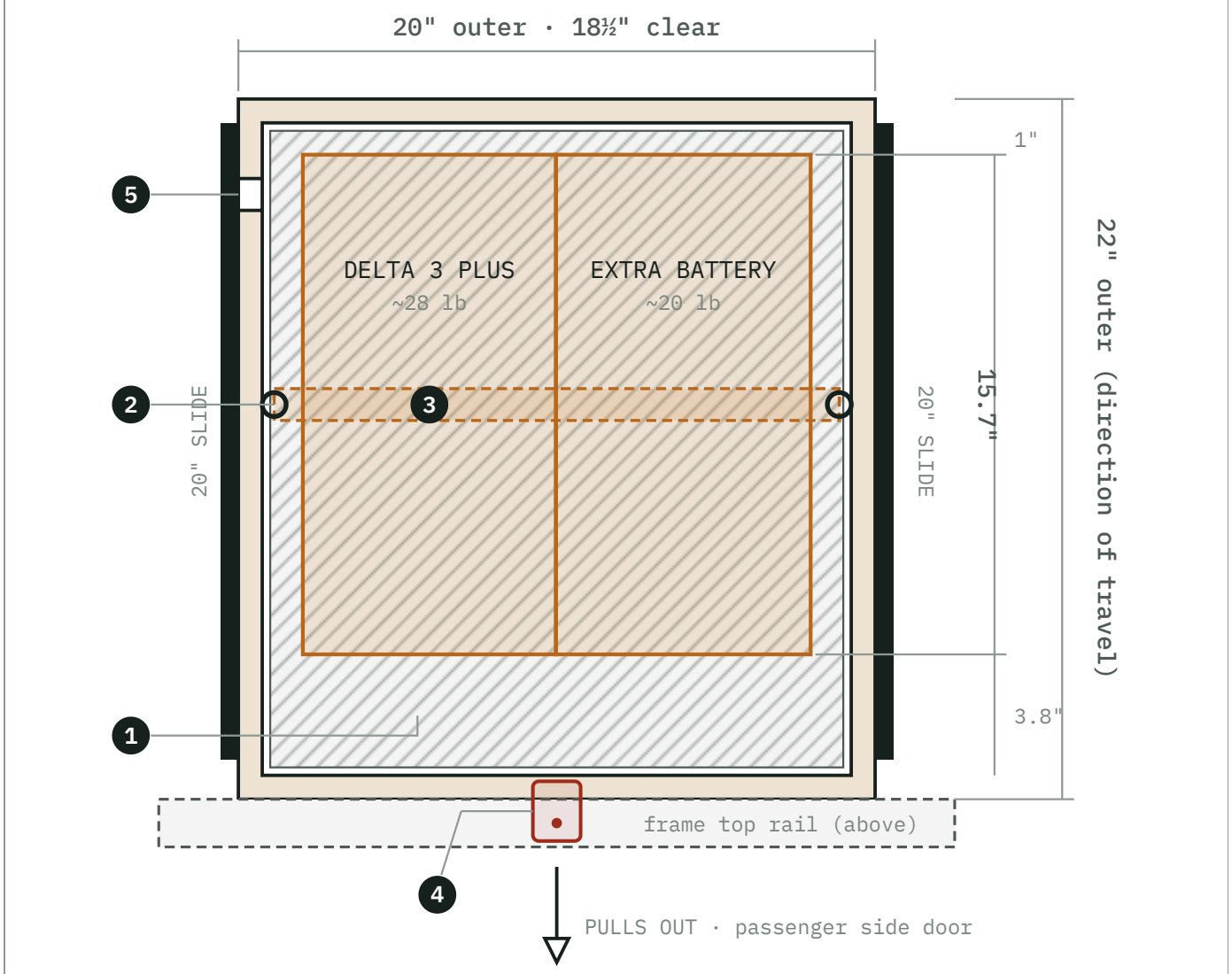



**FIG 1 — TOP VIEW**

Both units run the full way along the direction of travel. Nothing boxes them in now that the cleats are gone. The mat stops them sliding, and the strap pins them down across the width.

## B From the side door

cut through the box at the strap line

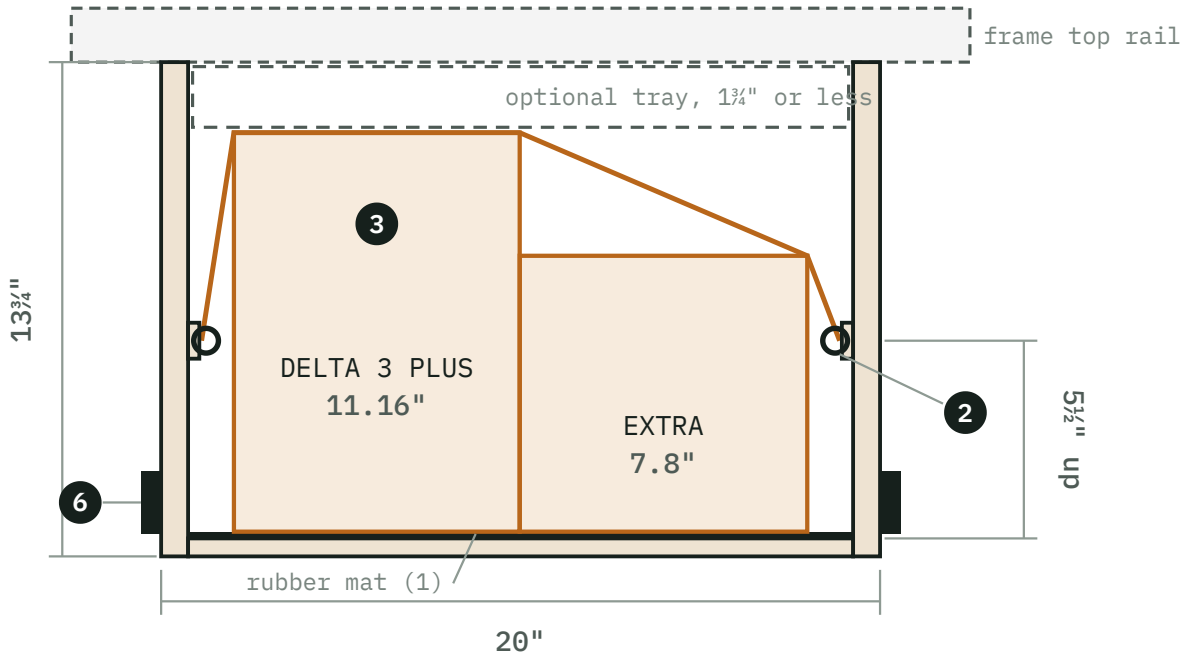



**FIG 2 — LOOKING IN FROM THE SIDE DOOR**

The units are different heights, so the strap pulls down on the whole DELTA top but only on the Extra Battery's outer top corner. The mat does the rest. Keep each D-ring high on its wall, at about 5½", where it's clear of the slide's screws on the other face.

## C From the side

cut along the direction of travel, through the DELTA 3

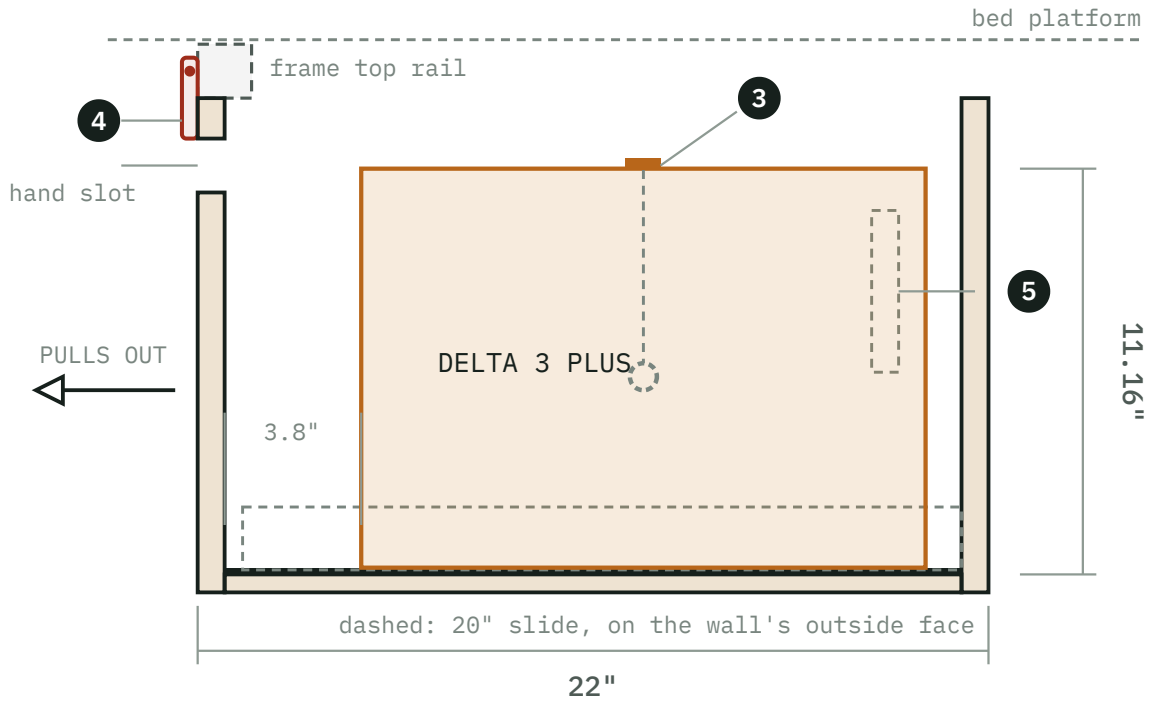



**FIG 3 — SIDE SECTION**

The drawer's own weight can't keep it shut, because the slides run free on a bend. The turn latch on the top rail's outer face swings down over the pull wall's top edge. Turn it 90° to open the drawer.

1. **Rubber mat**, full floor, as built. No cleats: the mat's grip, plus the strap's down-pull, keeps the units from walking.

2. **D-rings ×2**, one on the inside face of each side wall, about 5½" up: mark the screw center ~7½" down from the box's top edge. #8 × ⅝" screws, 3/32" pilot. The wall is ¾" and the slide is screwed to the other face, so keep the screws ⅝" or shorter and well above the slide's band.

3. **Cam strap**, across the 18½" width, over both units. Run it under the tray if you build one. Cinch it before the van moves.

4. **Turn latch** on the frame's top rail, tongue down over the pull wall. This keeps the drawer shut. It's separate from the strap and not optional: 78 lb rolling open on a bend takes the slides with it. A friction catch also works. Bought: Wright Products V20A 1¾" (screw ~⅜" above the rail's bottom edge so it reaches ~1" onto the wall), or make a 3" × ¾" ply tab on a #8 × 1½" screw.

5. **Cord slot**, as built, near the back of the left side wall. Leave slack for full extension.

6 **20" slides**, low on the side walls' outer faces. For reference only: they're the reason the D-rings go high.

---

Drawn 30 Oct 2026 from the bench photos (drawer-05 to drawer-08). The DELTA 3 is on the cord-slot side, as photographed. Which side wall faces the cab isn't recorded, so the views say "side wall" rather than fore or aft.

# Panel B Frame

The middle module: the full 2×2 cube over the rear axle, with no divider and no drawers. It carries the two lift-out bed halves on a center bearer and holds the spare tire and totes underneath.

| ASSEMBLY | FRAME, OUTER | RAIL TOP | BAY, CLEAR | STOCK | SCREWS |
|---|---|---|---|---|---|
| **Comp. 3 · step 1**<br>+ center bearer, Comp. 2 / Phase 6 | **46 × 29"**<br>W (across van) × L (fore-aft) | **17¾"**<br>floor to rail top, feet set · match Panel A | **13⅞" tall**<br>bottom-rail top → top-rail underside | **1-7/16" 2×2**<br>measured · sets every between-length | **32 × 2½"**<br>SPAX #8 Torx · from the 133 box |
| DRAWN FROM | STATUS | | | | |
| **Build plan**<br>Sections 3, 6 · params.scad · Panel A as built | **Built**<br>14 Oct: frame, braces, loops and drop-in bearer done · strap relief notches cut | | | | |

## A Check before you cut

2 methods · 1 carry-over

Panel B is Panel A's frame with bottom rails on all four faces and no divider, so most of it is a repeat of what you already built. Three things are new: the bearer goes in drop-in, the bottom side rails sit higher than Panel A's, and the rail top gets set against Panel A.

METHOD **The center bearer drops in, so the spare tire can get past it**

The center bearer (two 2×2s side by side, at the centerline) splits the top opening, the only way into Panel B, into two holes:

```
(46 − 2 × 1-7/16 − 2⅞ bearer) ÷ 2 = 20⅛" wide × 26⅛" fore-aft · each
```

The spare is a 28½" disc, 6.4" thick. To pass through a hole it has to go through edge-on, and at 6.4" thick the largest disc that fits a 20⅛ × 26⅛" hole at any angle is **27.0"**. The spare is **1½" too big**. With the bearer fixed in place, the spare couldn't be loaded or pulled out in an emergency, and a screwed bearer can't be undone in the van: its screws would go in through the end rails' outer faces, tight against Panels A and C.

**So the bearer is drop-in. It rests on four 1" steel corner braces (L-brackets) screwed to the end rails' inner faces, as in Fig 4. It carries the halves exactly as before, since the load is straight down. Take the halves off, lift the bearer out, and the full 43⅛ × 26⅛" opening is back. That passes discs up to 45". No tools needed.**

**Why 1" braces and not 1½":** the bearer sits on the brace's flat leg, so a brace set flush with the rail's underside leaves the bearer proud by the steel's thickness (about 1/16"). The bed halves would rock on it. And a 1½" brace's upright leg is taller than the 1-7/16" rail, so its tip can reach the rail top. Fit the braces with the frame **upside down** (step 09): the bearer's top and the rail tops both lie on the floor, so they come out flush whatever the steel is.

**While you're checking the spare:** lying flat it's 28½" against 26⅛" between the end rails. It fits only because it sits on its 3" cleats, above the bottom rails, and its edge can pass under the top end rails. That leaves ¼" to the frame's outer face at each end, so center the skid fore-aft.

METHOD

### The bottom side rails sit 1¾" up the legs, so their screws miss everything

Unlike Panel A, Panel B has side bottom rails as well as end ones. The side rails butt into the legs' inner faces between them, and their only fixing is through the leg from its outer face, fore-aft, the same direction as the end rails' screws.

At the leg bottom, four screws would share the lower 1-7/16" of each leg with the ½" leveling-insert bore. So the side pair sits with its underside **1¾" above the leg bottom**, just above the top of the bore. Its screws land about 2½" up, in solid wood, with nothing else nearby. The end rails stay at the leg bottom, exactly as on Panel A.

```
end-rail pair: ~¾" up, ⅜" either side of center (as Panel A) · side-rail pair: ~2½" up, ⅜" either side of center · any order
```

**Why not raise the 46" end rails instead:** the spare is 28½" across and there's only 26⅛" between the end rails, so its rim overhangs the bottom end rails at both ends. It clears them only because it rides on 3" cleats over their 2-7/16" tops. Any higher and the spare can't lie flat. The side rails are 4" outboard of the spare's edge, so raising them costs nothing.

**Set the side rails with a 1¾" spacer block under each end, and pilot every hole.**

CARRY-OVER

### Set the rail top to Panel A, not to the tape measure

Panel A reads 16" from the floor to the rail bottom as built, so its rail top is about 17½". The deck plane needs 17¾" so the bed lands at 18½", flush with Panel C. Both panels share one mattress, so **Panel B's rail top has to match Panel A's**, whatever that turns out to be. Set Panel B's feet with the two frames side by side on the garage floor and a straightedge across the seam. Then raise both together by the same turn.

The bay height doesn't change with the feet: `bay = L − 2 × 1-7/16 = 13⅞"` for the 16¾" legs. The spare stack is 15.3" from the floor, which clears the top rail's underside by about 1" with the feet at 1".

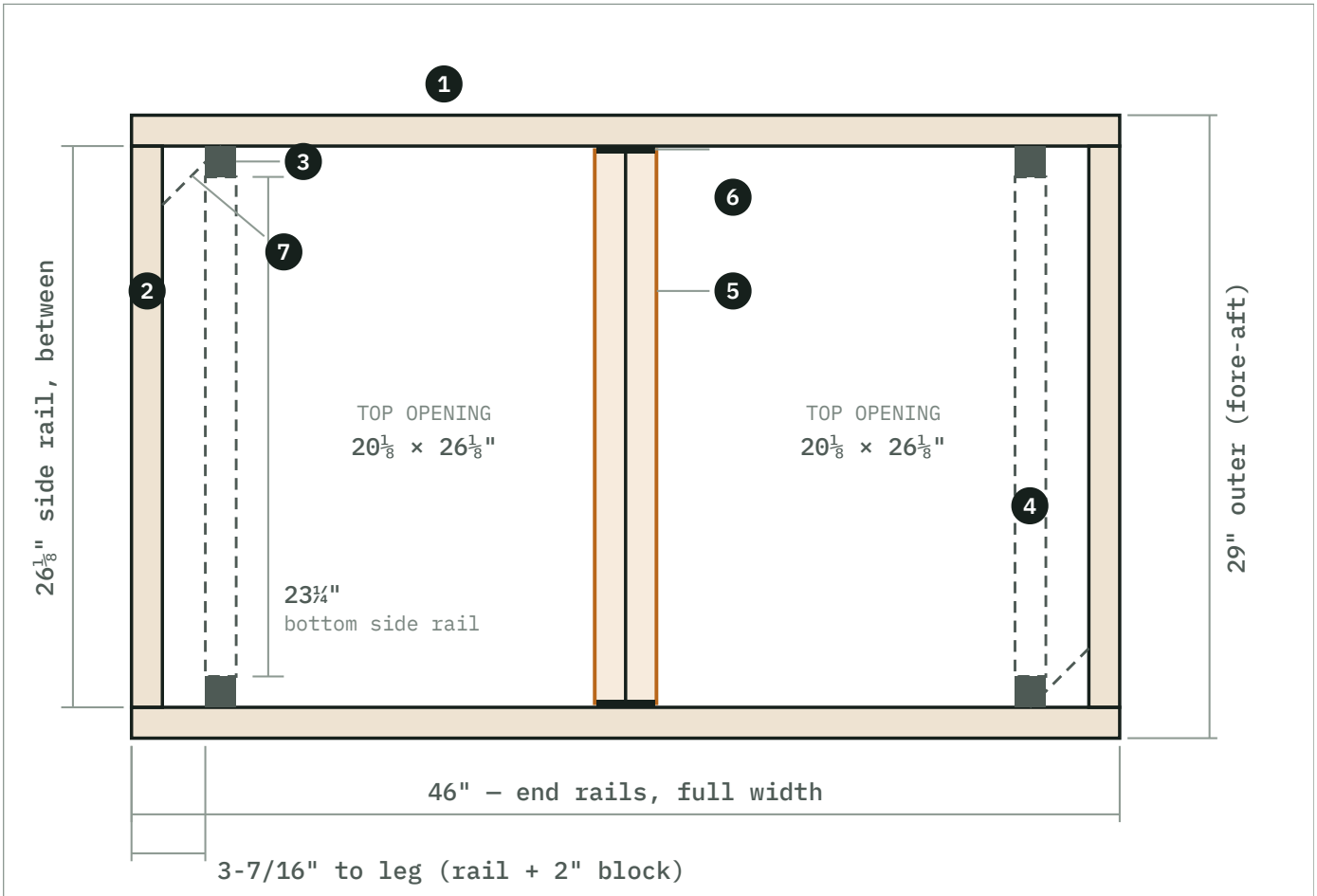



**FIG 1 — PLAN**

The end rails are the outboard pair and set the 46"; everything else fits between them and is cut to measure. The 46" bottom end rails sit directly under the top end rails, so they hide in this view. The dashed pair between the legs are the bottom side rails.

1. **End rail ×2 — 46"**, 2×2. The outboard pair: the legs lap against their inner face and the corner screws go through their outer face.
2. **Side rail ×2 — 26⅛"** (29 − 2 × 1-7/16). Flush with the frame's outer edge. Stand the end rails up at 29" outer and cut to the gap you measure.
3. **Leg ×4 — 16¾"**, already bored and footed. Lapped to the end rail's inner face, top flush with the rail top, and set with the 2" block against the side rail.
4. **Bottom side rail ×2 — 23¼"** (dashed). Between the legs, on the legs' line, underside 1¾" above the leg bottoms, clear of the foot bore. Cut to the measured gap once the legs are on.
5. **Center bearer — 2 × 2×2, ~25⅞"**, glued face to face into one 2⅞" beam. Top flush with the rail tops. Drop-in on the corner braces (Flag 1), cut to the measured gap between the end rails less ¼".
6. **Corner braces ×4 — 1" zinc L-brackets**, two at each end. Screwed to the end rails' inner faces with the flat leg pointing inboard, set against the bearer with the frame upside down. One under each 2×2 of the bearer.
7. **Diagonal braces ×2 — 4" flat strap** (dashed), on the underside at two opposite corners, as on Panel A. About 2¾" along each rail, with the screws clear of the corner bracket's.

## C End elevation

looking fore-aft at the B/C face · to scale

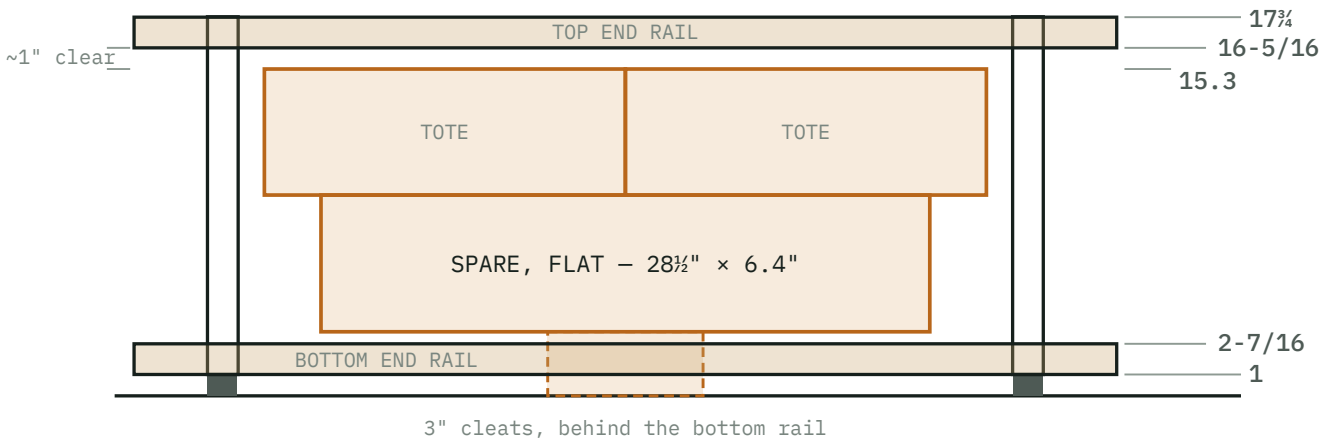



**FIG 2 — ELEVATION, FEET AT 1"**

Heights in inches from the van floor. The legs stand behind the end rails because they're lapped inboard. The spare's 3" cleats clear the 2-7/16" bottom-rail curb by about ½". If the feet go out much past 1¼", re-check that the curb still clears.

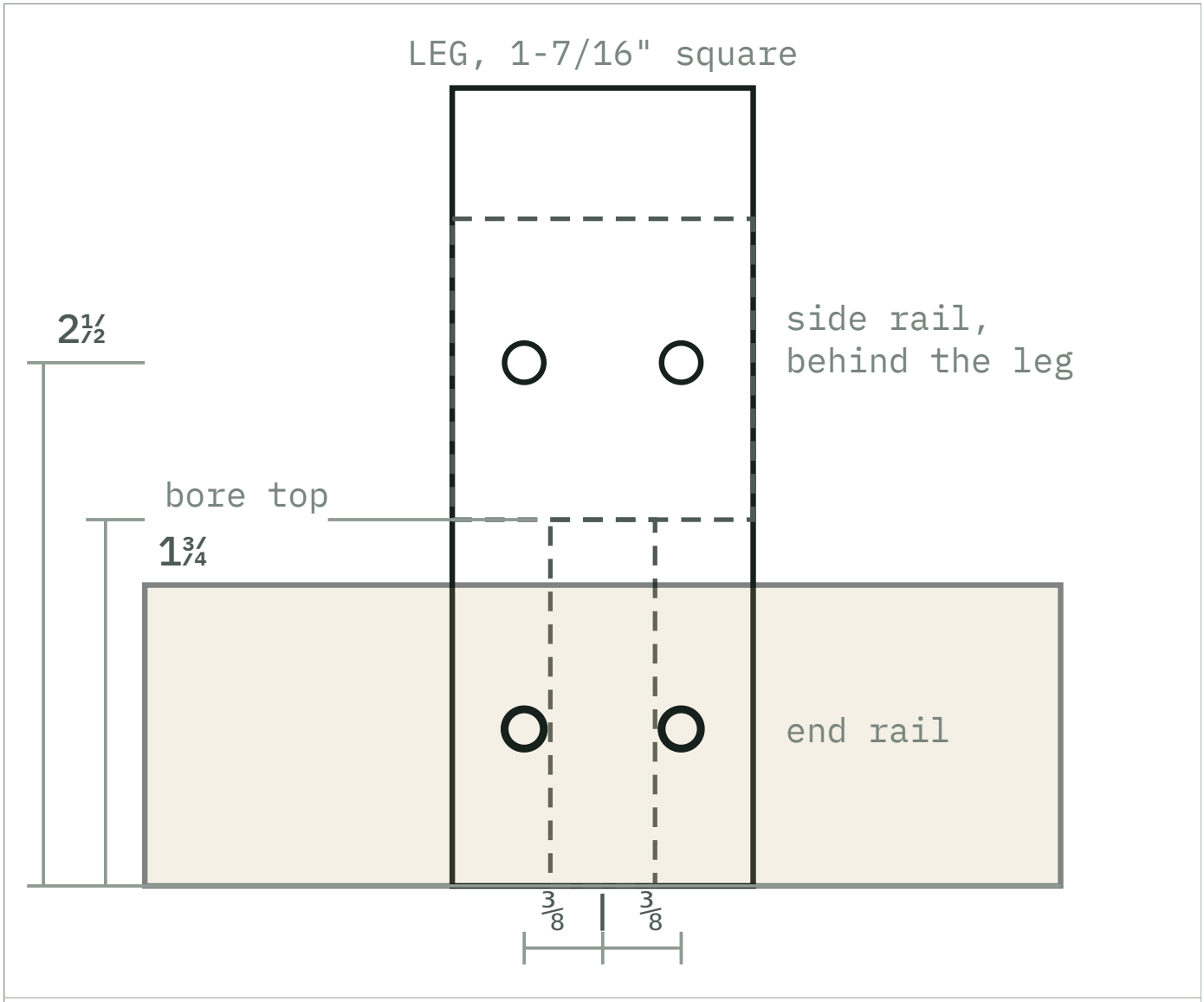



**FIG 3 — LEG BOTTOM, FROM OUTSIDE**

Heights from the leg's bottom end. The shaded band is the bottom end rail, drawn see-through, and its two screws sit either side of the ½" insert bore. The side rail butts the leg's far face from 1¾" up, just above the bore, and its two screws come through the leg about 2½" up. The two pairs are about 1¾" apart, so either can go in first.

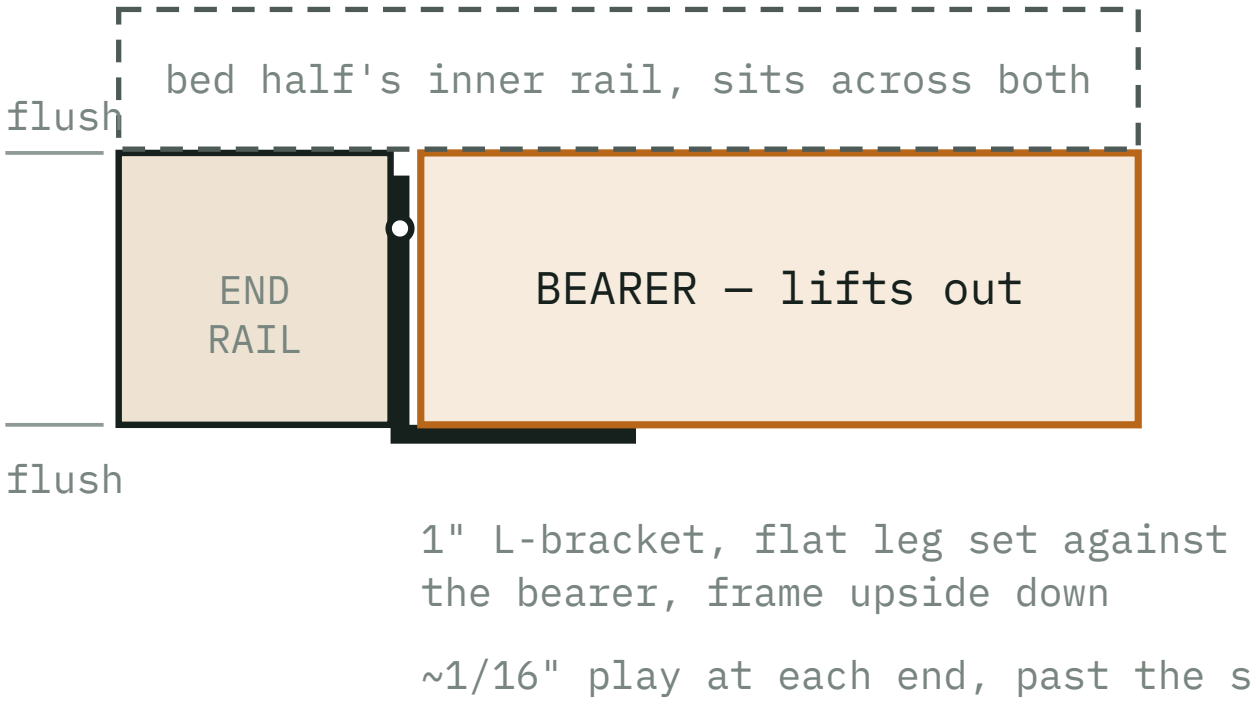



**FIG 4 — DROP-IN BEARER, SECTION AT AN END**

The bracket's upright leg is screwed to the end rail and its flat leg is the ledge. The bearer is cut to the measured gap less ¼". The braces' steel takes about ⅛" of that, so it drops in and lifts straight out. Nothing is screwed into it. The bed halves hold it down, and nothing else can move it.

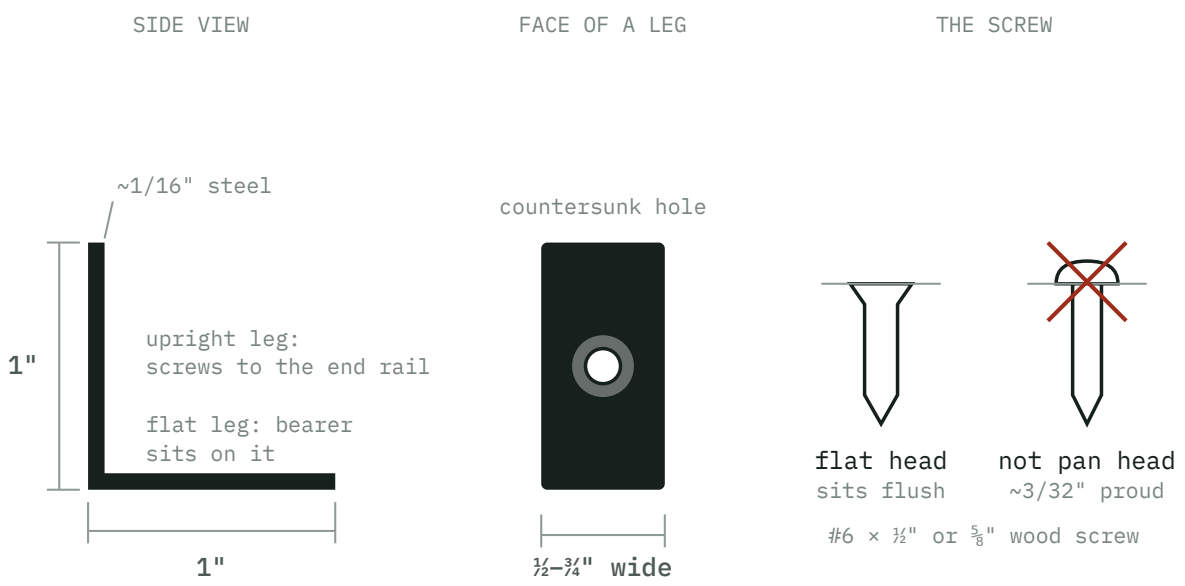



**FIG 5 — THE CORNER BRACE TO BUY**

A 1" zinc-plated corner brace. Only three things matter. The upright leg has to be shorter than the 1-7/16" rail (1" leaves about ½" to spare, and 1¼" still works). The hole has to be countersunk. And the screw has to be flat-head, because its head faces the bearer's end, which has only about 1/16" of play. The flat leg's hole stays empty.

| CHECK | BUY | NOT |
|---|---|---|
| Leg length | **1" (1¼" is fine)** | 1½" or 2": the upright reaches the rail top |
| Holes | **Countersunk, one or two per leg** | Plain punched holes: the screw head stands proud |
| Screw | **#6 flat-head wood screw, ½"–⅝"**<br>usually in the pack; ¾" is the most the rail takes | Pan-head or washer-head |
| Qty | **4 (one 4-pack)** | — |
| Where | Home Depot hardware aisle, with the mending plates and corner braces. About $4 a 4-pack, zinc or stainless. | |

## E Cut list

2×2 stock measured at 1-7/16"

| PIECE | QTY | LENGTH | NOTES |
|---|---|---|---|
| End rail | 2 | 46" | **Done.**<br>**Cut first. Everything else is measured off these.** |
| Side rail | 2 | 26⅛" | **Done.**<br>**Design figure. Cut to the gap between the end rails, standing at 29" outer.** |
| Leg | 4 | 16¾" | **Done: cut, bored, feet in.** |
| Bottom end rail | 2 | 46" | **Done.**<br>**Under the top end rails, across the legs' outer faces.** |
| Bottom side rail | 2 | 23¼" | **Done.**<br>**Design figure, 1¾" up the legs. Cut to the gap between the legs once they're on.** |
| Center bearer | 2 | ~25⅞" | **Done.**<br>**Glued face to face. Drop-in: cut the glued beam to the measured end-rail gap less ¼". Rough-cut the two pieces at 26⅛" first.** |

**Off the saw, four 8-ft boards.** Cut the fixed 46" pieces now. Rough-cut the fitted pieces a little long and trim each one to its measured gap as you reach it in the build order.

| BOARD | CUT INTO | USED | WHEN |
|---|---|---|---|
| 1 | 46" + 46"<br>top end rails | 92" | Now, off a stop block |
| 2 | 46" + 46"<br>bottom end rails | 92" | Now, off the same stop |
| 3 | 26⅜" + 26⅜" + 23½"<br>side rails + one bottom side rail, rough | ~76½" | Trim the side rails at step 01 and the bottom side rail at step 05 |
| 4 | 23½" + 26⅛" + 26⅛"<br>other bottom side rail + the bearer pair, rough | ~76" | Trim at step 05, and the bearer at step 09 |

## F Hardware

Panel B only

| ITEM | QTY | WHERE |
|---|---|---|
| SPAX #8 × 2½" Torx | 32 | 8 corners · 8 leg laps · 8 bottom side rails · 8 bottom end rails |
| Flat corner bracket + its short Philips screws | 4 | Inside each top corner, after the screws |
| 4" steel flat strap (diagonal brace) | 2 | Underside, two opposite corners |
| 1" zinc corner brace (L-bracket) + its ½" screws | 4 | **Done. For the drop-in bearer (Flag 1). Not the 2" stainless spares, which are taller than the rail.** |
| Wood glue | – | Every wood-to-wood joint, plus the bearer's two halves |
| 1" footman loop + #10 × ¾" screws | 2 | Spare-stack strap, on the inside faces of the bottom rails. Easier now, while the frame is upside down. |

## G Build order

same bench method as Panel A

**01** DONE

### Cut the end rails, then measure for the side rails

Two at 46". Stand them 29" apart outside to outside and cut the side rails to the gap, which should be about 26⅛". SPF varies board to board, so measure this panel's own gap.

**02** DONE

### Glue up the ring flat, in four corner clamps

Build it on the garage floor, like Panel A. Check the diagonals while the glue is still wet: both should read **54⅜"**. Equal diagonals are what let the A/B and B/C seams close flat later.

**03** DONE

### Two screws a corner, then the brackets

Pilot and countersink through the clamp, then drive 2 × 2½" SPAX through the end rail's face into the side rail's end, pulled below flush. The flat bracket goes inside the corner afterwards.

**04** DONE

### Lap the four legs

Put the 2" block against the side rail's inner face and the leg against the block. Lap the leg against the end rail's inner face with its top flush with the rail top. Clamp it, pilot, and drive 2 × 2½" through the rail face into the leg's side. The feet are already in, so skip that step.

| | |
|---|---|
| 05 | DONE<br>**Flip it, then fit the bottom side rails, 1¾" up**<br>Measure between the legs (about 23¼") and cut. Cut a 1¾" spacer block. With the frame upside down, stand the spacer on each leg's bottom end and rest the rail on it, on the legs' line. That puts its underside 1¾" up from the leg bottom. Glue it, then drive 2 screws through each leg's outer face into the rail's end, about 2½" up and ⅜" either side of center (Fig 3). |
| 06 | DONE<br>**The 46" bottom end rails**<br>Across the legs' outer faces, directly under the top end rails, with the underside flush with the leg bottoms, glued, exactly as on Panel A. Drive 2 screws per leg, about ¾" up and ⅜" either side of center, to miss the insert bore. They sit well below the side-rail screws, so this step and the last can go in either order.<br>**Optional relief notches.** If you\'ll run the backup straps to the 3rd-row striker loops, notch the end rail that meets Panel C before you fit it: ½" deep, 3" wide, at 12¾–15¾" and 30¼–33¼" from the driver end (Anchor Board sheet). Keep the leg screws and the footman-loop screws out of those two bands. |
| 07 | DONE<br>**Diagonal braces on the underside**<br>Two 4" straps at two opposite corners, on the underside of the top ring. Never on the rail tops, because the bed halves land there. |
| 08 | DONE<br>**Footman loops, while it's upside down**<br>Two 1" loops on the inside faces of the bottom rails, one each side of where the spare stack sits, for its cam strap. Optional now, but it's awkward to reach later. |
| 09 | DONE<br>**The center bearer, drop-in**<br>Glue the two 2×2s face to face and let them cure clamped. Measure the gap between the end rails at the centerline, and cut the beam to that **less ¼"**, about 25⅞".<br>With the frame still upside down, sit the bearer in the gap at the centerline, top face down on the floor, so it's flush with the rail tops. Set a 1" brace in each corner where the bearer meets an end rail: upright leg on the end rail's inner face, flat leg on the bearer's underside. Pilot and screw the braces to the rail only, never to the bearer. Lift the bearer out and back in once to check it drops freely. |
| 10 | DONE<br>**Set the feet against Panel A**<br>Stand it right way up beside Panel A and put a straightedge across the seam. Turn Panel B's feet until its rail tops match Panel A's, then take both to 17¾" together. The alignment pins and seam latches come later (Phase 7), once the three panels are mated. |

Geometry from `params.scad` ( `panel_b_length` , `panel_width` , `leg_inset` , `bed_bearer_*` , the spare dims) and `platform.scad` 's `module_frame` , with lengths set to the measured 1-7/16" stock per Section 3, and Panel A's bench photos for method.

Where the model and this sheet differ: `module_frame` draws the bottom end rails as 39" pieces on the legs' own line, which passes through the legs. The cut list's 46" and Panel A as built put them across the legs' outer faces, and this sheet follows the build. `params.scad` checks that the spare passes the open top with the bearer out.

# Panel C Frame

The tailgate module: the only panel with a fixed deck and a wall. The fridge and kitchen unit slide out underneath it, and the rear pantry sits on its deck. Two leg lengths: lapped at the front, butted under the rail at the rear corners.

| ASSEMBLY | FRAME, OUTER | DECK TOP | UNDER THE REAR RAIL |
|---|---|---|---|
| **Comp. 4 · steps 1–3**<br>frame, deck, front wall | **46 × 35¾"**<br>W (across van) × L (fore-aft) | **18½"**<br>= rail top · flush with the bed on A/B | **17"**<br>floor to underside · fridge needs 16.67 |
| **STOCK** | **SCREWS** | **DRAWN FROM** | **STATUS** |
| **1-7/16" 2×2**<br>measured · sets every between-length | **26 × 2½" + ~54 × 1¼"**<br>SPAX #8 Torx · Everbilt #8 flat head · braces: 12 × 3" | **Build plan**<br>Section 3, Comp. 4 · params.scad · A/B as built | **Built to the deck**<br>13 Oct: frame, wall, cleats, deck. Side braces, height and pantry mat next |

## A Check before you cut

3 methods · 1 check · 2 carry-overs

The frame is Panels A and B again with different lengths. What's new is the deck, the wall and the rear corners.

METHOD

### The front wall hangs on the legs' back faces, clear of the floor

The legs lap against the front rail's inner face, and the wall goes on the legs' *back* faces, on the fridge side. That puts the wall 1-7/16" behind the rails (Fig 2), out of reach of both: the top rail is above the wall and in front of it, and the bottom rail is in front of the legs.

**So glue three 6" 2×2 offcuts behind the bottom rail, between the legs. They're the same thickness as the legs, so they come out flush with the legs' back faces, and the wall screws to the legs and the blocks: 7 screws.**

**Keep the wall off the floor.** At 17" tall on the floor it's ¾" longer than it needs to be, and it would touch the floor before the feet do, so the panel would rock on it. Set its bottom edge **½" below the leg bottoms**. With the feet at 1" that's ½" of air under it, and its top lands ¼" under the deck.

**The grommet goes 2¼" from the driver edge.** The leg starts at 3-7/16" on the other face, so the grommet's flange needs the hole well outboard of that. At 2¼" it clears by more than ½".

CHECK

### The low intake is three 3" round vents

The vents on hand are a 4-pack of 3" round snap-in soffit vents, so the passive intake under the fan is three 3" holes rather than one 9 × 2" slot.

**Cut three 3" holes in a row (Fig 3): centers 7⅜", 11⅛" and 14⅞" from the driver edge, 4⅛" up from the wall's bottom edge. That's about 21 sq in open, more than the slot's 18 and the fan's 17.7, with ¾" of ply between holes. Check the hole size on the vent's package first, and that its clips hold in ½" ply. If they don't, glue the vents in.**

**Done: the three holes are cut and the wall is hung.**

METHOD

### The deck, the cleats and the corner brackets all want the rails' inner faces

The deck's edge covers the top ¾" of every rail's inner face, and the cleat under it covers the next ¾". A 2" corner bracket fitted the way it went on Panels A and B lands on both.

**Fit each corner bracket with its top edge ¾" below the rail top, level with the cleat tops, so the deck's corner rests on it. Stop every cleat 2⅛" short of each corner so the bracket fits in the gap. That makes the cleats 36¼" (front, between the legs), 38⅞" (rear) and 28⅝" (each side), all cut from the strips you already ripped.**

CARRY-OVER

### Match the deck to the bed by raising A and B, never by lowering C

Panel C's deck has to be flush with the bed platform on A and B, which is A/B's rail top plus ¾". As built, A's rail top is about 17½", so its platform lands at 18¼", ¼" under Panel C's 18½".

Don't close that by lowering Panel C. The fridge stack is 16.67" tall and slides out under the rear end rail, whose underside is 17" with the feet at 1". Lower Panel C ¼" and there's under ⅛" left. **Turn A's and B's feet out ¼" instead.** Also, the rear legs are butted under the rail, so with all four feet at 1" the rear rail sits 1/16" lower than the front. Turn the rear feet out 1/16" more than the front.

**As built, the rear rail's underside sets anywhere from 17" to 17¾" on the feet.** The fridge slides under it bare today, but it's only 15.79" tall on the floor. On its slide tray (not built yet) it tops out at 16.67", and it needs ¼" of running clearance for bounce. That still leaves 0.33" at 17" and 1.08" at 17¾", so the whole range works. The slides bolt to the anchor board, not to Panel C, so turning C's feet moves only the rail above the fridge. Use the range to meet A and B, but **never set it below 17"**, and measure the finished tray before you settle on C's setting.

CARRY-OVER

### The control cluster's drawn site overlaps a leg

The electrics plan puts the control cluster on the wall's Panel B face at 40.4–45.4" from the driver edge, "over the passenger front leg". The wall sits behind that leg, so the leg stands 1-7/16" proud of the wall right in the middle of the cluster's footprint. None of this changes what you cut today. Just don't put anything in that column. The cluster gets re-sited when the electrics go in (Component 6).

METHOD

### Side braces tie the rear corner legs to the front, 12½" off the floor

The rear legs are butted under the rail and held by 2 screws and a bracket, and below the deck nothing ties them to the front of the panel. Tipped or racked, they carry the load on those 2 screws alone. A 2×2 brace on each side, from the rear leg's forward face to the front wall, closes that gap (Section F).

**They run in the rear legs' band**: the outer 1-7/16" under each side rail, flush with the side rails' outside faces. Inboard of that band is the fridge and kitchen slide path. The front legs are 2" further in, so a brace can't reach them directly. Each one stops at a 4⅞" block glued to the wall's fridge face, and the block screws through the wall into the leg.

**Not at the floor.** The bottom crossbar at the wall end is only 1-7/16" tall, and a brace that low would sit on the fridge riser's flange, the H1 eye bolt and the kitchen strap's footman loop. The kitchen's rear strap climbs the passenger band to the unit's lid, which is about 11⅞" up if the unit sits on the board's rungs. So the brace's bottom goes at **12½"** (11½" above the leg bottoms). That's about ½" over the strap and still under the rear legs' L-brackets, which start at 15".

**The side trim from 26" forward of the hatch comes 2" inside Panel C's edges but stands no more than 2" tall, so the braces and blocks clear it by over 10".**

## B Plan view

looking down, deck off · to scale

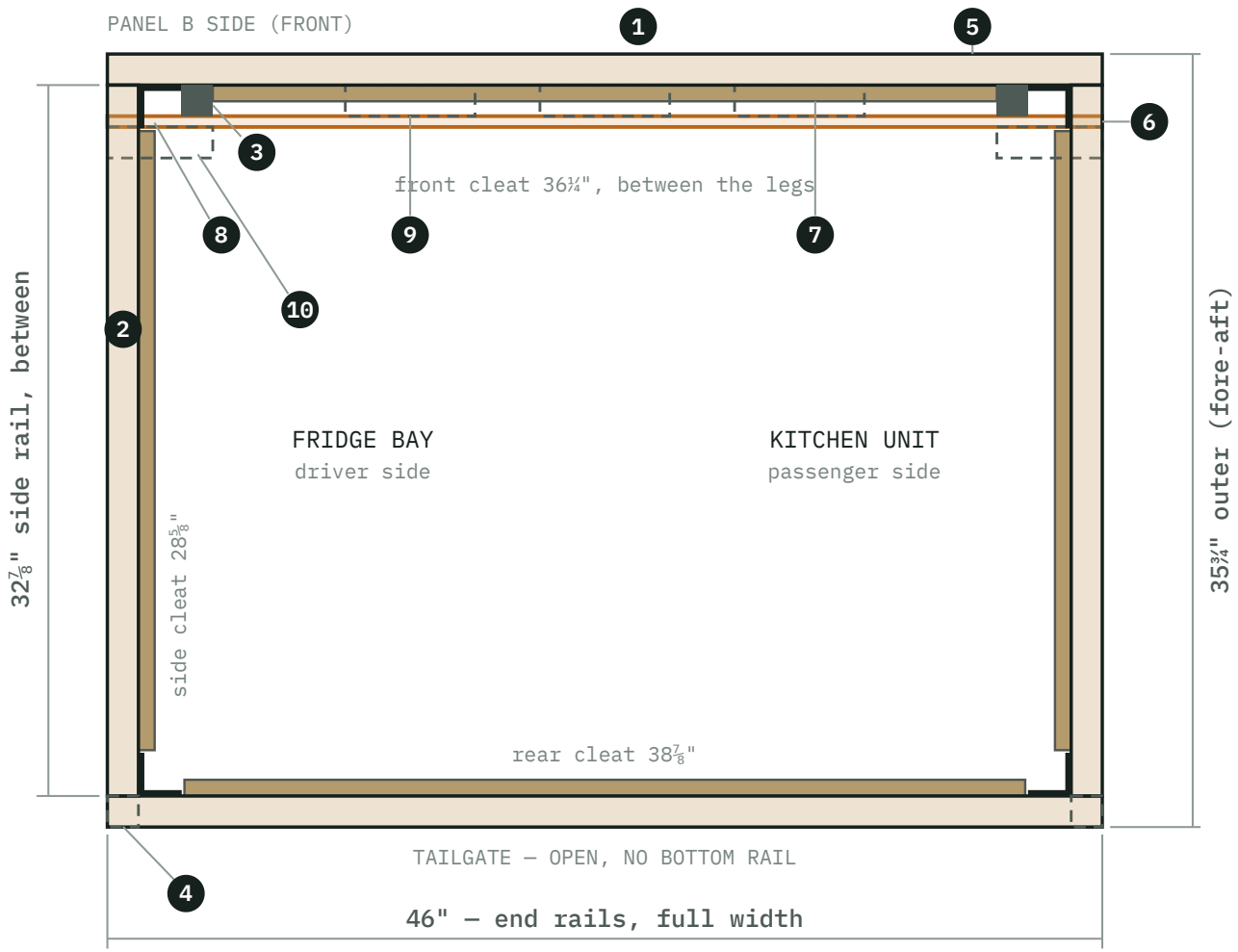



**FIG 1 — PLAN, DECK OFF**

The end rails are the outboard pair and set the 46". Everything else is fitted between them. The front bottom rail sits directly under the front top rail, so it's hidden in this view. The back blocks are dashed because they sit low, behind it. The rear legs are dashed because they stand under the rear rail's ends. The side braces run directly under the side rails, 12½" off the floor, so they're hidden too. Their blocks are dashed at the wall's two ends.

1. **End rail ×2 — 46"**, 2×2. The outboard pair. The front legs lap against the front one's inner face, and the rear one sits on the rear legs.

2. **Side rail ×2 — 32⅞"** (35¾ − 2 × 1-7/16). Stand the end rails at 35¾" outer and cut to the gap you measure.

3. **Front leg ×2 — 17½"**, already bored and footed. Lapped to the front rail's inner face, top flush with the rail top, 2" block against the side rail, as on A and B.

4. **Rear leg ×2 — 16"**, already bored and footed. Butted under the rear rail at the true corners, flush with both outside faces. Held by 2 screws down through the rail plus an L-bracket (Fig 4).

5. **Front bottom rail — 46"** (hidden). Across the front legs' outer faces, directly under the front top rail, underside flush with the leg bottoms, as on A and B. The tailgate face gets none; the appliances come out that way.

6. **Front wall — 46 × 17", ½" ply**, already cut. On the front legs' back faces, bottom edge ½" below the leg bottoms. Fan hole, 3 vent holes and 1 grommet hole (Fig 3).

7. **Deck cleats — ¾ × ¾"**, already ripped. Front 36¼" between the legs, rear 38⅞", sides 28⅝". Tops ¾" below the rail top, stopping 2⅛" short of each corner.

8. **Corner bracket ×4 — 2" stainless L**. Inside each corner, top edge ¾" below the rail top, level with the cleats, so the deck's corners rest on them.

9. **Back block ×3 — 6" of 2×2**, from offcuts (dashed). Behind the bottom rail, between the legs, flush with the legs' back faces. The wall's bottom edge screws to them.

10. **Side brace ×2 — 29½", block ×2 — 4⅞"**, 2×2 (hidden, dashed). Each brace runs under its side rail from the rear leg's forward face to a block on the wall's fridge face, bottom 12½" off the floor (Section F).

## C Section at the front wall

cut fore-aft between the legs · to scale

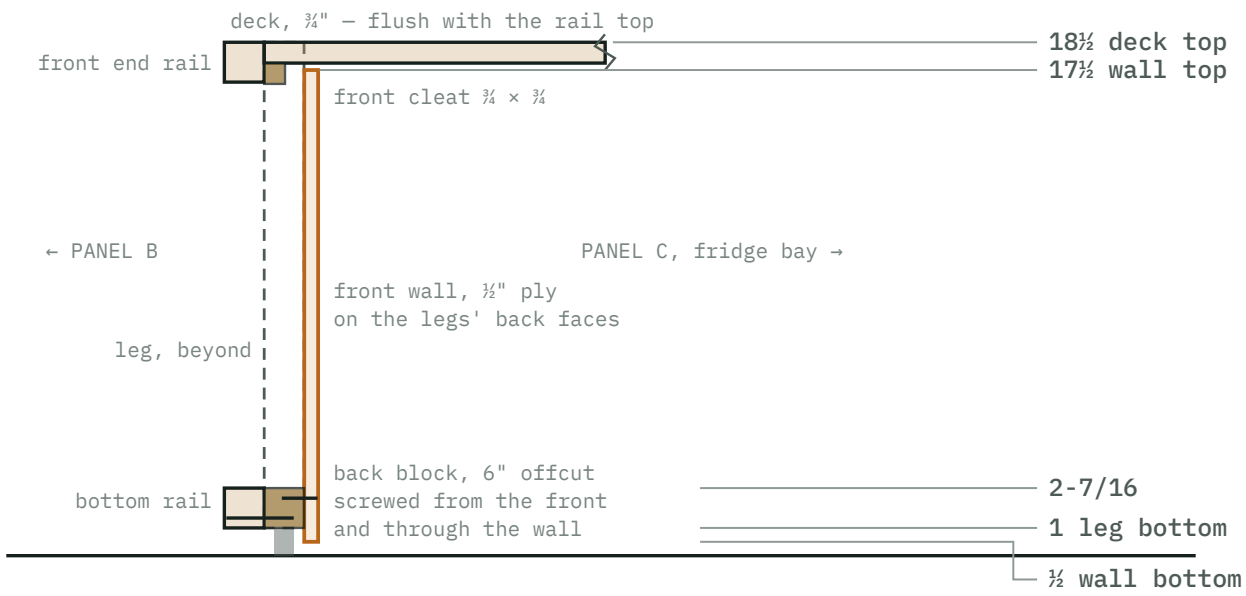



**FIG 2 — SECTION, FEET AT 1"**

Heights in inches from the van floor. The leg is beyond the cut, dashed. The wall is 1-7/16" behind the rails, so it can only fix to the legs and to blocks that fill that same 1-7/16". The ¼" between the wall top and the deck and the ½" under the wall are both clearances. Neither should touch.

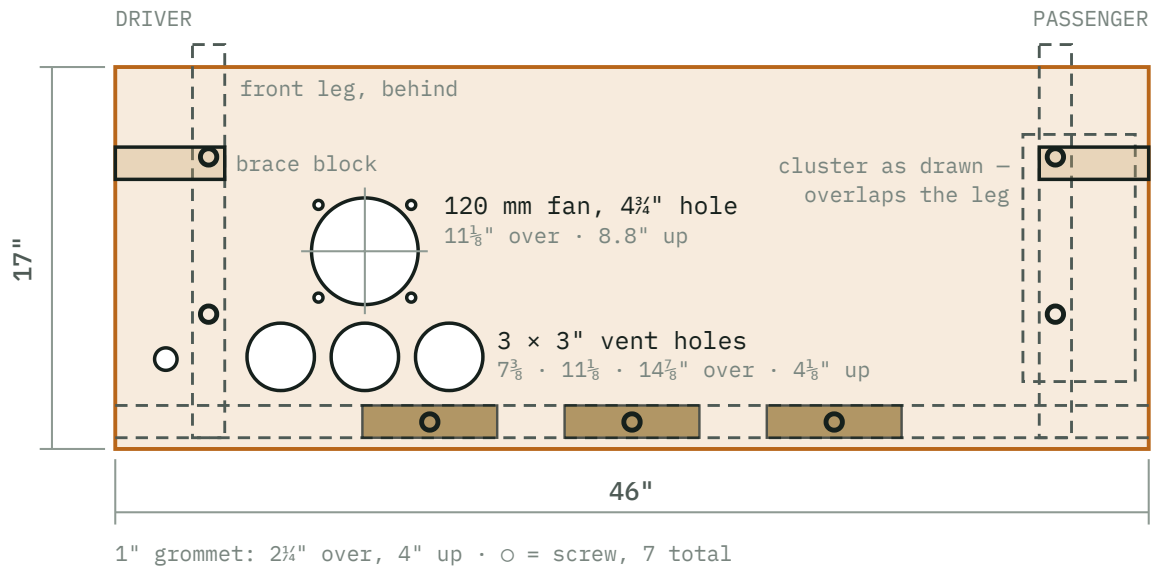



**FIG 3 — FRONT WALL, FROM THE FRIDGE SIDE**

Driver on the left, the same view as the plan's flat-pattern render. Every height is from the wall's own bottom edge. Screws: 2 into each leg at 4⅛" from each end, about 6" and 13" up, and 1 into each block about 1¼" up, clear of the block's own screws near its ends. Drive them from this side, before the deck goes on. The two side-brace blocks (Section F) go on this face at the wall's ends, 12" up from its bottom edge, over the upper leg screws. Those heads have to be flush.

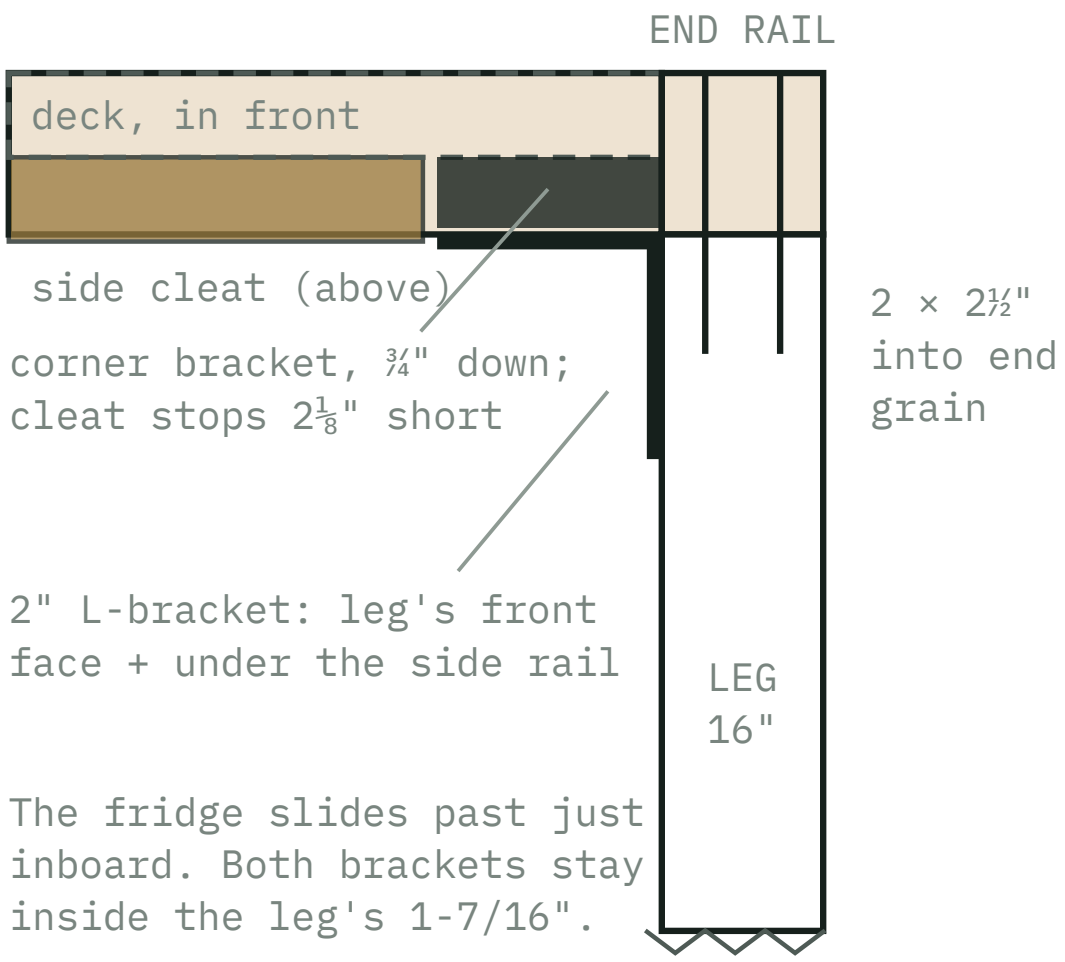



**FIG 4 — REAR CORNER, FROM INSIDE, LOOKING OUTBOARD**

The screws down into end grain only locate the leg, and the L-bracket is what holds it. Put the bracket on the leg's forward face with its flat leg under the side rail, not on the leg's inboard face. The fridge and kitchen unit slide along that inboard face, and screw heads there would snag them.

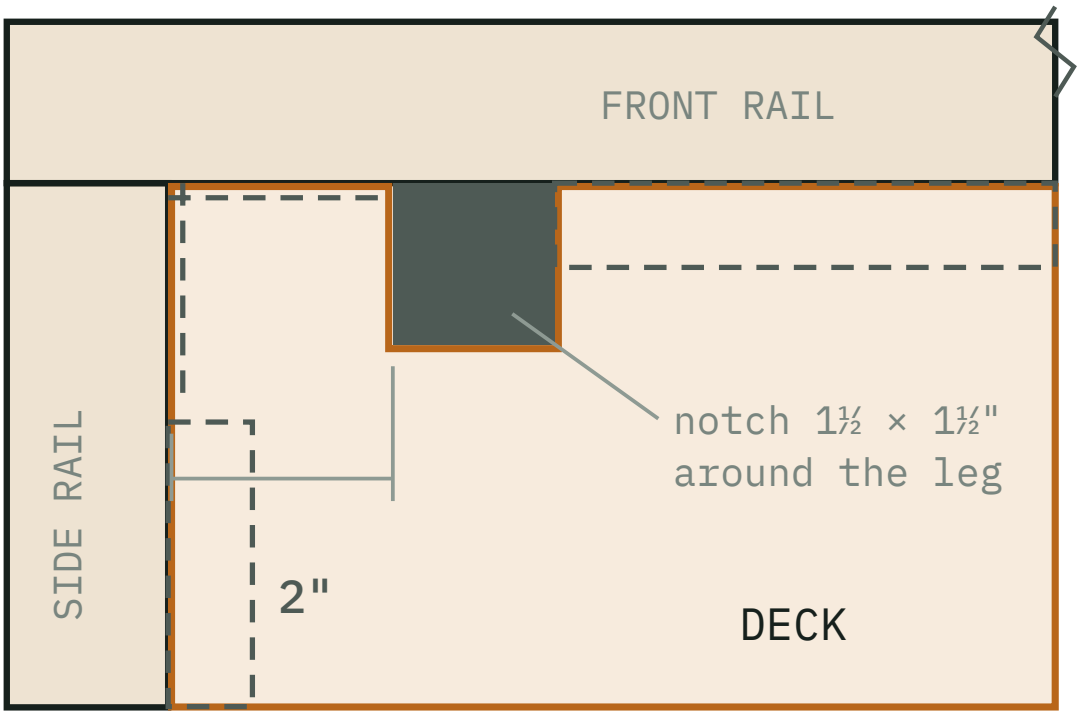



**FIG 5 — DECK NOTCH, FRONT CORNER**

The lapped front legs come up into the deck plane, so the deck gets a 1½" square notch at each front corner, starting 2" in from its side edge (the same 2" block). Mark it off the real legs with the deck in the opening. The rear legs are under the rail and need no notch. The cleats and bracket are dashed.

## F Side braces

rear leg → wall block, both sides · to scale

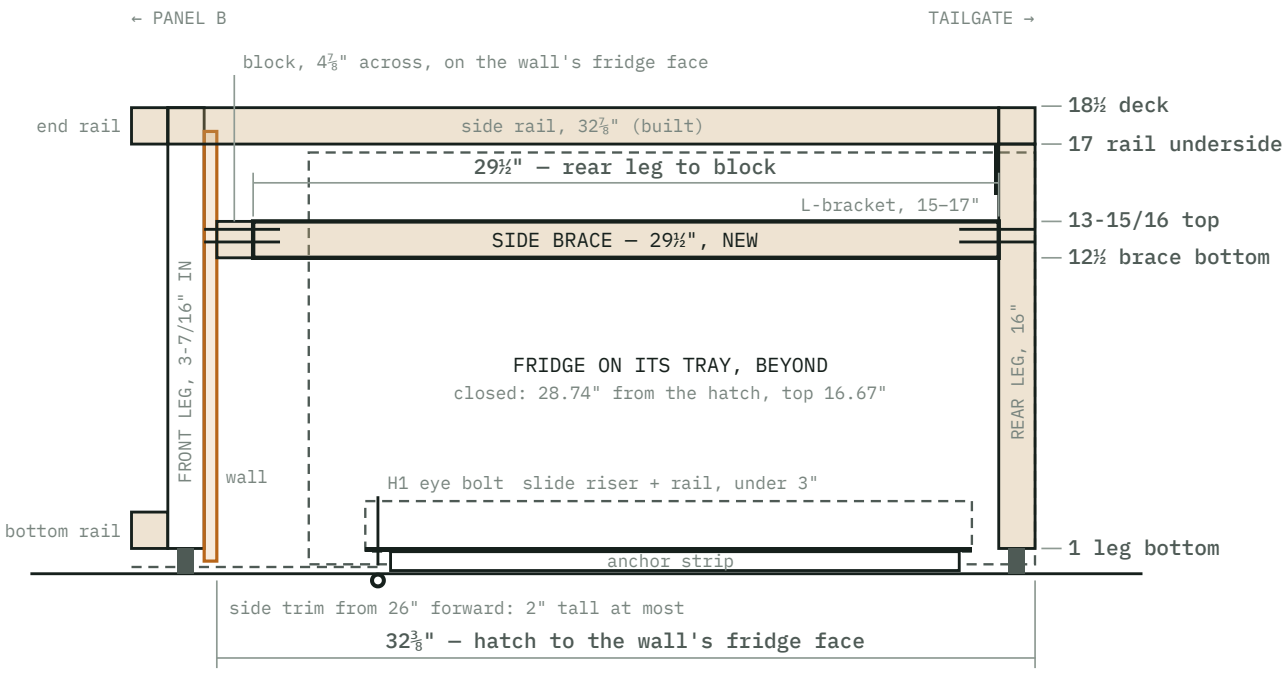



**FIG 6 — DRIVER SIDE, FROM OUTBOARD**

Heights in inches from the van floor, feet at 1". The passenger side is the mirror image. The brace sits in the rear leg's 1-7/16" band, flush with the side rail's outside face above it, so the fridge (dashed, beyond) slides past its inboard face. Each end takes 2 × 3" SPAX into the brace's end grain, drawn inside the wood: at the rear through the leg from its tailgate face, at the front from the Panel B side through the wall's end and the block. The wall's end covers the band just past the block, which is why the brace can't continue to the bottom crossbar.

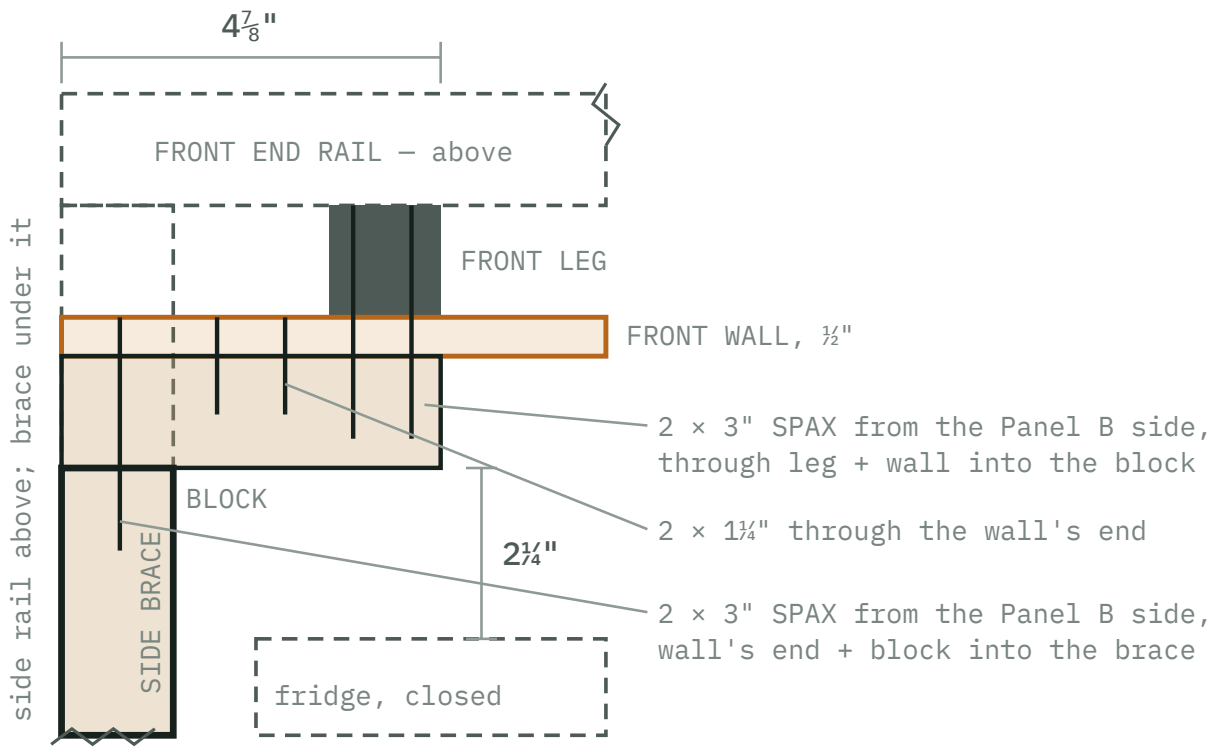



**FIG 7 — FRONT CORNER, CUT AT 13", FROM ABOVE**

Panel B at the top, driver edge on the left, about 1:2.5. The leg's back face is covered by the wall, so the block on the wall's fridge face picks up the brace and the screws reach it through the leg. The brace's own 2 screws come straight through the wall's end and the block, outboard of the leg, where the Panel B side is open. The 3" screws go ⅜" in from each side of the leg and ⅜" up from the block's bottom, below the wall screw already in the leg. The passenger corner is the mirror image.

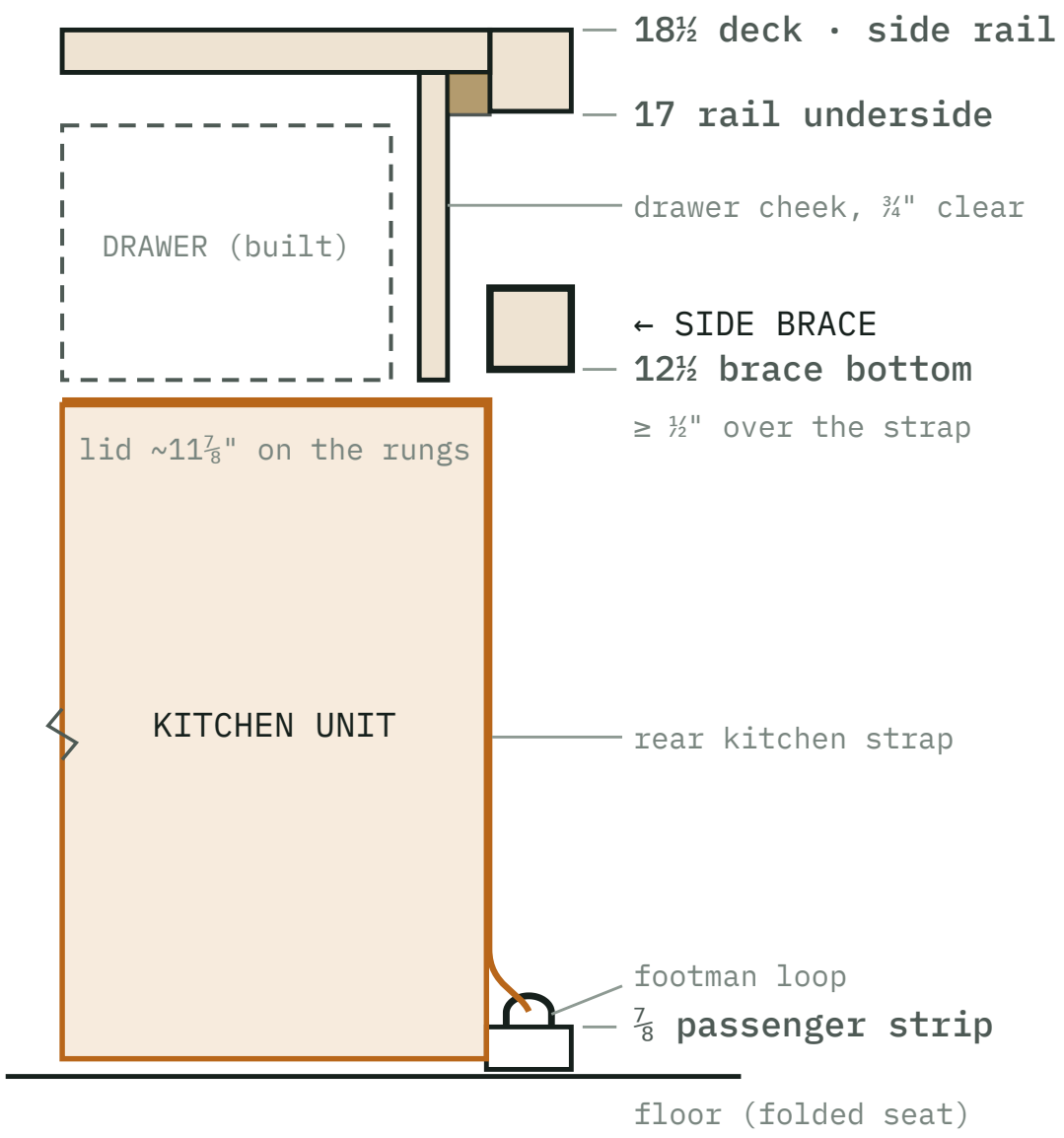



**FIG 8 — PASSENGER SIDE, LOOKING FORWARD**

Cut about 10" from the hatch, through the footman loop, about 1:5. The rear strap climbs the 1-7/16" band beside the kitchen unit, which is why the brace can't sit lower here. If the unit turns out to sit straight on the seat rather than on the rungs, its lid is 11.55" and the gap only gets bigger. The drawer's outer cheek hangs inboard of the side cleat, ¾" from the brace.

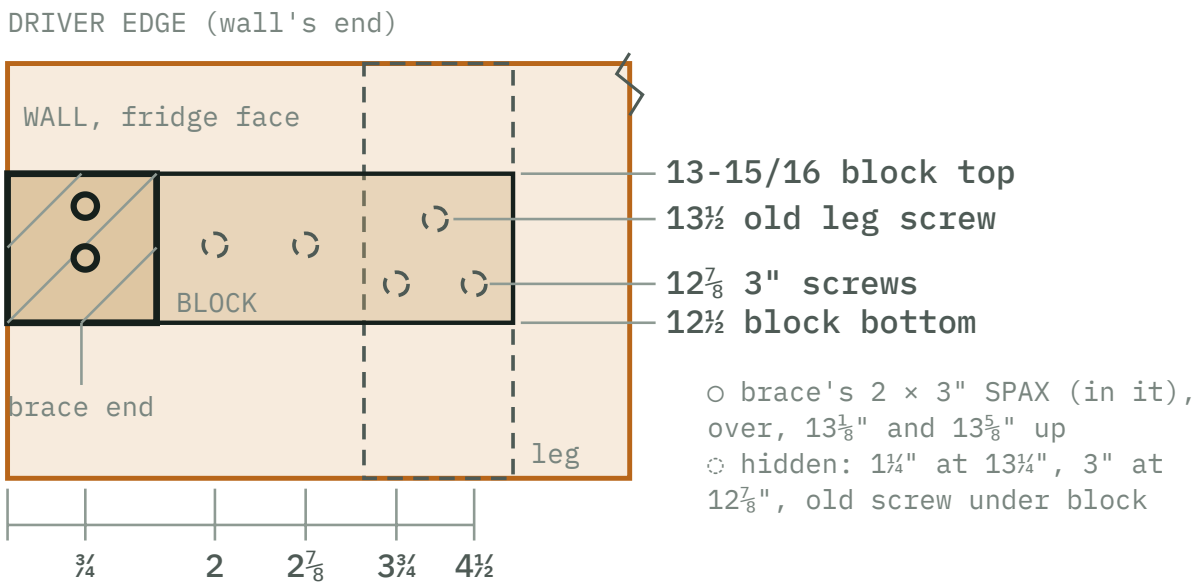



**FIG 9 — THE JOINT, FROM THE FRIDGE SIDE**

Driver end, about ½ full size, every dimension from the wall's end or the floor. The brace's end covers the block's first 1-7/16". Its two 3" SPAX (○) are driven from the Panel B side, ¾" from the wall's end, through the wall and the block and 1-1/16" into the brace's end. The 1¼" wall screws stay at 2" and 2⅞", clear of them. The 3" screws come through the leg from the Panel B side and stop ⅜" short of the block's back face. The passenger end is the mirror image.

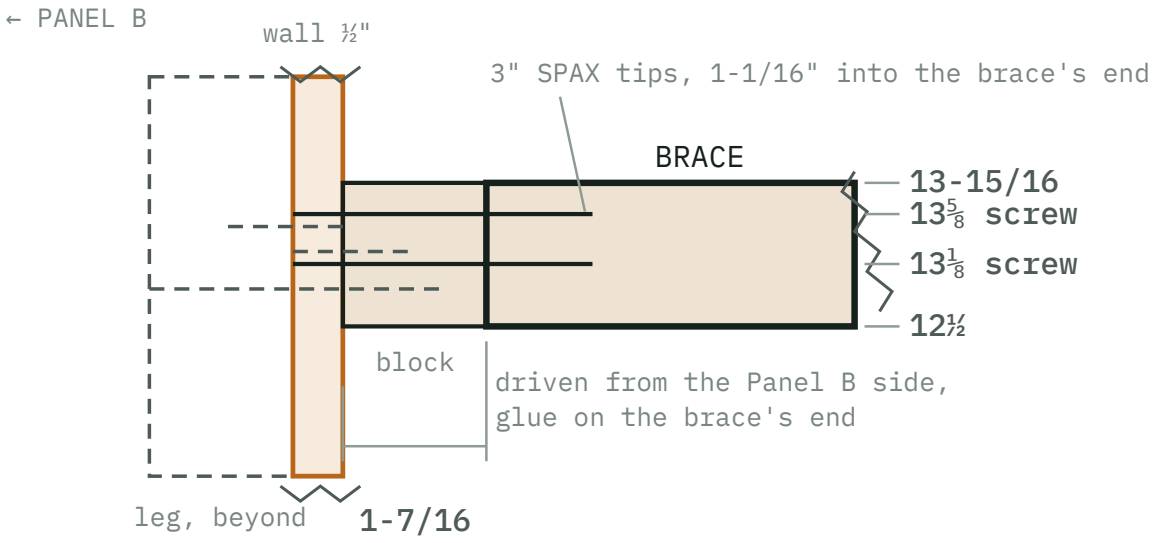



**FIG 10 — SECTION ALONG THE BRACE, LOOKING INBOARD**

Cut through the brace's middle, about ½ full size, Panel B on the left as in Fig 6. The brace's two 3" SPAX run straight in from the wall's Panel B face, ¾" from its end: ½" of wall, 1-7/16" of block, then 1-1/16" into the brace's end grain. The rear end is the same with the leg in place of the wall and block: 1-7/16" of leg, 1-9/16" into the brace. Everything dashed is inboard of the cut: the leg, the 3" screws through it into the block, the 1¼" wall screws, and the old screw from the fridge side into the leg, whose head the block now covers.

## G Cut list

2×2 stock measured at 1-7/16"

| PIECE | QTY | LENGTH | NOTES |
|---|---|---|---|
| End rail | 2 | 46" | **Done.** |
| Side rail | 2 | 32⅞" | **Done.** **Cut to the gap between the end rails, standing at 35¾" outer.** |
| Front bottom rail | 1 | 46" | **Done.** **Under the front top rail, across the legs' outer faces.** |
| Back block | 3 | 6" | **Done.** **From 2×2 offcuts.** |
| Legs | 4 | 17½" / 16" | **Done: cut, bored, feet in.** |
| Deck, ¾" birch | 1 | 33 × 43" | **Done: trimmed, notched, screwed down.** |
| Front wall, ½" ply | 1 | 46 × 17" | **Done: holes cut, hung.** |
| Deck cleats, ¾ × ¾" | 4 | 36¼ · 38⅞ · 28⅝ × 2 | **Done: fitted.** |
| Side brace | 2 | 29½" | To cut. Cut each to the gap you measure at 12½", from the rear leg's forward face to its block (Fig 6). |
| Brace block | 2 | 4⅞" | To cut, from an offcut. From the wall's end to the front leg's inboard face (Fig 7). |

**Off the saw, three 8-ft boards**: the one you have plus two more.

| BOARD | CUT INTO | USED | WHEN |
|---|---|---|---|
| 1 | 46" + 46"<br>end rails | 92" | Now, off a stop block |
| 2 | 46" + 33⅛"<br>bottom rail + one side rail, rough | ~79¼" | Trim the side rail at step 01 |
| 3 | 33⅛"<br>the other side rail, rough | ~33" | About 63" left over: both side braces (59") come out of it |

## H Hardware

Panel C only

| ITEM | QTY | WHERE |
|---|---|---|
| SPAX #8 × 2½" Torx | 26 | 8 corners · 4 front-leg laps · 4 down into the rear legs · 4 bottom rail · 6 back blocks |
| 2" stainless L-bracket | 6 | 4 ring corners + 2 rear legs.<br>A, B and C's corners use 12 of the 16 bought, and the rear legs take 2 of the 4 spares. |
| Everbilt #8 × 1¼" flat head | ~54 | **7 wall · ~22 cleats at 6" · ~20 deck at 8" · 4 brace blocks. The first box went to brackets and braces, so plan on a second 100-ct box.** |
| 3" snap-in soffit vent | 3 | From the 4-pack you have (Flag 2) |
| 1" grommet | 1 | Fridge DC line, from the EASYEAH pack |
| SPAX #9 × 3" Torx | 12 | **To buy, a small box. 4 for the brace blocks through the front legs and the wall, 8 for the braces: 2 into each end (Figs 7, 9, 10).** |
| Wood glue | — | Every wood-to-wood joint, the cleats, the back blocks and the side braces |
| Black non-slip mat | 14¼ × 43" | Pantry end of the deck, under the tote and the microwave; no cutouts |
| VELCRO Extreme Outdoor, 1" × 10 ft | 2 rolls | **To buy, about $11 a roll: heat-rated adhesive. ~115" of edge loop, hook on the mat and loop on the deck. Amazon** |

## I Build order

same bench method as A and B

| | |
|---|---|
| 01 | DONE<br>**Cut the end rails, then measure for the side rails**<br>Two at 46". Stand them 35¾" apart outside to outside and cut the side rails to the gap, about 32⅞". |
| 02 | DONE<br>**Glue up the ring upside down, in four corner clamps**<br>Rail tops on the garage floor, as on A and B. Check the diagonals while the glue is wet: both should read **58¼"**. |
| 03 | DONE<br>**Two screws a corner, then the corner brackets, ¾" off the floor**<br>2 × 2½" SPAX through the end rail's face into the side rail's end. Then set each bracket on a ¾" spacer (a cleat offcut) so its edge sits ¾" below the rail top, level with where the cleats will go (Flag 3). |
| 04 | DONE<br>**Lap the two 17½" front legs**<br>2" block against the side rail, leg against the block, lapped to the front rail's inner face with its top flush with the rail top. Clamp, pilot, 2 × 2½" through the rail face into the leg. |
| 05 | DONE<br>**Front bottom rail and the three back blocks**<br>Still upside down: the 46" rail across the front legs' outer faces, underside flush with the leg bottoms, glued. 2 screws per leg, ⅜" either side of center and about ¾" up, to miss the foot bore. Then glue the three 6" blocks behind it, between the legs, centered about 14", 23" and 32" from the driver end and flush with the legs' back faces. Drive 2 screws through the rail's outer face into each block, about 1" in from each end of the block.<br>**Optional relief notches.** If you\'ll run the backup ratchet straps forward to the 3rd-row striker loops, they pass under this rail: cut two notches in the rail and the two outer blocks on the bench first: ½" deep, 3" wide, at 12¾–15¾" and 30¼–33¼" from the driver end (Anchor Board sheet). It's a few saw cuts now and a chisel job once the frame is built. Keep every screw out of those two bands: the block screws 1" in from each block end already miss them. |

06

DONE

### Turn it over onto the rear legs

Stand the frame up on its front legs, and stand the two 16" legs under the rear corners, flush with both outside faces. Glue the leg tops, then pilot and countersink 2 × 2½" down through the rear rail's top into each leg. Then the 2" L-bracket on each leg's forward face, flat leg under the side rail (Fig 4).

07

DONE

### Cut the wall's openings, then hang it

On the bench: the 4¾" fan hole, the three 3" vent holes and the 1" grommet hole at 2¼" (Fig 3). Then stand the wall against the front legs' back faces, on the fridge side, with its bottom edge ½" below the leg bottoms. With the feet at 1" that's two ½" ply offcuts under it. Clamp, pilot, and drive 7 × 1¼" from the fridge side: 2 into each leg and 1 into each block.

**Optional relief notches.** For the backup straps, cut two notches with the other openings, 3" wide by 1" up from the bottom edge, at 12¾–15¾" and 30¼–33¼" from the driver end. The block screws go at least 1¼" up from the wall's bottom edge, above the notch tops.

08

DONE

### The cleats

Trim them to fit (Flag 3) and glue and screw them to the rails' inner faces every 6", tops ¾" below the rail top. Use a deck offcut as the gauge. The front cleat goes between the legs only.

09

DONE

### Fit, notch and fix the deck

Measure the opening and trim the deck 1/16" under it. Drop it in, mark the two front legs off the real thing, lift it out and cut the 1½" notches (Fig 5). Drop it back in. It should be flush with the rail tops all round. Screw it into the cleats every 8", countersunk flush, because the mattress lies on it.

| | |
|---|---|
| 10 | **Side braces, 12½" off the floor**<br>Work with Panel C standing on its feet at 1" on a flat floor. Don't tip it onto its rear legs to reach in.<br>**Blocks first.** Glue a 4⅞" block to the wall's fridge face at each end, its outer end flush with the wall's end and its bottom edge 12" above the wall's bottom edge, which is 12½" off the floor. It covers the upper wall screw in each leg, about 13" up. That head has to be countersunk flush. If it stands proud, relieve the block over it. From the Panel B side, drive 2 × 3" SPAX through the leg's front face and the wall into the block, ⅜" up from the block's bottom and ⅜" in from each side of the leg. Then drive 2 × 1¼" through the wall into the block at 2" and 2⅞" from the wall's end, 13¼" off the floor, clear of where the brace's own screws will go (Figs 7 and 9).<br>**Then the braces.** Measure each gap at 12½", rear leg's forward face to the block, about 29½", and cut to it. Glue both ends, set the rear end on a 12½" spacer (the block sets the front), keep the brace flush with the side rail's outside face above it, and clamp it. At the rear, pilot and drive 2 × 3" SPAX through the leg's tailgate face into the brace's end. At the front, from the Panel B side, pilot and drive 2 × 3" SPAX through the wall's end and the block into the brace's end. Both pairs go ¾" in from the outside face, at 13⅛" and 13⅝" off the floor (Figs 9 and 10). These go into end grain, so the glue matters: let it cure before the panel moves. Nothing goes into the inboard face, because the fridge and kitchen slide past it. |
| 11 | **Set the height against A and B**<br>In the van, or on a flat floor beside them: Panel C's deck at 18½", the rear feet 1/16" further out than the front. Bring A and B up to meet it, not C down to meet them (Flag 4). The rear rail's underside sets anywhere from 17" to 17¾", and all of that clears the fridge on its tray. Never set it below 17". The bare fridge already slides under it. Check again once the slide tray is built. |
| 12 | **Non-slip mat on the pantry end, held with Velcro**<br>The mattress covers only the front 20" of Panel C, leaving about **14¼ × 43"** of deck at the tailgate end, where the rear pantry sits. The pantry is now a tote beside the microwave, strapped to footman loops on the outside of the side rails (Strap Anchors sheet), so nothing goes into the deck here. Cut black non-slip mat to 14¼ × 43", no cutouts, stopping at the mattress end.<br>Hold it with Velcro so it can be replaced: soft loop side on the deck, hook side on the mat, a strip all round about 1" in from the edge, and nothing across the middle (a ridge under the mat rocks the tote and the microwave). Sand and alcohol-wipe the deck, and staple each end of every deck strip. Test one short strip on the mat's back for a day first: if the adhesive won't hold on the rubber, glue the hook strips on with contact cement. Keep rubber out from under the mattress, and if the deck is painted, let it cure a week first. |

Geometry from `params.scad` (`panel_c_length`, `leg_inset`, `leg_lap`, `pcwall_*`, `intake_*`, the fridge stack) and `platform.scad`'s `module_frame` and front-wall placement, with lengths set to the measured 1-7/16" stock per Section 3, and Panels A and B as built for method.

Where the model and this sheet differ: `module_frame` draws the front bottom rail as a 39" piece on the legs' own line, which passes through the legs. The cut list's 46" and A/B as built put it across the legs' outer faces, and this sheet follows the build. The plan's front-wall render matches Fig 3. The side braces and their blocks (Section F, added 24 Oct) are in the model as `pc_brace_*`, drawn at 1½" stock (29¼" and 5"); this sheet sizes them to the measured 1-7/16" (29½" and 4⅞").

# Tote + Microwave Strap Anchors

One cam strap goes over the tote lid and the microwave, side to side across the van. Each end hooks to a footman loop (or a clip D-ring) screwed to the outside face of Panel C's side top rail, just under the deck. Mirror the driver-side drawings for the passenger side.

| ANCHORS | WHERE | HEIGHT | SCREWS |
|---|---|---|---|
| **2 footman loops**<br>stainless, from the spares | **7½" in**<br>from the tailgate end, both sides | **¾" down**<br>centered on the 1½" rail face | **#10 × ¾"**<br>the loops' own screws · pilot first |
| STRAP | | | |
| **1" cam · 8 ft**<br>snap hooks · buckle on the tote lid | | | |

## A The whole end, from the tailgate

looking forward · hatch open · 12 px = 1"

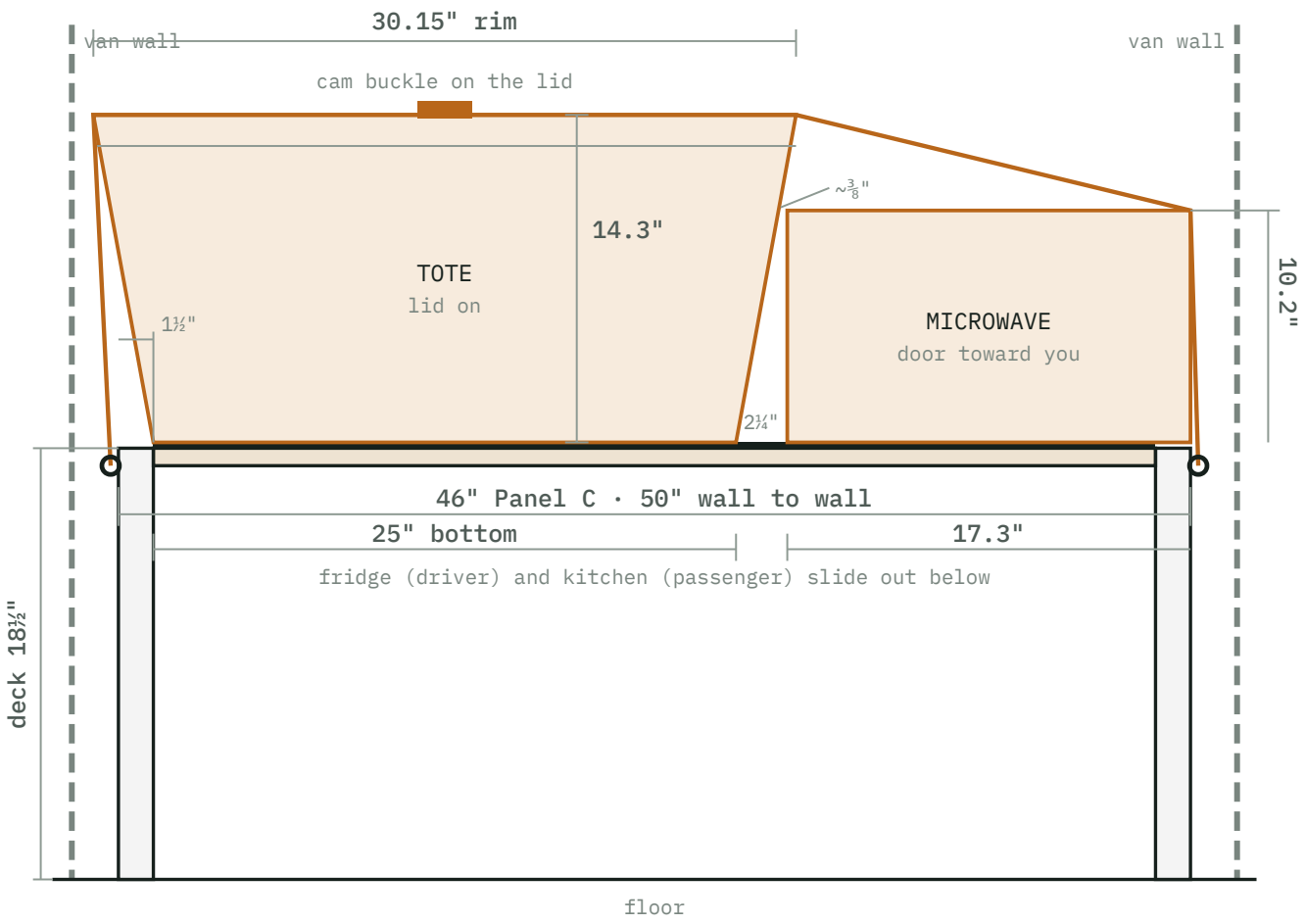



**FIG 1 — THE WHOLE END, LOOKING FORWARD**

Drawn to scale. The tote stands 1½" in from the driver edge and the microwave sits flush with the passenger edge. The tote's side flares 0.18" for every inch of height, so the gap between them closes from 2¼" at the deck to about ⅜" at the microwave's top corner. The strap runs from the driver rail's anchor up to the lid, across it, then drops to the microwave's outer top corner and down to the passenger rail's anchor. It bears on the whole lid but only that one corner of the microwave.

## B Cut across the side rail

driver side, looking forward from the tailgate · 40 px = 1"

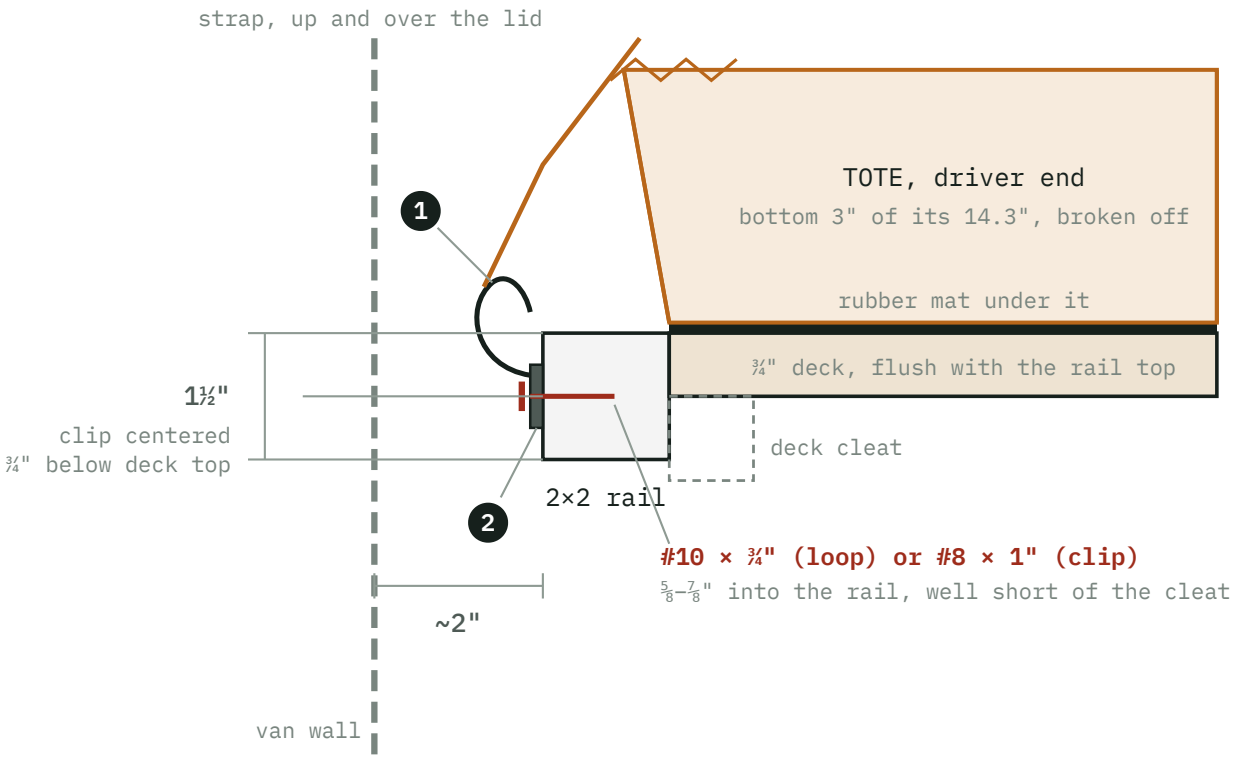



**FIG 2 — CUT ACROSS THE RAIL**

The clip screws to the rail's outside face, below the deck, so it takes none of the deck's width. With the strap tight, the ring swings up and lies close to the rail. It sticks out about ¾", well inside the 2" to the van wall. The footman loop's ¾" screws go about ⅝" into the 1½" rail (a clip D-ring's 1" screws about ⅞"), both well short of the deck cleat behind it.

## c From outside the panel

driver side · tailgate at right · 14 px = 1"

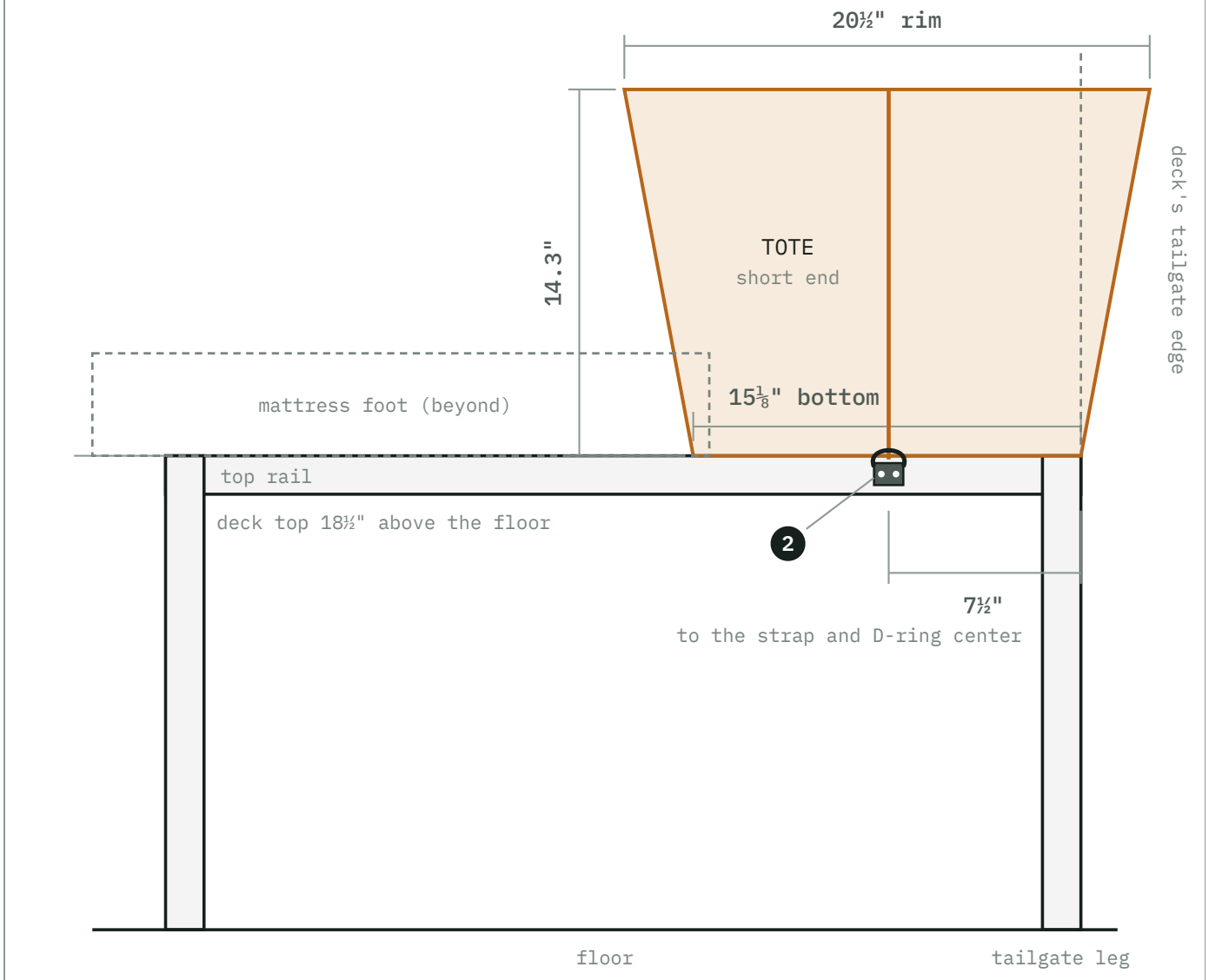



**FIG 3 — PANEL C'S SIDE FROM OUTSIDE**

Put the strap and both D-rings 7½" from the tailgate end. That's the middle of the tote's 15⅛" bottom and close to the microwave's middle, so the strap crosses square. It's well clear of the tailgate leg, so the screws go into the rail alone. The passenger side is the mirror image, under the microwave.

## D The anchor, face on

full size-ish · 80 px = 1"

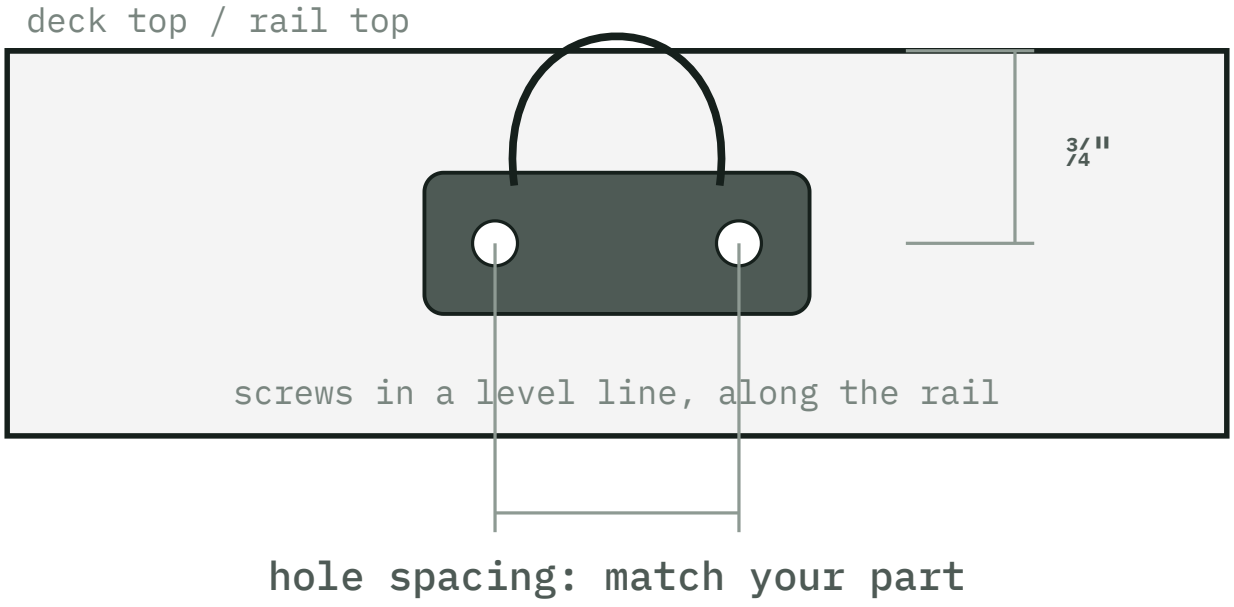



**FIG 4 — CLIP ON THE RAIL FACE**

Set the screws in a level line along the rail, both on the ¾" centerline, so each has the most wood above and below it. A clip up to about 1⅜" tall fits on the 1½" face.

1 Mark 7½" from the tailgate end, both rails

2 Hold the clip centered on the face, mark holes

3 3/32" pilots, ¾" deep (tape flag on the bit)

4 #8 × 1" screws, snug, don't strip

5 Mats down, tote + microwave in place

6 Hook one end, over the tote lid and microwave

7 Hook the other end, cinch at the tote lid

Touch up the black paint over the screw heads if you like.

**INSTALL ORDER**

Both D-rings go in before the van, on the bench. The only thing you do in the van is the strap.

1. **Anchor** — the stainless footman loops on hand (6 spares) are the anchor: the strap's snap hook clips into the loop. A clip D-ring drawn here works the same; rate either for 400 lb or more. Not the recessed deck D-rings from the old pantry kit.

2. **2-hole mount** on the rail's outer face — the footman loops' own #10 × ¾" screws, or #8 × 1" for a clip D-ring. Either stops well short of the deck cleat behind the rail.

Drawn 30 Oct 2026. Panel C's rail and deck are from the plan (deck flush between the rails). The tote and microwave sizes are the owner's measurements. The ~2" to the van wall is from V7 (50" wall to wall vs the 46" panel).

# Fridge Tray

The ½" ply tray the Rocky 40 rides on, with a 2½" apron along each long edge. It hangs between the two slides with nothing underneath it, so the slides add width beside the fridge and no height under it.

| ASSEMBLY | TRAY, OUTER | PANEL | APRONS |
|---|---|---|---|
| **Comp. 7 · step 3**<br>the tray only · slides come later | **19⅛ × 28¾ × 2½"**<br>as built: the aprons measure ⅝", not ¾" · drawn at 19⅜" | **17⅞ × 28¾", ½" ply**<br>fridge base + ⅛" · measure yours first | **2 × 28¾"**<br>1×4 ripped to 2½" · bottoms flush with the panel |
| **CARRIES** | **STACK HEIGHT** | **DRAWN FROM** | **STATUS** |
| **Rocky 40**<br>40.6 lb empty · ~90 lb loaded | **16.67"**<br>⅜ air + ½ tray + 15.79 fridge · rail is 17-1/16" | **Build plan**<br>Comp. 7, Section 8 · params.scad · Panel C as built | **Glued**<br>13 Oct: aprons glued and screwed. D-rings next · slides wait on the anchor board |

## A Check before you cut

1 check · 1 method · 2 carry-overs

The tray is three pieces and a dozen screws. What matters is its width, where the aprons go, and where it ends up in Panel C.

**CHECK** **Size the panel to your fridge**

The aprons' top edges are the fridge's side stops, so a panel exactly the fridge's width leaves no room to set it in.

**Measure the fridge across its base and add ⅛". For the Rocky 40 that's 17⅞". The length is 28¾", the fridge's length over its end handles.**

METHOD

### The aprons go on the panel's edges, not under it

The fridge has almost no headroom. On the tray it tops out at 16.67", and Panel C's rear rail is 17-1/16" off the floor at deck 18½". That leaves 0.39", and the plan wants at least ¼" for bounce. Anything under the panel, aprons or slides, raises the fridge into the rail. Undermount slides alone would add about 1.2".

**So the aprons are glued and screwed to the panel's long edges, with their bottoms flush with its underside (Fig 3). The panel hangs ⅜" off the floor, and each slide's moving rail screws to the outside face of an apron.**

CARRY-OVER

### The tray sits 2½" in from the driver edge

Panel C's driver rear leg stands in the corner. Pulled out, the driver slide's moving rail runs out the tailgate, so it has to be clear of the leg's footprint.

**The driver apron's outer face sits 2½" from Panel C's driver edge, against the edge of the anchor board's driver strip, so the slide sits on that strip just inboard of the leg (Figs 1 and 2). That leaves about 1-3/16" between the kitchen-side slide and the kitchen unit at floor level.**

CARRY-OVER

### The slides go on with the anchor board

The fixed rails screw to the two riser angles (Riser Angles sheet), which bolt to the anchor board's driver strip and middle strip. The tray hangs between those two strips (Fig 2).

**Build the tray and stop before the slides. The moving rails screw to the aprons' outer faces at a height set by the board and risers, with at least 1½" of each rail on the apron.**

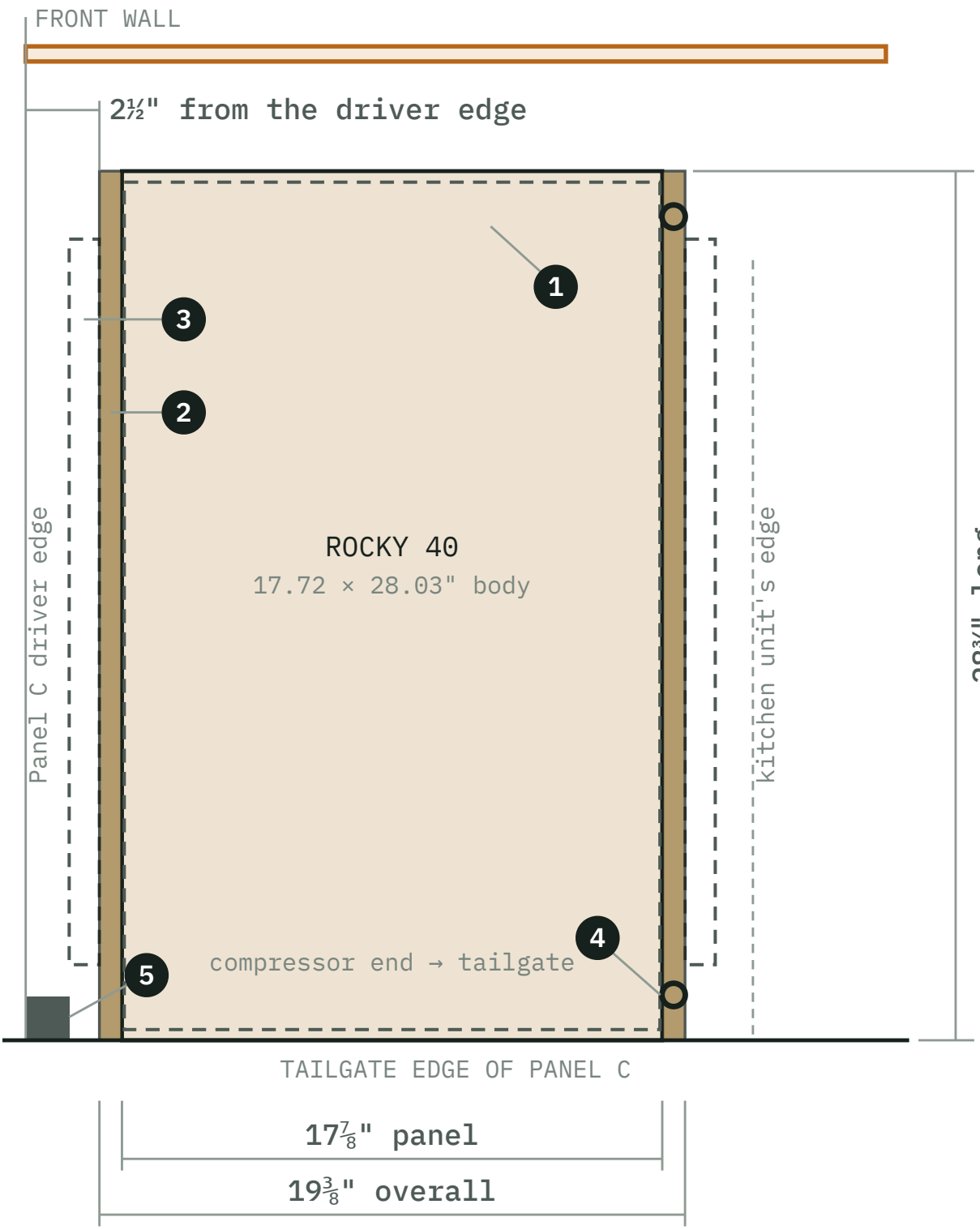



**FIG 1 — PLAN, CLOSED**

The tray sits with its end at Panel C's tailgate edge and its driver apron 2½" in, so the driver slide (dashed) is just inboard of the rear leg and slides out past it. The slides and D-rings go on later. Only the D-ring spots are marked now.

1. **Tray panel — 17⅞ × 28¾", ½" ply.** The fridge's base width plus ⅛". Face grain can run either way: the aprons carry the long span.
2. **Side apron ×2 — 2½ × 28¾"**, a 1×4 ripped. Glued and screwed to the panel's long edges, bottoms flush (Fig 3). Its top edge stands 2" above the panel and stops the fridge sliding sideways.
3. **VADANIA 24" slide ×2**, on hand (dashed, fit later). Each stands on edge outside an apron: the fixed rail on a riser angle on the anchor board, the moving rail screwed to the apron.
4. **Footman loop ×2**, stainless, on hand. Lengthwise on the kitchen-side apron's top edge, one under each of the fridge's end handles. A cam strap runs from one, through both handles and over the lid, to the other.
5. **Panel C's driver rear leg** (built). Stands in the corner, 1-7/16" square. The driver slide sits inboard of it so the tray can pull out past it.

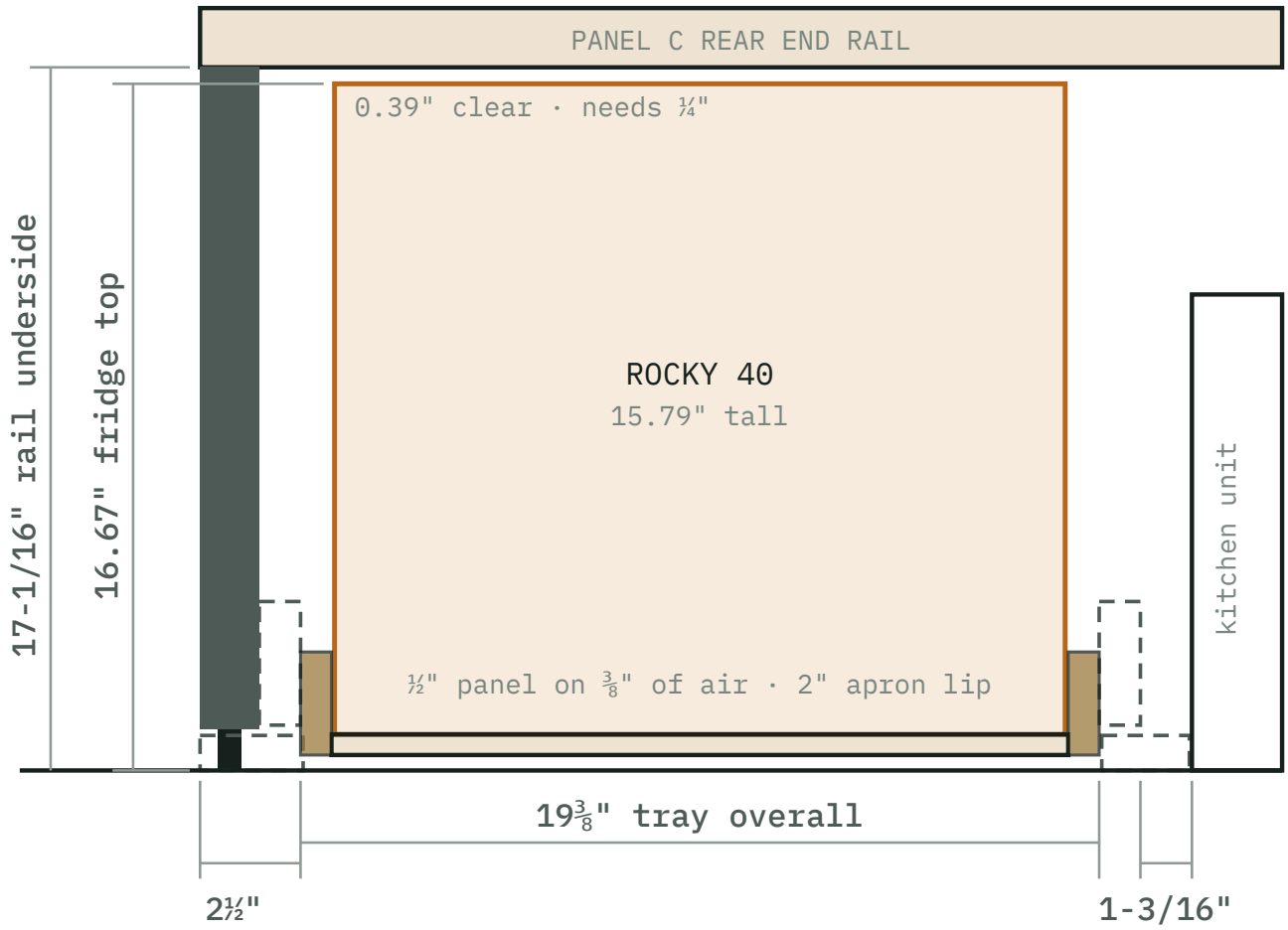



**FIG 2 — SECTION AT THE TAILGATE, CLOSED**

The fridge's height is set entirely by the tray: ⅜" of air, the ½" panel and the 15.79" fridge come to 16.67", under a rail at 17-1/16" with the deck at 18½". Raising Panel C in its 17–17¾" range adds clearance. The slides (dashed) stand beside the aprons and add no height. The low dashed blocks are the anchor board's driver strip and its middle strip, ripped to 2⅛". They start 2" in from the tailgate, behind this cut.

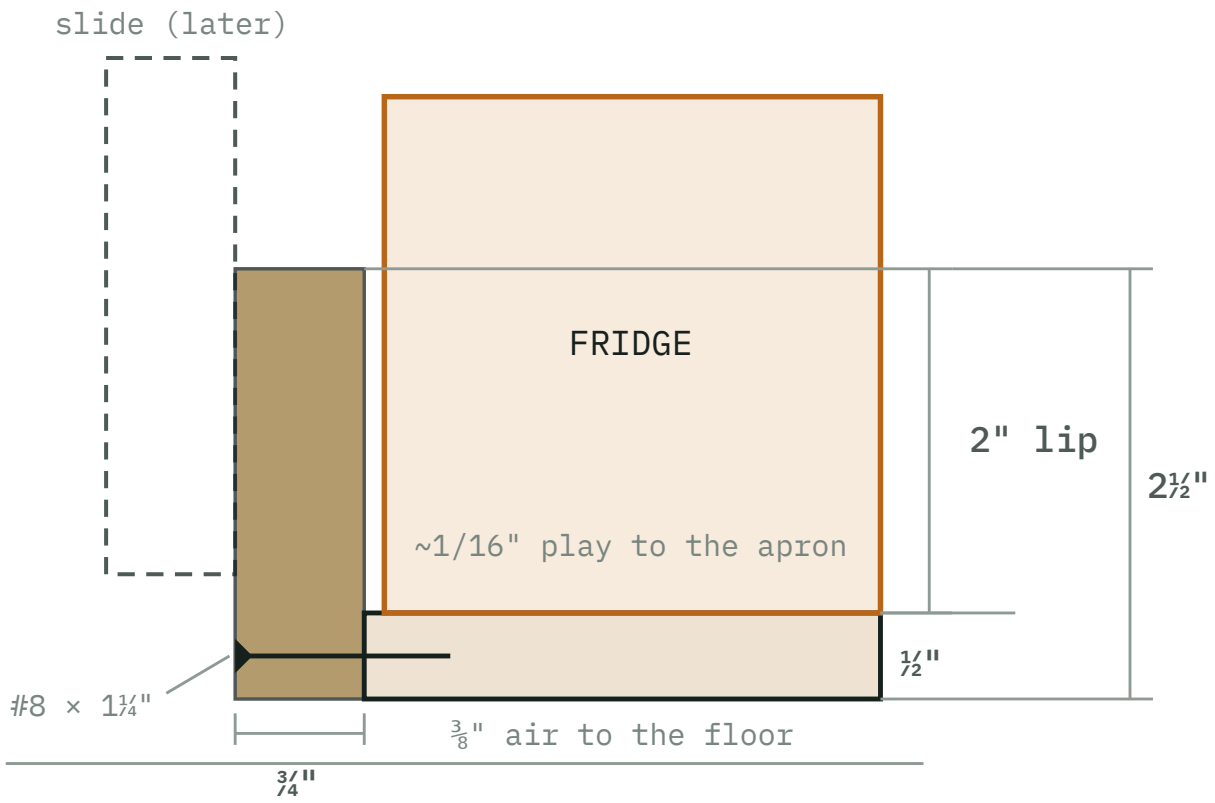



**FIG 3 — APRON JOINT**

Glue the full edge, then drive #8 × 1¼" flat heads through the apron's face into the panel's edge, ¼" up from the bottom. Pilot 3/32" into the ply edge and keep the bit centered on it. Countersink each head flush, because the slide's moving rail lies flat against this face.

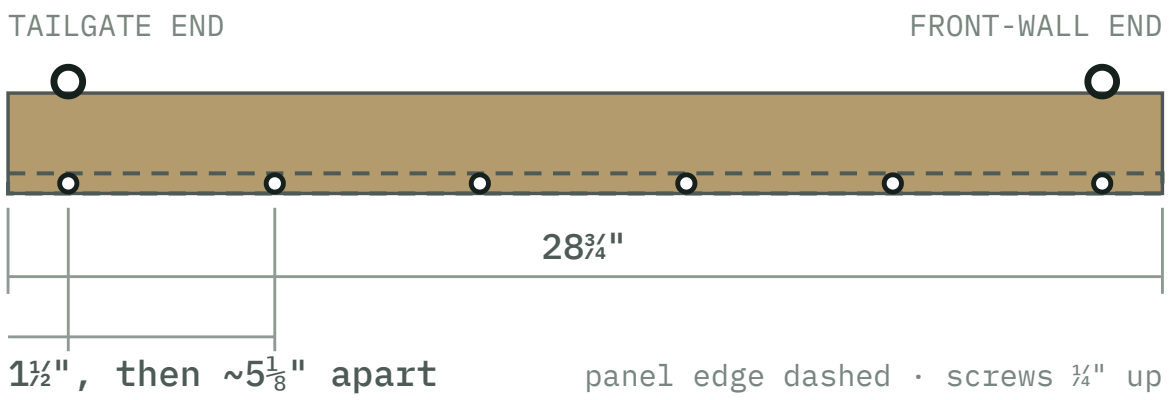



**FIG 4 — APRON, OUTER FACE**

Six screws per apron. The two D-ring circles are for the inboard apron only, and are nominal: mark them from the real handles with the fridge on the tray.

## E Fridge hold-down

2 footman loops + 1 cam strap

Two stainless footman loops screwed along the top edge of the kitchen-side apron, one under each of the fridge's end handles. One 1" cam strap runs loop to handle, along the lid, handle to loop, and pulls the fridge down onto the tray.

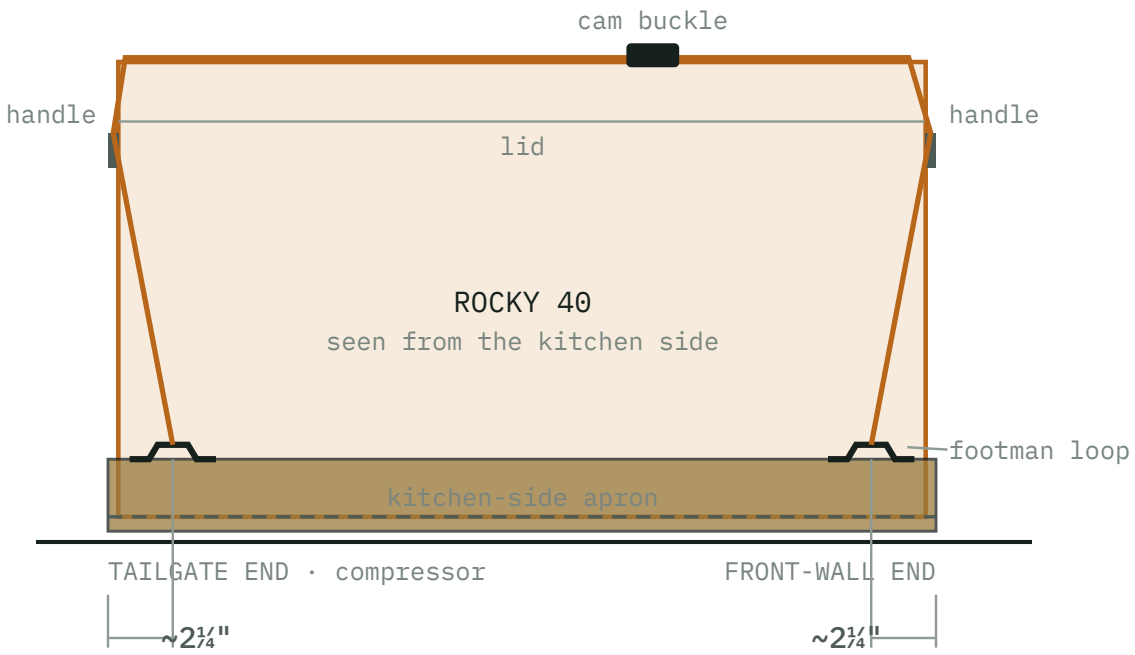



**FIG 5 — HOLD-DOWN, FROM THE KITCHEN SIDE**

The loops are centered about 2¼" in from each end of the apron, under the handles, with the end screw at least ¾" from the apron's end grain so it can't split out; mark them from your fridge's real handles. Keep the buckle on top of the lid where you can reach it with the tray pulled out. Unbuckle to open the lid.

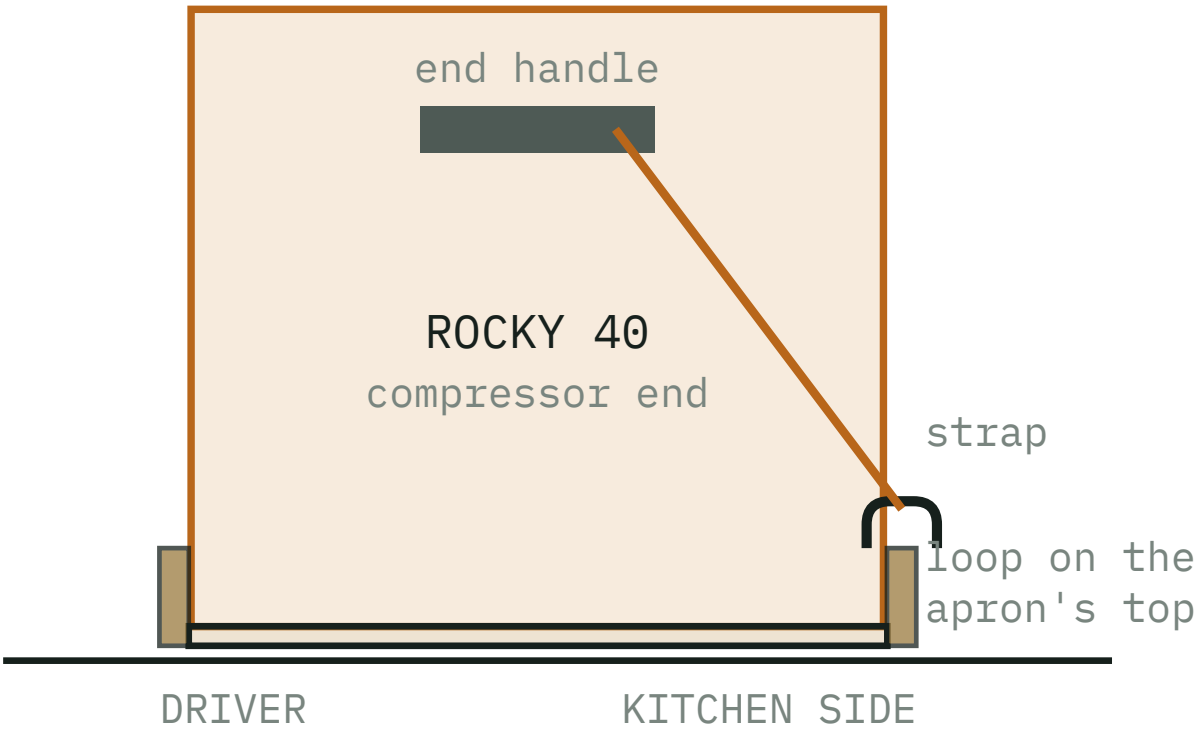



**FIG 6 — FROM THE TAILGATE**

The handle is in the middle of the fridge's end, so the strap angles inboard from the loop up to it. That angle also pulls the fridge against the driver-side apron.

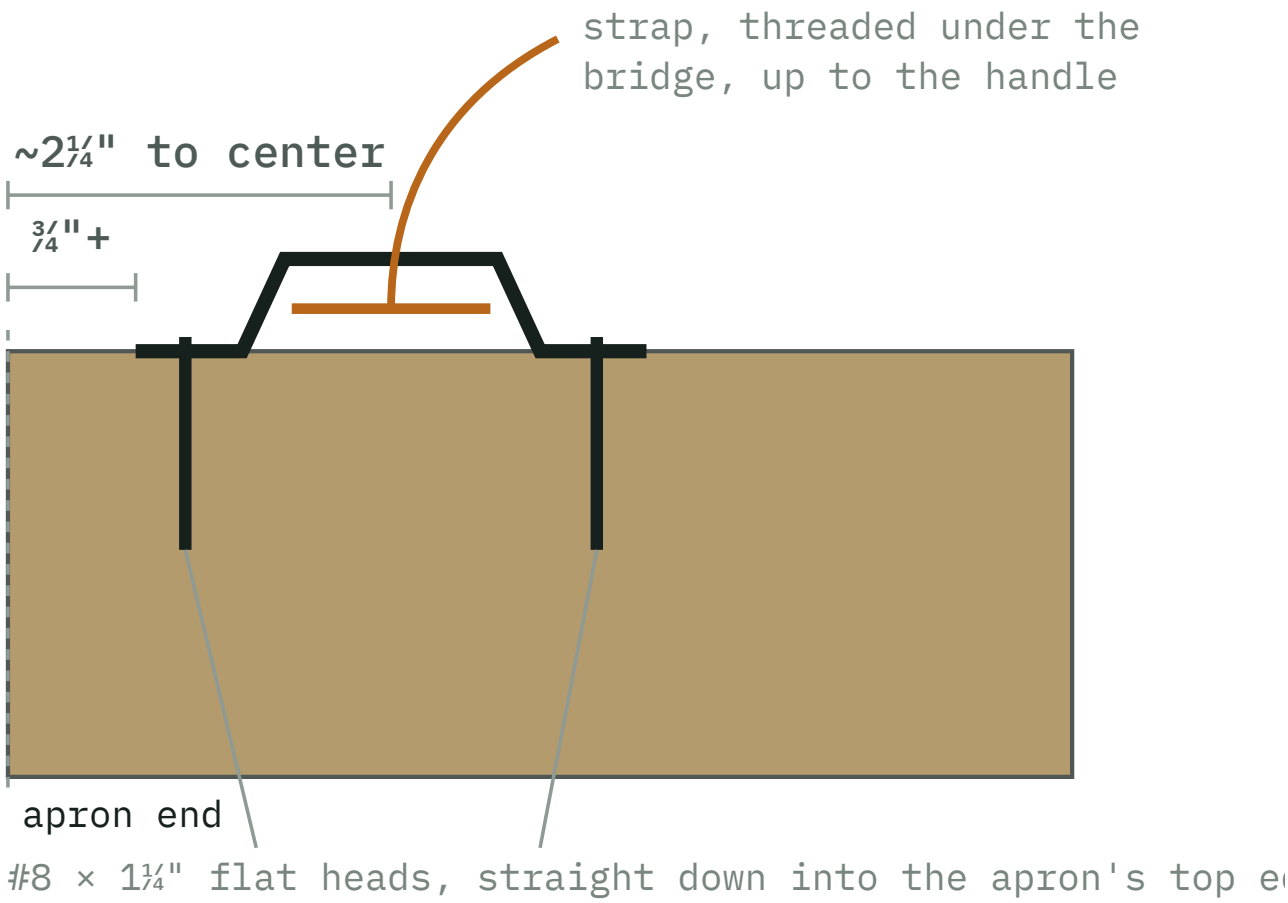



**FIG 7 — FOOTMAN LOOP ON THE APRON**

Lengthwise on the ¾" top edge, its tabs centered on the edge. Drill 3/32" pilots straight down, 1¼" deep, and drive #8 × 1¼" flat heads into the countersunk holes. Thread the strap under the bridge; no hooks.

## F Cut list

1 panel · 2 aprons

| PIECE | QTY | SIZE | NOTES |
|---|---|---|---|
| Tray panel, ½" ply | 1 | 17⅞ × 28¾" | From the ½" sheet's leftover.<br>Width = the fridge's measured base + ⅛". |
| Side apron, 1×4 | 2 | 2½ × 28¾" | Ripped from one 1×4.<br>57½" of board plus kerf. The bed's 1×4s cover it if you buy seven. |

## G Hardware

tray only

| ITEM | QTY | WHERE |
|---|---|---|
| Everbilt #8 × 1¼" flat head | 12 | 6 per apron, into the panel's edges (Fig 4). **On hand.** |
| Wood glue | — | Both apron joints, full length. **On hand.** |
| Stainless footman loop + 1" cam strap, #8 × 1¼" screws | 2 + 1 | Fridge hold-down: loops along the kitchen-side apron's top edge, strap through the end handles. **To buy**, about $10. On the shopping list. |
| VADANIA 24" locking slide pair | 1 | Later, with the anchor board. **On hand.** |
| Screw-mount cord clips | 3 | Later: the fridge's DC line slack, clipped to the driver-side fixed rail, never to the tray. **On hand.** |

## H Build order

about an hour, plus glue time

**01** DONE

### Measure the fridge

Across its base, along its length over the handles, and where each end handle sits. The panel is the base width plus ⅛".

**02** DONE

### Cut the panel

17⅞ × 28¾" (or your width) from the ½" leftover. Check it's square: the diagonals should match, about 33⅞" at 17⅞".

**03** DONE

### Rip and cut the two aprons

Rip a 1×4 to 2½", then crosscut two pieces at 28¾", the same length as the panel. Sight down each one and put any bow facing up, so the bow sits away from the floor.

| | |
|---|---|
| 04 | DONE<br>**Dry-fit on a flat floor**<br>Panel flat on the floor, an apron against each long edge, ends flush. The floor makes the bottoms flush. Set the fridge in: it should drop between the aprons with a little play each side. |
| 05 | DONE<br>**Glue and screw the aprons**<br>Glue each edge, clamp the apron on, and check the bottoms are still flush. Pilot 3/32" into the ply edge, ¼" up from the bottom, at 1½" from each end and five more evenly between (about 5⅛" apart). Countersink and drive 6 × 1¼" per side, heads flush (Fig 3). |
| 06 | **Clean up**<br>Wipe off the squeeze-out on the inside corners before it sets, so the fridge sits flat on the panel. When the glue has cured, ease the edges with sandpaper. A coat of finish is optional. |
| 07 | **Mark and fit the D-rings**<br>Set the fridge on, compressor end toward the tailgate end of the tray. Mark the kitchen-side apron's top edge under each end handle, pilot 3/32" and screw the two footman loops down lengthwise with #8 × 1¼" flat heads (Figs 5–7). Thread the cam strap loop to handle, along the lid, handle to loop, buckle on top, and cinch it. |
| 08 | STOP HERE<br>**Slides wait for the anchor board**<br>The moving rails screw to the aprons' outer faces, at a height set by the anchor board and its risers, with at least 1½" of each rail on the apron. They go on with the anchor board. |

---

Geometry from `params.scad` ( `fridge_ext_*` , `fridge_tray_t` , `fridge_tray_gap` , `fridge_apron_t` , `fridge_rail_t` , `fridge_riser_t` , `fridge_stack_top` ) and the Panel C Frame sheet as built: rear legs 1-7/16" square at the true corners, deck 18½", rear rail underside 17-1/16".

Where the model and this sheet differ: `params.scad` sizes the panel at the fridge's 17.72" and puts the driver apron against the rear leg ( `x_fridge_module` ). This sheet adds ⅛" to the width and sets the tray 2½" in from the driver edge, so the driver slide clears the leg at full extension and the tray hangs between the anchor board's strips. The model's open issue about the apron's ¾" thickness is settled here the way the plan's section V2 draws it: rail outside the apron, apron against the panel.

# Riser Angles

Two 24" steel angles that carry the fridge slides on the anchor board. Each slide's fixed rail screws to an angle's upright leg; the angle's flat leg, the flange, bolts down to an anchor-board strip.

| ASSEMBLY | STOCK | DRIVER RISER | KITCHEN RISER |
|---|---|---|---|
| **Section 8 · Comp. 7**<br>anchor board · fridge slides | **2 × 2 × 3/16" angle**<br>2 pieces, 24" each | **As bought**<br>2" flange, on the driver strip | **Flange 1"**<br>ripped full length · on the middle strip |
| **POSITION** | **FIXING** | **DRAWN FROM** | **STATUS** |
| **2½" → 26½"**<br>from Panel C's tailgate edge | **4 × ¼-20 each**<br>down into T-nuts in the strip | **Build plan**<br>Section 8 · params.scad · Fridge Tray sheet | **Cut**<br>14 Oct · holes wait for the anchor board sheet |

## A Check before you cut

1 method · 2 checks · 1 carry-over

Angle steel is an L: two legs, each 2" wide and 3/16" thick. On a riser one leg stands upright and the other, the flange, lies flat on the board. Only one flange gets cut.

METHOD

### The flanges point away from the tray

The tray's aprons hang ⅜" off the floor, below the board's top, right beside each riser's upright. A flange pointing toward the tray would sit where the apron runs.

**So both flanges point outboard: the driver one toward the van's side wall, the kitchen one toward the kitchen unit.**

CHECK

### The driver riser goes in as bought

Its full 2" flange reaches ¼" past the board's driver edge, about 1" off the floor. Nothing is there: the rear leg is behind where the riser starts, and the van's side wall is farther out. All four bolts still land in the strip.

```
upright back at 1¾" from the driver edge − 2" flange = ¼" past the edge
```

CHECK **The kitchen riser's flange is cut to 1"**

A full 2" flange would reach 24⅝" from the driver edge and run under the kitchen unit, whose edge sits at about 24". The unit can't move; it already stands against Panel C's rear leg.

```
upright back at 22⅝" + 1" flange = 23⅝" · middle strip ends at 24" · ⅜" inside it
```

CARRY-OVER **Drill with the anchor board**

**The flange bolt holes and the slide's screw holes are laid out on the anchor board sheet, with the T-nuts. Don't drill the angles before then.**

## B The angle

end view · about 3× full size

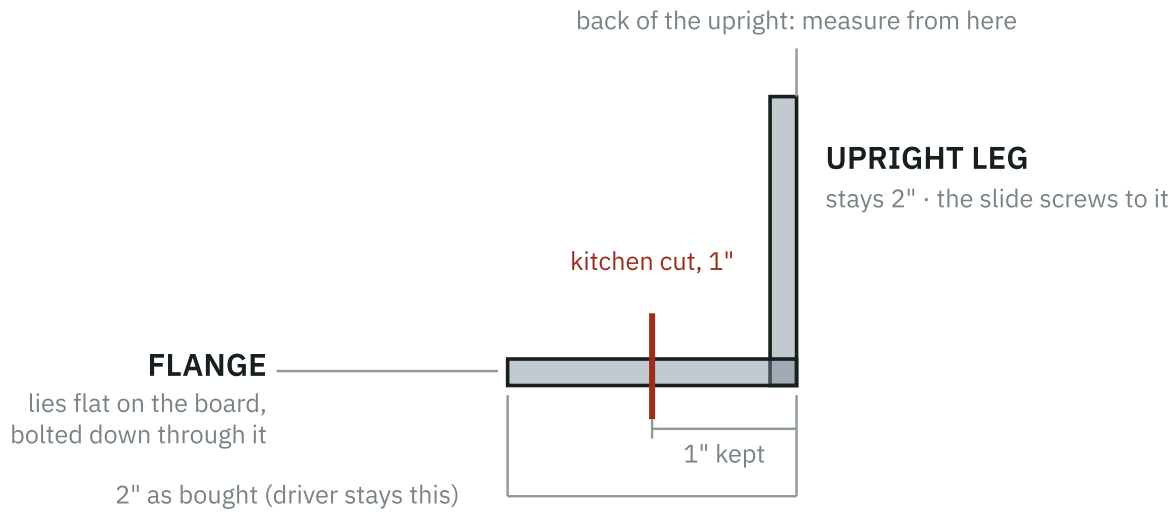



**FIG 1 — END VIEW**

Measure the cut across the flange from the back of the upright leg, the outside face of the L. Everything past the line comes off. The driver riser stays at 2".

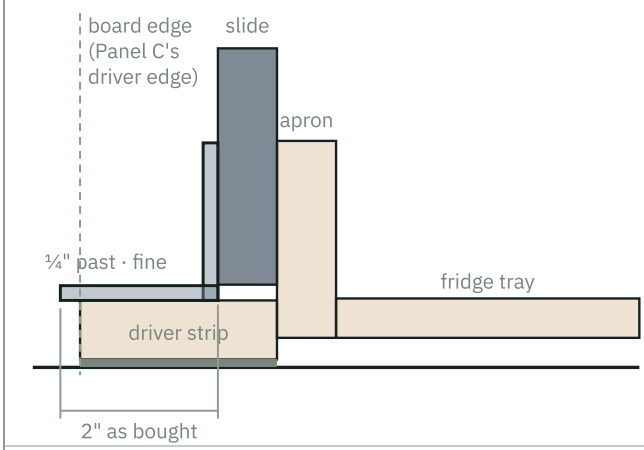



**FIG 2 — DRIVER SIDE**

The full 2" flange sticks ¼" past the board's edge, in clear air. The slide stands against the upright on the tray side, and the apron beyond it.

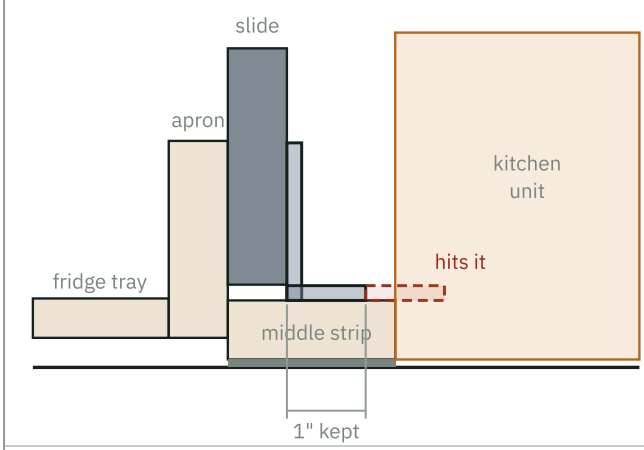



**FIG 3 — KITCHEN SIDE**

Red dashed is the strip that comes off. At 2" the flange would run under the kitchen unit; at 1" it stops ⅜" inside the middle strip.

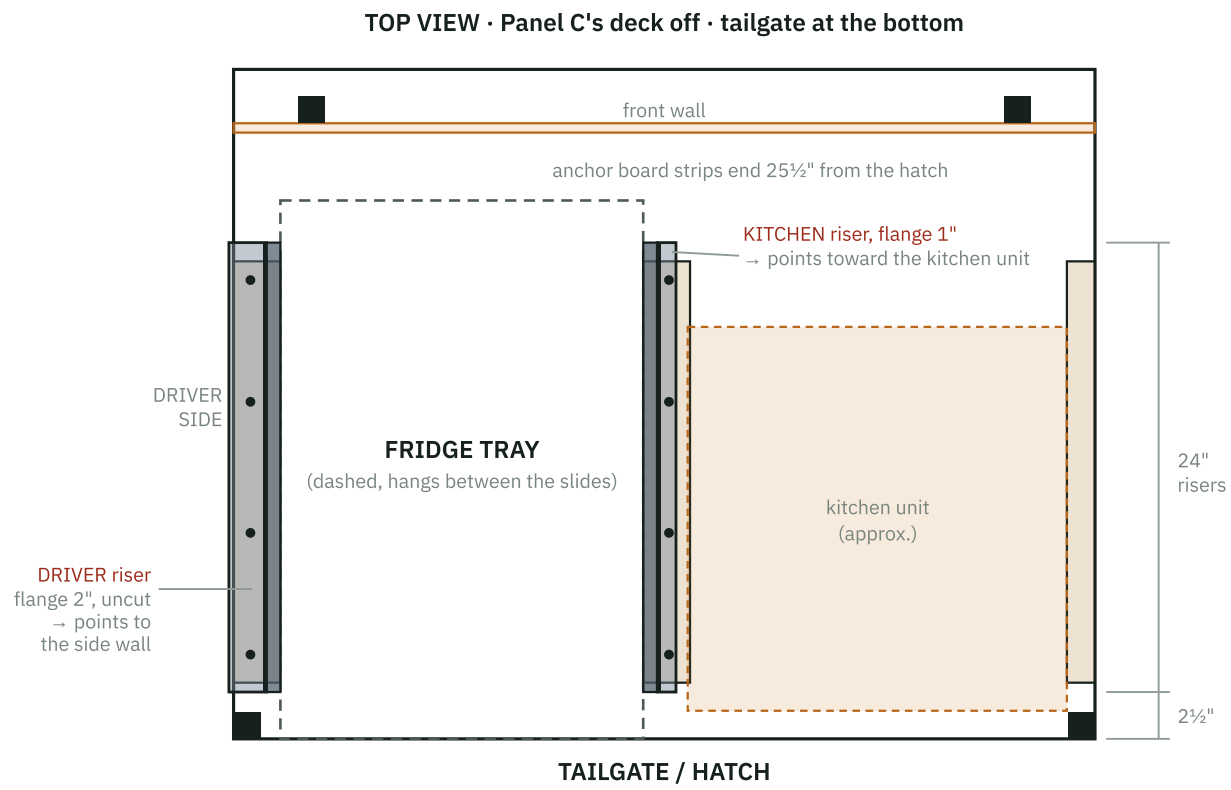



**FIG 4 — RISERS IN PANEL C**

Both risers run 24" fore-aft from 2½" in from the tailgate, each on an anchor-board strip, with the slide against its upright and the flange pointing away from the tray. The dots are the 4 bolts per riser; their exact spots go on the anchor board sheet.

## E The cut from above

front of van to the right · to scale

**FIG 5 — BOTH RISERS FROM ABOVE**

The dark edge is the upright leg seen end-on; measure from it. The kitchen cut takes a 1" strip off the kitchen-unit edge, the full 24", parallel to the upright. Round the two corners at each end of the cut edge.

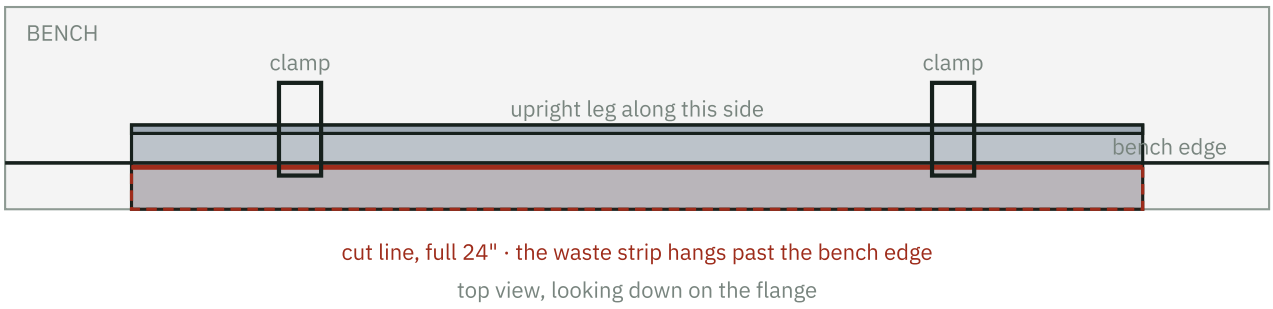



**FIG 6 — ON THE BENCH**

Flange flat, the line just past the bench edge so the waste hangs free, clamps near the upright.

## F Cut list

steel

| PIECE | QTY | SIZE | NOTES |
|---|---|---|---|
| Riser, driver | 1 | 24" × 2 × 2 × 3/16" | **Cut 14 Oct.** **Flange as bought.** |
| Riser, kitchen | 1 | 24", flange 1" | **Cut 14 Oct.** **Flange ripped to 1" from the back of the upright.** |

## G Build order

grinder · cut-off wheel

| | |
|---|---|
| 01 | DONE<br>**Cut both angles to 24"**<br>Square cuts, burrs ground off. Mark one DRIVER and the other KITCHEN. |
| 02 | DONE<br>**Mark the kitchen line**<br>Upright leg up, flange flat. Measure 1" across the flange from the back of the upright at both ends and the middle, and join the marks with a straightedge. |
| 03 | DONE<br>**Rip the kitchen flange**<br>Clamped flat with the waste hanging past the bench edge (Fig 6). A shallow scoring pass the full length, then light passes 6–8" at a time, moving the clamps along. Cut dry, wheel square, face shield and gloves, sawdust swept away first. |
| 04 | DONE<br>**Clean up and check**<br>Grind the cut edge straight and smooth, round the corners at both ends, and check 1" along the full 24". |
| 05 | LATER<br>**Drill with the anchor board**<br>Four ¼-20 clearance holes per flange and the slide's screw holes in each upright, laid out on the anchor board sheet. |

Geometry from `params.scad` ( `fridge_rail_t` , `fridge_riser_t` , `aboard_*` ) and the Fridge Tray sheet: driver apron's outer face 2½" from Panel C's driver edge, tray 19⅜" overall, slide stack ¾", kitchen unit's edge about 24" from the driver edge. Anchor board strips as cut: driver strip 0–2½", middle strip 21⅞–24".

# Bed Platform

The slatted platform over Panels A and B: three pieces, each a pair of flat 1×4 rails with slats pocket-screwed between them, all in one flush ¾" plane at 18½". Panel A's piece screws down; Panel B's two halves lift out.

| ASSEMBLY | OVERALL | PANEL A PIECE | PANEL B HALVES |
|---|---|---|---|
| **Comp. 2**<br>bed platform | **58 × 49"**<br>fore-aft × across · ends at the B/C seam | **29 × 49"**<br>2 rails + 5 × 42" slats · screwed down | **2 × 29 × 24½"**<br>2 rails + 5 × 17½" slats each · loose |
| **GAPS** | **JOINT** | **DRAWN FROM** | **STATUS** |
| **2⅞" · 3⅛"**<br>Panel A · Panel B halves | **Glue + 2 pocket screws**<br>per slat end · 1¼" coarse · 60 in all | **Build plan**<br>Comp. 2 · Bed Platform Detail · parts as cut | **Cut**<br>14 Oct · all 21 parts cut · assembly next |

## A Check before you assemble

2 methods · 2 checks

Every piece is the same idea: two 29" rails lying flat, slats butted between their inner edges, the first and last slat flush with the rail ends.

METHOD

### Assemble face down on a flat floor

The pocket holes go on the slats' underside, so the mattress side never sees a screw. Lay every part mattress-side down on a flat floor, drill and screw from above, and the floor makes the top faces flush for you.

**Every slat end is glued and screwed: glue on the slat's end, then the two pocket screws, which clamp the joint while the glue cures.**

METHOD

### Two jigs: the pocket-hole jig and spacer blocks

**Pocket-hole jig** (Kreg 320): it clamps to a board and guides a stepped bit at 15°, drilling an angled pocket into the slat's underside that a screw runs through into the rail's edge. Set the drill guide and the bit's depth collar for ¾" stock.

**Spacers**: strips of plywood ripped to the gap width and cut 13" long, laid flat between slats so every gap comes out equal without measuring. The 13" length keeps them from twisting or tipping. Rip two at **2⅞"** for Panel A and two at **3⅛"** for Panel B.

CHECK

### The narrow slat goes in the middle of each B half

One 1×4 went to the fridge tray's aprons, so two of Panel B's ten slats are 2½"-wide 1×3. Put one in the middle of each half. That half's four gaps open to 3⅛"; Panel A's stay at 2⅞".

```
Panel A: (29 − 5 × 3½) ÷ 4 = 2⅞" · Panel B half: (29 − 4 × 3½ − 2½) ÷ 4 = 3⅛"
```

CHECK

### Square before you screw

**Measure both diagonals before the last screws go in: about 56-15/16" on Panel A's piece and 37-15/16" on each B half. Equal diagonals mean it's square; push the long corner in until they match.**

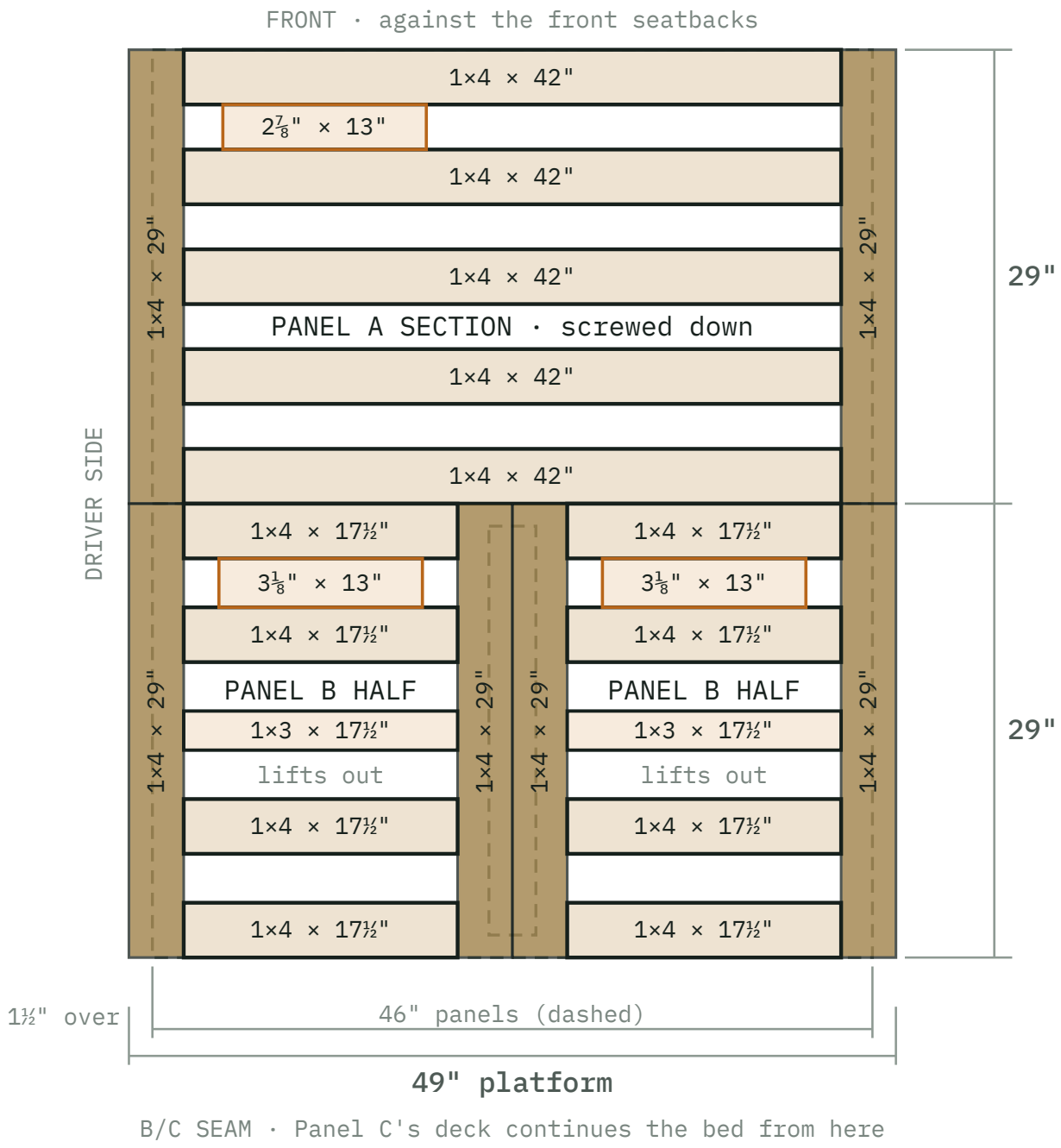



**FIG 1 — THE PLATFORM ON PANELS A AND B**

Panel A's piece spans Panel A; the two halves split Panel B down its centerline. All three overhang the 46" panels 1½" each side. Every piece is labeled with its lumber and length; the narrow 1×3 slats are shaded amber. The amber strips are the 13" plywood spacers:

**2⅞" IN PANEL A'S PIECE, 3⅛" IN BOTH B HALVES**

, because the halves' narrow middle slat leaves more room to share out. Panel B's center bearer is dashed under the halves' inner rails.

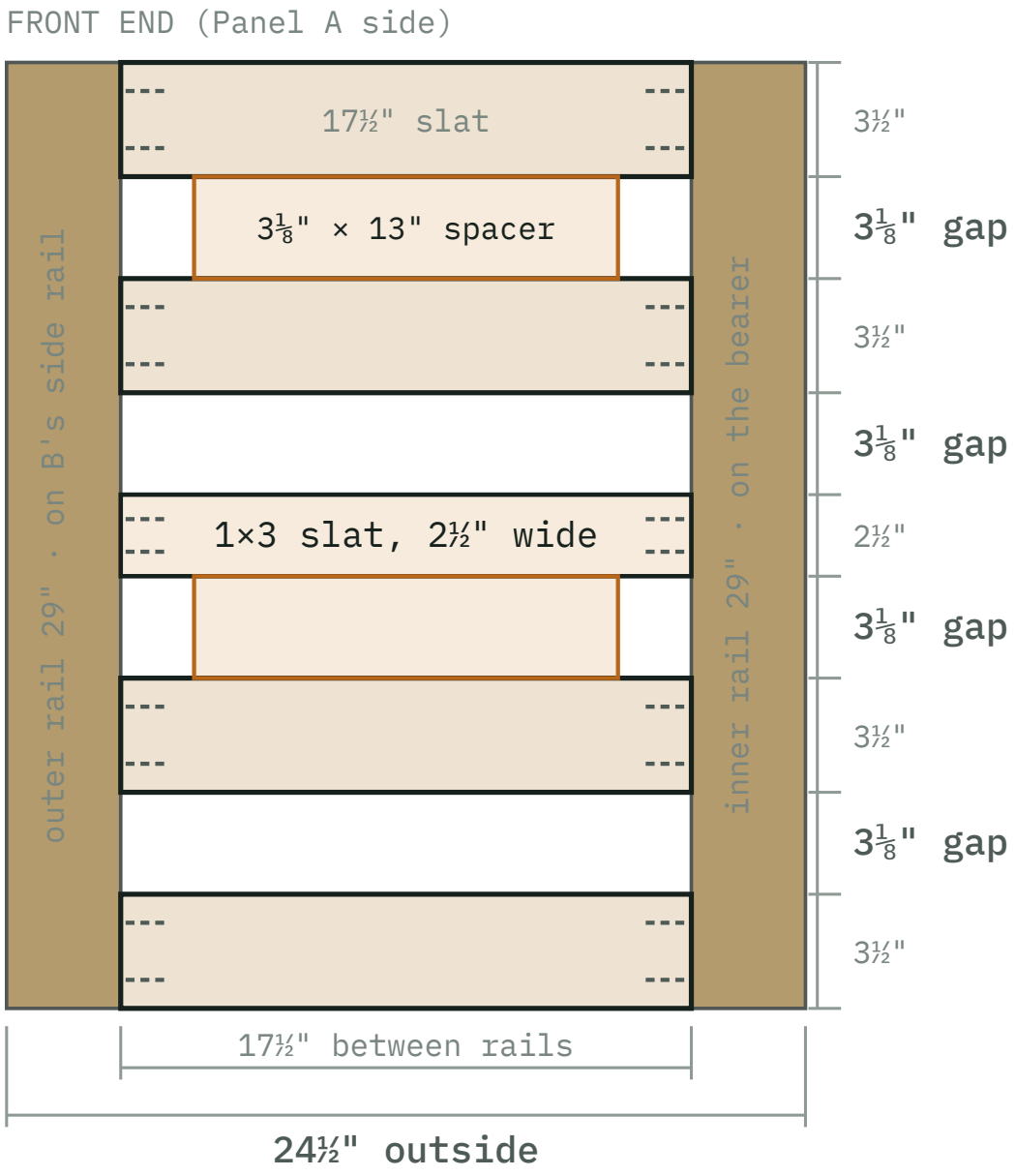



**FIG 2 — A B HALF, FACE DOWN**

The end slats sit flush with the rail ends. Each slat end gets two pocket holes (dashed), ⅞" in from its edges on a 3½" slat and ¾" on the 2½" one. The 13" spacers set the 3⅛" gaps. Panel A's piece is the same with 42" slats, all 3½" wide, and 2⅞" gaps.

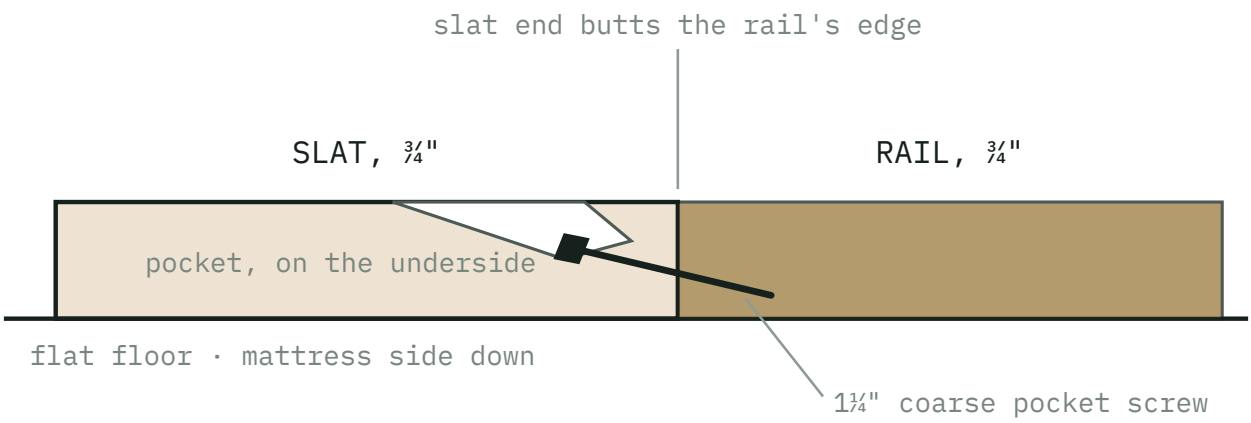



**FIG 3 — POCKET-SCREW JOINT**

About 4× full size, shown face down the way it's built. Glue the slat's end, then the screw runs from the pocket in the slat's underside, through the slat's end, into the rail's edge, clamping the glue line.

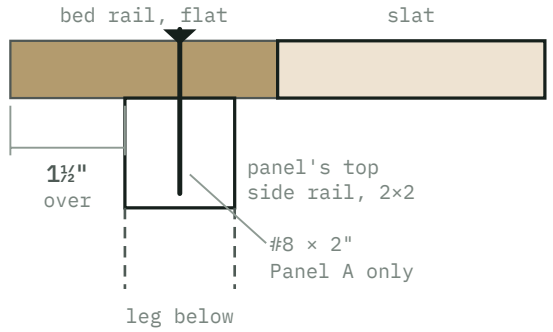



**FIG 4 — OUTER EDGE**

The bed rail rests on the panel's top side rail and overhangs it 1½". On Panel A only, a countersunk #8 × 2" screw goes down through the bed rail into the frame rail.

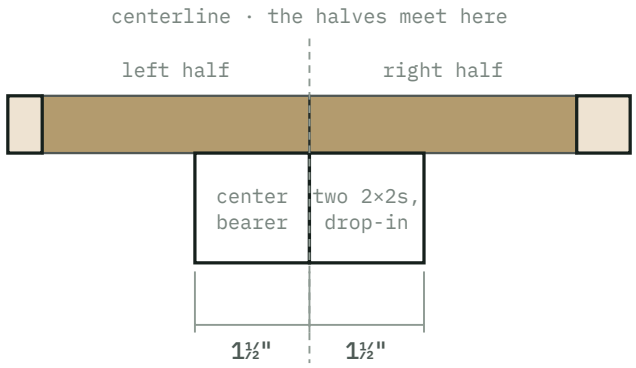



**FIG 5 — PANEL B CENTERLINE**

Each half's inner rail bears 1½" on the 3" drop-in center bearer. Nothing is screwed, so either half lifts straight out.

## E Parts

all cut 14 Oct

| PIECE | QTY | SIZE | GOES TO |
|---|---|---|---|
| Rail, 1×4 | 6 | 29" | 2 per piece, lying flat |
| Long slat, 1×4 | 5 | 42" | Panel A's piece |
| Short slat, 1×4 | 8 | 17½" | 4 per B half |
| Short slat, 1×3 | 2 | 17½" × 2½" | The middle slat of each B half |
| Spacer, ½" or ¾" ply scrap | 2 + 2 | 2⅞ × 13" · 3⅛ × 13" | Ripped on the table saw. Assembly only, not part of the bed |

## F Hardware

bed only

| ITEM | QTY | WHERE |
|---|---|---|
| 1¼" coarse-thread pocket screws | 60 | 2 per slat end, 15 slats. Kreg SML-C125-100, 100-pack. **To buy**, about $10. |
| #8 × 2" flat head wood screws | 18 | Panel A's piece only: 4 down through each rail into the side rails, 5 through each end slat into the end rails, countersunk flush.<br>The 1¼" #8s are too short here and the 2½" SPAX poke through the 2×2. |
| Wood glue | — | Every slat end, two thin coats on the end grain. **On hand.** |
| Kreg Pocket Hole Jig 320 + stepped bit | 1 | Set for ¾" stock, held with one of your clamps. **To buy**, about $40. Add a long #2 square-drive bit if the box has none. |

## G Build order

about 2 hours

**01**

### Sort and face the parts

Three piles: Panel A (2 rails, 5 × 42"), and two B halves (2 rails, 4 × 17½" 1×4 and 1 × 17½" 1×3 each). Sight down each slat: if it bows, the crown (the hump) goes **up**, on the mattress side, so weight flattens it instead of adding to the sag. Otherwise pick the better face. Pencil an X on the other face, the underside.

| | |
|---|---|
| 02 | **Drill the pocket holes**<br>Jig and bit collar set for ¾". Two holes in each end of every slat, on the X face: ⅞" in from each edge on the 1×4s, ¾" on the 1×3s. 60 holes in all. Run a test hole in a scrap first. |
| 03 | **Rip the spacers**<br>**Measure your slats first.** A 1×4 is often 3¼–3⅜" wide, not 3½", and five slightly narrow slats leave about 1" extra in the last gap. Work out the gaps from your boards: Panel A = (29 – 5 × slat width) ÷ 4; B half = (29 – 4 × 1×4 width – 1×3 width) ÷ 4. The 2⅞" and 3⅛" below are for full 3½" and 2½" boards. Then from ½" or ¾" ply scrap: rip two strips at 2⅞" and two at 3⅛", then cut all four to 13". Pencil the width on each so they don't get mixed up. |
| 04 | **Lay out Panel A's piece face down**<br>On a flat floor, mattress side down: the two rails 49" apart outside to outside, the five 42" slats between them, X faces up. First and last slat flush with the rail ends; 2⅞" spacers in each gap. |
| 05 | **Glue and screw it, squaring as you go**<br>Wood glue on each slat end before it goes in. End grain soaks glue up, so brush on a thin coat, wait a minute, then add a second coat. Clamp a rail and the end slat, drive that end's two pocket screws, then the other end slat. Check the diagonals (56-15/16"), then fill in the middle slats against the spacers. A bowed slat touches the floor in the middle with its ends lifted: press each end flat beside the rail while you drive its screws. Snug, not stripped: pine strips easily. Wipe off squeeze-out, and leave it flat for an hour before moving it. |
| 06 | **Build the two B halves the same way**<br>Glue and pocket screws at every slat end, the same way. Rails 24½" apart outside, the 1×3 slat in the middle, 3⅛" spacers. Diagonals about 37-15/16". |
| 07 | **Ease the edges**<br>Sand the top edges and corners round so nothing wears the mattress cover. |
| 08 | **Fit Panel A's piece and screw it down**<br>Mattress side up on Panel A, 1½" over each side, ends flush with Panel A's frame. Pilot and countersink, then #8 × 2": 4 through each rail into the side rail, about 2½" from each end and two between, and 5 through each end slat into the end rail, about 8" apart. |

| | |
|---|---|
| 09 | **Drop in the B halves**<br>Center bearer in place first. Each half sits with its outer rail on B's side rail and its inner rail on the bearer, against Panel A's piece at one end and Panel C's deck at the other. No fasteners. |
| 10 | LATER<br>**Bubble level, with the height setting**<br>Stick the bar level on the outer edge of Panel A's driver-side rail at mid-span when the panels are leveled. |

---

Geometry from the build plan's Bed Platform Detail and `params.scad` (`bed_frame_length` 58", platform 49" across, 1×4 at ¾ × 3½"), Panels A and B as built (46" wide, 1-7/16" rails, deck plane 18½"), and the bed parts as cut on 14 Oct, including the two 1×3 slats.

# Fridge Slides

How the fridge tray hangs: each apron on a slide, each slide on a riser angle, each riser bolted into T-nuts in an anchor-board strip. Every hole in the steel, the board and the aprons.

| ASSEMBLY | SLIDES | RAIL HEIGHT | FORE-AFT | STEEL HOLES |
|---|---|---|---|---|
| **Comp. 7 · Section 8**<br>fridge slides on the anchor board | **VADANIA VD2576**<br>24" · 76 mm rail · locking · side-mount | **15/16" → 3-15/16"**<br>bottom 1/16" above the board · from the floor | **2½" → 26½"**<br>from Panel C's tailgate edge | **9/32"**<br>8 in the flanges · ~8 in the uprights |
| **BOARD** | **DRAWN FROM** | **STATUS** | | |
| **8 T-nuts**<br>¼-20, flush from below | **Build plan**<br>Riser Angles · Fridge Tray · params.scad | **Ready to drill**<br>risers cut · tray built | | |

## A Check before you drill

2 checks · 2 methods

The tray's right side, the kitchen side, hangs from the kitchen-side slide: apron → moving rail → fixed rail → kitchen riser → middle strip. The left side is the mirror image on the driver strip.

### CHECK The slides don't come apart, and may bring their own bolts

VADANIA's directions mount each slide in one piece: press the lock button and pull the rail out to expose the screw holes, screw the outer (fixed) rail to the cabinet, then set the drawer in place and screw the inner (moving) rail to it, sliding the rail along to reach each hole. They call for "bolts, nuts, or other suitable fasteners from the accessory pack".

**The slides came with their bolts and nuts: use those for the fixed rails into the steel risers. Into the wood aprons: #12 × ⅝" flat-head wood screws, unless the pack has short wood screws of its own.**

### CHECK Keep the apron screws ⅝" or shorter

The aprons are ¾" thick and the fridge sits about 1/16" inside them. A ¾" screw would come through into the fridge's side.

### METHOD The rails hang 1/16" above the board

Resting on the board, the moving rail would scrape the strip every time the tray slides. Set the fixed rail's bottom edge 1/16" above the flange's underside (the board's top) with a shim while you mark and drill: a piece of cereal-box cardboard or two playing cards is about 1/16".

```
board top 0.85" + 1/16" = rail bottom 15/16" · rail 3" · tray hangs ⅜" · apron overlap ≈ 1⅞" (needs 1½")
```

METHOD **Drill the steel slow and wet**

**Center-punch every mark and clamp the angle down. Pilot with the ⅛" Thunderbolt, then open the hole with step bit #1 until the 9/32" step (the sixth from the tip) finishes cutting. The 3/16" steel is taller than one step, so flip the angle and touch the hole from the other side to straighten it, then kiss both faces with the next step to deburr. Low speed, firm pressure, a drop of Tap Magic in each hole.**

## B What holds the tray

section at floor level · looking forward · to scale

**FIG 1 — WHAT HOLDS THE TRAY**

Looking forward from the tailgate. Each apron screws to a slide's moving rail; the moving rail rides in the fixed rail; the fixed rail is screwed to a riser's upright; the riser's flange bolts down into T-nuts in an anchor-board strip. The board lies on its mat, strapped forward and down to the 3rd-row seats' latch hoops.

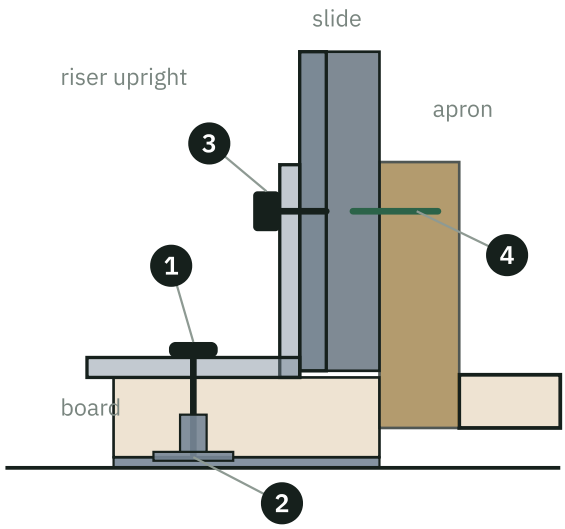



**FIG 2 — LEFT (DRIVER) SIDE**

1 ¼-20 × 1" button head through the flange. 2 ¼-20 T-nut, flush in the strip's underside. 3 ¼-20 flat-head machine screw through the fixed rail and upright, lock nut on the outboard face. 4 #12 × ⅝" flat-head wood screw through the moving rail into the apron. Rail bottom 15/16" and screw line 2-7/16" above the van floor.

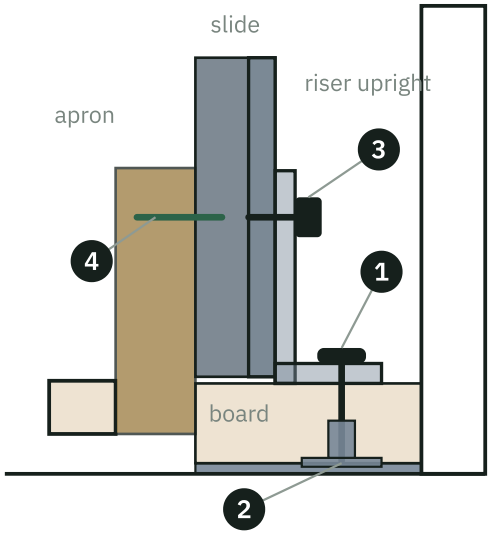



**FIG 3 — RIGHT (KITCHEN) SIDE**

The mirror of Fig 2. The flange is 1" here, so the bolt sits ⅝" from the back of the upright, and the rail screw's nut is on the kitchen-unit side, about 1¼" short of the unit.

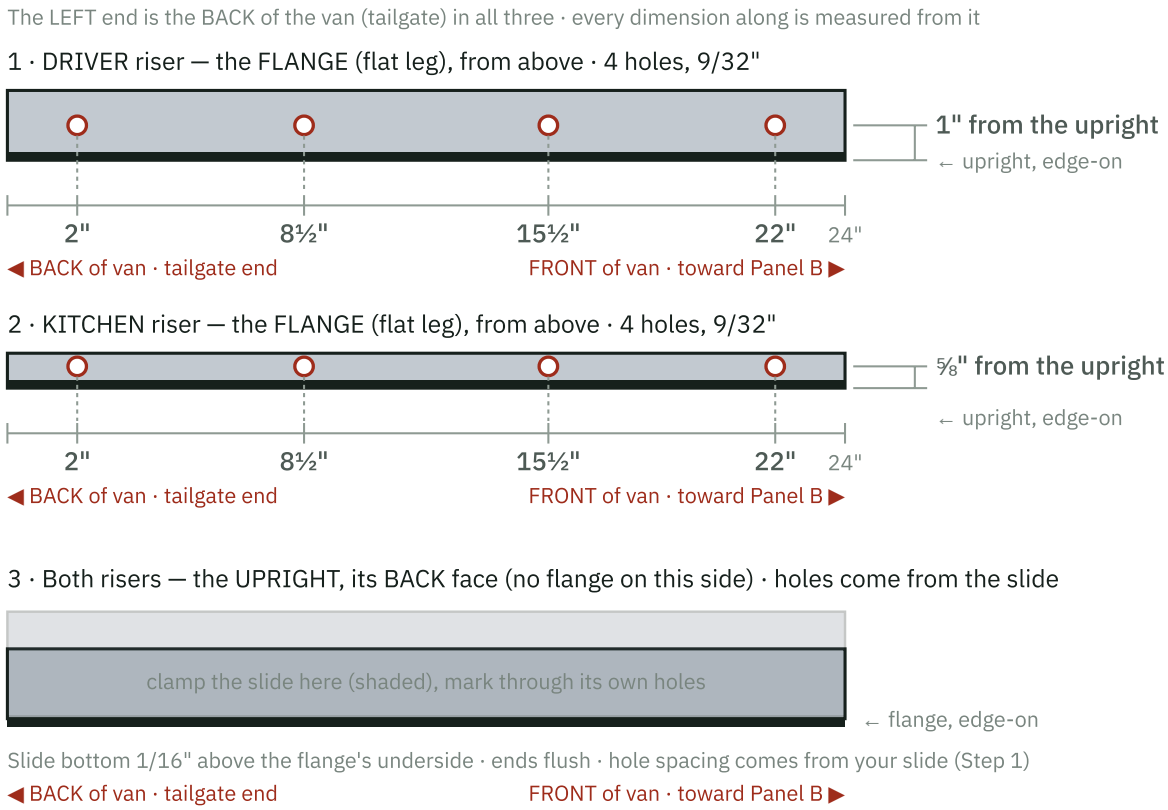



**FIG 4 — DRILLING THE RISERS**

Parts 1 and 2 are the flanges, the legs that lie flat and bolt down to the board: four holes each, measured along from the tailgate end (the back of the van) and across from the upright. Part 3 is the upright, the leg the slide bolts to: its holes have no measurements because they come from the slide. Clamp the slide on, 1/16" above the flange's underside with its ends flush, and mark through its holes.

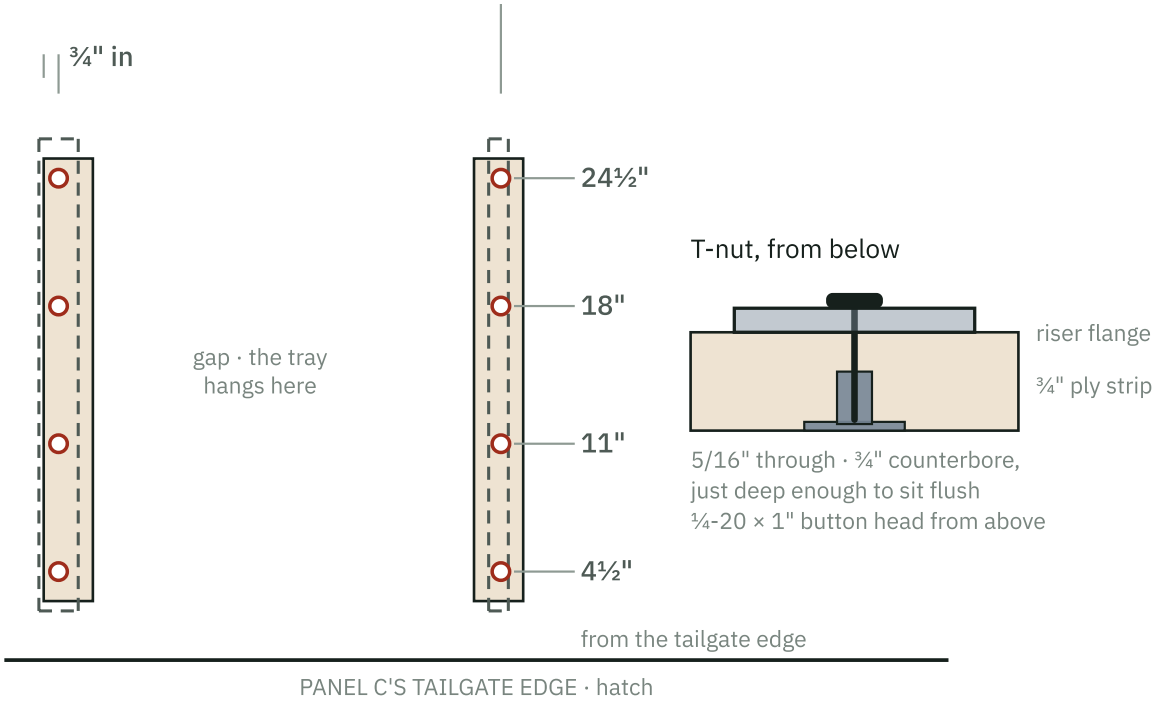



**FIG 5 — T-NUTS IN THE BOARD**

Eight T-nuts, four per strip, on the lines the riser holes make. Transfer them from the drilled risers: set each riser on its strip with its tailgate end ½" behind the strip's back end (the strips start 3" from the hatch, the risers 2½") and its flange edge where shown, and mark through. Every T-nut goes in before the board goes in the van.

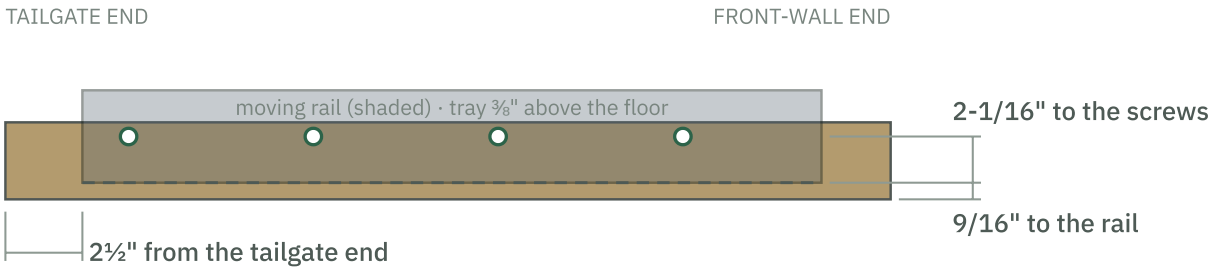



**FIG 6 — MOVING RAIL ON THE APRON**

Both aprons the same way. With the rail's bottom 9/16" above the apron's bottom the tray hangs ⅜" off the floor and about 1⅞" of rail sits on the apron. Mark through the rail's holes; ⅝" screws stop ⅛" short of the fridge side.

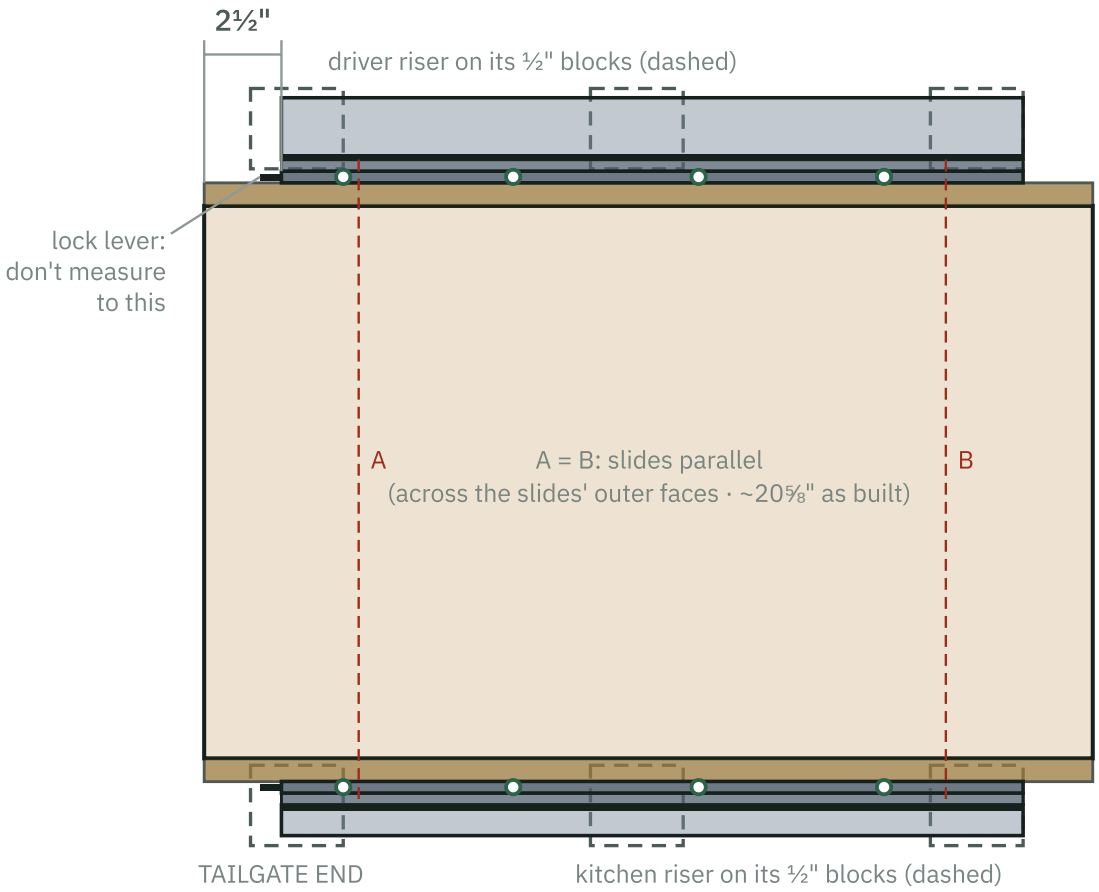



**FIG 7 — LINE IT UP, SLIDES CLOSED**

Driver riser on the left, kitchen on the right, looking from the tailgate end; lock levers at that end. Each riser sits on three separate single-layer ½" blocks (one near each end, one in the middle), never stacked. The tray sits flat on the floor. Measure the 2½" to the square end of the slide's rail, not to the lock lever. Push each riser in until its slide lies flat on the apron, then check A = B.

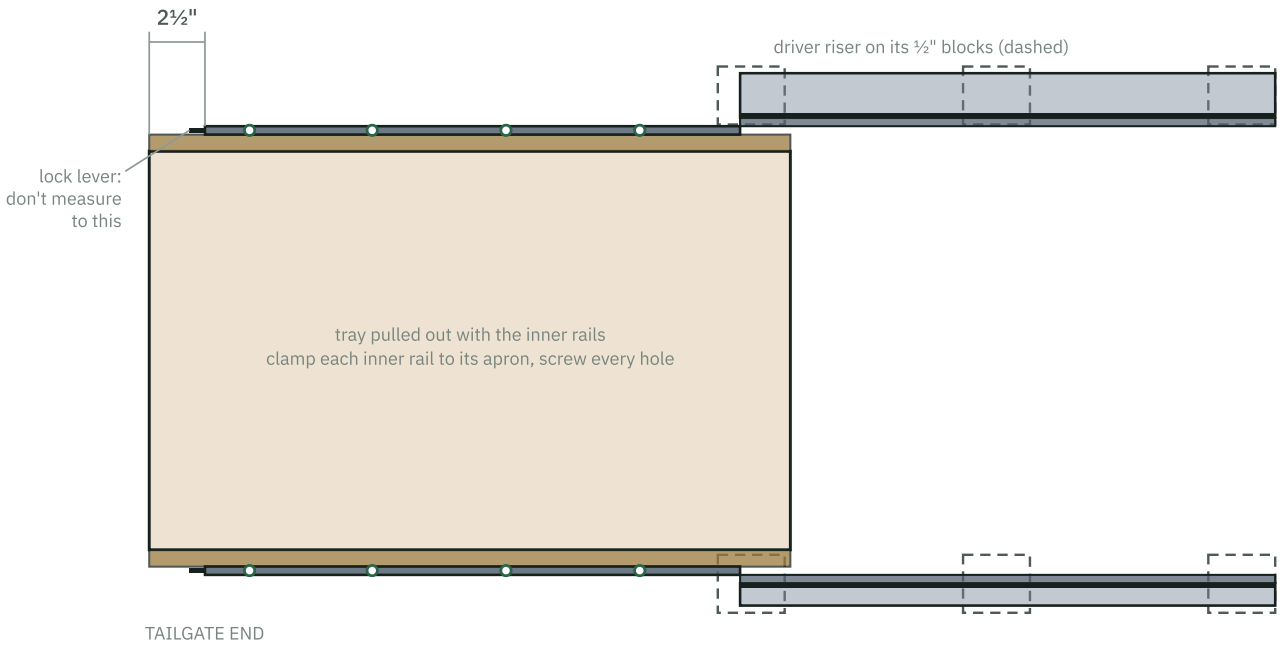
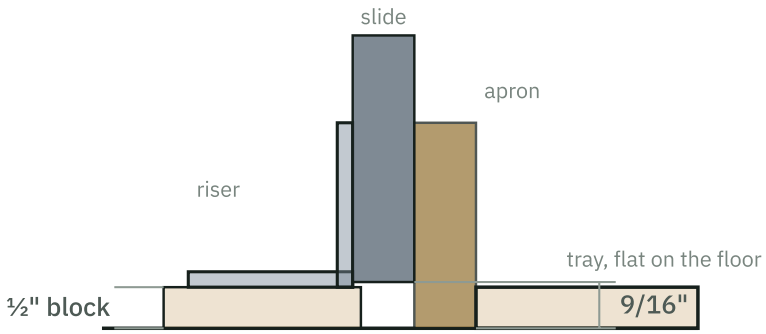



**FIG 8 — PULL OUT TO SCREW**

Pull the tray and both inner rails out together. Keep each inner rail's end 2½" from its apron's end (clamp it there) and screw every exposed hole. Push it closed and recheck the 2½" at the fixed rails.




The slide sits 1/16" above its riser's base, so ½" of block puts it 9/16" up

**FIG 9 — THE HEIGHT**

Driver side, from the tailgate end, about twice full size. One ½" layer under the riser, not a stack; the tray flat on the floor. The kitchen side mirrors it.

## F Hardware

both slides

| ITEM | QTY | WHERE |
|---|---|---|
| ¼-20 × 1" button-head machine screw | 8 | Down through the riser flanges into the T-nuts (Fig 2, item 1) |
| ¼-20 four-prong T-nut | 8 | Flush in the strips' underside (Fig 5) |
| Slide bolts + nuts, from the VADANIA accessory pack | 8–10 | Fixed rails to the risers' uprights, heads inside the rail (item 3). ½" long. **Came with the slides.** Through about 1/16" of rail and 3/16" of steel they leave about ¼" for the nut: enough only if the bolt's end comes flush with the nut. If not, use M6 × 16 mm. |
| #12 × ½" flat-head wood screw (¾" if the tips are ground flush) | 8–10 | Moving rails to the aprons (item 4). **Bought.** As built the aprons are ⅝", so a ¾" screw comes through about ⅛" on the fridge side: use ½", or grind every tip flush with a Dremel cut-off wheel. Pilot ½" into the wood (tape 9/16" from the bit's tip). |
| Milwaukee Thunderbolt bits + step bits #1 and #4 | – | ⅛" pilots in steel and the aprons · 3/32" footman-loop pilots · step bit #1 to 9/32" for every steel hole · step bit #4 to 5/16" for the T-nut holes, then to ¾" from below, about 1/16" deep, for the counterbores. **On hand.** |
| Center punch, cutting fluid | – | The steel holes. **Tap Magic on hand.** |

## G Build order

bench first, van last

**01** **Check the hardware and the lock end**

Open the slides' accessory pack: bolts and nuts for the steel. Press the lock button and pull one slide out to full length to see the holes. The lock button goes at the **tailgate end** on both slides, so you can reach it with the tray out. Try a bolt in a hole of each rail. Measure the bolt's threads, not its head: about ¼" means M6, which the 9/32" holes suit.

**02** **Drill the flange holes**

Both risers, four each at 2", 8½", 15½" and 22" from the tailgate end: 1" from the back of the upright on the driver riser, ⅝" on the kitchen riser (Fig 4). Punch, ⅛" pilot, 9/32", deburr.

**03** **Fixed rails onto the risers**

Riser flange-down on a flat bench. Stand the whole slide against the back of the L: the upright's flat face that has no flange on that side, so the flange points away from the slide. Put a 1/16" shim under the slide (a piece of cereal-box cardboard or two playing cards) so its bottom edge sits 1/16" above the bottom of the L (the bench). Ends flush with the riser's ends, lock button at the tailgate end. Clamp it. Slide the moving rail along to expose each fixed-rail hole that lands on the upright, mark, drill 9/32", and bolt: heads inside the rail, nuts outside. Run the slide its full travel to check it clears every head.

**04** **Drill and seat the T-nuts**

Lay each riser on its strip (Fig 5), mark through the flange holes, and drill 5/16" straight through. Turn the board over, counterbore each hole ¾" just deep enough for the T-nut's flange, and tap the T-nuts in flush. Test each with a bolt.

| | |
|---|---|
| 05 | **On the garage floor: board on its mat, risers loose**<br>Flat floor, the rubber mat down, the board on it top face up: the mat sets the board at its van height. Both risers on their strips, flanges pointing outward, slides facing each other across the gap, lock buttons at the strip tips. Start the eight ¼-20 × 1" bolts finger-tight so the risers can still shift. |
| 06 | **Set the tray between the slides**<br>Two ⅜" spacers on the floor in the gap (scraps that stack to ⅜"), the tray on them. Slide it fore-aft until the slides' tailgate ends sit 2½" in from the aprons' tailgate ends; the tray's tailgate end then sits about 2" past the strip tips. Push the risers in until each slide lies flat on its apron, and check its bottom edge is about 9/16" above the apron's bottom (Fig 6). |
| 07 | **First screws, tray closed**<br>Tape a depth stop on the ⅛" bit ⅝" from its tip. At a moving-rail hole you can reach near the front of each slide, pilot and drive a #12 × ⅝" flat head. The tray now hangs on both slides. |
| 08 | **Work along the rest of the holes**<br>Pull the spacers. Draw the tray out a few inches at a time; more moving-rail holes show through the openings in the other rail sections. Pilot and screw each one that lands on the apron. |
| 09 | **Square it up, then tighten**<br>Run the tray out to the lock and back to the lock several times so the loose risers settle where the tray wants them. Close it and tighten the eight riser bolts. Run it again: smooth, no binding, locked at both ends. |
| 10 | **Unbolt it for the van**<br>Undo the eight riser bolts: the tray, slides and risers lift off as one unit. In the van it goes neoprene, board, the H1 strap hooked in, then the unit back on its T-nuts and bolted down. Test full travel there: clear of Panel C's rear leg and under its rear rail with the fridge on, then load and strap the fridge. Re-check the riser bolts after the first drive. |

Geometry from `params.scad` (`fridge_rail_t`, `fridge_tray_gap`, `aboard_*`) and the Riser Angles and Fridge Tray sheets: driver apron's outer face 2½" from Panel C's driver edge, tray 19⅜" overall, board top 0.85" with a 0.1" mat. Where the model and this sheet differ: the model stands each rail on its riser's flange, which puts the rail bottom at 1.1"; with the flanges pointing outboard the rail stands beside the upright instead, 1/16" above the board, which gives the apron screws about ⅜" more band.

# Anchor Board

Three plywood strips under Panel C, tied by two steel rungs, carrying the fridge slides' risers and the kitchen unit's straps, and chained forward and down to the 3rd-row seat-latch hoops. No crossbar, no tongues, no holes in the van.

| ASSEMBLY | STRIPS | RUNGS | ANCHORS |
|---|---|---|---|
| **Section 8**<br>anchor board · as redesigned Oct 2026 | **3 × 22½" long**<br>2½" · ~2½" · 1½" wide · 3" → 25½" from the hatch | **2 × 46"**<br>3 × 3/16" steel flat bar · in grooves underneath | **H1 · H2**<br>seat-latch hoops, 28" from the hatch |
| **KITCHEN** | **T-NUTS** | **DRAWN FROM** | **STATUS** |
| **2 straps**<br>front → H2, H4 · rear → loops on the strips | **8 · ¼-20**<br>for the two risers | **Van survey 18 Oct**<br>hoops, trim, seat surface · Fridge Slides sheet | **In progress**<br>assembled on the bench 30 Oct · van fit pending |

## A Check before you cut

3 checks · 3 methods

### CHECK Why there's no crossbar

From 26" forward the side trim closes the van to 42", and the fridge tray fills the space between the driver and middle strips for its whole length. A plywood crossbar could only reach the driver strip through a ½" neck. Steel rungs set into the strips' undersides pass under the tray (it hangs ⅜" over the seat; the rung is 3/16") and under the kitchen unit, so the strips stay one-piece and the board stays one unit.

### CHECK The hoops are the seat latches

The four white hoops in the headrest trough, 28" from the hatch, with their ⅝" bars 1⅞" below the seat surface when it's loaded (2⅜" unloaded), are the 3rd-row seats' latch strikers: built to hold the seats in a crash. Measured from the driver line their centers are at 4¾", 24¾", 31¼" and 41¾" (H1–H4), each 3" wide.

**H1 sits under the fridge tray's front corner: wrap its chain on before the tray assembly goes in. The hoops' bars measure ⅝", too thick for a ¼" shackle, which only closes around about ⅜".**

CHECK

### The strips end past the seat

**The seat surface is solid to 24¾"; the strips run to 25½", overhanging the trough by ¾" and leaving the riser's last ½" clear for the eye bolt. Every T-nut, the last at 24½", is over solid seat.**

METHOD

### Chains from the risers' front ends

From each riser's front end down to its hoop is only about 5": too short for any ratchet strap, whose mechanism alone is longer. So each riser gets a forged ¼-20 shoulder eye bolt hanging down through a 5/16" hole ½" from its front end, just ahead of the strip. A ¼" quick link joins the eye to a short length of ¼" chain, and the chain wraps once around the hoop's ⅝" bar and a second ¼" quick link closes it back on itself: H1 for the driver riser, H2 for the kitchen riser. Turning the eye bolt's nut on top of the flange draws the chain tight; a second nut locks it. **Tension it with the fridge loaded on its tray.** The seat squashes about ½" under weight (the hoop bar is 2⅜" below the bare seat top, 1⅞" loaded), so a chain tightened on an empty board goes slack once the fridge is in. Turn the eye to face its hoop. The load goes hoop → chain loop → quick link → chain → eye bolt → riser → four bolts → strip → rungs.

METHOD

### Backup straps and the relief notches

**The optional backup straps run from the board forward to the 3rd-row striker loops on the 2nd-row floor, under Panel C's front edge and Panel B's end rail. If you run them, cut 3"-wide relief notches there at 12¾–15¾" and 30¼–33¼" from the driver line: 1" up in Panel C's front wall, ½" in its front bottom rail and outer back blocks, ½" in Panel B's end bottom rail.**

METHOD

### Strips from the old comb

**Cut the crossbar off the old comb; each of its three strips is already one piece. Trim each to 22½". New ¾" plywood only if you'd rather start fresh.**

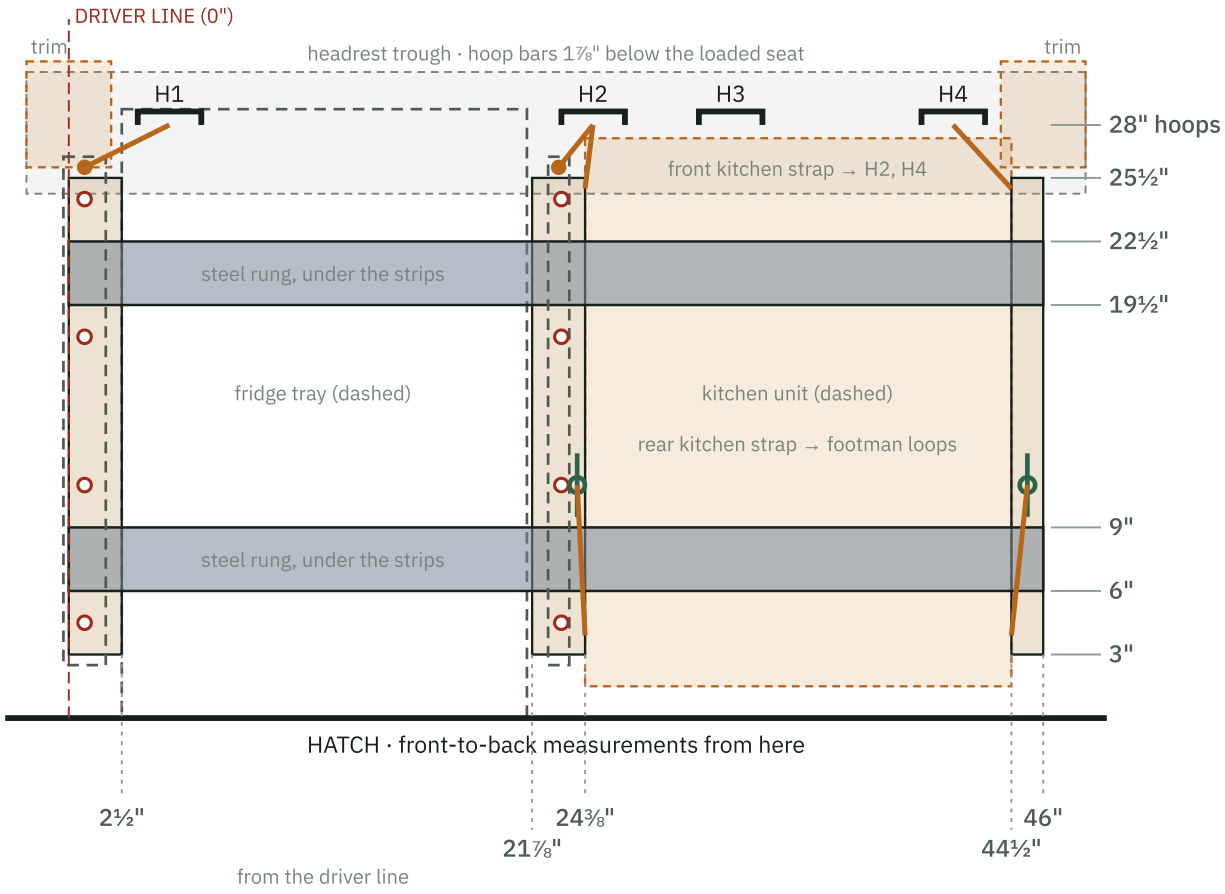



**FIG 1 — THE BOARD IN THE VAN**

To scale, front of the van at the top. Three one-piece plywood strips, tied only by the two steel rungs underneath. Red circles: the 8 riser T-nuts (mark them from the risers). Amber dots: eye bolts in the risers' front ends, with short chains to H1 and H2. Other amber lines: the kitchen's two straps. Green: the rear strap's anchors, an eye bolt in the kitchen riser's second T-nut and a footman loop on the passenger strip. H1–H4 are the 3rd-row seat-latch hoops in the headrest trough.

# C Section and rung detail

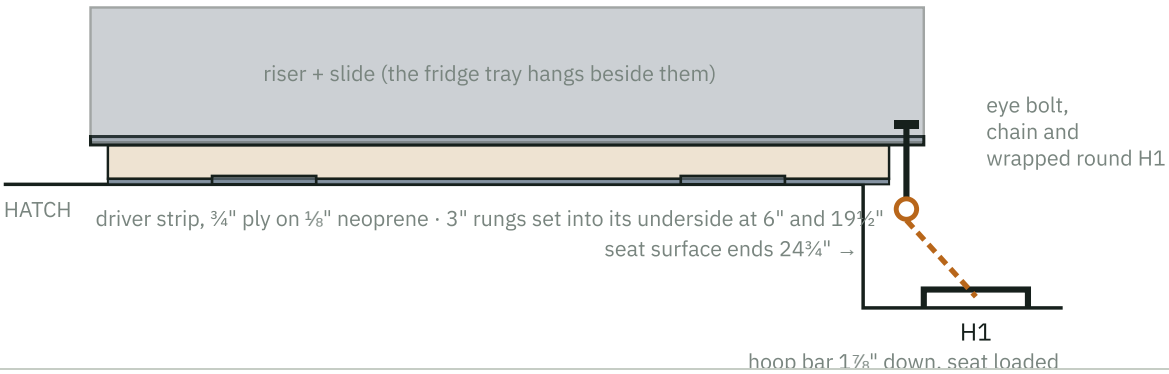



**FIG 2 — ALONG THE DRIVER STRIP**

Side view, heights drawn larger than lengths. The strip runs ¾" past the end of the seat surface; its last T-nut, at 24½", is still over solid seat. The eye bolt hangs down through the riser's front end, just ahead of the strip, and its chain runs forward and down to H1: a hard stop pulls it tight, and it also holds the board down.

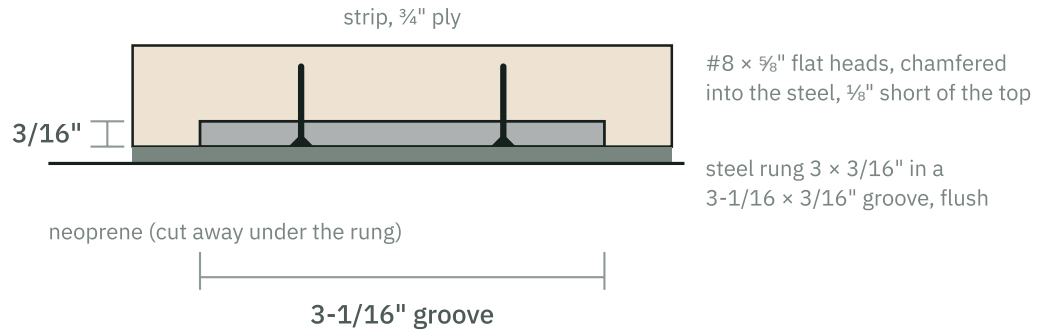



**FIG 3 — A RUNG CROSSING A STRIP**

About 4× full size. Four screws per crossing on the driver and middle strips (a 2 × 2 square), two on the 1½" passenger strip. The neoprene is cut away under each rung so the strip still sits flat.

## D T-nuts

8, for the risers

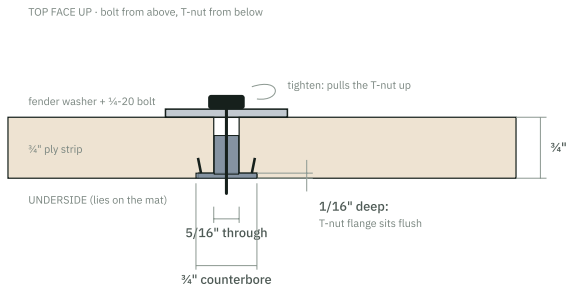



**FIG 4 — PULLING A T-NUT IN**

Same method as before: 5/16" through from the top, ¾" counterbore about 1/16" deep from below, the T-nut pulled flush with a bolt and fender washer.

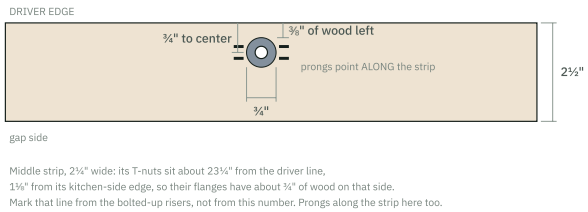



**FIG 5 — WHICH WAY THE PRONGS GO**

Prongs along the strip, never at an edge.

## E Cut list

| PIECE | QTY | SIZE | NOTES |
|---|---|---|---|
| Driver strip, ¾" ply | 1 | 2½ × 22½" | From the old comb. 3"–25½" from the hatch, on the driver line. |
| Middle strip, ¾" ply | 1 | 2¼ × 22½" | From the old comb. Only ½" of wood outboard of the kitchen riser's flange, too narrow for a footman loop, so that strap anchor is on the riser (Hardware). |
| Passenger strip, ¾" ply | 1 | 1½ × 22½" | From the old comb. 44½"–46". |
| Rung, steel flat bar 3 × 3/16" | 2 | 46" | **Two 48" bars (Steelworks 3 × 3/16", Lowe's), cut to 46".** |
| Neoprene strips, ⅛" | 3 | to each strip | ⅛" inside the edges, cut away under the rungs. |

## F Hardware

| ITEM | QTY | WHERE |
|---|---|---|
| ¼-20 T-nuts + ¼-20 × 1" button heads + washers | 8 | Risers to the strips. **On hand.** |
| #8 × ⅝" flat-head wood screws, zinc-plated (Everbilt 825381, or a 100-pack) | 20 | Rungs up into the strips, chamfered into the steel. Not ¾": it would come out the top of the strip. **To buy.** |
| Forged ¼-20 × 2" shoulder eye bolt + nut (Hardware Essentials 851885, 5-pack) | 2 | Down through each riser's front end; a 3rd from the pack, cut short, anchors the kitchen rear strap (below). 500 lb rated in line. **To buy.** |
| ¼-20 hex nut | 2 | Jam nut on top of each eye bolt's nut. **To buy.** |
| ¼" grade 30 galvanized chain (RELIABILT at Lowe's, or Everbilt 806666) | 2 ft | Two pieces, eye bolt to hoop plus about 3" each to wrap the ⅝" bar; cut to the links that come out just taut. 1,250 lb rated. **To buy.** |
| ¼" stainless quick link (Everbilt 43394) | 4 | 2 per chain: one to the eye bolt, one closing the loop around the hoop's ⅝" bar (no shackle: a ¼" one only closes on about ⅜"). 1,200 lb rated. **To buy.** |
| Ratchet strap, 1" (Erickson) | 2 | Kitchen front (H2–H4) and rear (loops). **On hand.** |
| Stainless footman loop + #8 × 1¼" screws | 1 | Kitchen rear strap, passenger end: on the passenger strip, 9½"–12½" from the hatch. **On hand.** |
| ¼-20 eye bolt, cut to ~1" (a spare from the 851885 5-pack) | 1 | Kitchen rear strap, middle end: goes in the kitchen riser's second T-nut in place of a button head, 11" from the hatch, clamping the flange the same way. Thread a nut on, cut, back the nut off to clean the threads; it must reach the T-nut through the flange and ply but not poke out below the strip. Washer or jam nut to set the eye toward the strap. **On hand.** |
| Rusty-metal primer + enamel, spray (e.g. | 1 + 1 | Both rungs, all faces and edges, before they're screwed on (step 4). Plus a wire brush or flap disc and a degreaser wipe. **To buy.** |

| ITEM | QTY | WHERE |
|---|---|---|
| Rust-Oleum Stops Rust, 12 oz each) | | |
| Ryobi ⅝" countersink bit (wood / plastic / metal) | 1 | Countersinks the rung holes so the #8 heads sit flush in the steel. Low speed, oil, stop when the head is flush. Used in place of the ½" twist-bit chamfer. **Bought and used.** |
| E6000 adhesive | 1 tube | Beds each rung in its groove: metal to plywood, under the screws. Let it cure (24–72 h) before the board takes load. **Bought and used.** |

## G Build order

**01** **Cut the strips**

Cut the crossbar off the old comb. Trim the three strips to 22½". Chamfer the top edges at both ends.

**02** **Groove the undersides**

Lay the strips upside down side by side, back ends even. Mark two 3-1/16"-wide bands across all three: 3"–6" and 16½"–19½" from their back ends (6"–9" and 19½"–22½" from the hatch). Router or table saw each band 3/16" deep. Check a rung sits flush.

**03** **Cut and drill the rungs**

Cut both bars to 46". Lay the strips at their spacing (Fig 1: 0–2½", 21⅞–24⅜", 44½–46" from the driver line) and the rungs in their grooves. Mark four screw holes per crossing on the driver and middle strips, in a square: ⅝" in from each edge of the strip, and ¾" and 2¼" across the rung. Mark two on the passenger strip, centered on it, at ¾" and 2¼" across the rung. That's 10 per rung. Drill 3/16" through the steel (step bit), countersink each hole with the ⅝" countersink bit at low speed until the head sits flush, and deburr the back of each hole. Number each rung and its crossings so they go back where they were marked.

04

### Rust-proof the rungs, then screw them on

The bars are hot-rolled steel. Their gray mill scale is no rust protection, and the drilling and chamfering leave bare steel. Under the board they sit on the seat's fabric, which holds damp, and rust there stains the seat. So finish them before they go on.

Wire-brush or scuff each rung all over (a flap disc or 80-grit), paying most attention to the cut ends and the holes. Wipe it with mineral spirits or denatured alcohol and let it dry. Spray one coat of rusty-metal primer, then one or two light coats of enamel. Do both faces and the edges, and don't load paint into the chamfers: the screw heads have to seat flush. Let it dry to the touch, about an hour, before handling, and give it a day before the screws go in so the paint doesn't tear around the heads. If the riser angles are still bare steel, do them in the same session.

Then set each rung back in its grooves, drill 3/32" pilots about 7/16" deep into the ply (tape flag; there's only 9/16" of wood above the groove), and screw up into the strips with #8 × ⅝" flat heads. **As built:** each rung was bedded in its groove with E6000 (metal to plywood) before the screws went in, so the rungs are now glued as well as screwed. Let it cure before the board takes load. Touch up any paint the screws chip.

05

### T-nuts for the risers

Set the tray assembly on the board, risers on their strips, back ends ½" behind the strips' back ends. Mark through the 8 flange holes, then drill, counterbore and seat the T-nuts (Figs 4–5).

06

### Eye-bolt holes in the risers

Drill 5/16" (step bit) through each riser's flange ½" from its front end, centered across the flange's flat. It lands just ahead of the strip's end, so the eye bolt hangs free. Eye bolt up from below, washer and nut on top, eye turned toward its hoop.

07

### Strap anchors and neoprene

The passenger strip's footman loop lengthwise at 9½"–12½" from the hatch. On the middle strip, a short ¼-20 eye bolt in the kitchen riser's second T-nut, 11" from the hatch, in place of that bolt's button head: the strip's ½" outer band is too narrow for a loop. Cut the neoprene to each strip, ⅛" inside the edges, with gaps under the rungs.

08

IN THE VAN

### Install

Neoprene, then the board, back ends 3" from the hatch. Wrap a chain around H1's bar, close it with a quick link through the end link and a link of the standing chain, and lay it forward before the tray goes in. Bolt the tray assembly down. Wrap the second chain around H2 the same way. Quick-link each chain to its eye bolt on the link that's just taut, with the fridge loaded on its tray, turn the nut down until the chain is tight and the board won't move forward, then jam a second nut on top. Re-check after the first drive. Kitchen unit in; front strap in its front notches to H2 and H4, rear strap in its rear notches to the riser's eye bolt and the passenger footman loop.

---

Hoops, trim and seat surface from the owner's survey, 18 Oct 2026: hoops 28" from the hatch, 3" wide, bars ⅝" thick and 1⅞" below the seat surface loaded (2⅜" unloaded), at 1¼", 21¼", 27¾" and 38¼" (driver-side edges) from the driver-side trim; trim 42" apart and the walls 50" at that line, so each trim edge is 2" inside Panel C's 46" lines. Riser positions from the Fridge Slides sheet; kitchen unit 20⅛ × 25⅞", 1½" in from the hatch and from the passenger line, notches 1½–3⅛" and 22⅝–24⅛" from its hatch end.

# Kitchen Drawer

A shallow ½" plywood drawer on 24" soft-close slides, hung from Panel C's deck on two ½" cheeks over the kitchen unit. It pulls out the tailgate end like everything else in Panel C.

| ASSEMBLY | BOX, OUTER | CHEEKS | SLIDES |
|---|---|---|---|
| **Comp. 7**<br>kitchen drawer under Panel C | **18 × 26 × 4½"**<br>W × L × H · ~17 × 25 × 3½" clear inside | **2 × 26 × 5-7/16"**<br>½" ply · 19" apart inside | **Liberty 942405**<br>24" soft-close · ½" per side |
| **HEIGHTS** | **POSITION** | **DRAWN FROM** | **STATUS** |
| **12-5/16 → 16-13/16"**<br>box bottom → top, from the floor | **2-3/16" in**<br>from the tailgate edge and the passenger edge | **Build plan**<br>params.scad kdrawer_* · Panel C as built | **In progress**<br>box glued 18 Oct |

## A Check before you start

2 checks · 2 methods

METHOD

### The front and back cover the sides' ends

The 18" front and back run the box's full width, and the sides fit between them. The front is then one clean face with no end grain showing, and it covers the ½" slide gaps when the drawer is shut.

**Size the sides from your ply's real thickness: 26" − 2 × thickness, so 25" for ½" ply and 25-1/16" for 15/32".**

CHECK

### The slides want exactly ½" per side

The Liberty slides need ½" between the box and each cheek: 18" box, 19" between the cheeks. Too tight and they bind; too loose and they fall out of their tracks.

**The safest way to hold 19" is to let the drawer set it: put the slides on the box and the cheeks, run the box into the cheeks, and hang all of it as one unit.**

METHOD

### The bottom goes under the walls

The 18 × 26" bottom is the box's full outside size, so it glues and screws on underneath the four walls; there's no rabbet. #8 × 1¼" screws up through it into the walls' edges, 3/32" pilots centered on the ½" edge.

METHOD

### Build the whole unit square, then hang it

Lining two loose cheeks up under the deck, 19" apart, parallel and square, is close to impossible from below. So the cheeks, slides and box are built into one rigid unit on the bench first: three ¾" spreaders tie the cheeks together across their tops, set flush with the cheek tops and 3/16" above the box (Fig 6). The box in its slides holds the 19"; the bench holds the tops flat; the diagonals prove it square.

**Then hang it with Panel C standing on its own feet: carry the unit in low from the open tailgate end, lift it to the deck, prop it, and push it against the rear and side cleats. Don't tip Panel C over to reach under it.**

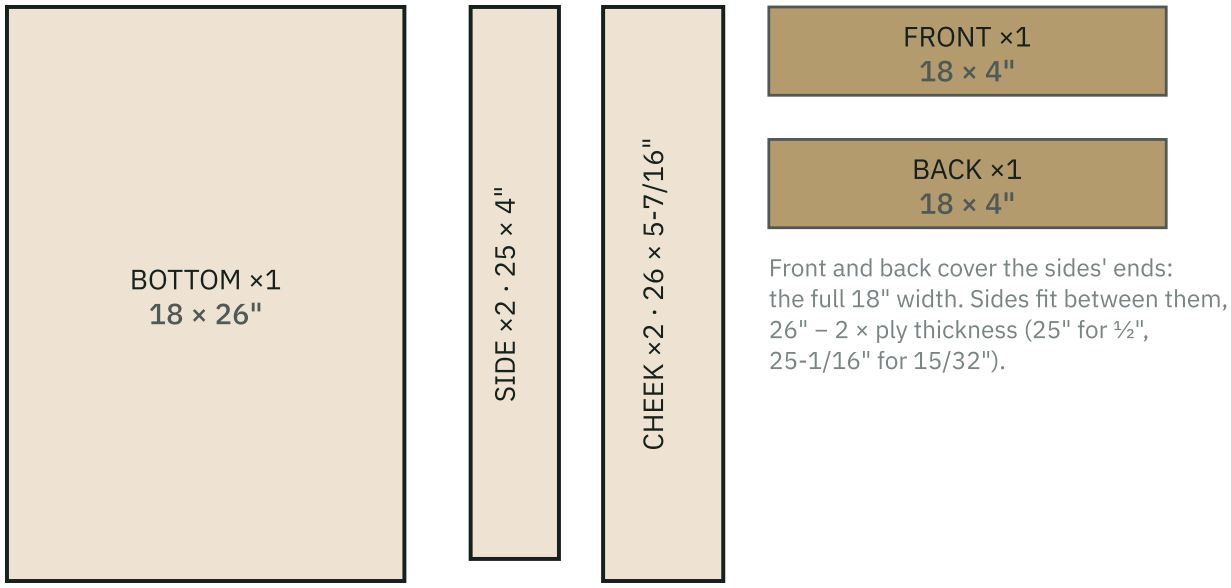



**FIG 0 — THE PARTS**

Every piece, to scale.

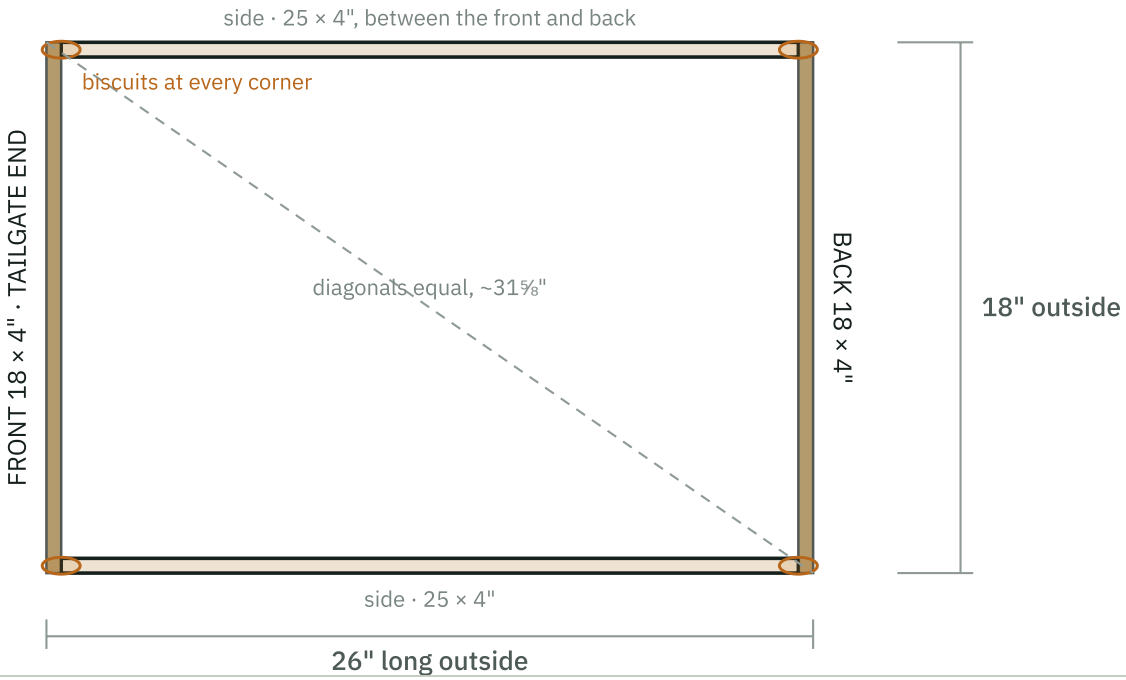



**FIG 1 — THE BOX**

The front and back run the full 18" width and cover the sides' ends. Pencil an X on every inside face and TOP on every top edge before slotting.

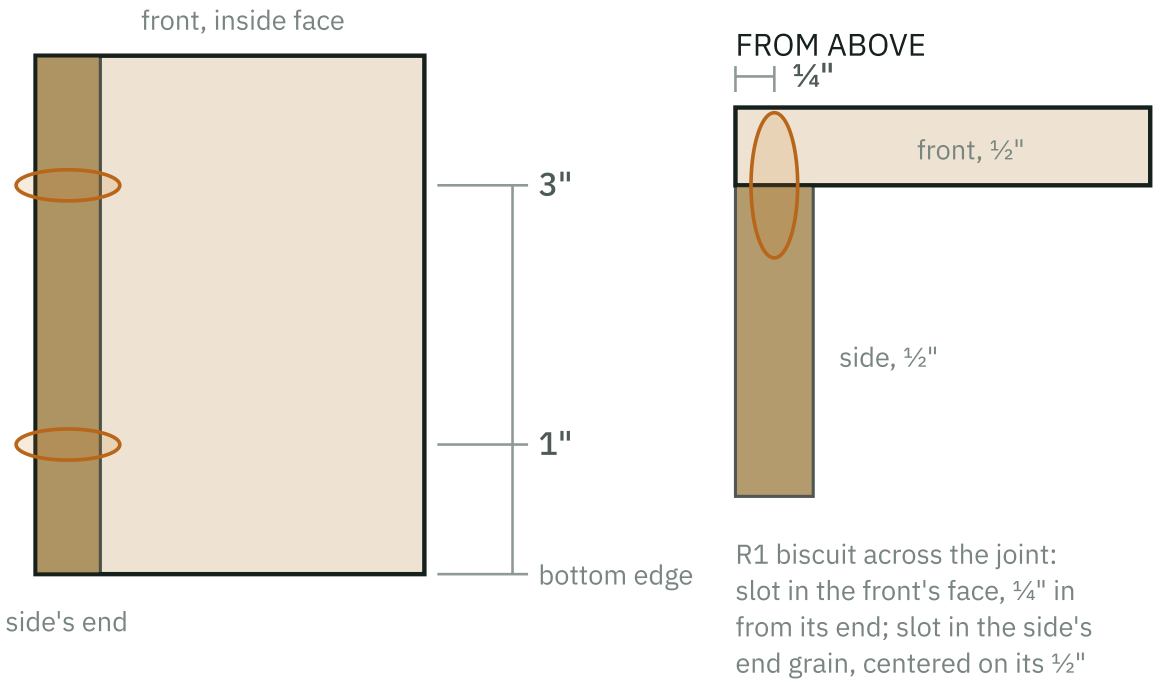



**FIG 2 — A CORNER**

Two R1 biscuits per corner at 1" and 3" up from the bottom edge. Test the joiner setting on scrap ½" ply first: the faces must come out flush.

## C Under the deck

to scale

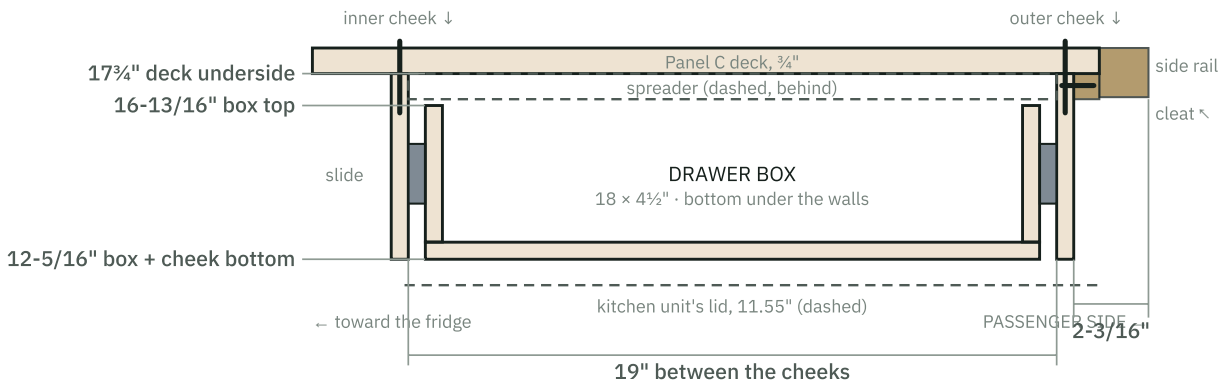



**FIG 3 — SECTION UNDER THE DECK**

Looking forward from the tailgate, about 1:5. The box bottom and the cheeks' bottoms line up at 12-5/16" off the floor, ¾" over the kitchen unit's lid. The box top passes under Panel C's rear rail (17-1/16") with about ¼" to spare.

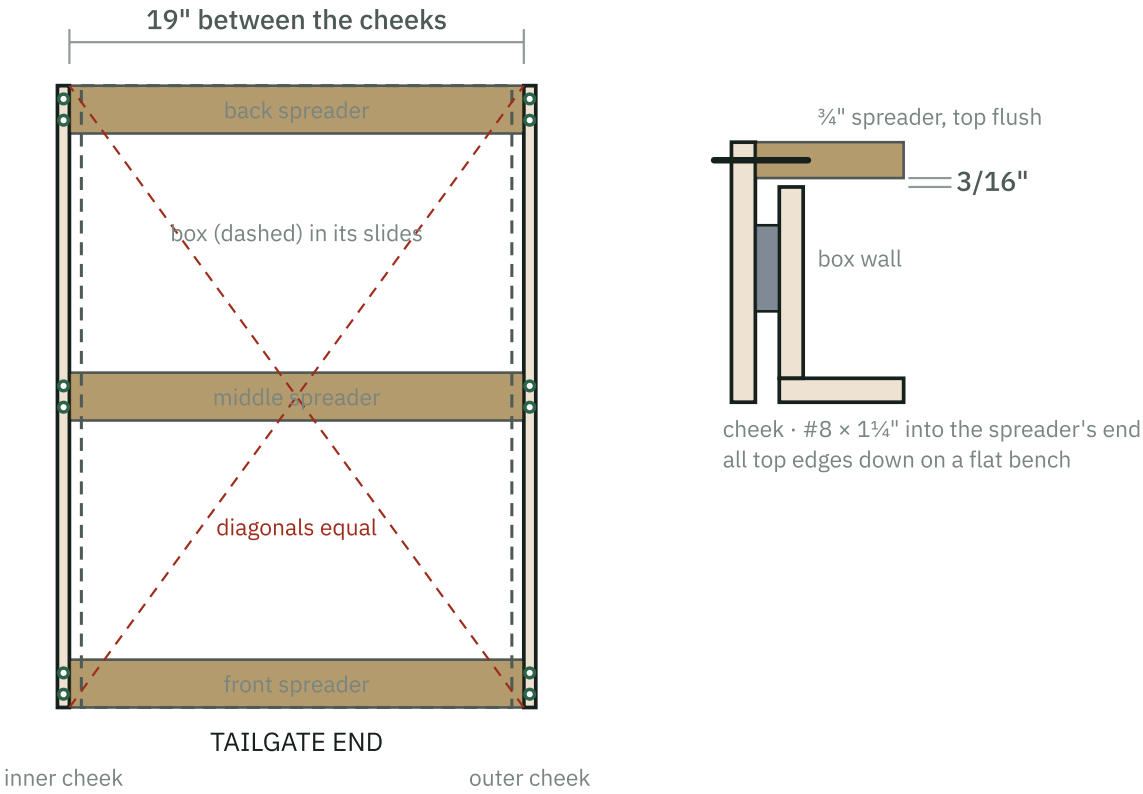



**FIG 6 — BUILD IT SQUARE, THEN HANG IT**

From above, about 1:9. Lay the cheeks top edges down on a flat bench with the box in its slides, glue and screw the three ¾" spreaders between them flush with the cheek tops, and check the diagonals before the glue sets. The unit then goes up as one piece. Inset: end section at a spreader, about 1:2.

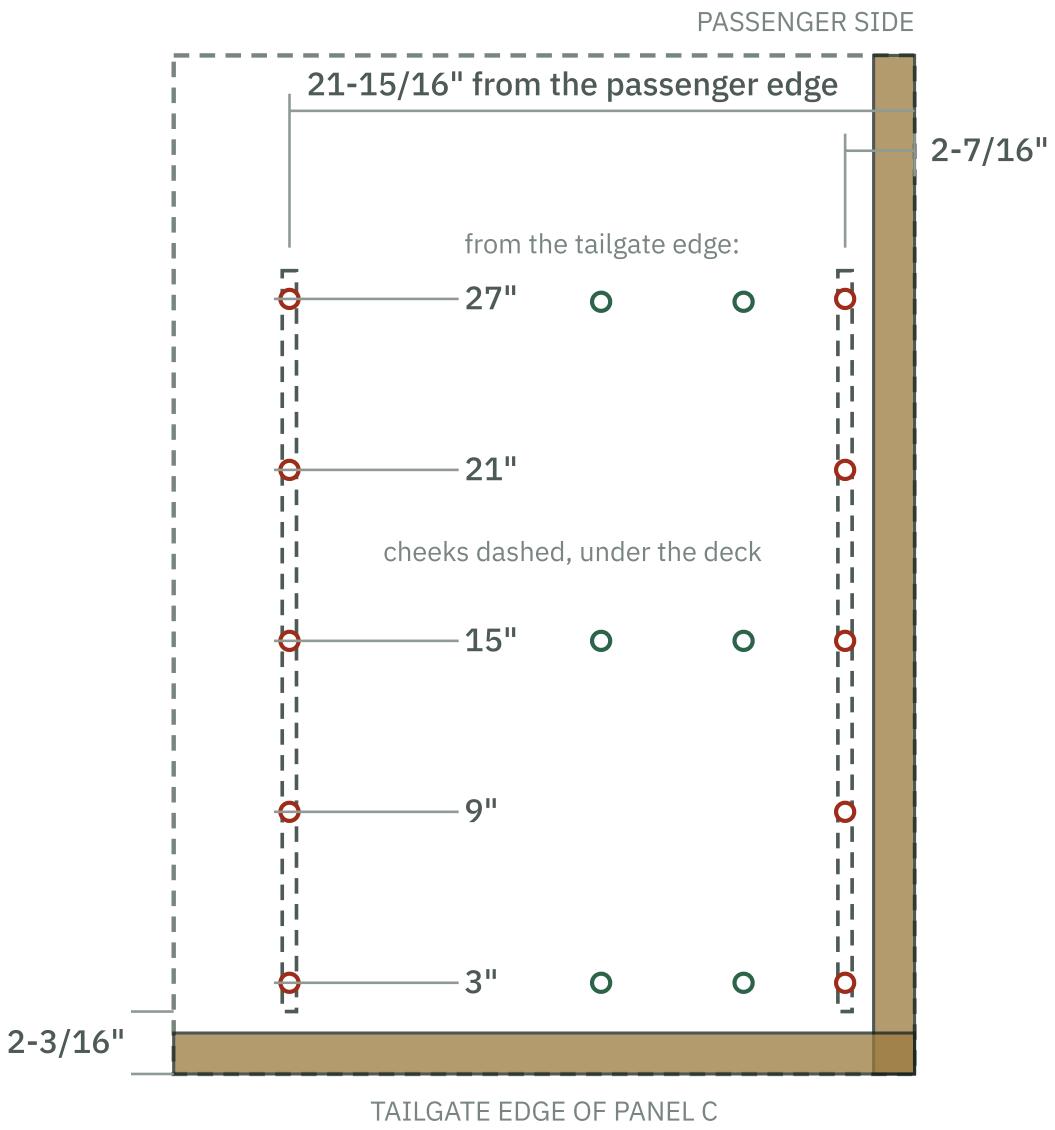



**FIG 4 — SCREWS DOWN THROUGH THE DECK**

Red: ten #8 × 2" flat heads, five per cheek on its centerline. Green: two more into each spreader, 6" and 11" from the passenger edge. All countersunk flush so the pantry drawers sit flat. Drill the ⅛" pilots down through the deck once the unit is propped in place.

## D The slides

to scale

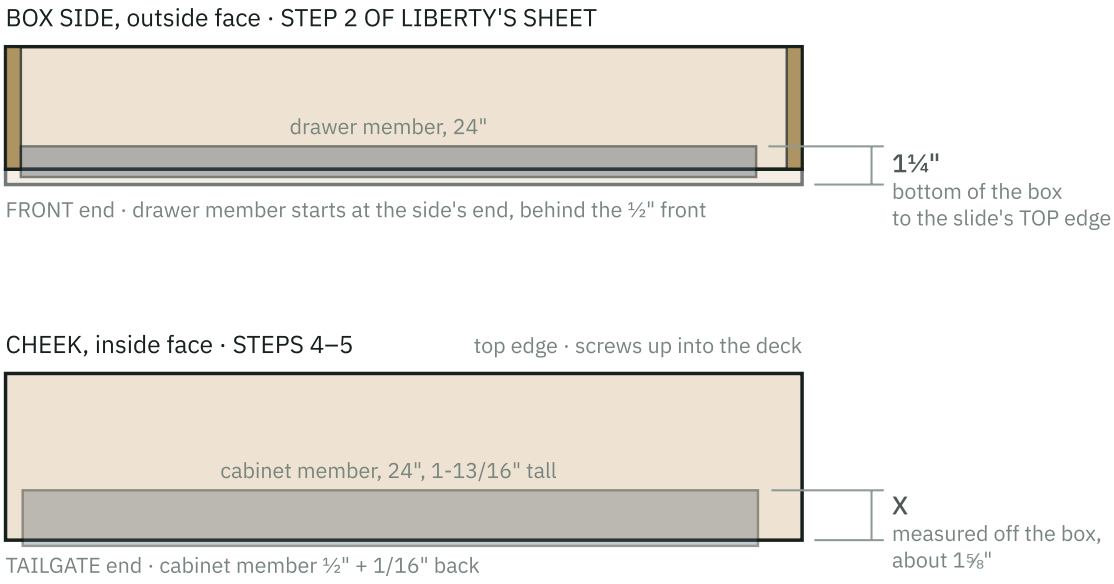



**FIG 5 — WHERE THE SLIDES GO**

Liberty's method. Drawer member: top edge 1¼" above the bottom of the box, starting at the side's front end. Cabinet member: clip it onto the mounted drawer member, measure X from the box's bottom to the cabinet member's top edge, and mark X up from the cheek's bottom, which lines up with the box's. Set it 1/16" further back than the drawer member so the front closes flush. Liberty's #6 × ½" pan heads suit the ½" cheeks.

## E Cut list

½" ply

| PIECE | QTY | SIZE | NOTES |
|---|---|---|---|
| Bottom | 1 | 18 × 26" | Under the walls. |
| Side | 2 | 25 × 4" | Between the front and back.<br>26" − 2 × ply thickness (25-1/16" for 15/32" ply). |
| Front, back | 2 | 18 × 4" | Cover the sides' ends.<br>Best face out on the front. |
| Cheek | 2 | 26 × 5-7/16" | Hang from the deck. |
| Spreader, ¾" ply | 3 | 19 × 2" | Between the cheeks, flush with their tops: front, middle, back.<br>From the ¾" offcut. Cut to the cheeks' real spacing with the box running in them. |

## F Hardware

| ITEM | QTY | WHERE |
|---|---|---|
| Liberty 942405 24" soft-close slides | 1 pair | Box sides and cheeks, with the 12 #6 × ½" pan heads in the box. Mount per Liberty's sheet. **On hand.** |
| R1 biscuits + wood glue | 8 | Box corners, 2 each. **On hand.** |
| #8 × 1¼" flat head | ~30 | Bottom up into the walls, about every 6" (~18); through the cheeks into the spreaders' ends, 2 per end (12). **On hand.** |
| #8 × 2" flat head | 16 + 3 | Down through the deck into the cheeks (10) and the spreaders (6); the outer cheek into the side cleat (3, use 1¼"). **On hand from the bed.** |
| Magnetic catch | 1 | Inside the back of the box, keeps it shut on the road. **On hand.** |

## G Build order

**01 — Mark and slot the corners**

X on every inside face, TOP on every top edge, TAILGATE on the front. Two R1 slots per corner at 1" and 3" up: in the sides' end grain, and in the front and back's inside faces ¼" in from their ends (Fig 2). Test on scrap first.

**02 — Dry-fit**

Biscuits in, no glue. Corners tight, tops flush, diagonals equal (~31⅝").

**03 — Glue up the walls**

Glue in the slots and on the joints, clamp, check the diagonals, wipe off squeeze-out. Let it set.

**04 — Bottom on**

Glue the walls' bottom edges, set the box on the bottom with the edges flush all round, then #8 × 1¼" screws up through the bottom into each wall about every 6", 1" from the corners. 3/32" pilots centered on the ½" edges.

05

### Slides onto the box and the cheeks

Follow Liberty's sheet (Fig 5). Press the release lever and pull each drawer member off. Mark 1¼" up from the bottom of the box on each side and screw the drawer member on with its top edge on the line and its front at the side's front end, behind the ½" front. Clip a cabinet member onto it, measure X from the box's bottom to the cabinet member's top edge, unclip, and mark X up from each cheek's bottom edge. Screw the cabinet members on with their top edges on that line, set ½" + 1/16" back from the cheeks' tailgate ends. #6 × ½" pan heads, slotted holes first.

06

### Assemble the unit

Run the box into the cheeks. It should glide and soft-close, the box front flush with the cheeks' ends and its bottom level with theirs. Tune the slides in their slotted holes now, then add the round-hole screws.

07

### Square it with the spreaders

Turn the unit upside down: cheeks' top edges flat on the bench, box in its slides between them. Measure between the cheeks at each end and cut the three ¾" spreaders to that length (19"). Glue each spreader's ends and set it between the cheeks, flush with their top edges: one flush with the tailgate ends, one in the middle, one flush with the back ends. Check the two diagonals across the cheeks are equal, then drive 2 #8 × 1¼" through each cheek into each spreader end (3/32" pilots). Let it set, turn it over and run the drawer again (Fig 6).

08

### Mark the deck

Mark sixteen screw spots on the deck top (Fig 4): 2-7/16" and 21-15/16" from the passenger edge at 3", 9", 15", 21" and 27" from the tailgate edge (the cheeks), plus 6" and 11" from the passenger edge at 3", 15" and 27" (the spreaders). Measure from the rear cleat's inner face first: if it isn't 2-3/16" from the tailgate edge, shift every mark by the difference.

09

### Hang it

Panel C standing on its feet, kitchen unit out. Measure from the floor to the deck's underside and subtract 5-7/16": cut two props to that height (about 12-5/16"), from 2×2 offcut or stacked scrap. Carry the unit in low from the open tailgate end, lift it against the deck and stand the props under the box. Slide it toward the tailgate until the cheeks' ends touch the rear cleat, outer cheek tight against the side cleat. Drill the ⅛" pilots down through the deck at the marks and countersink, drive the 16 #8 × 2" screws, then 3 #8 × 1¼" through the outer cheek into the cleat. Pull the props.

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|---|---|
| 10 | **Catch and test**<br>Slide the kitchen unit back in. Magnetic catch at the back. Pull out to full extension and back: clears the kitchen unit's lid, passes under the rear cleat and the front spreader, closes soft. |

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Geometry from `params.scad` (`kdrawer_*`, `kitchen_box_height`) and Panel C as built: deck top 18½", deck underside 17¾", side rail 1-7/16", cleats ¾ × ¾" with their tops at the deck's underside, rear rail underside 17-1/16". The cheeks and box are 26" long; the box's front and back cover its sides' ends. Spreaders ¾" between the cheek tops leave 3/16" over the box's 16-13/16" top.